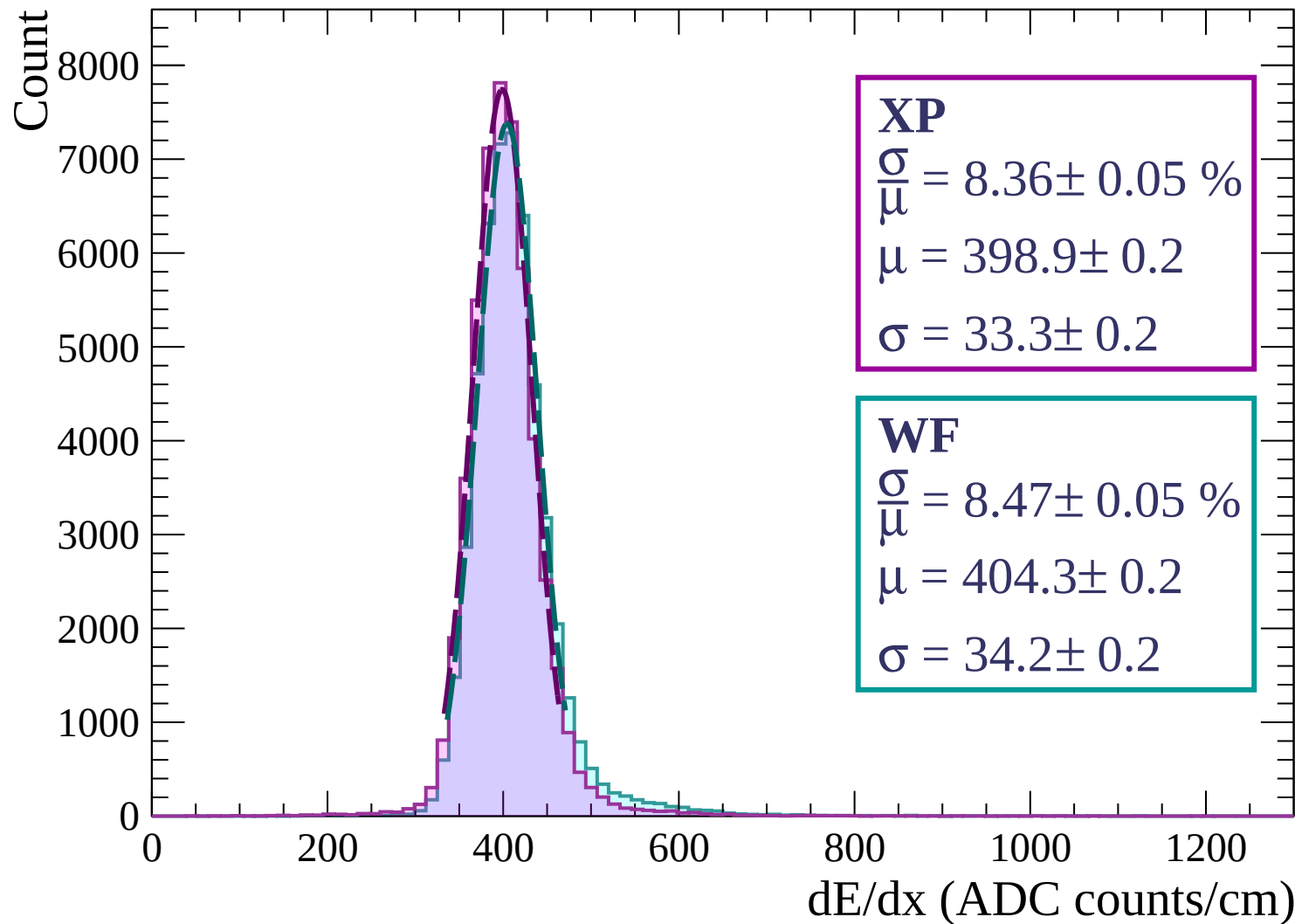
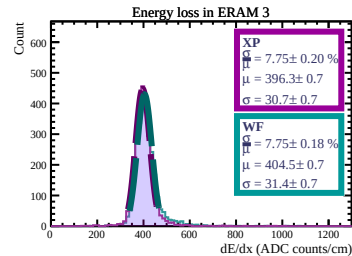
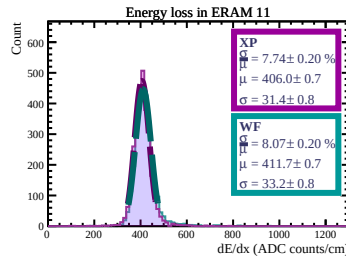
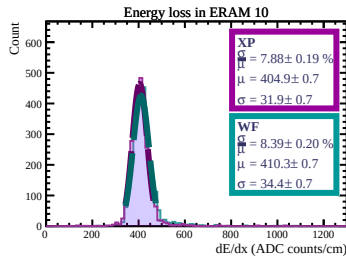
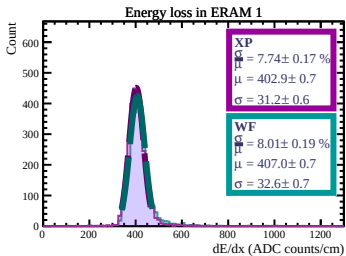
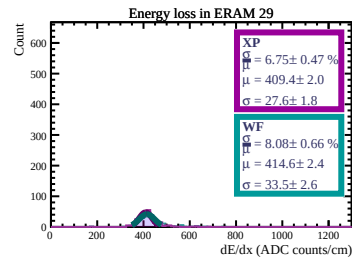
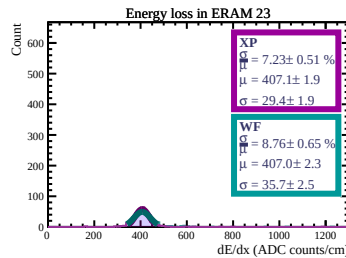
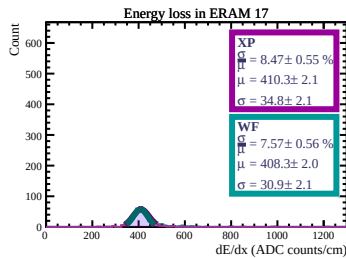
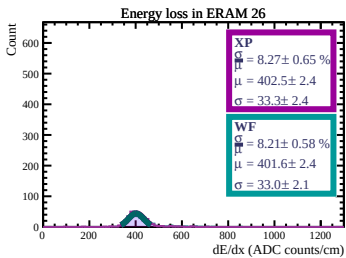
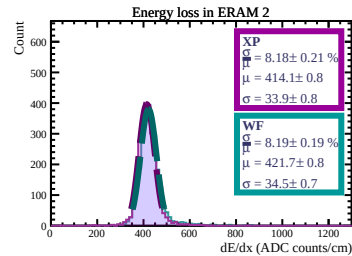
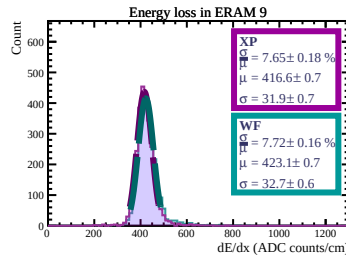
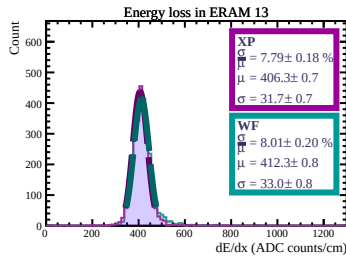
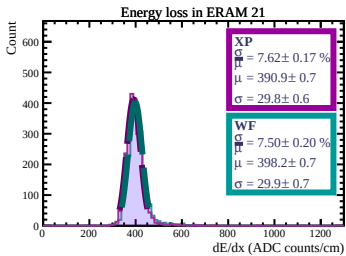
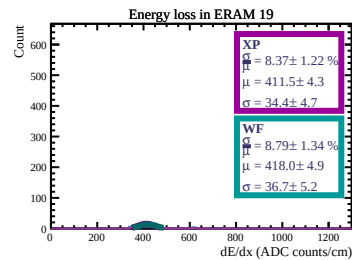
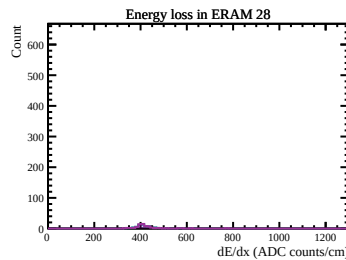
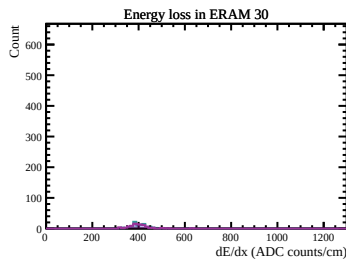
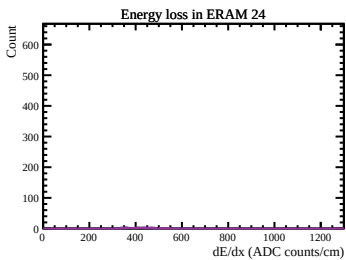
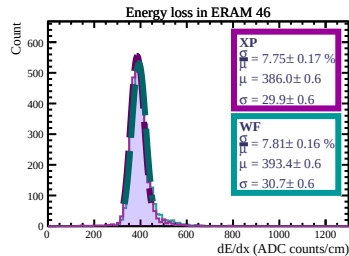
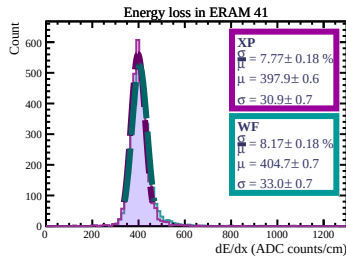
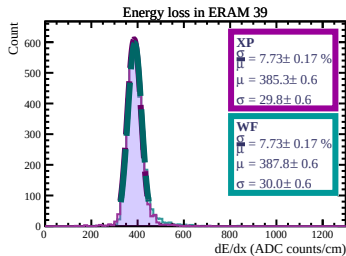
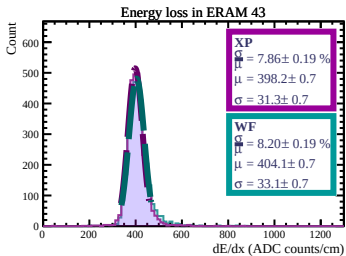
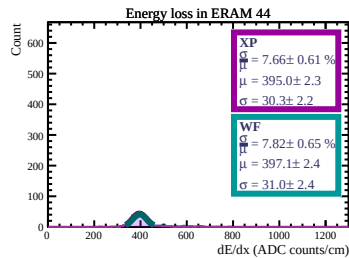
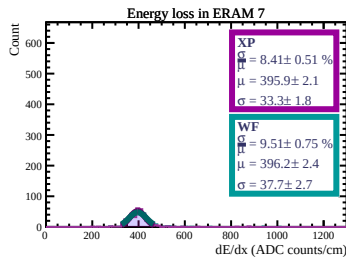
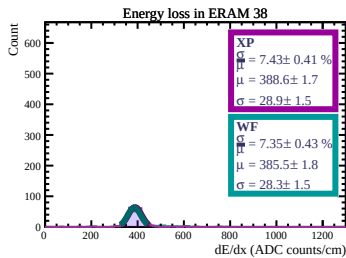
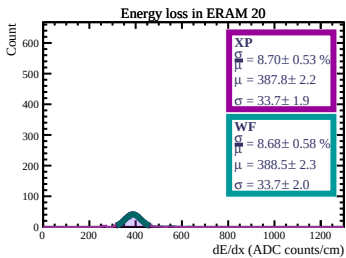
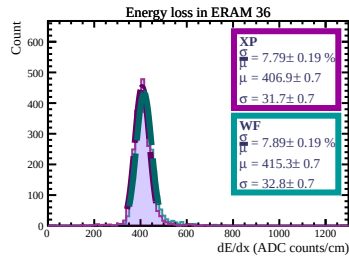
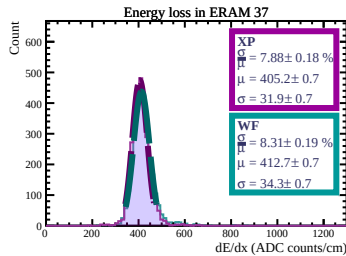
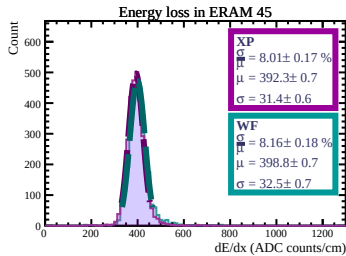
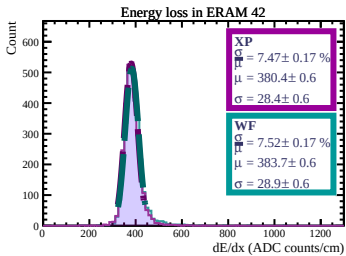
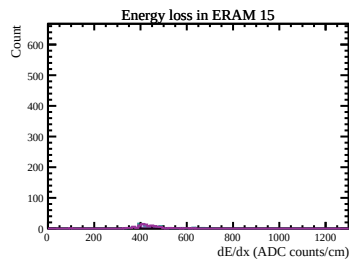
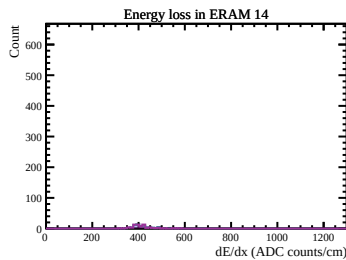
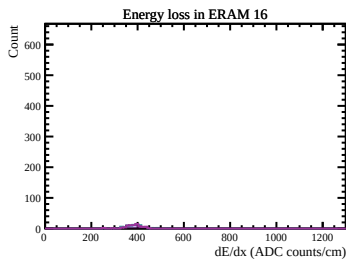
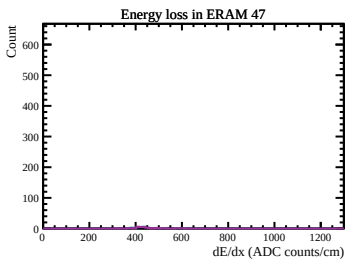
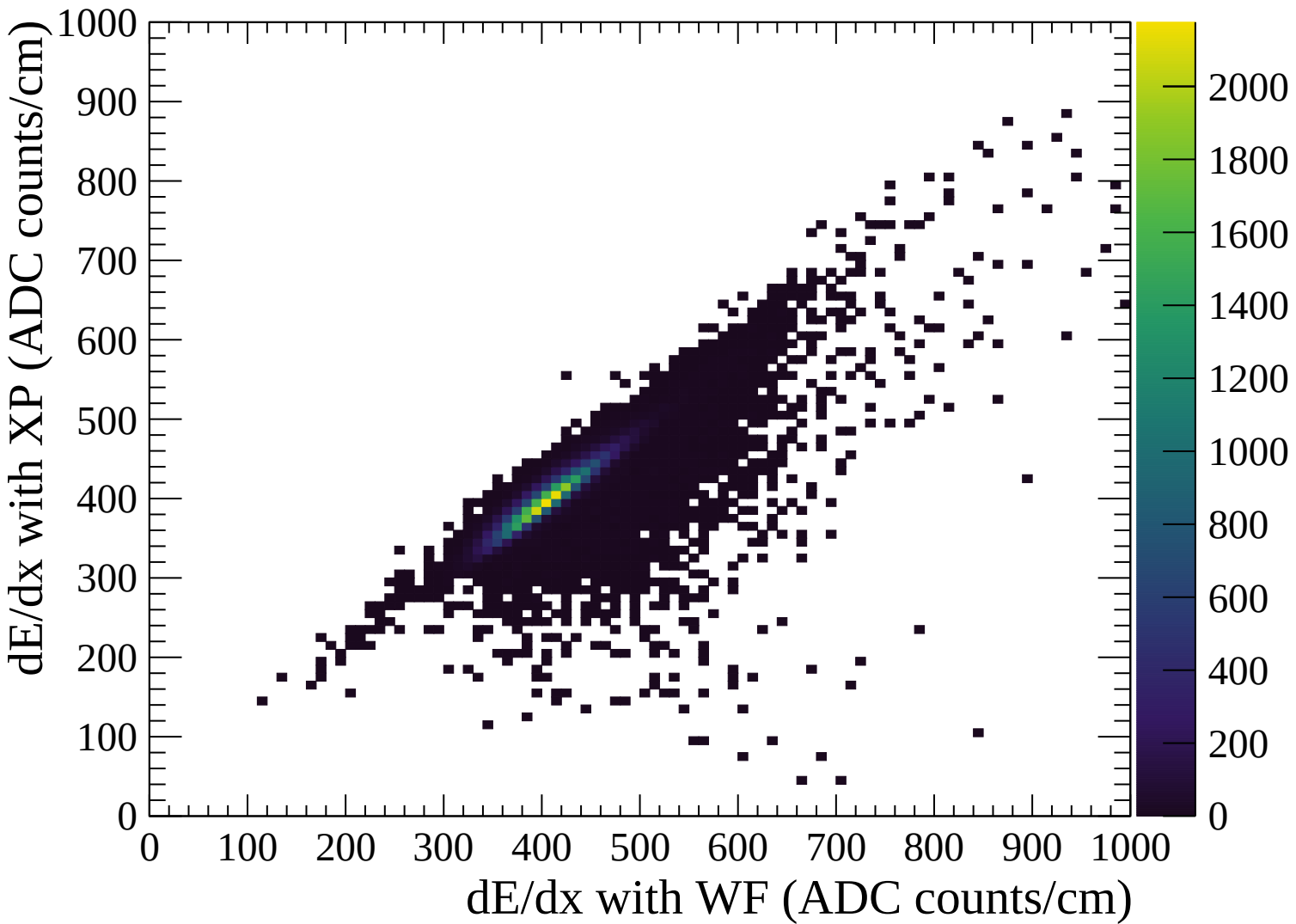
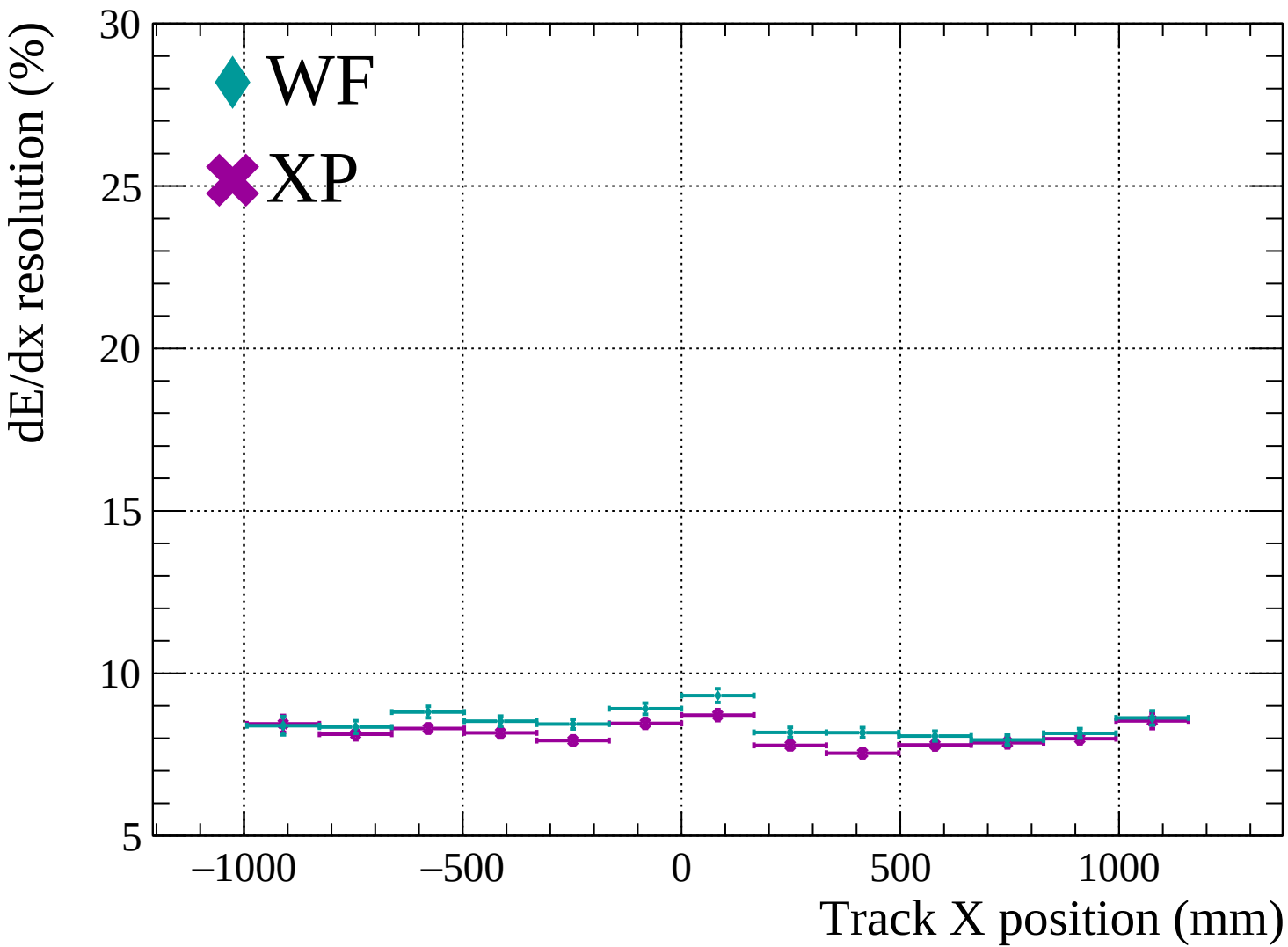


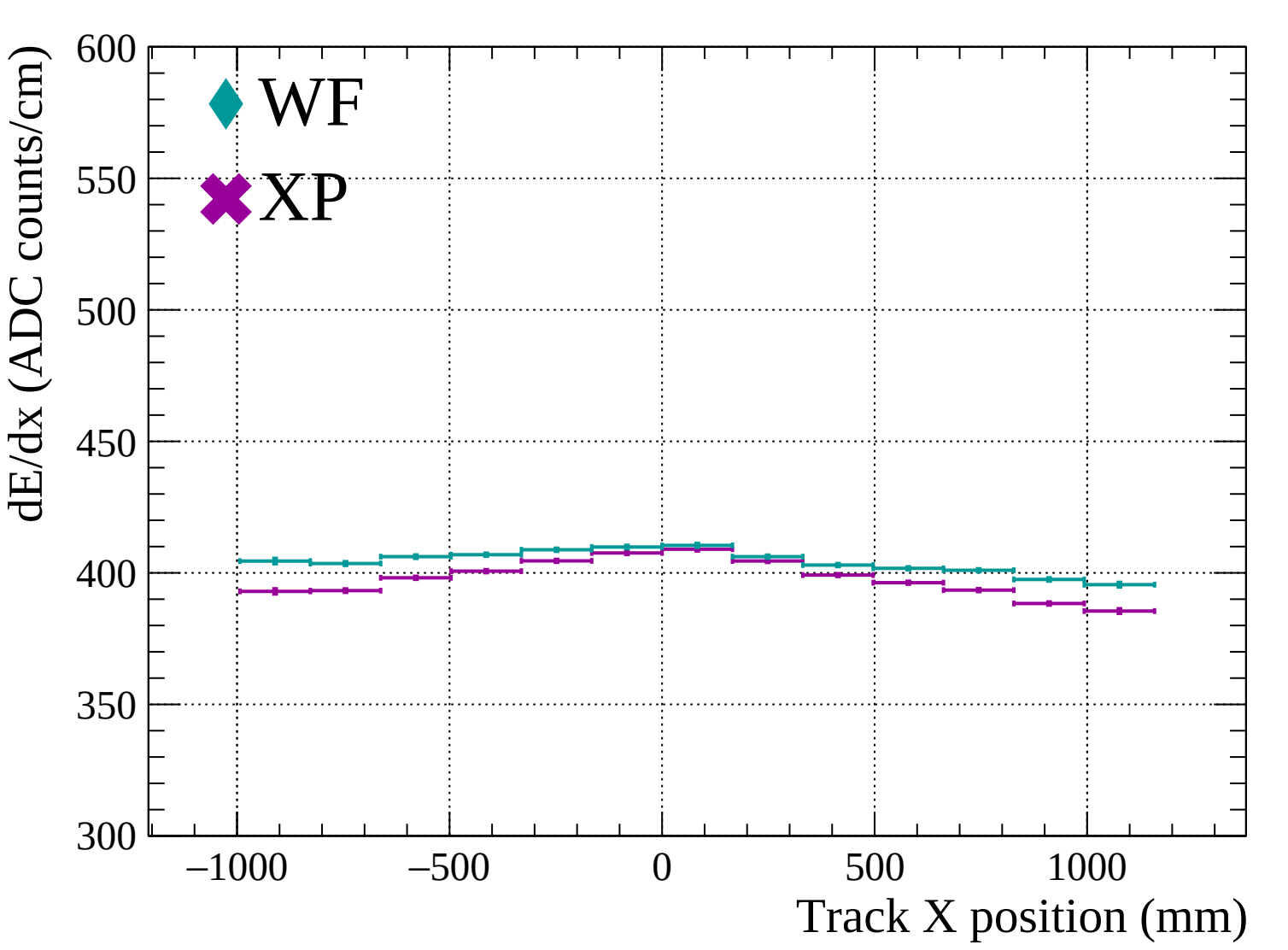


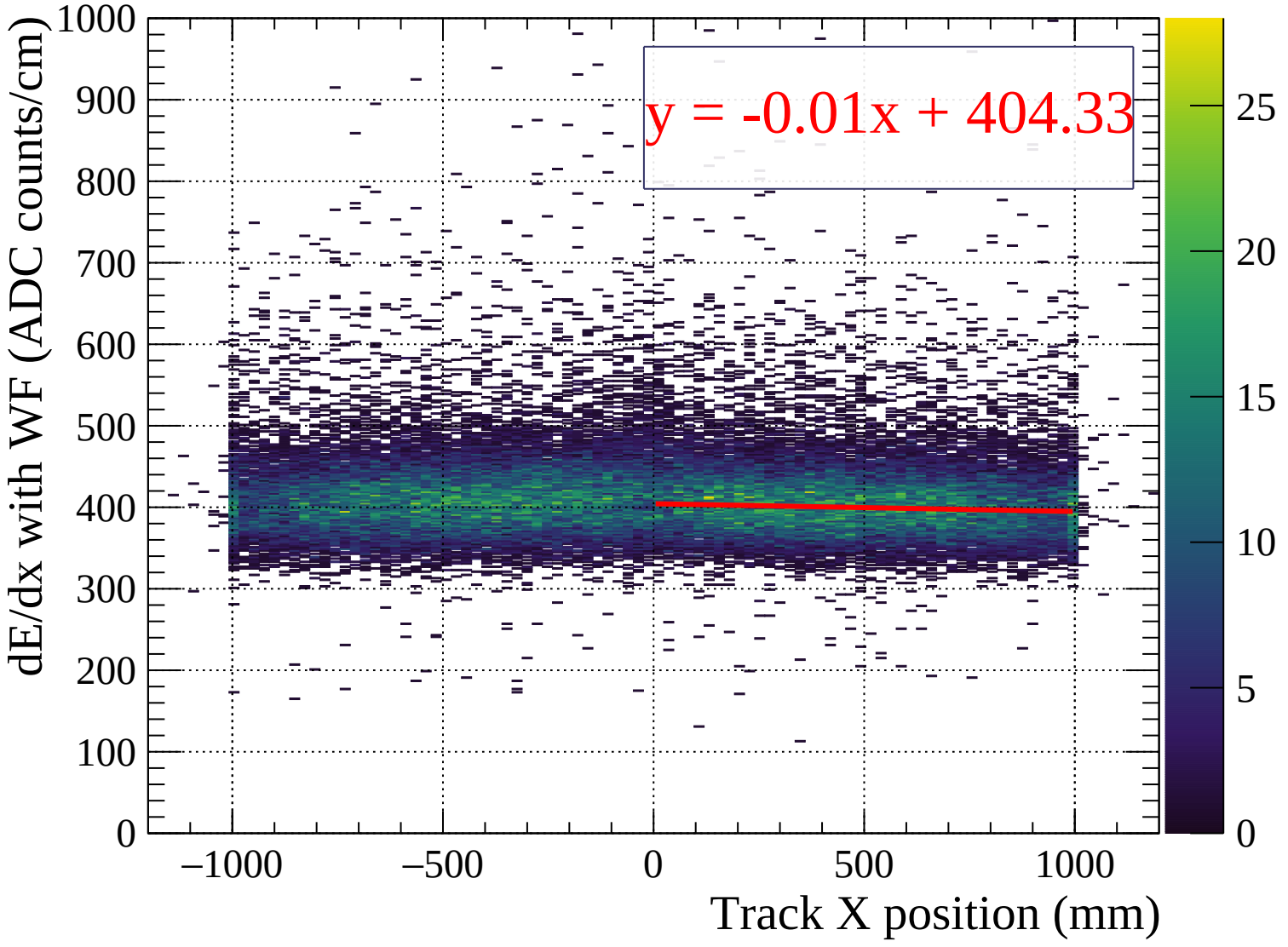


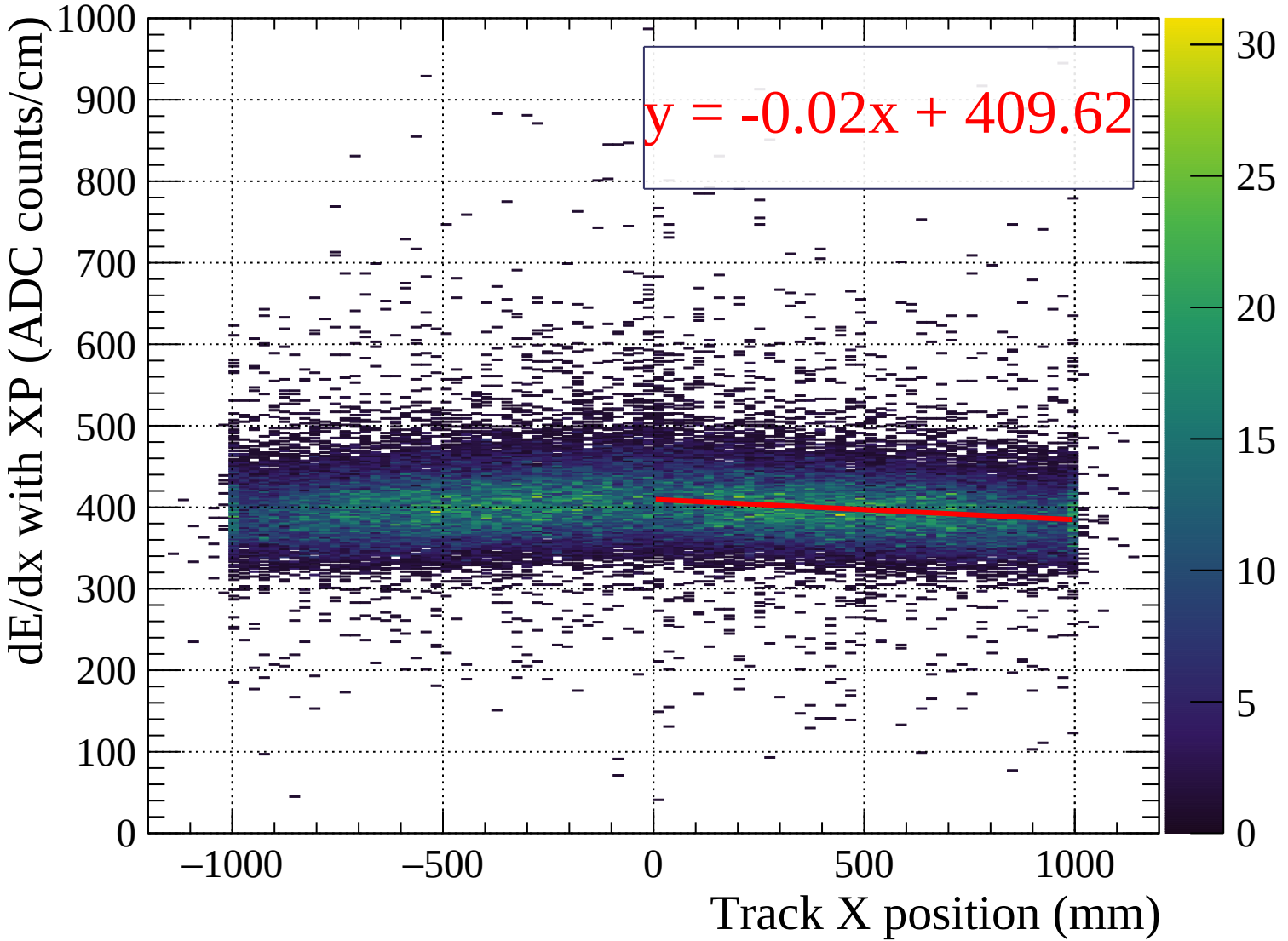


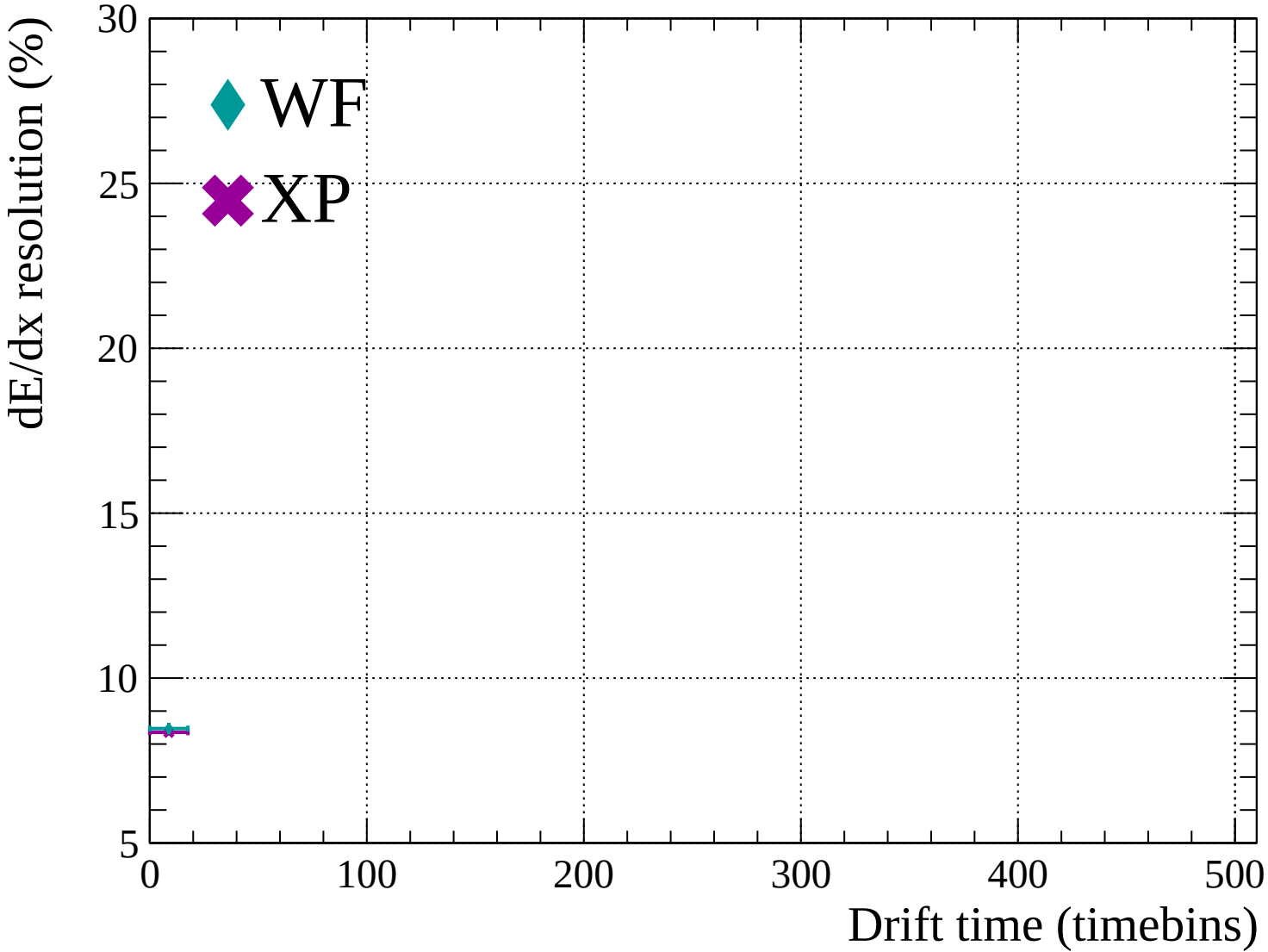


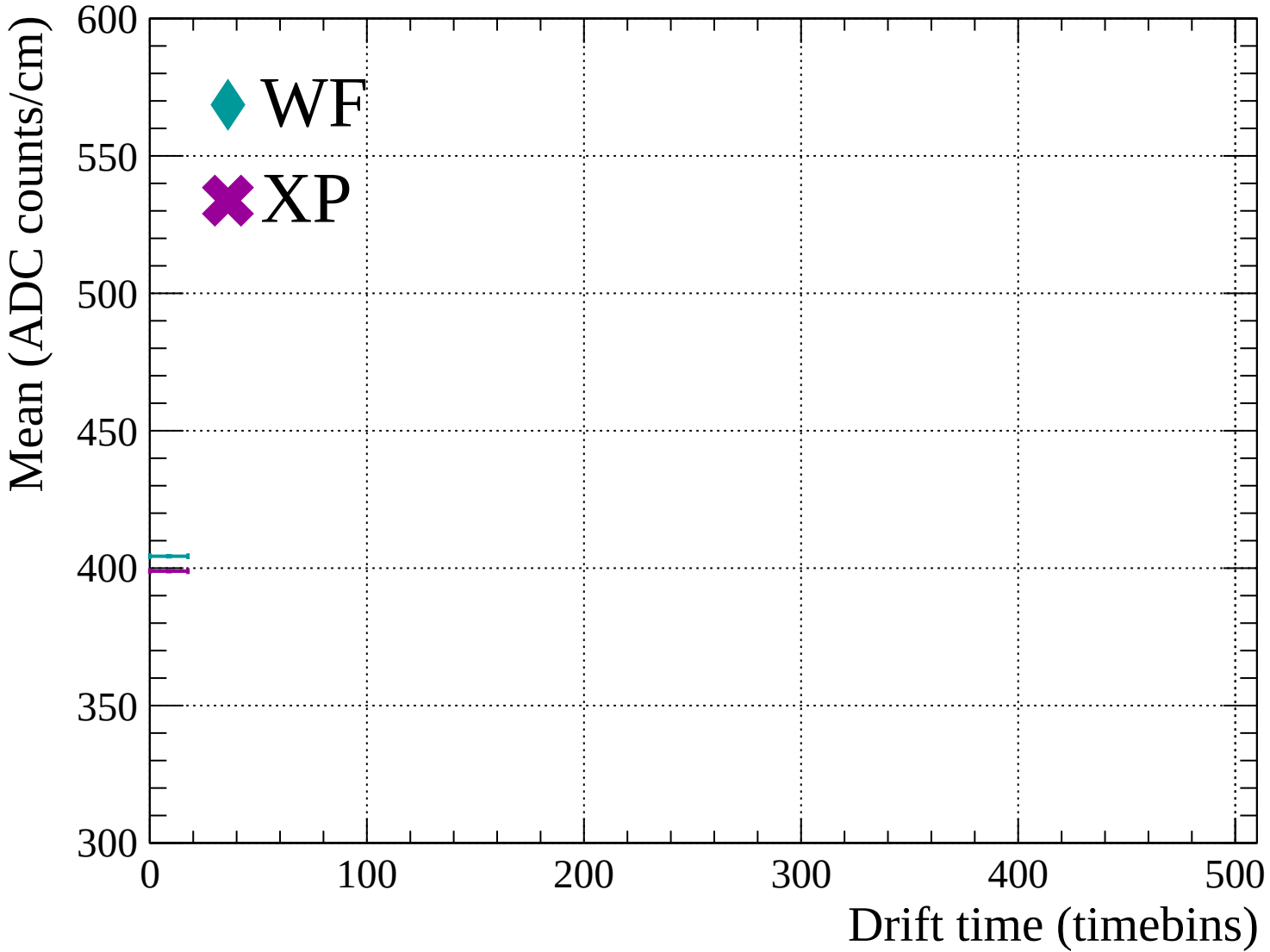


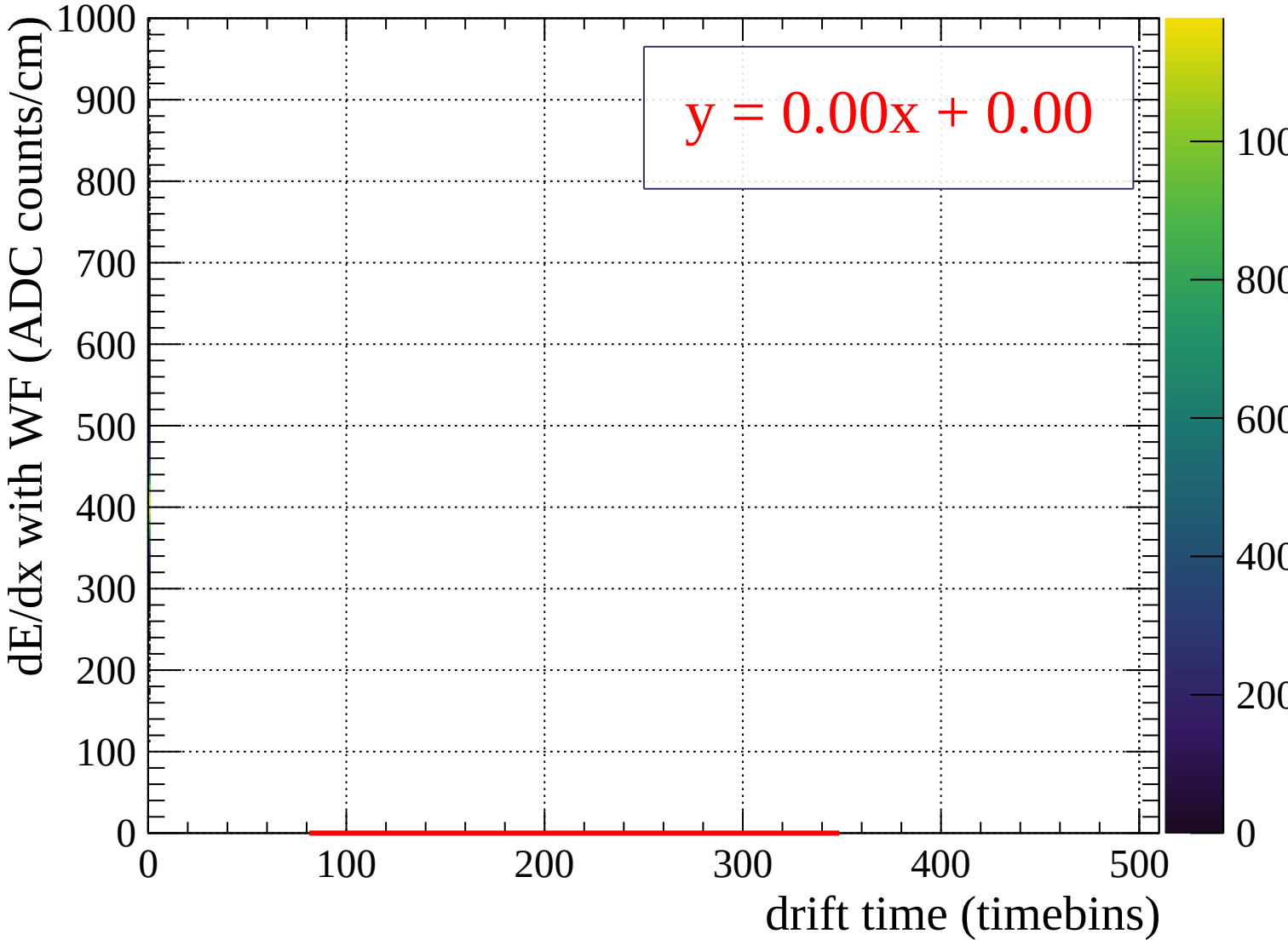


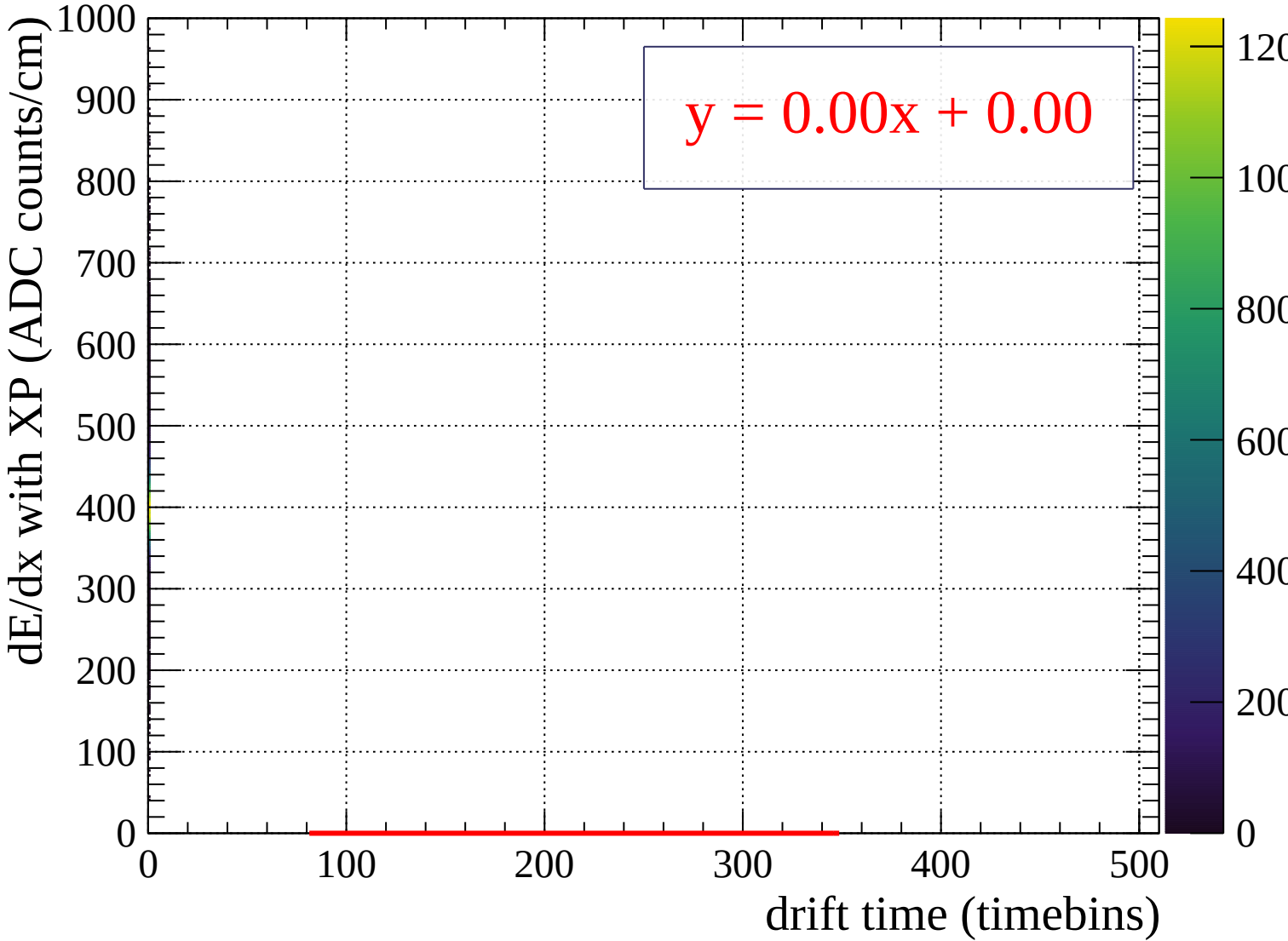


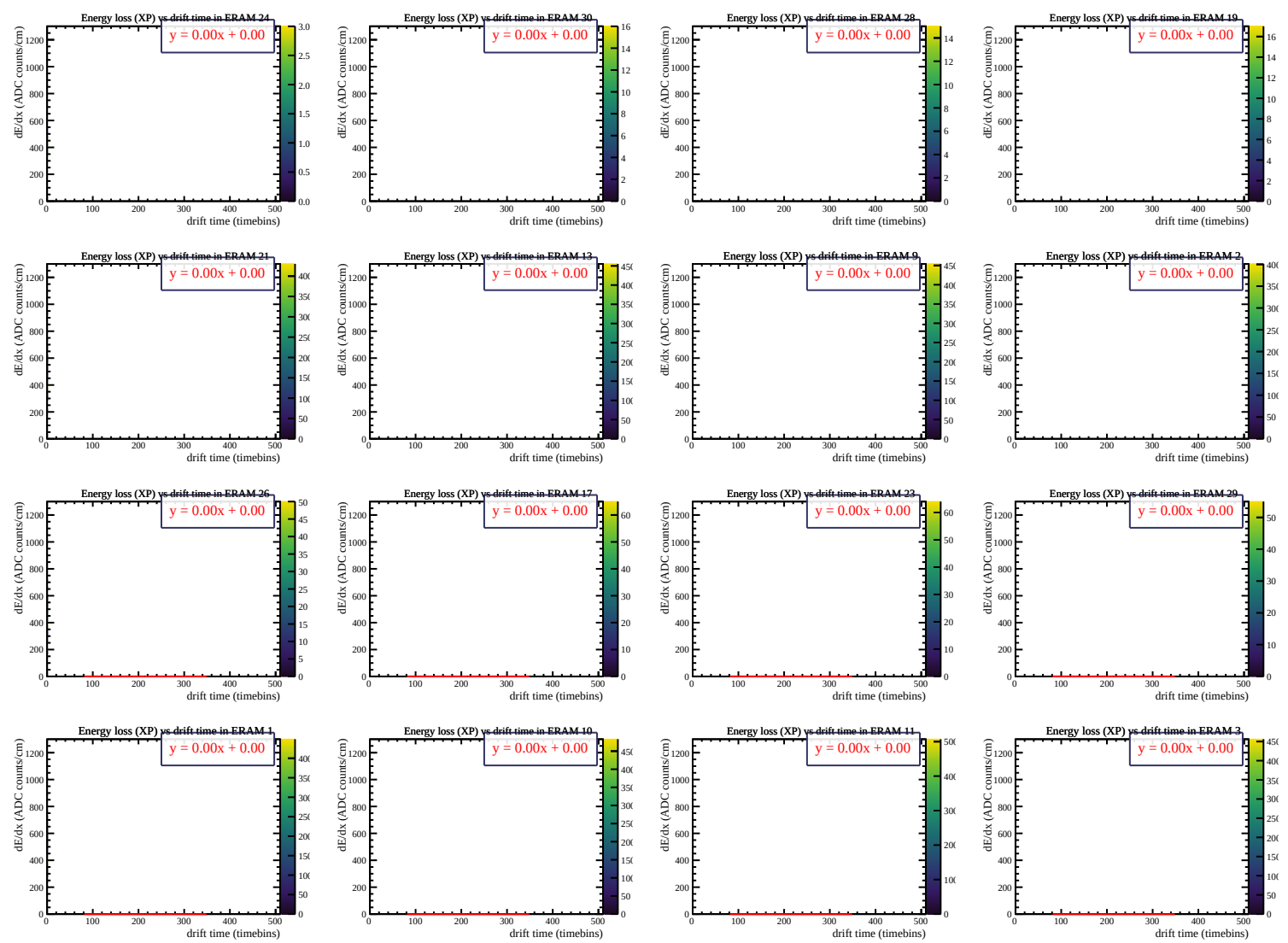


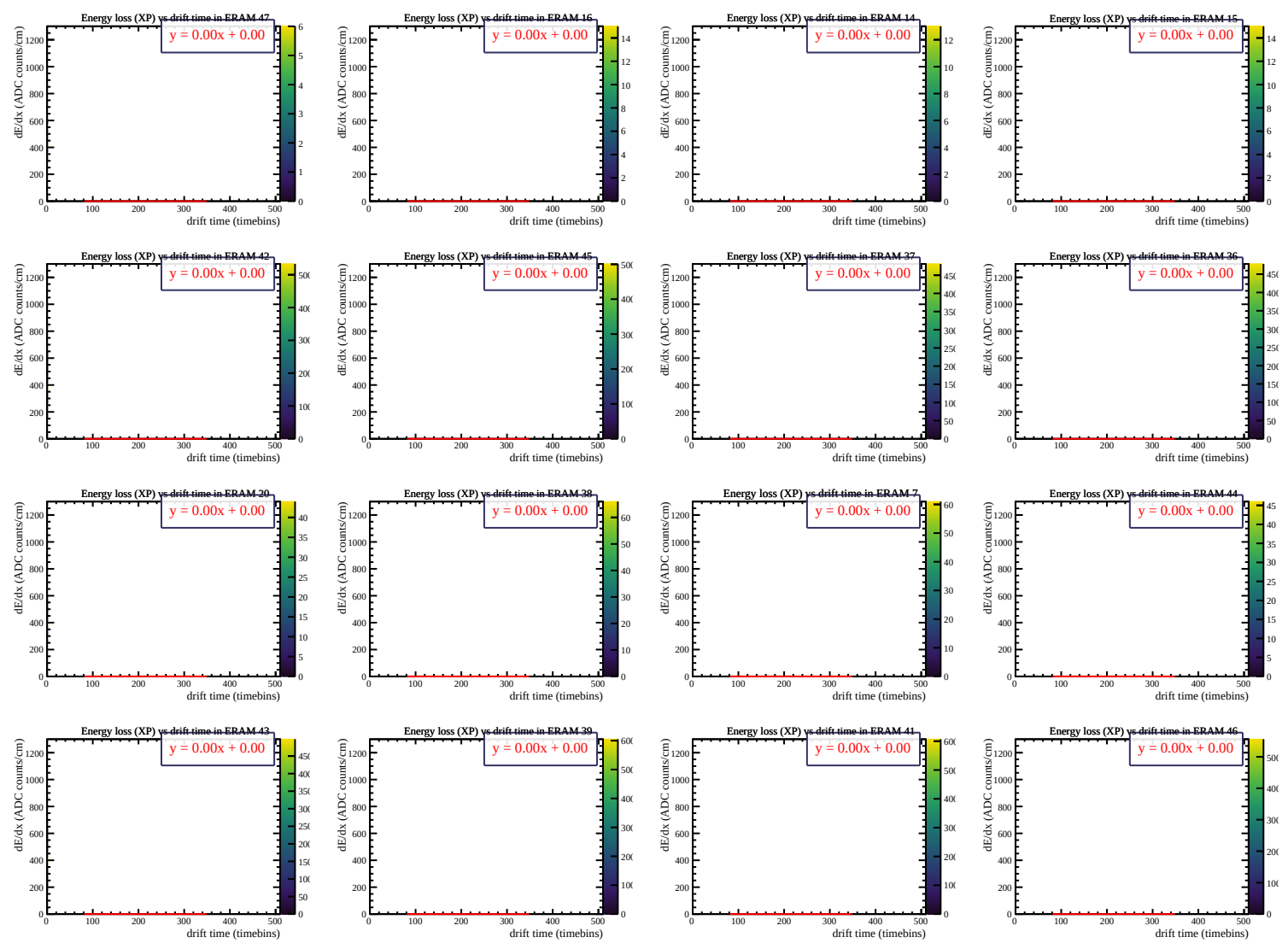


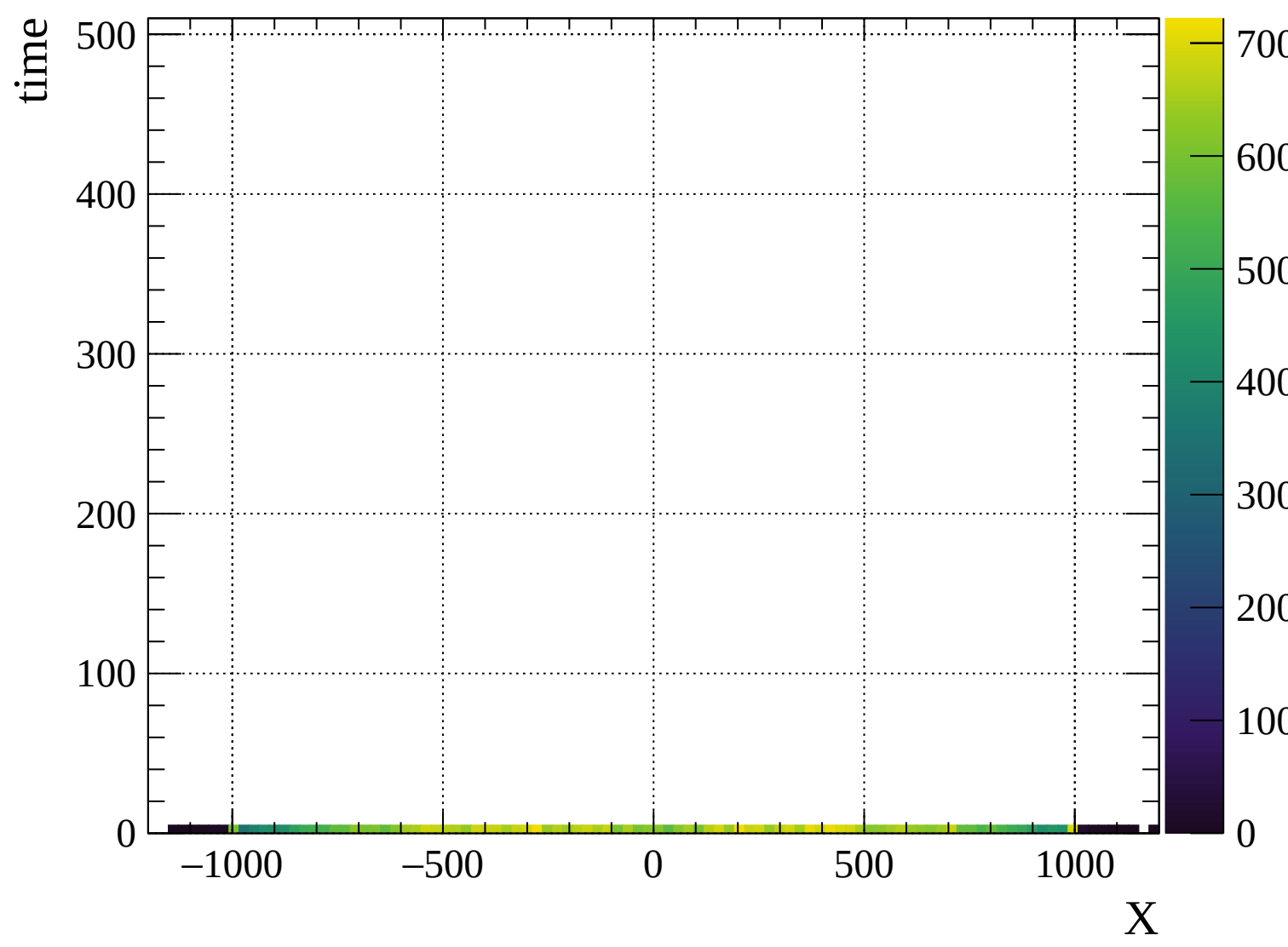


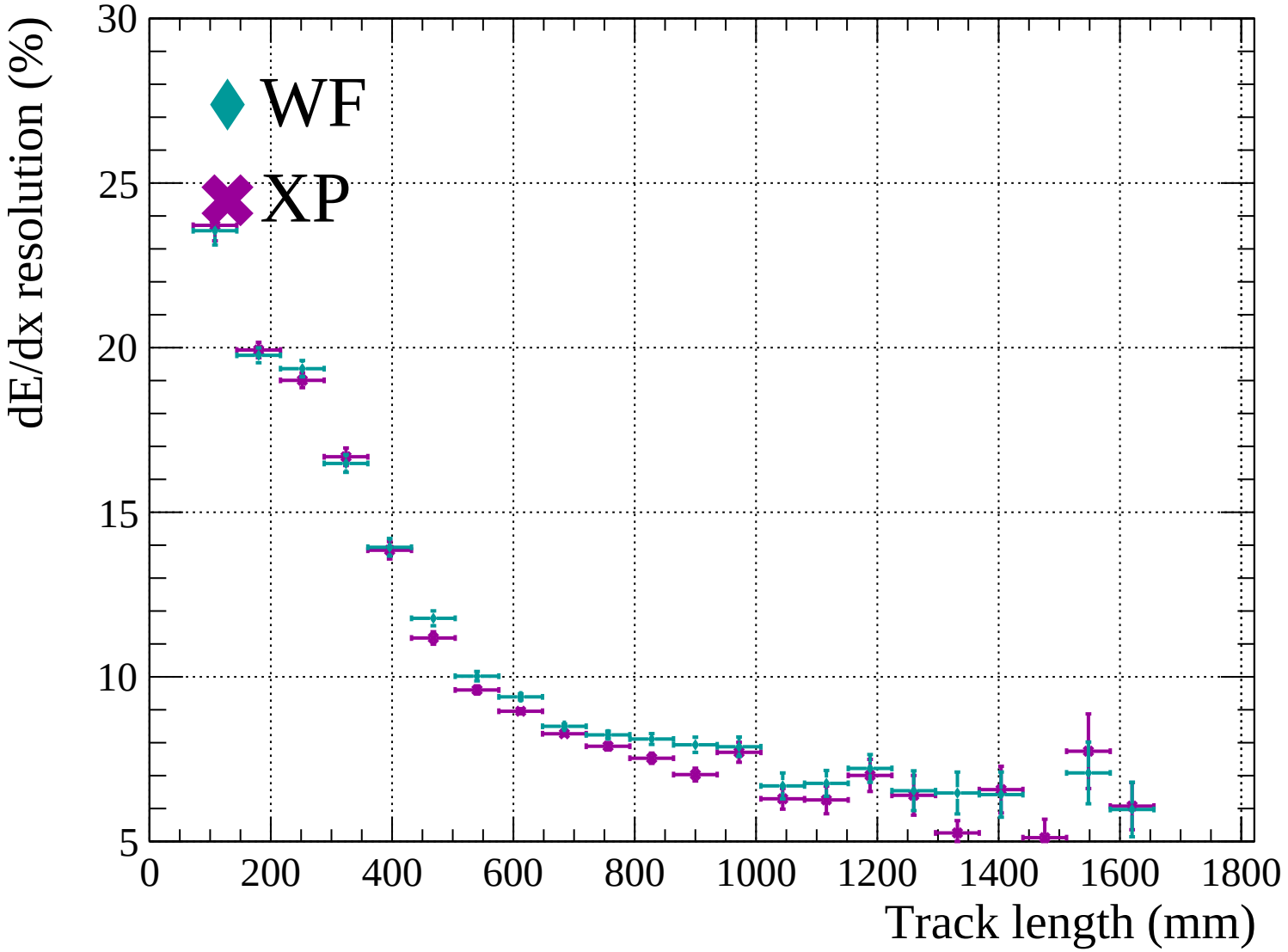


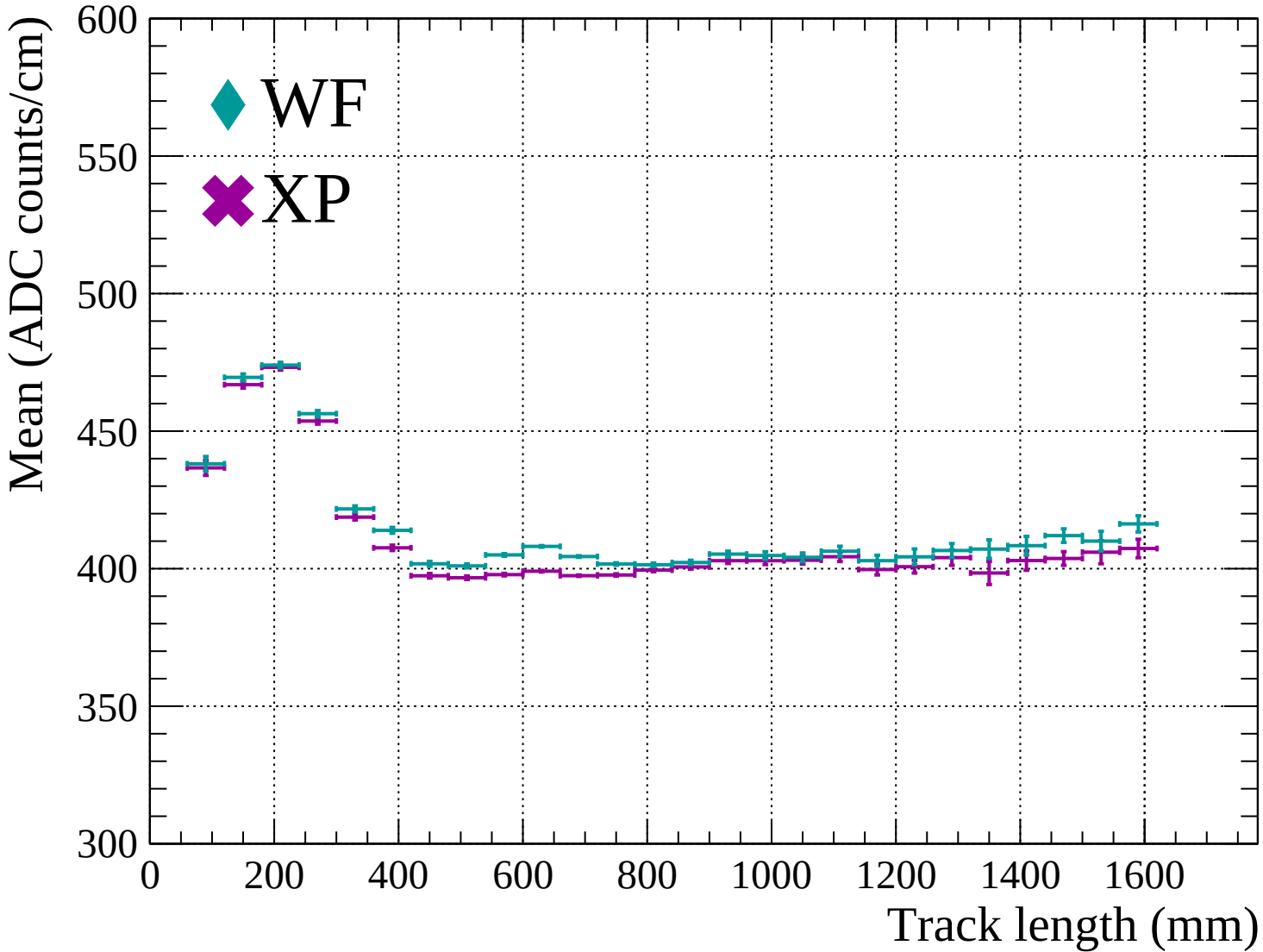


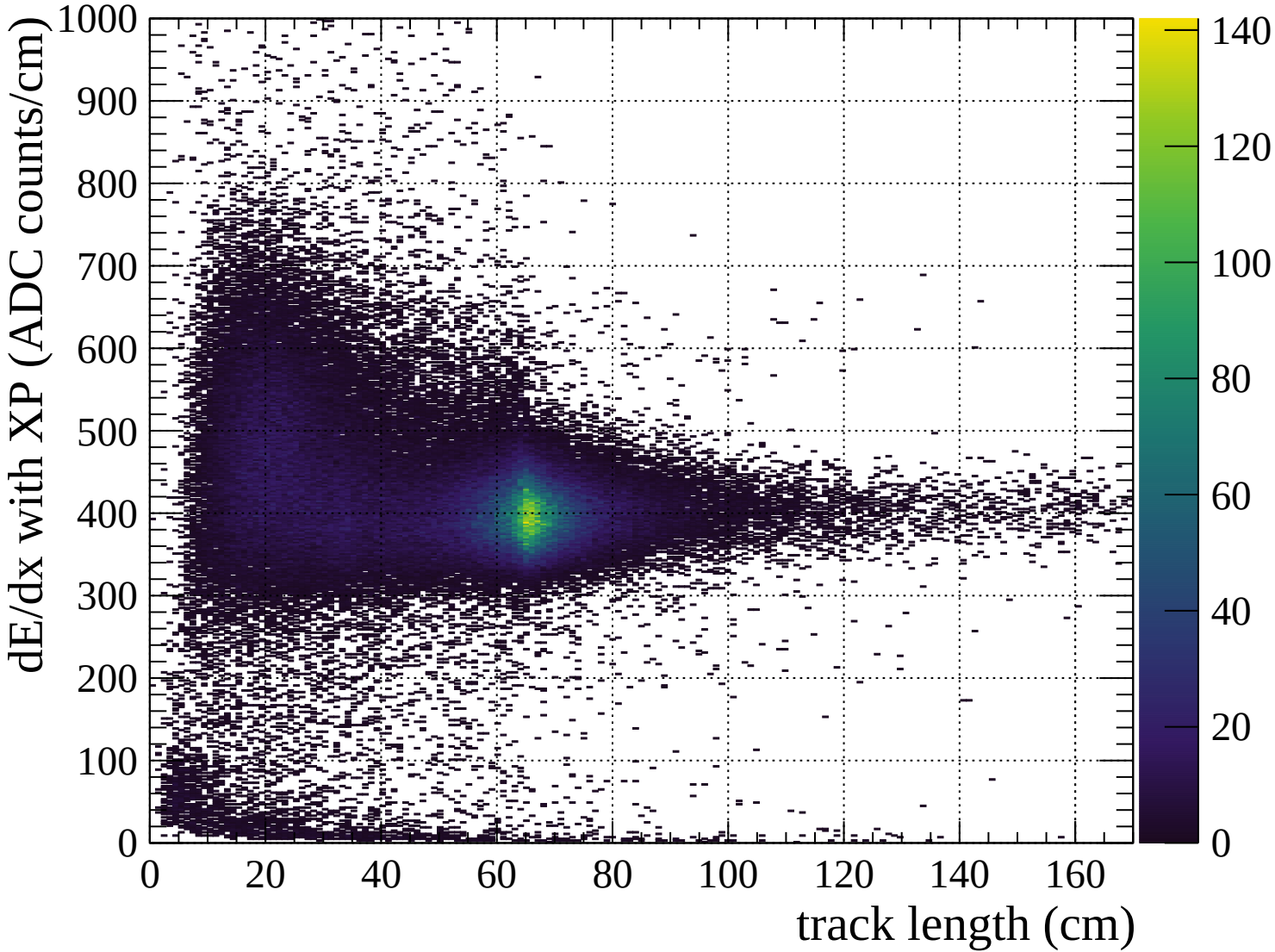


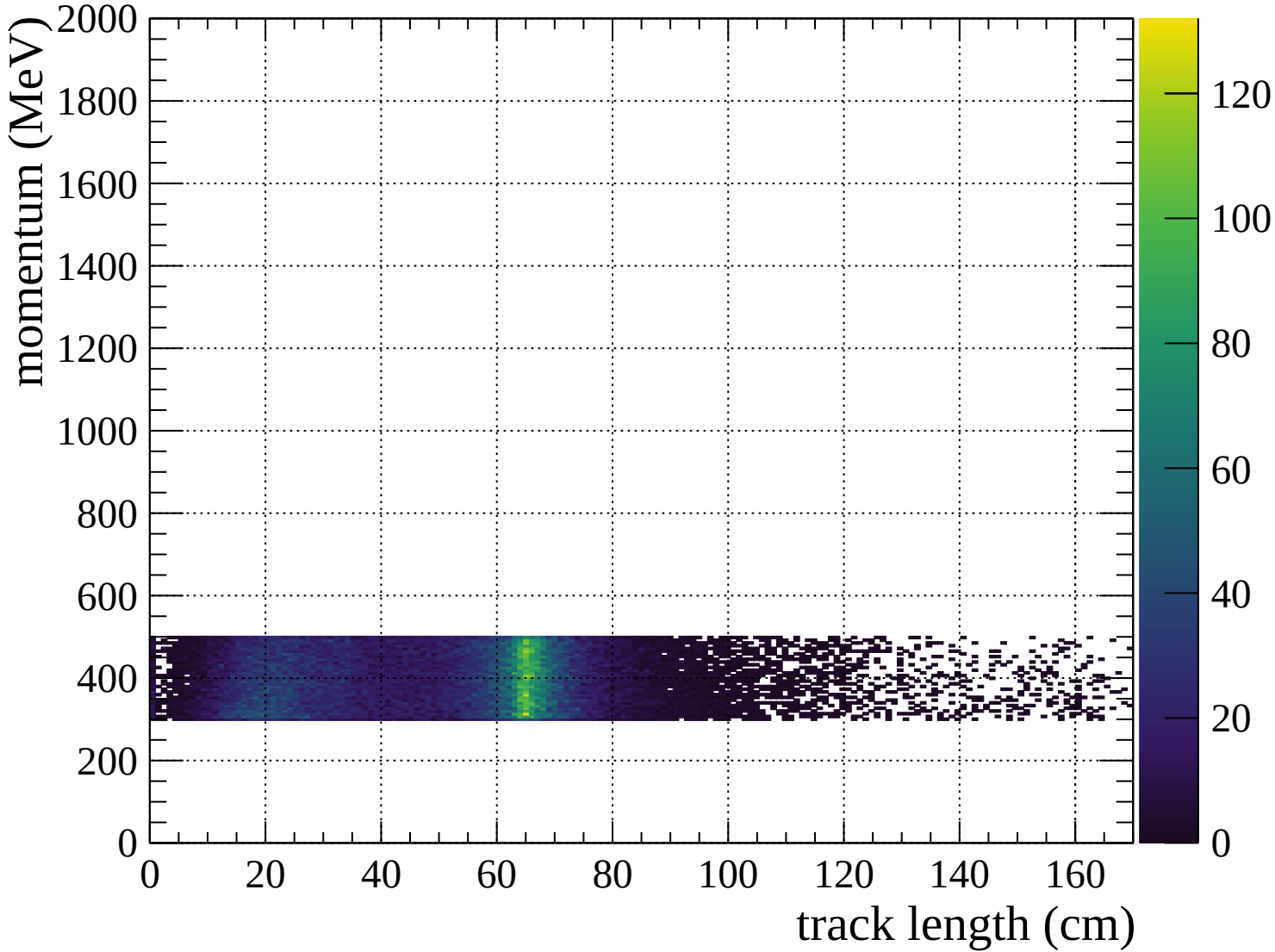


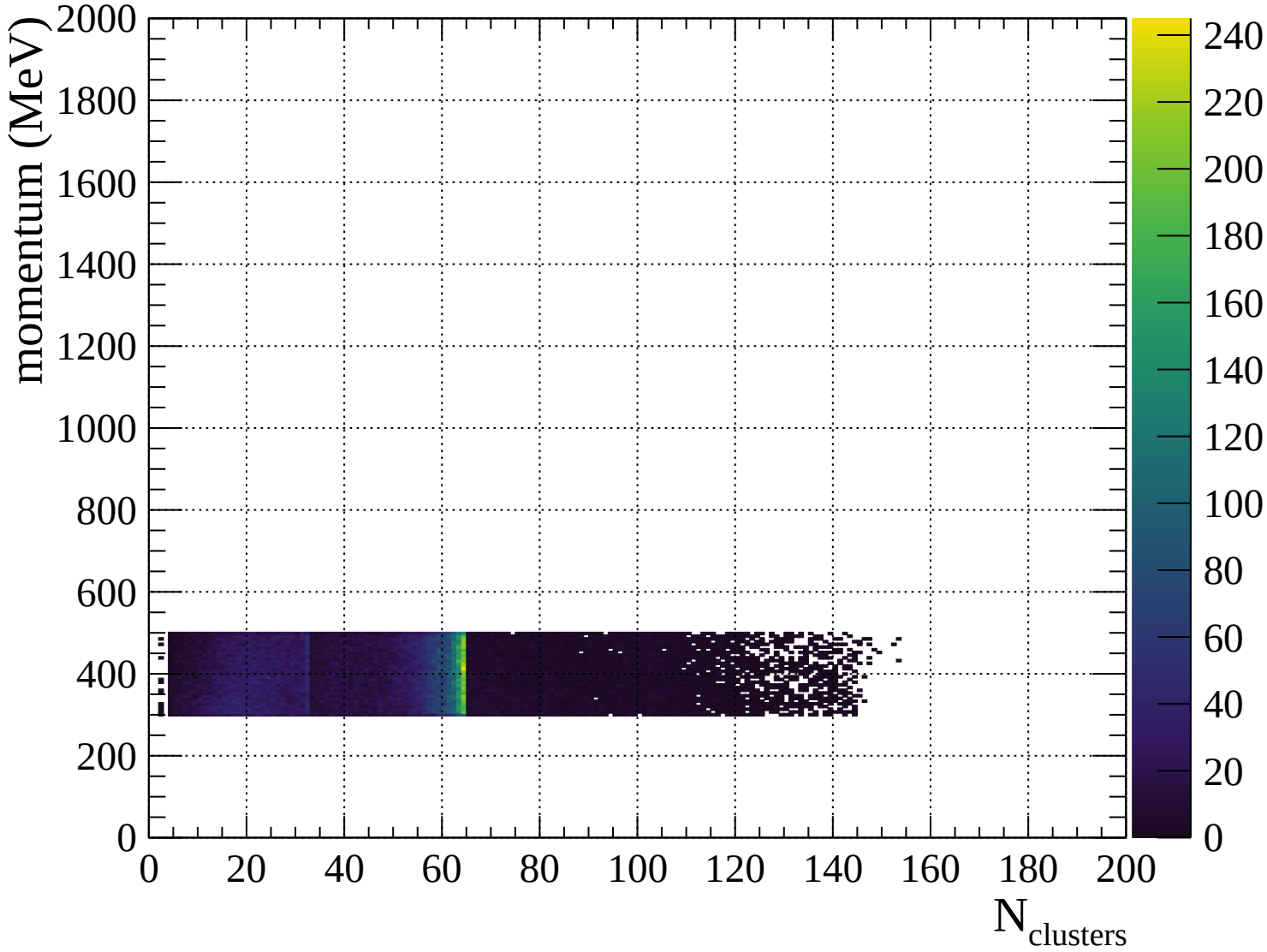


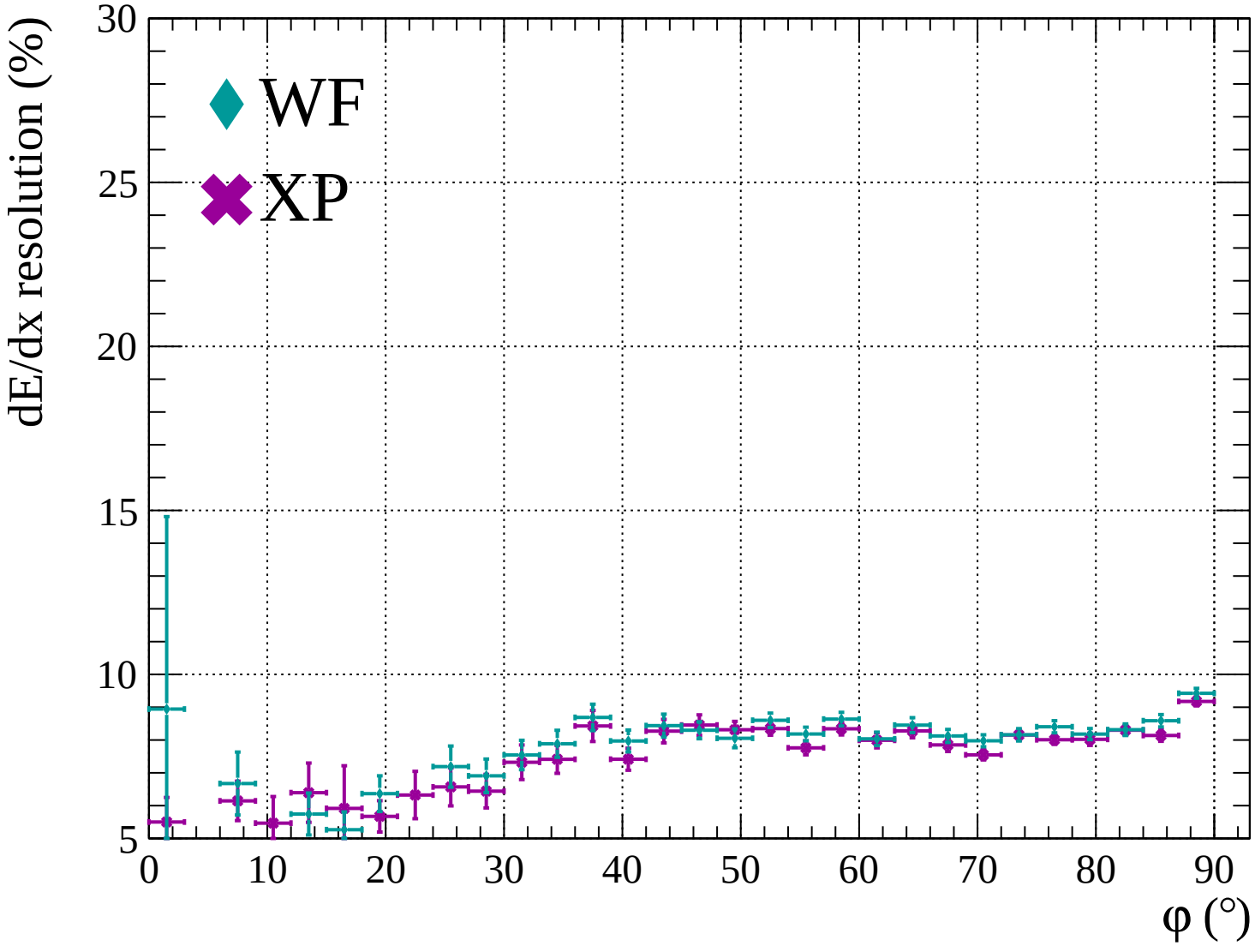


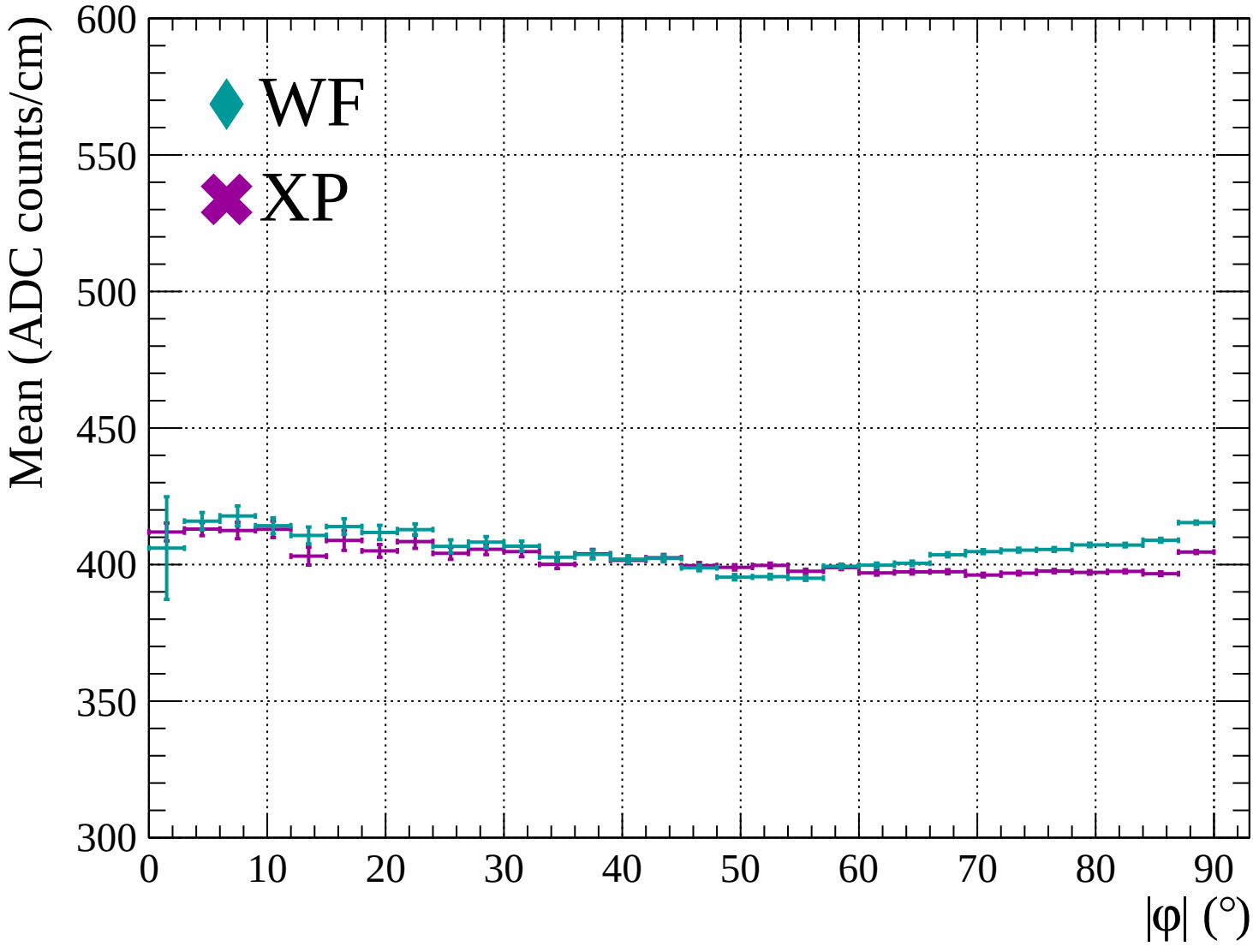


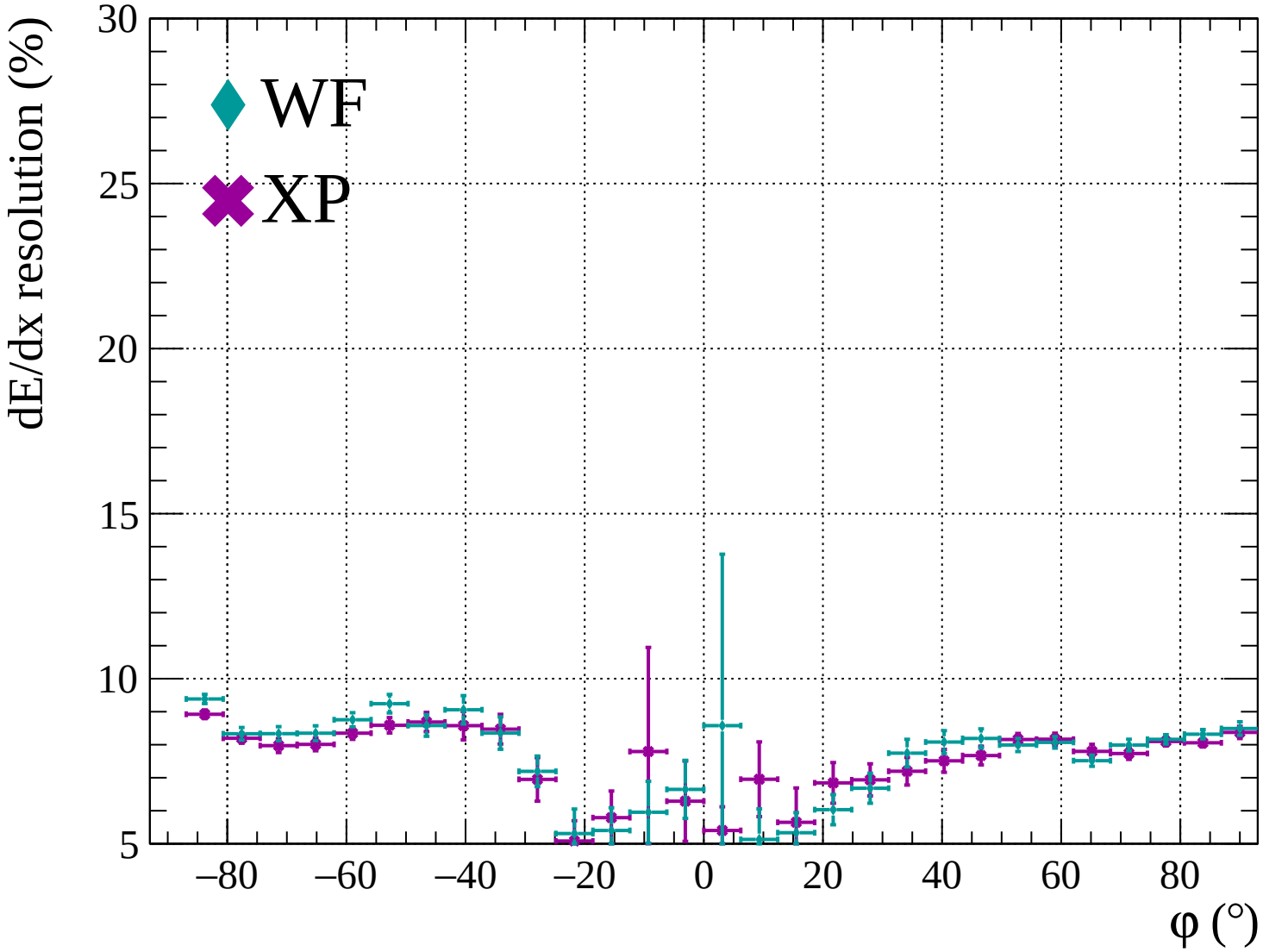


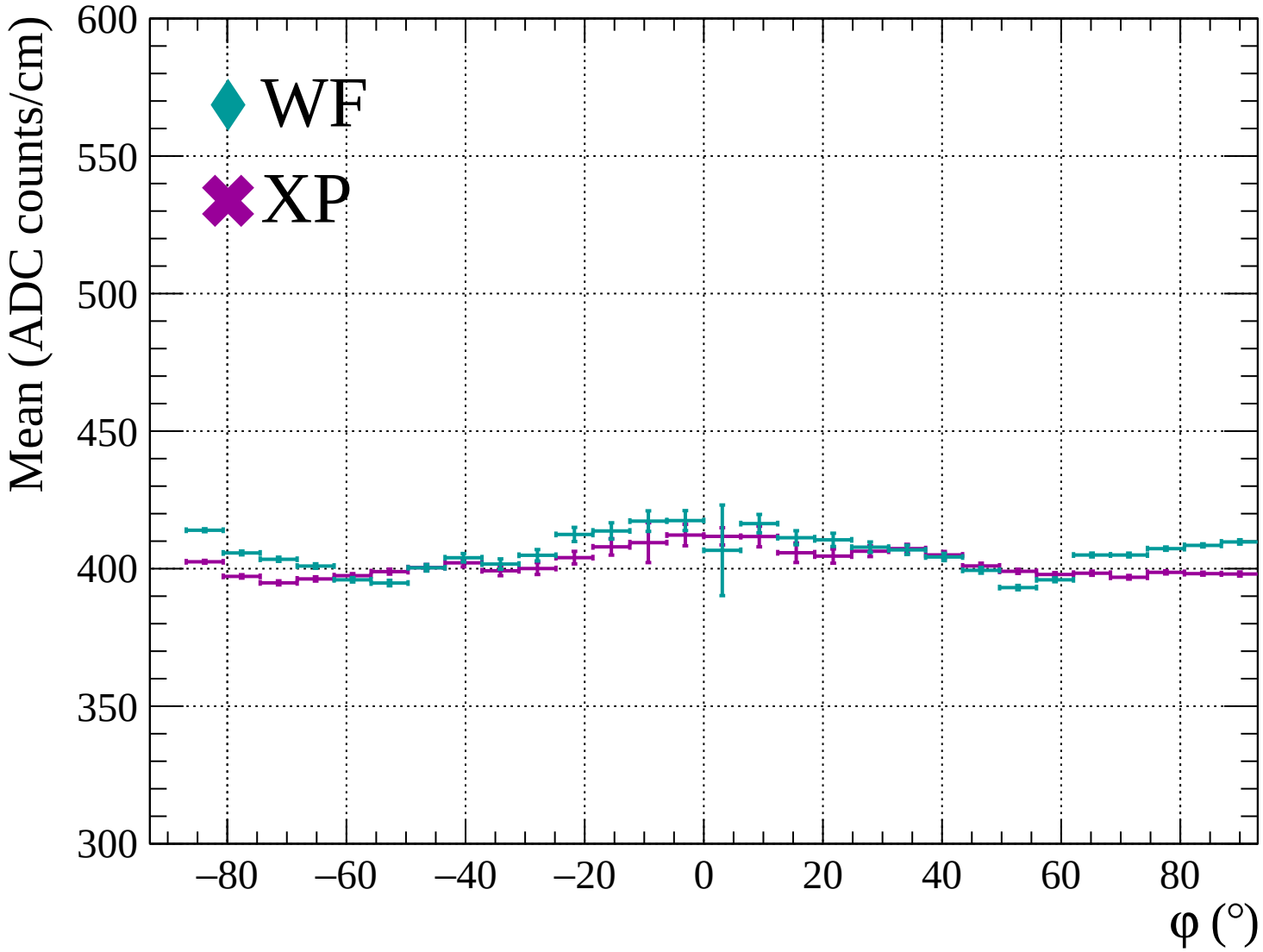


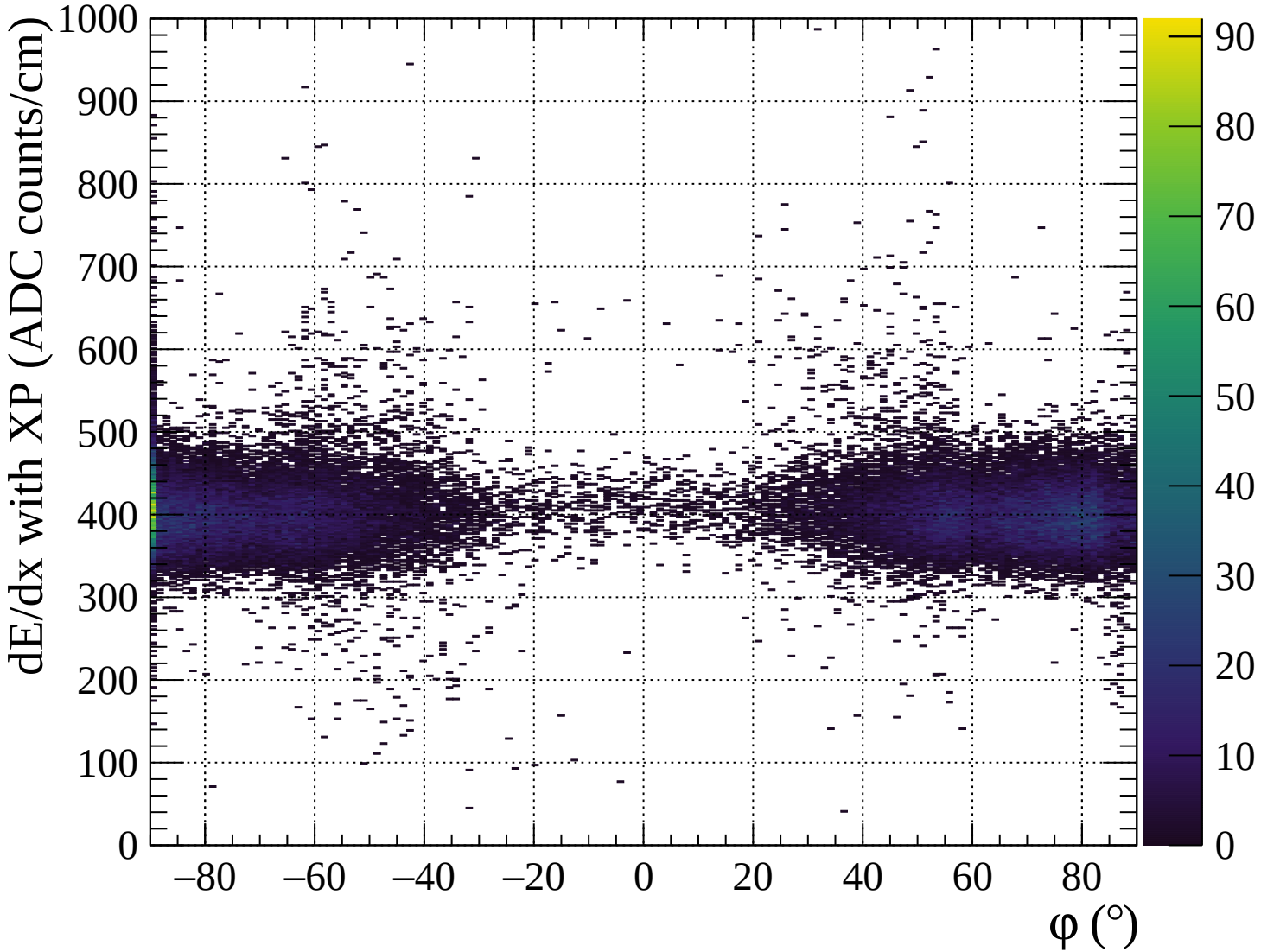


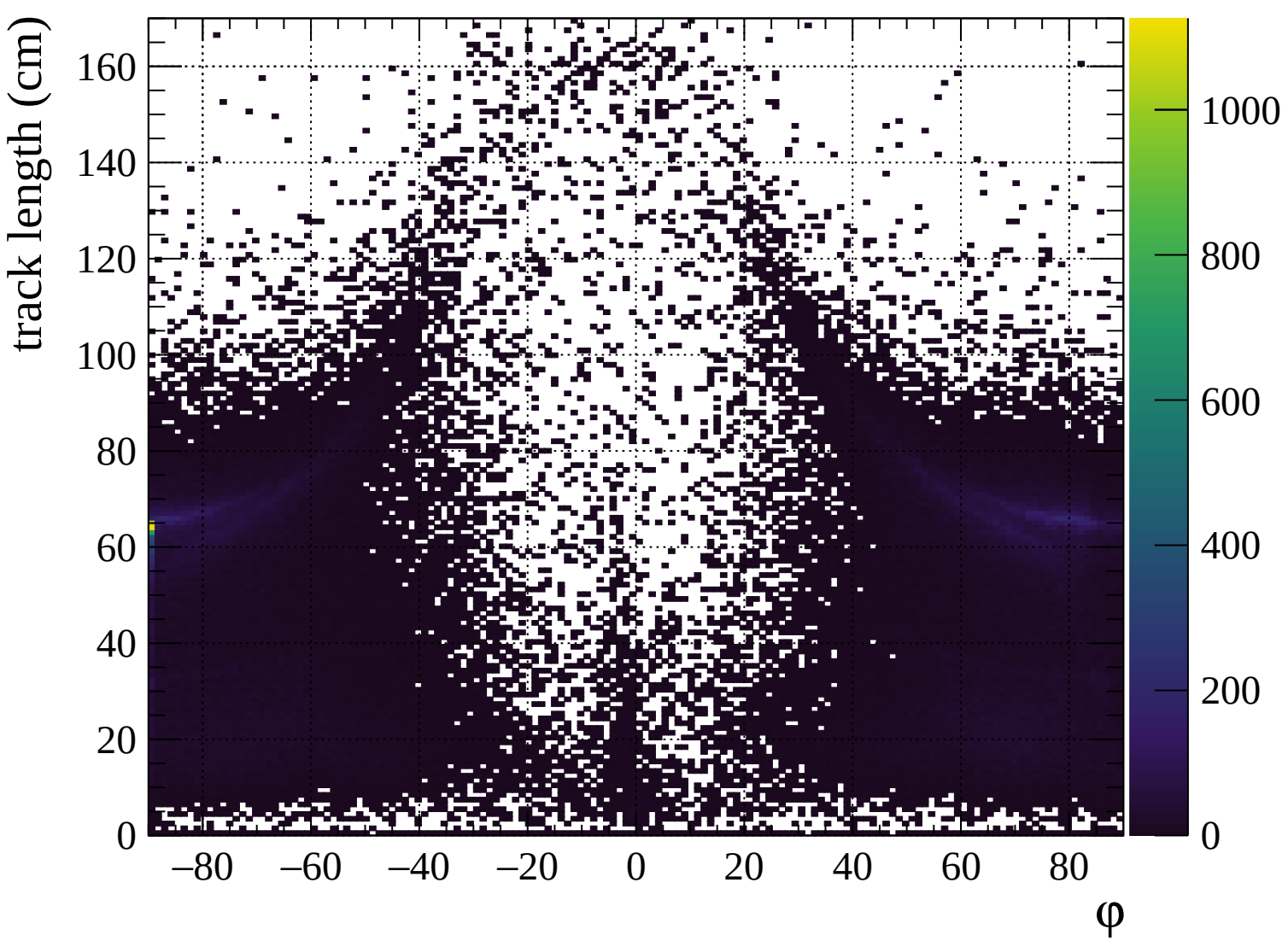


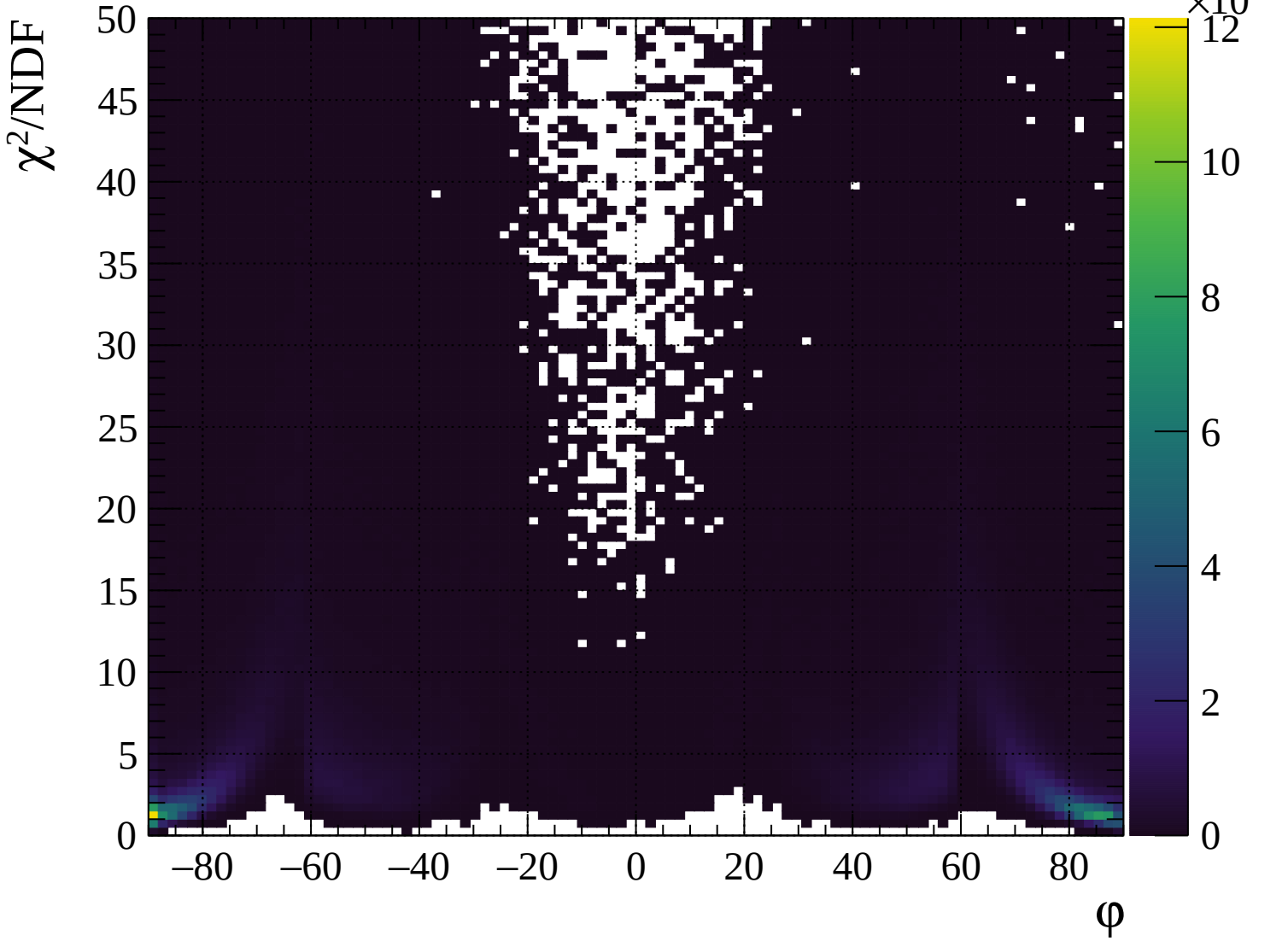


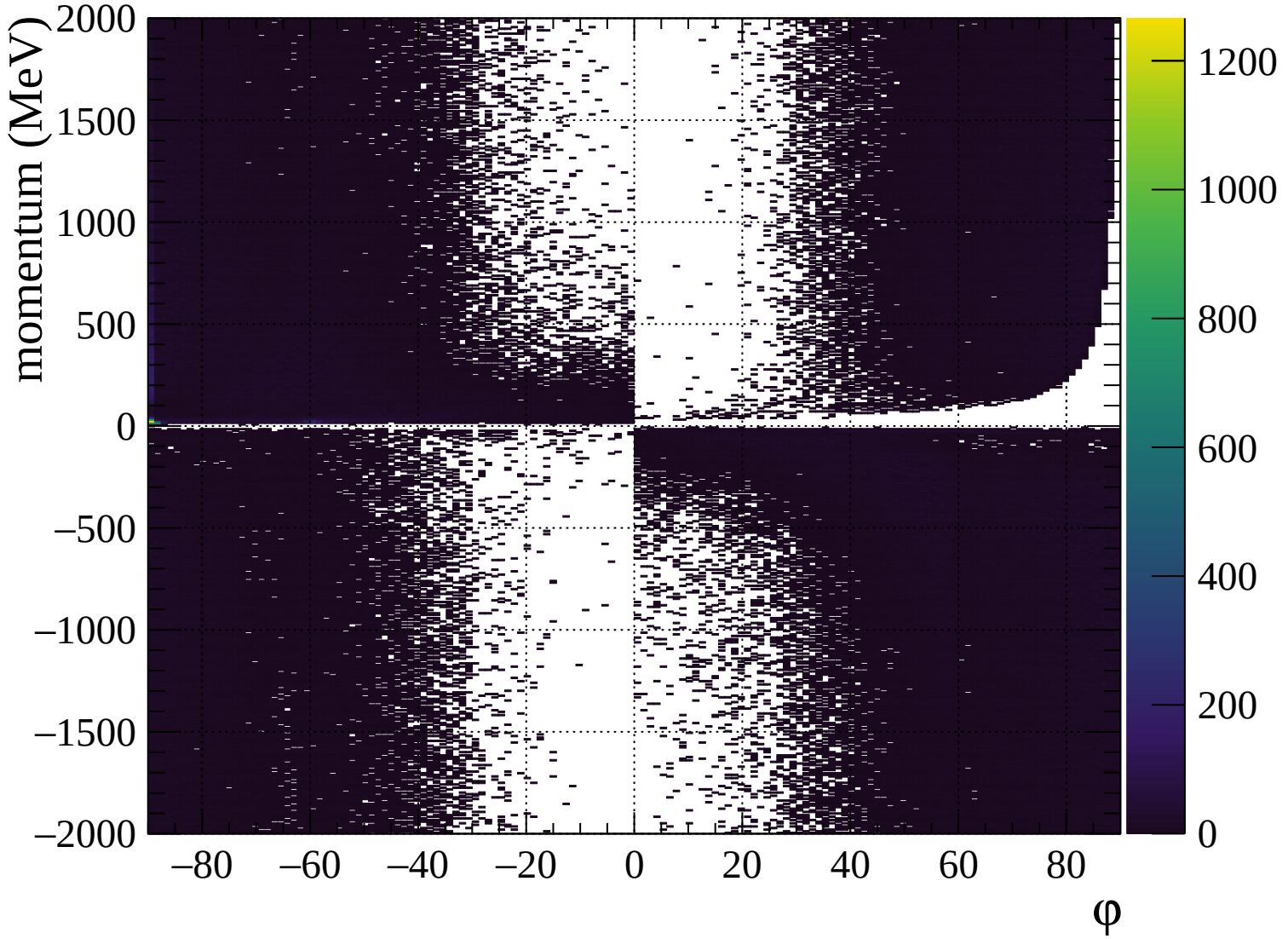


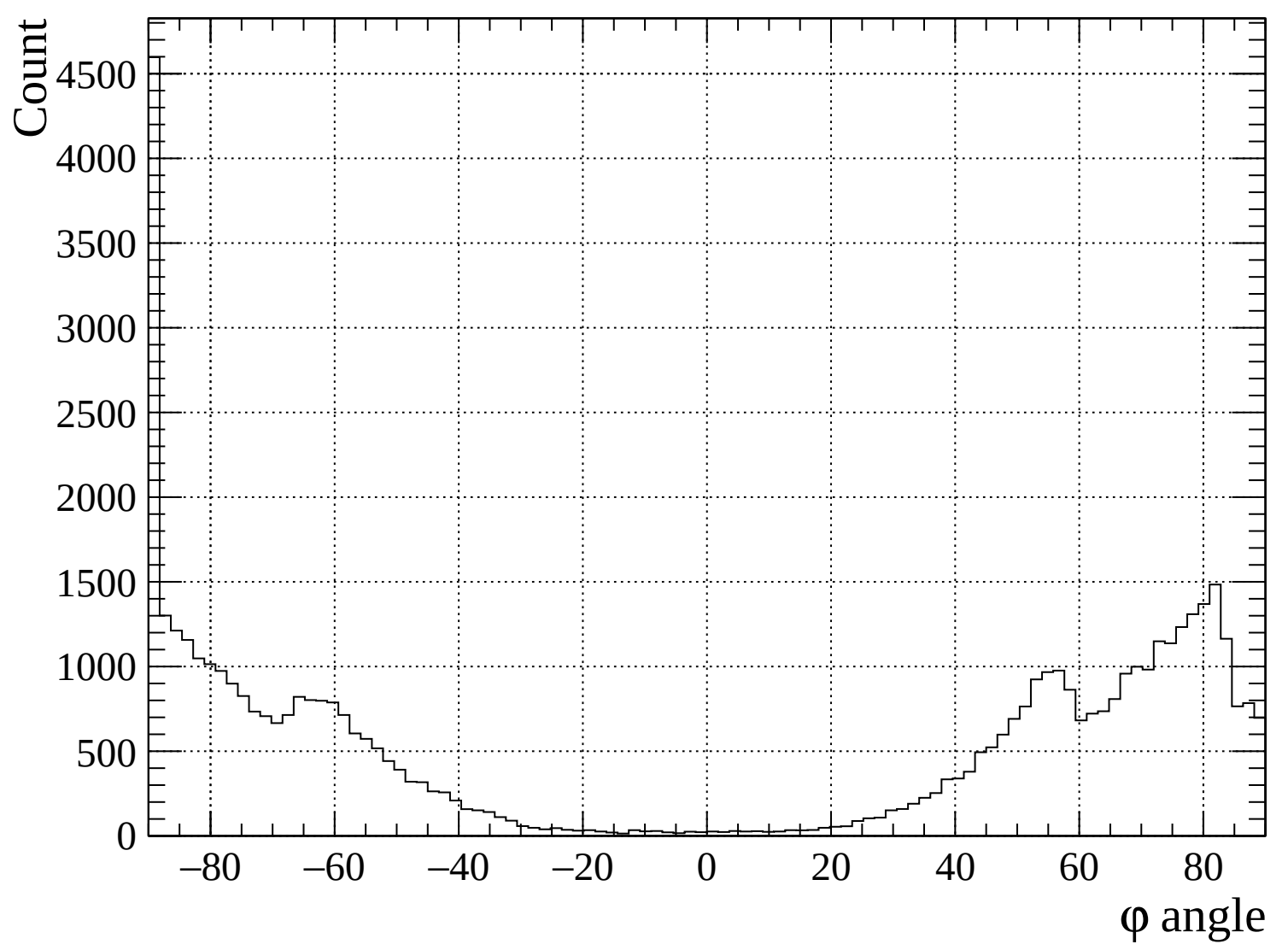


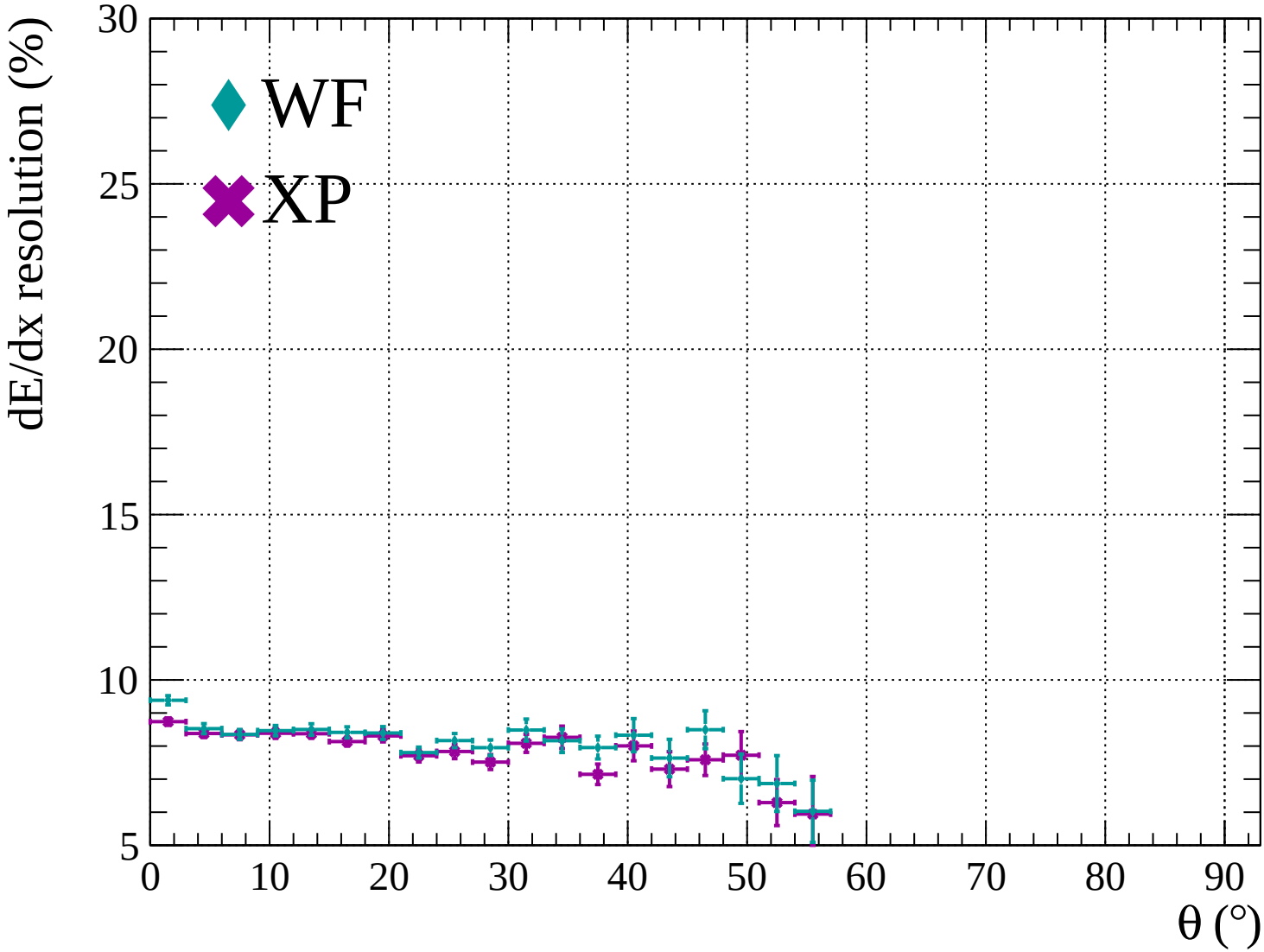


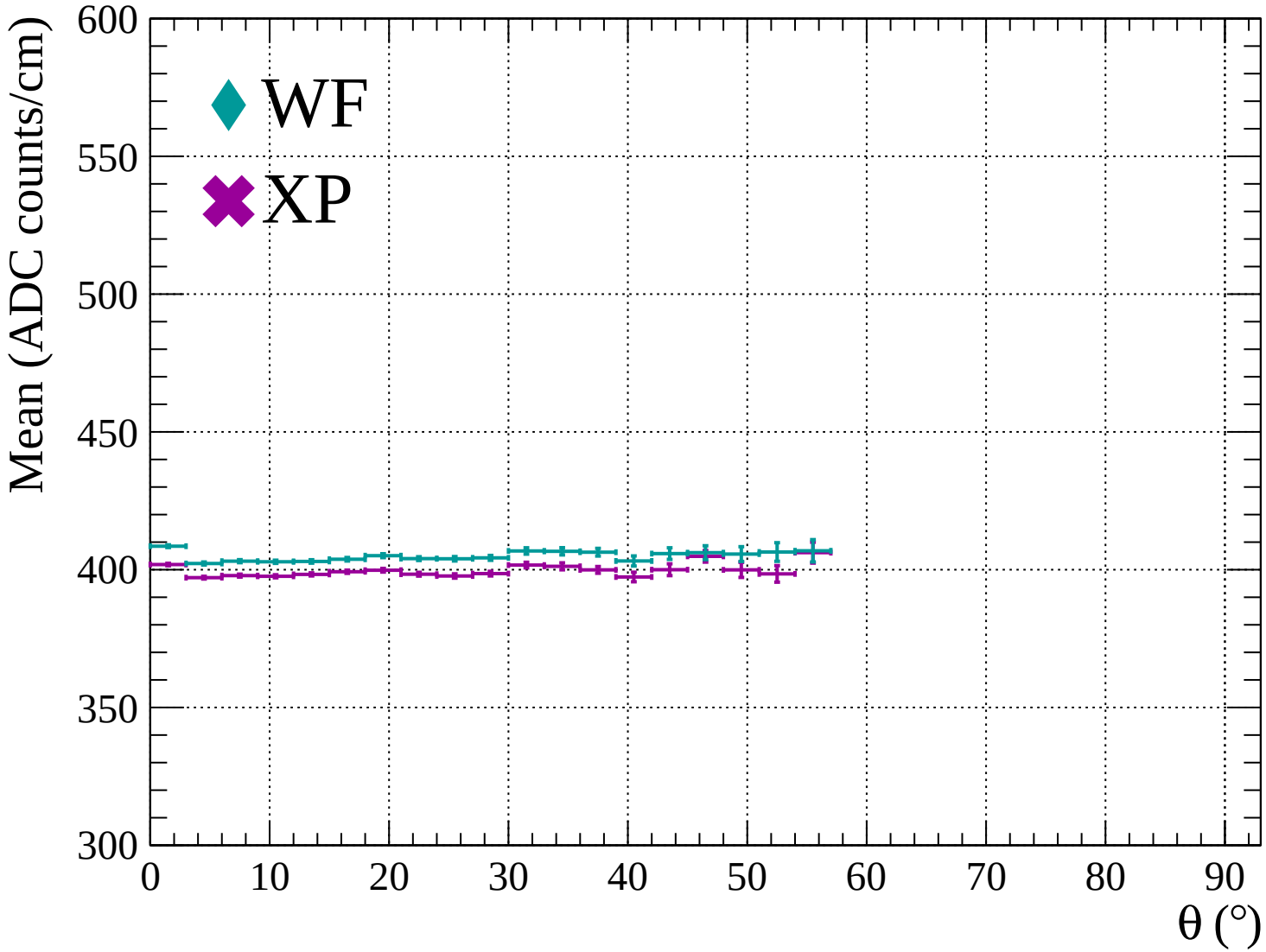


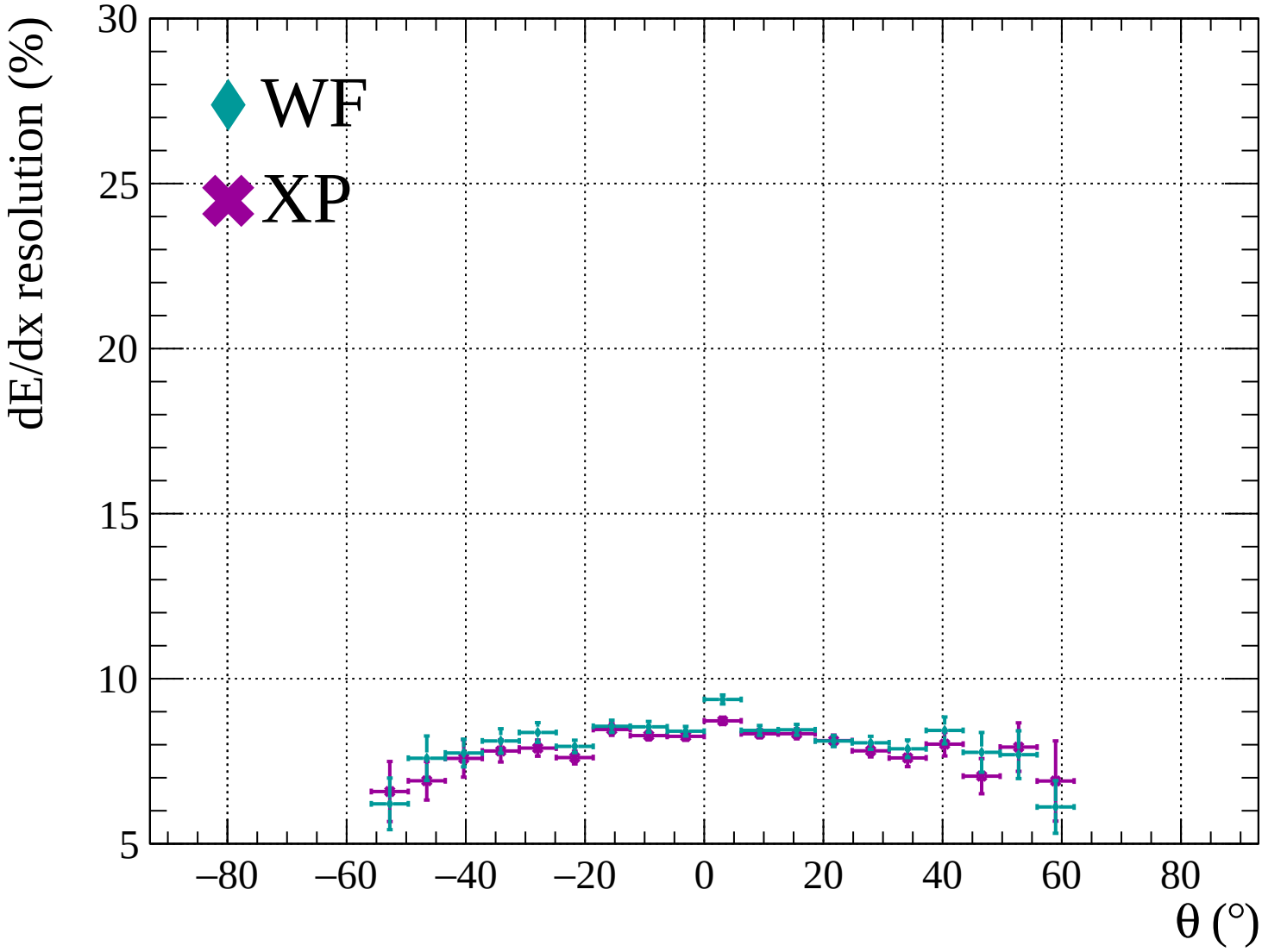


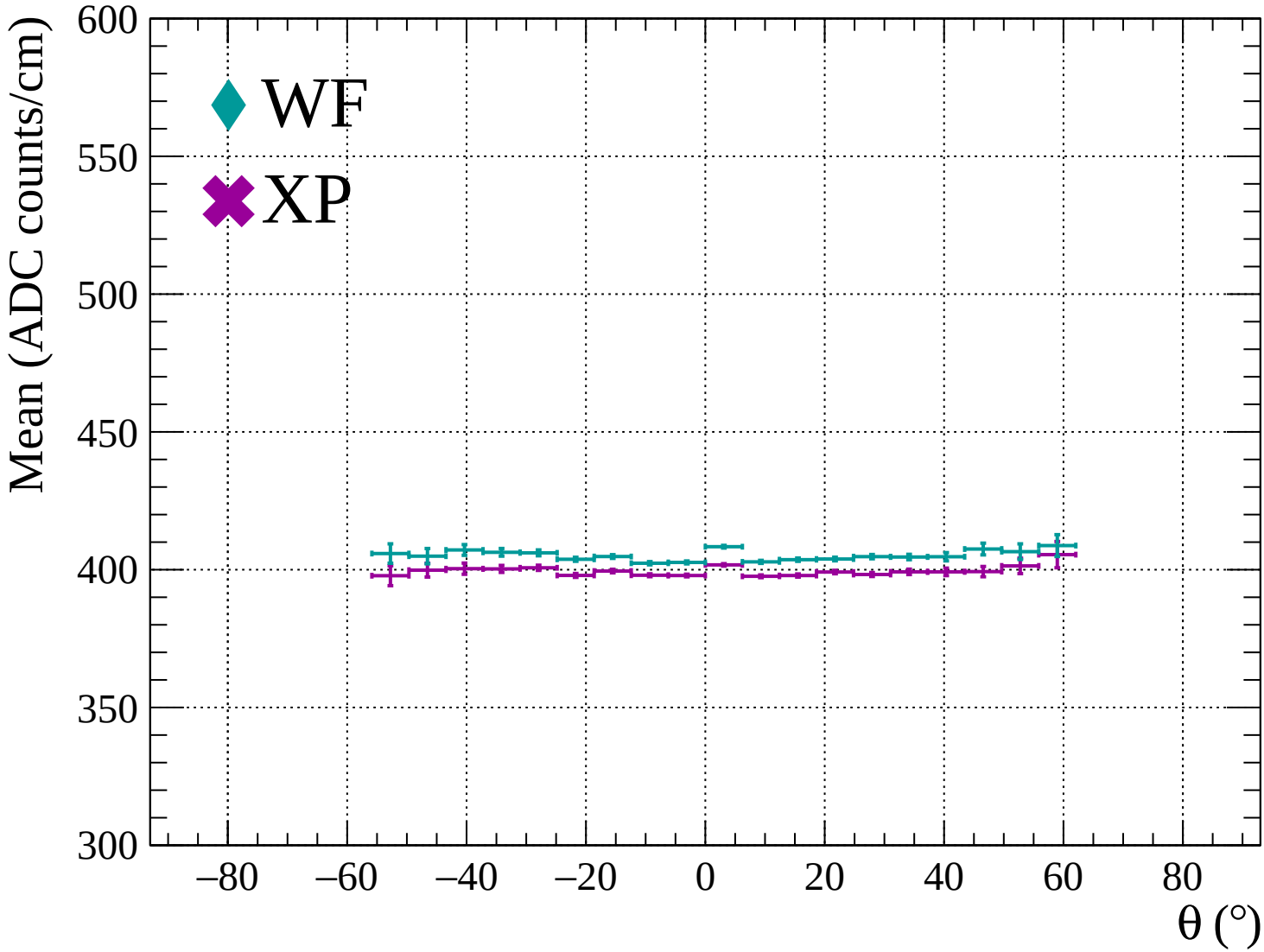


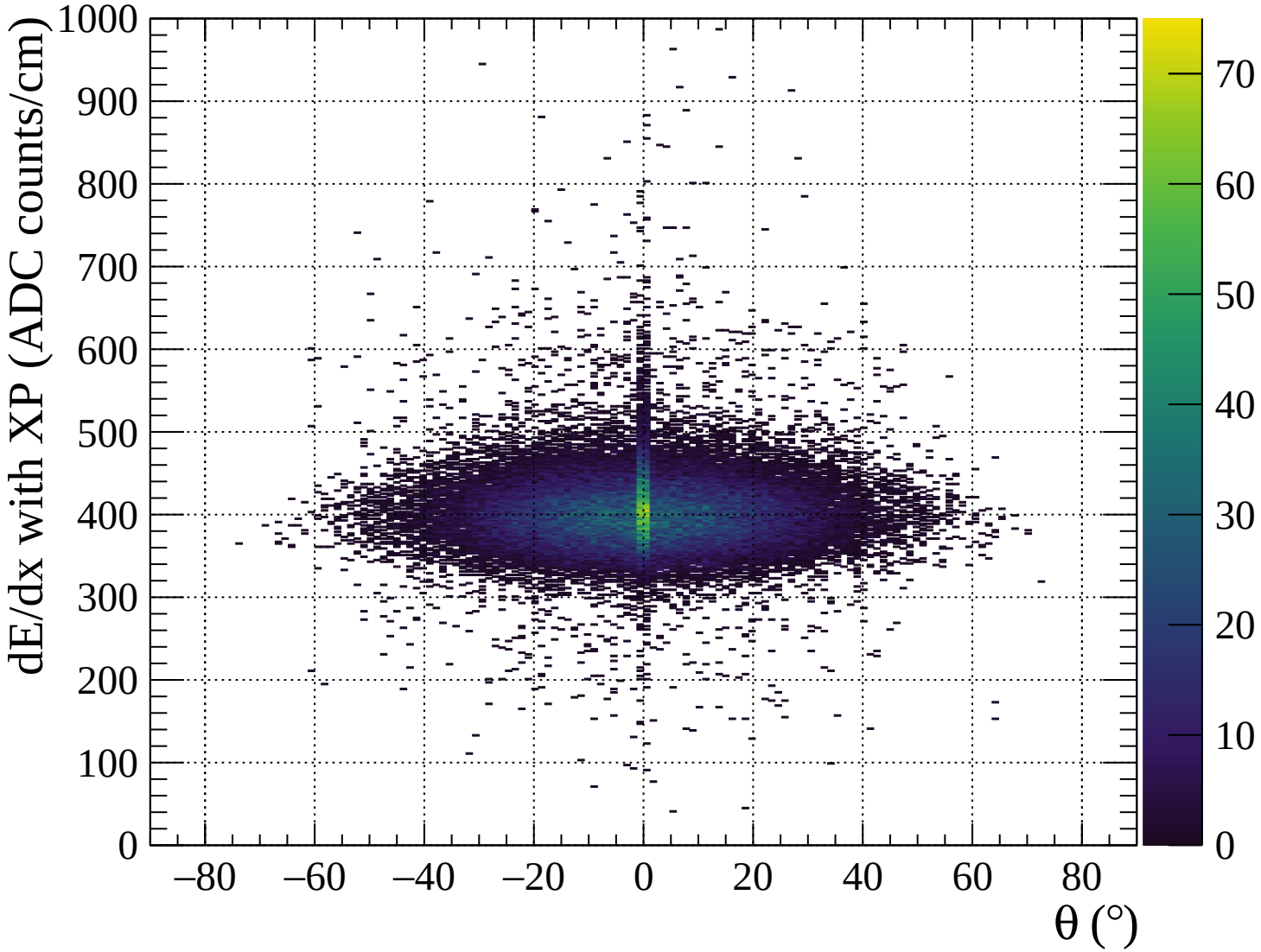


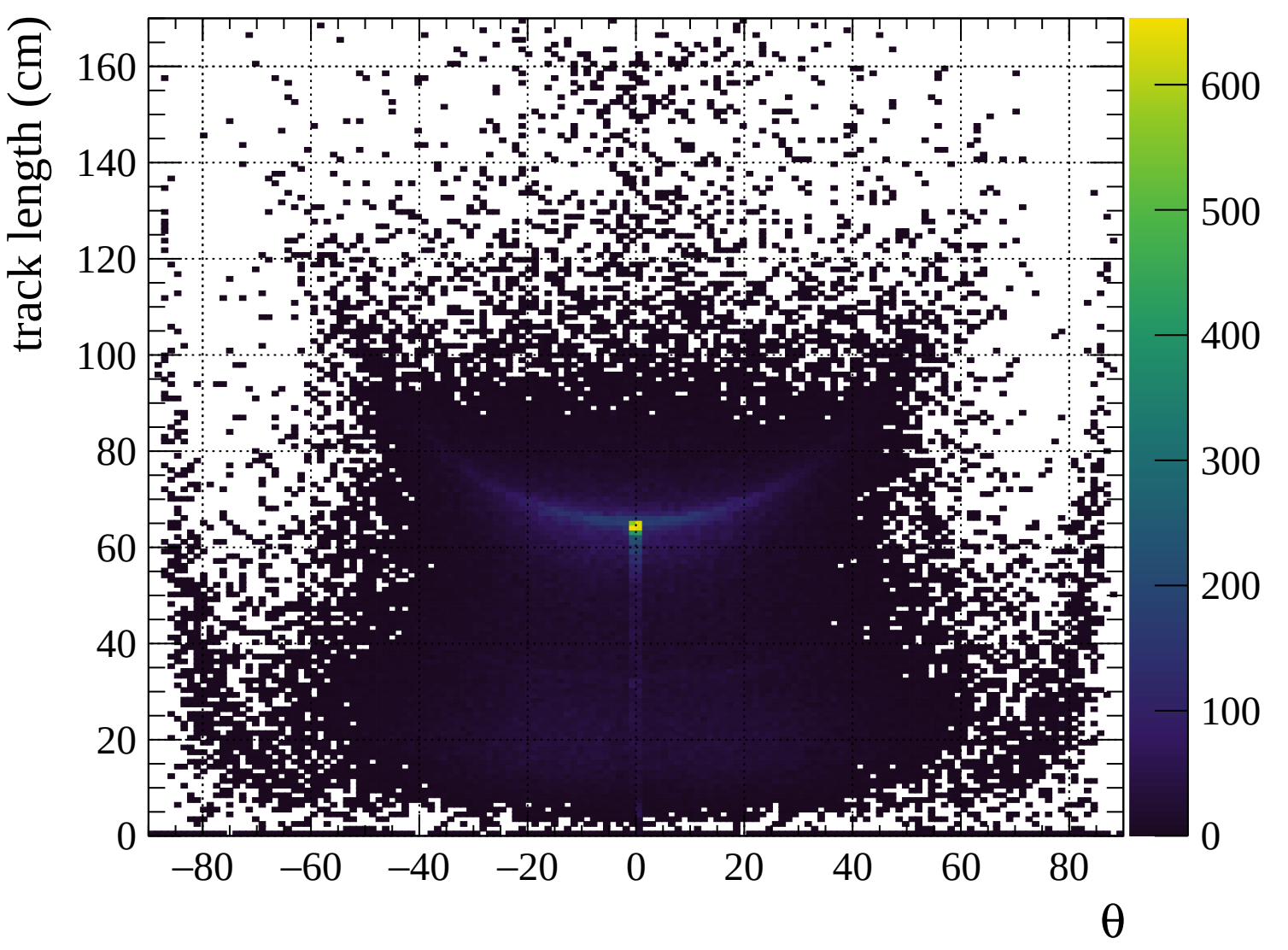


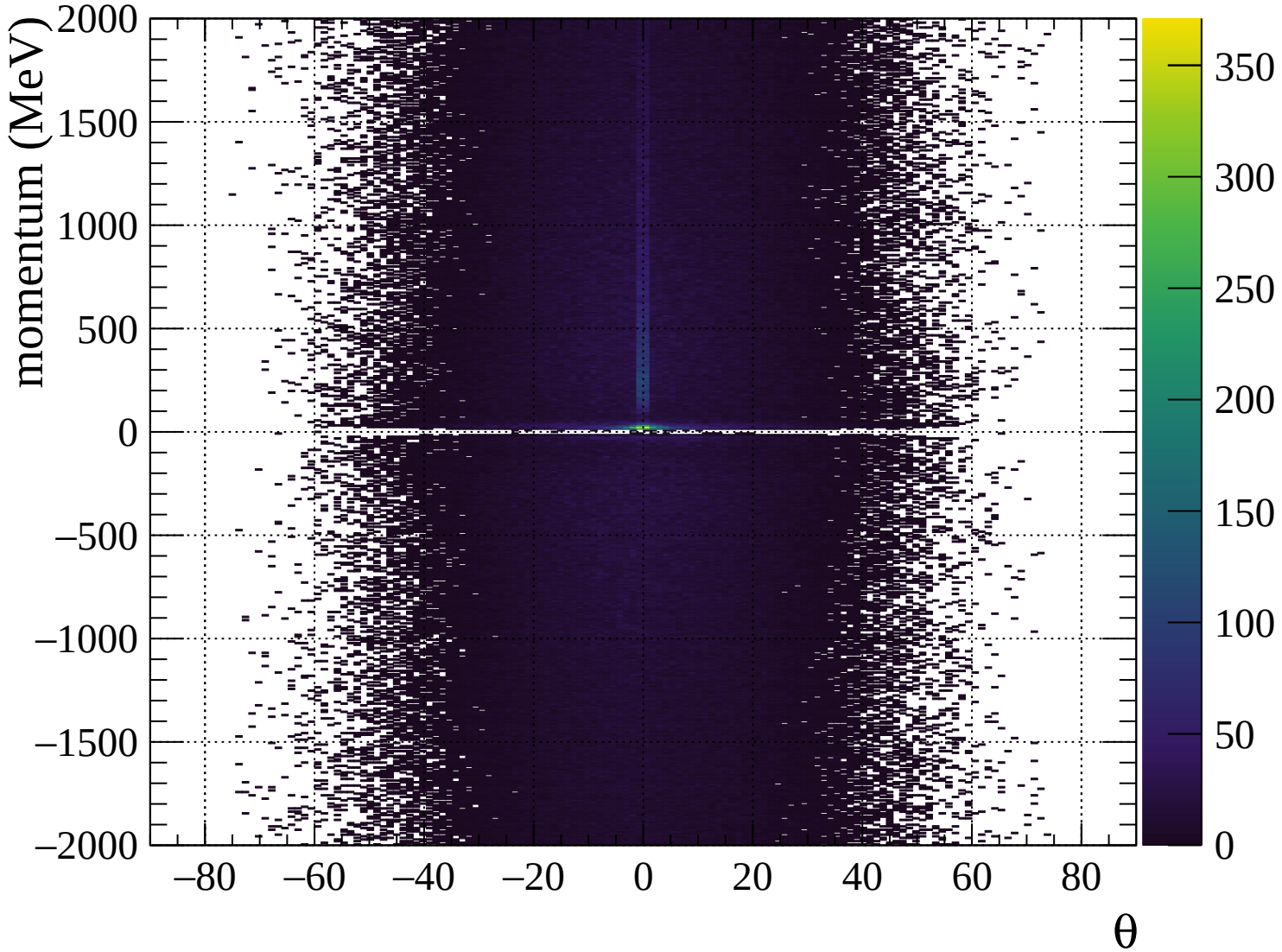


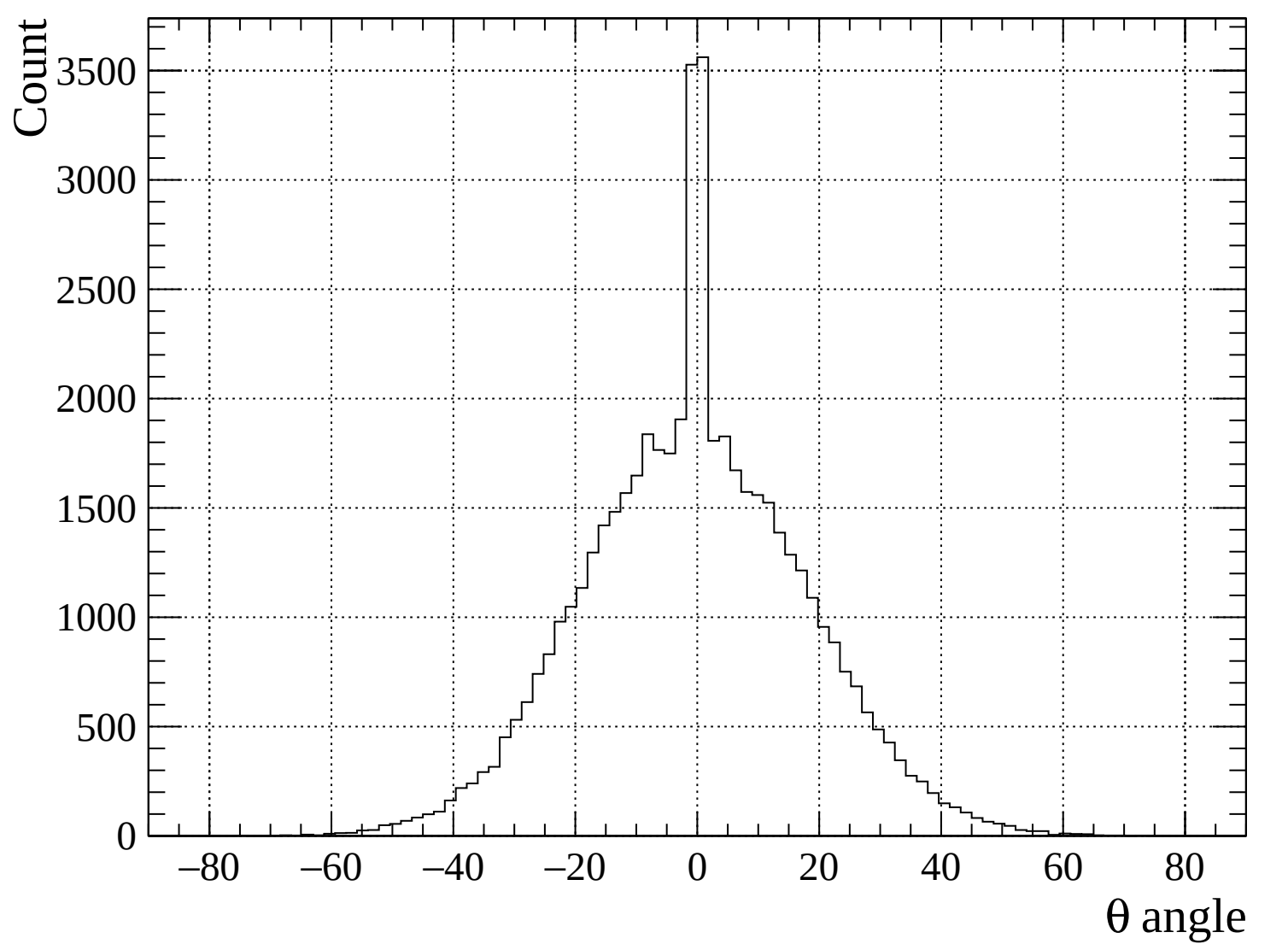


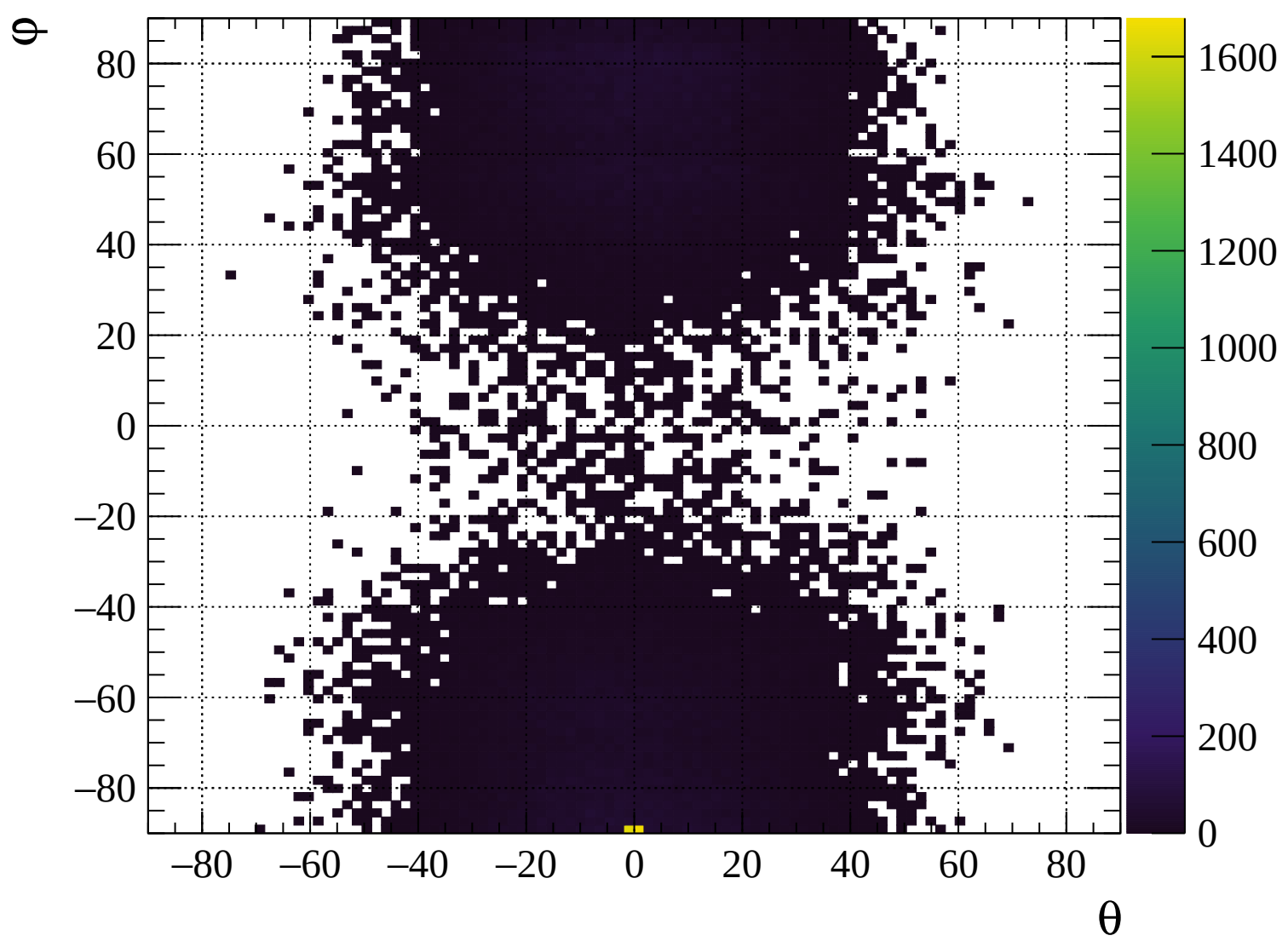


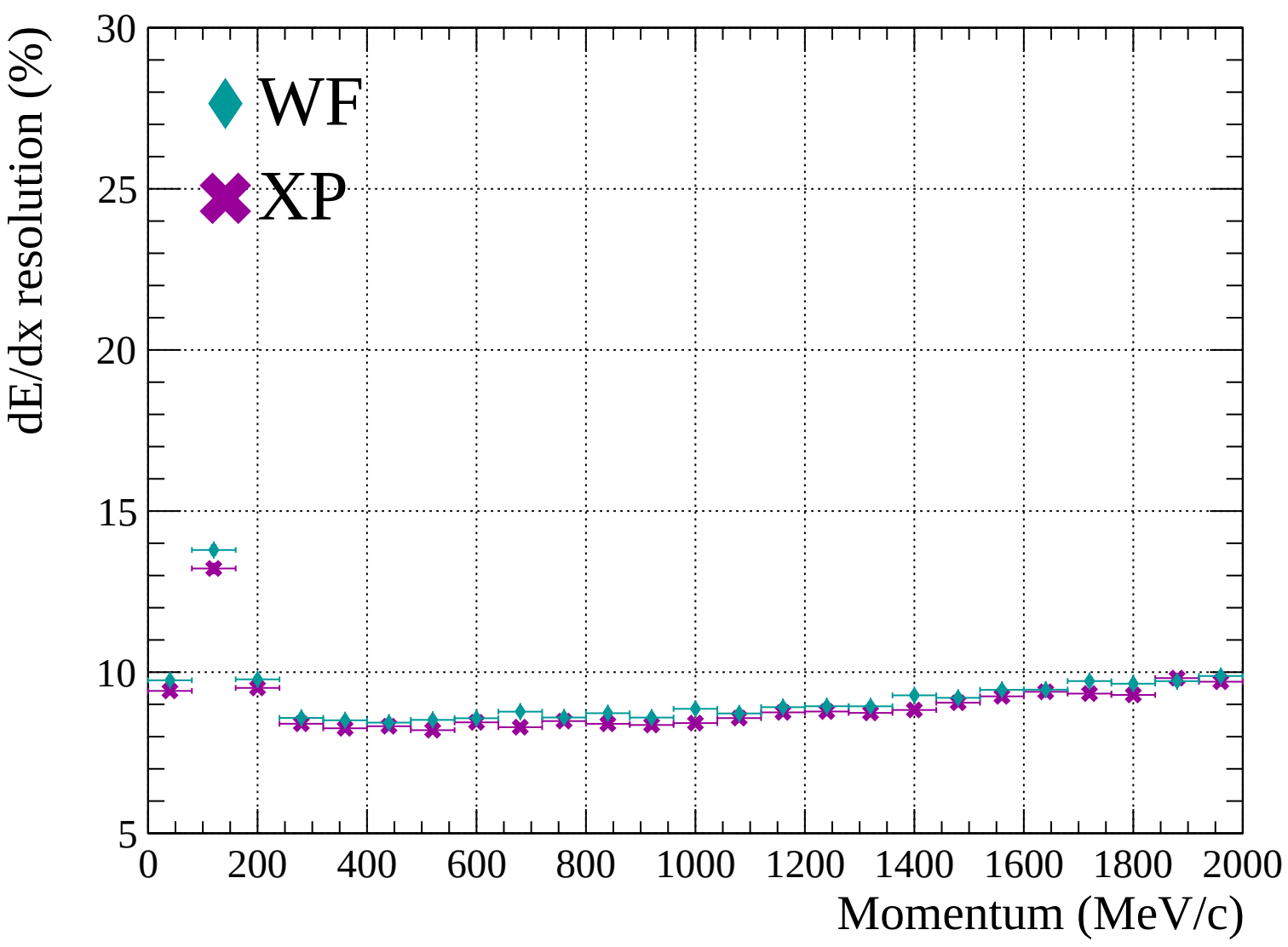


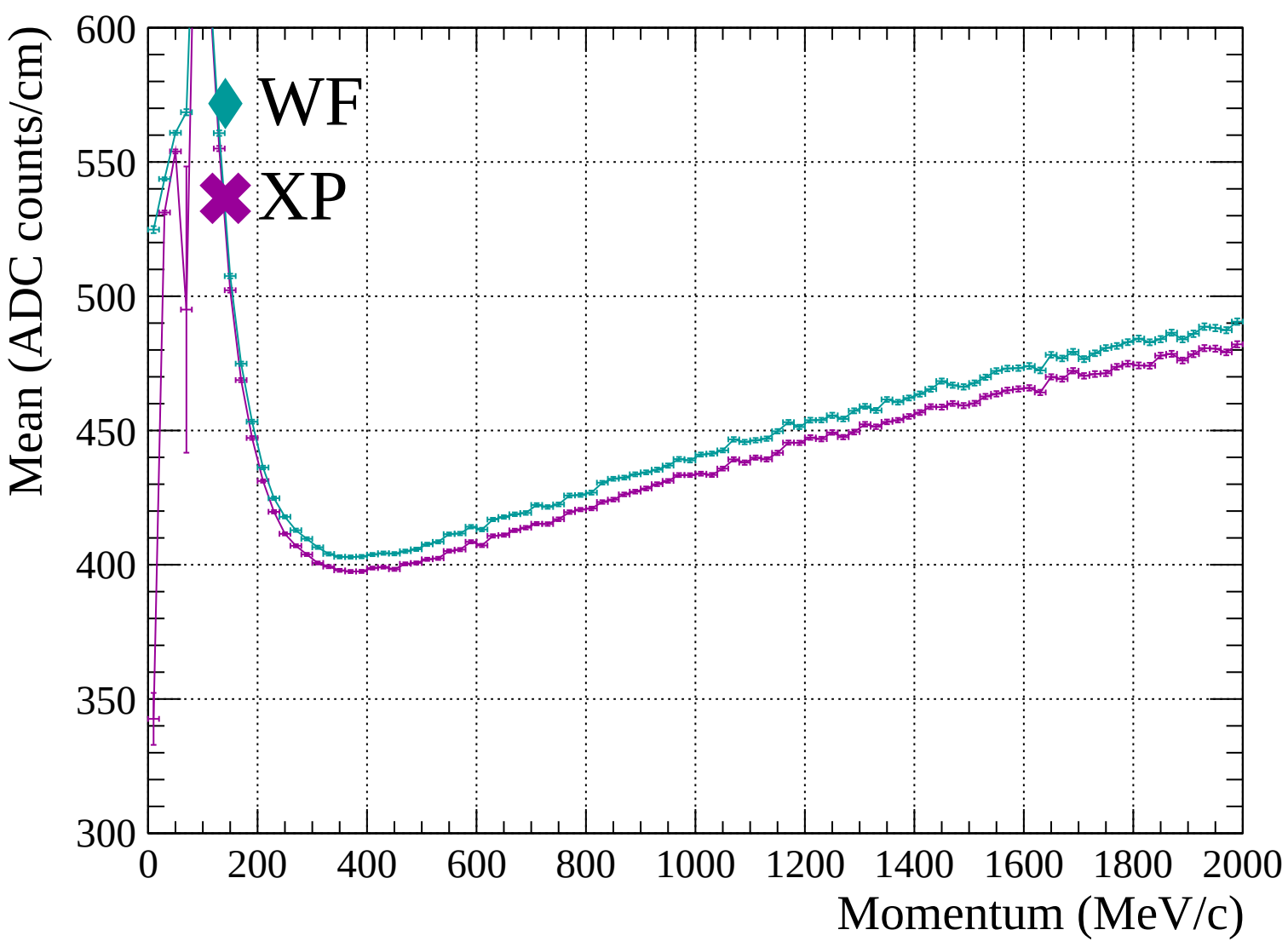


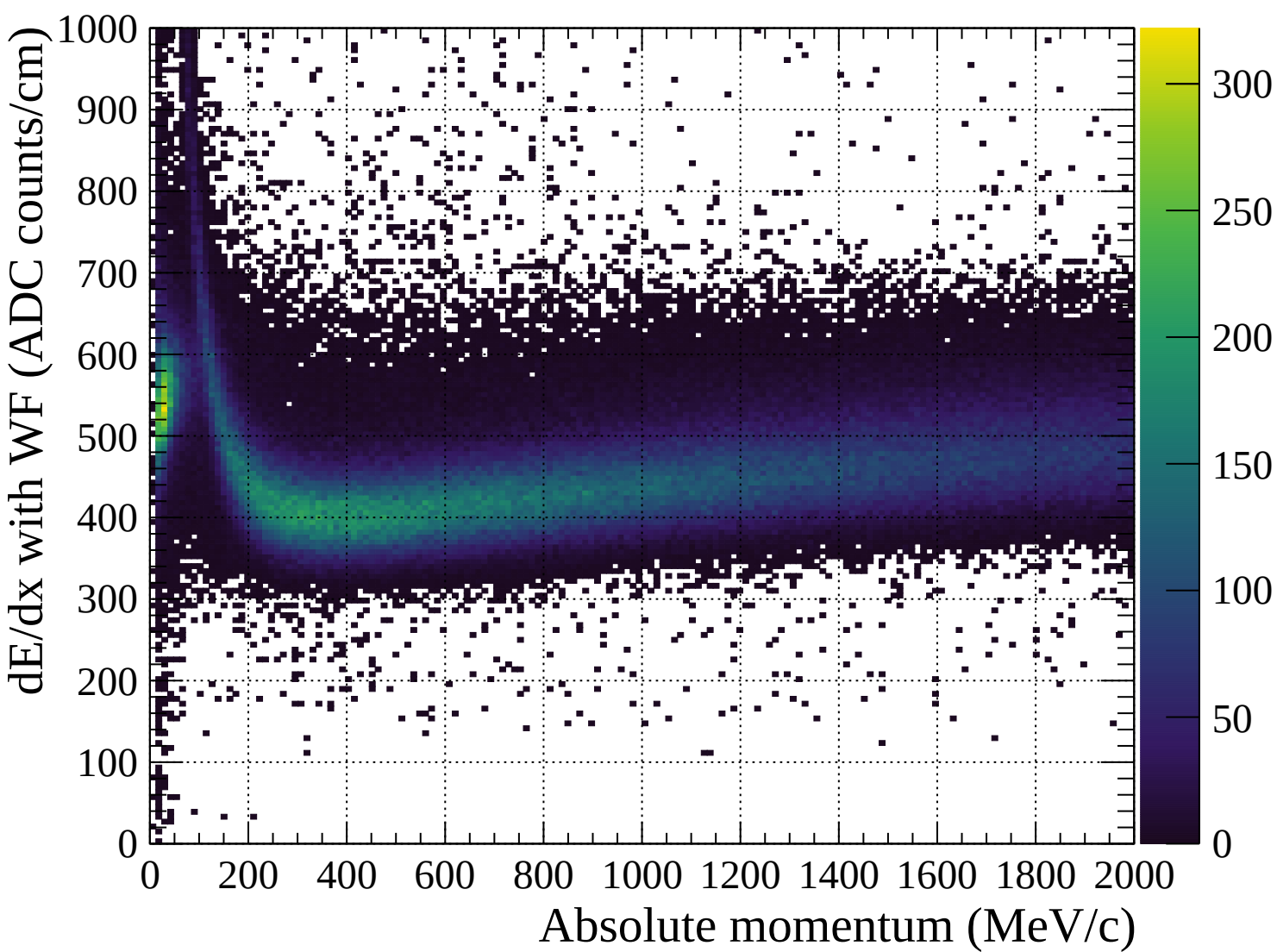


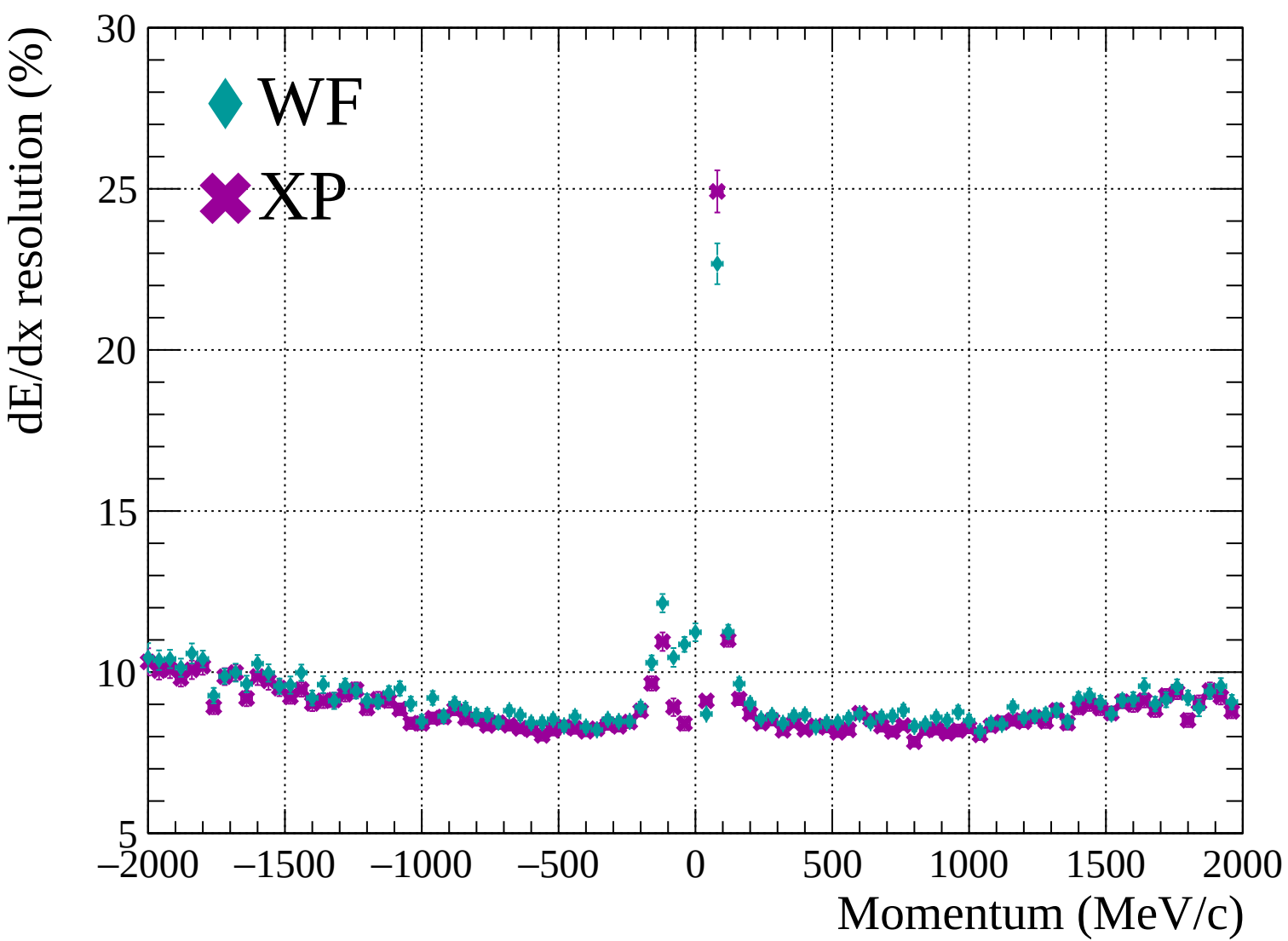


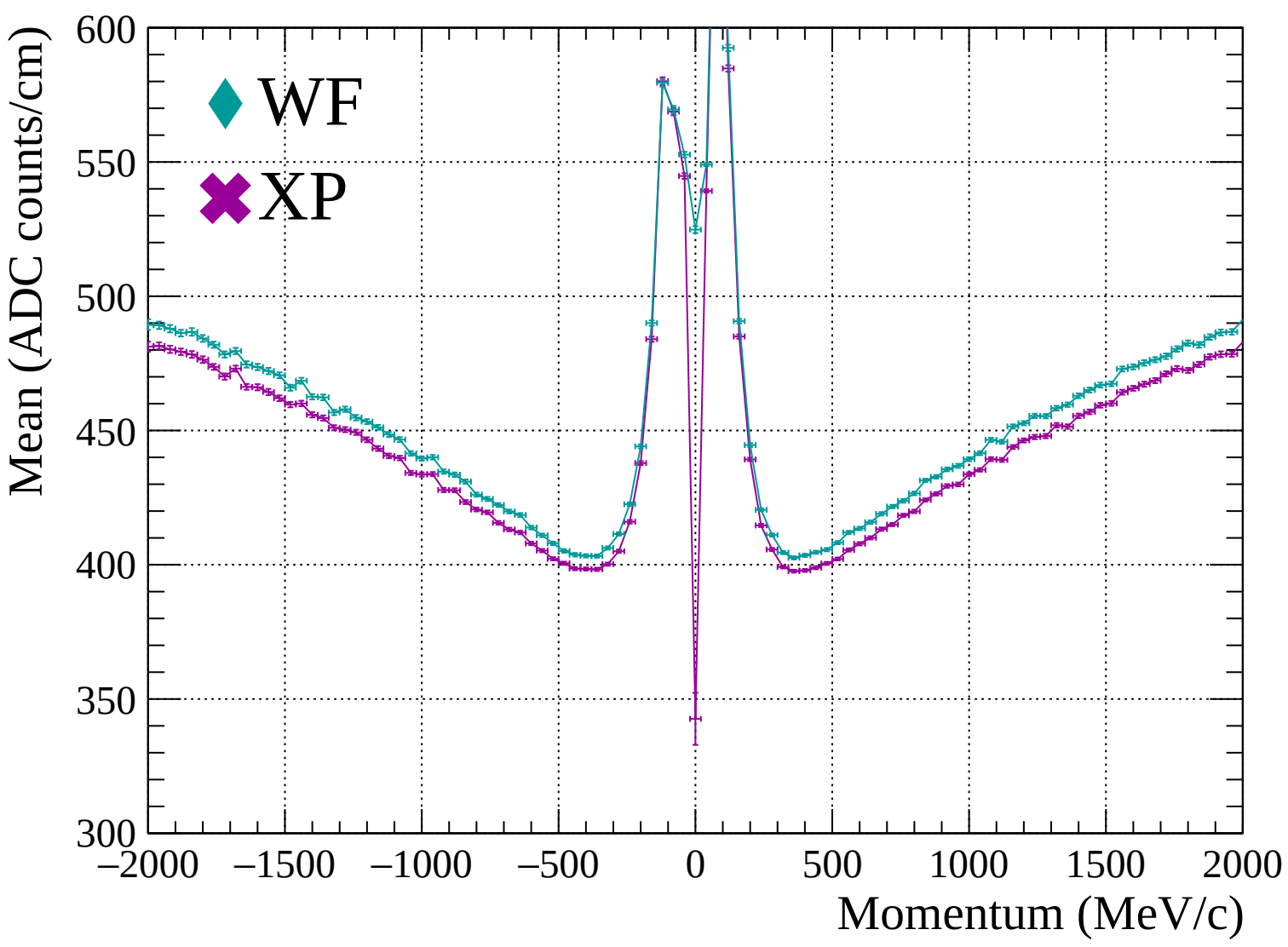


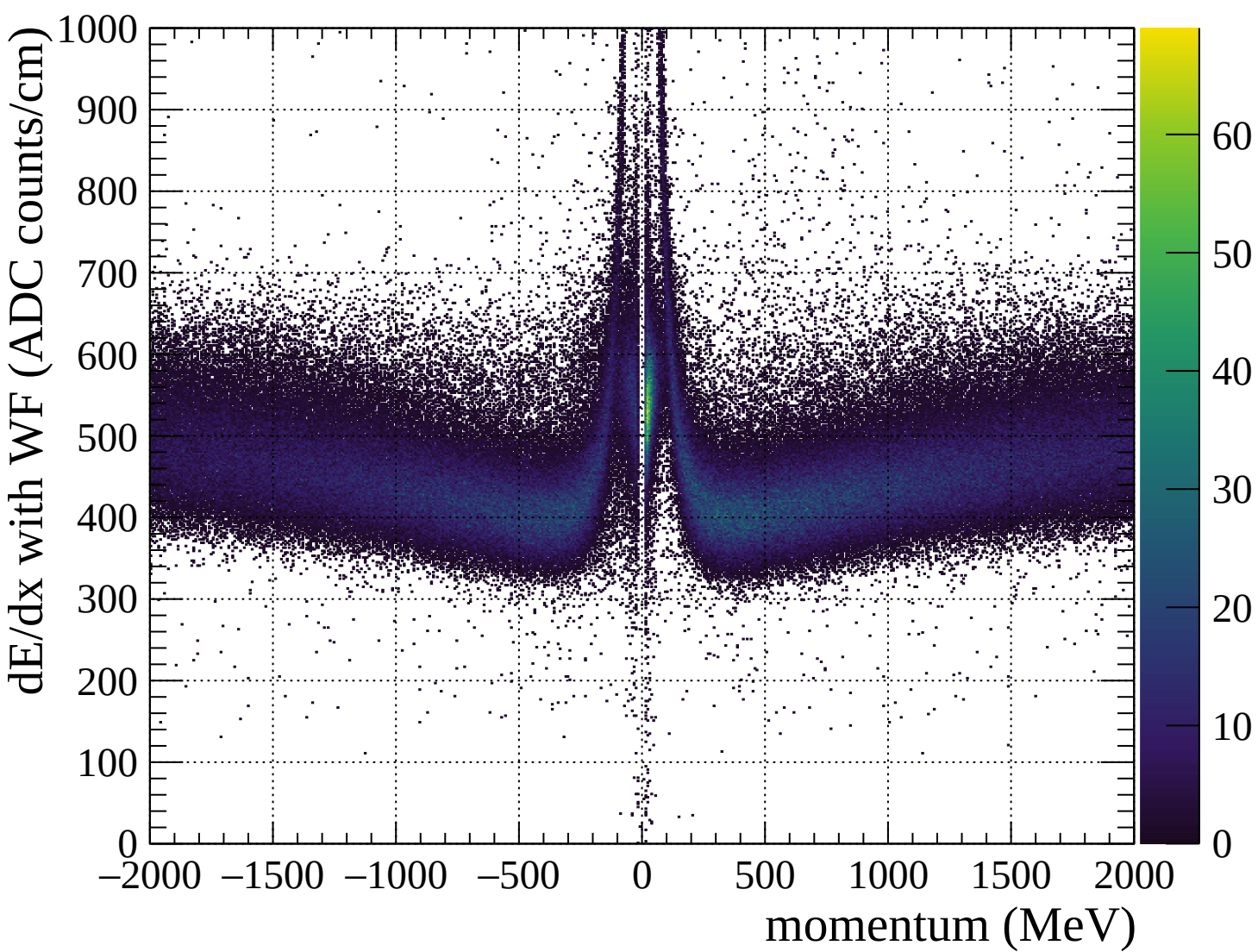


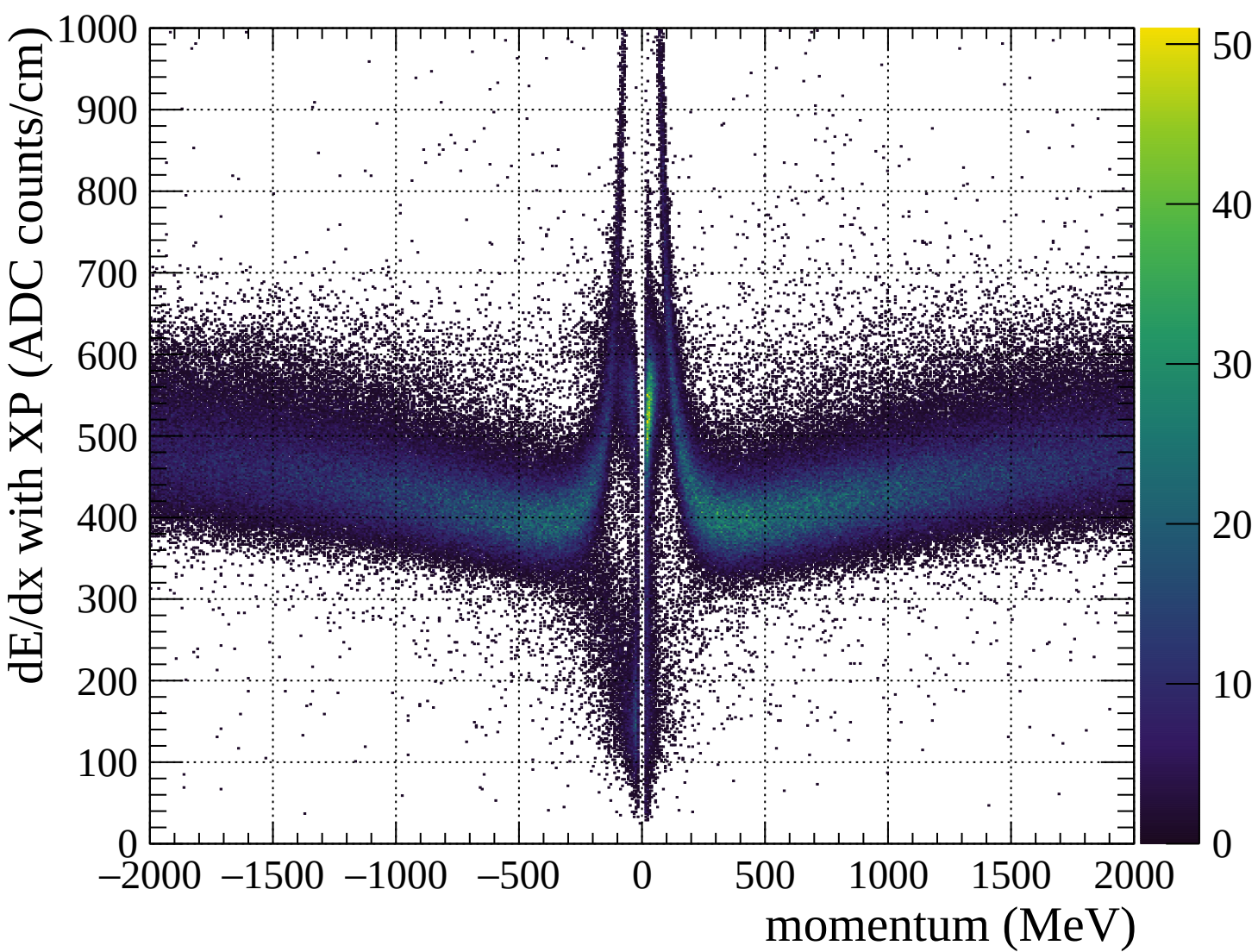


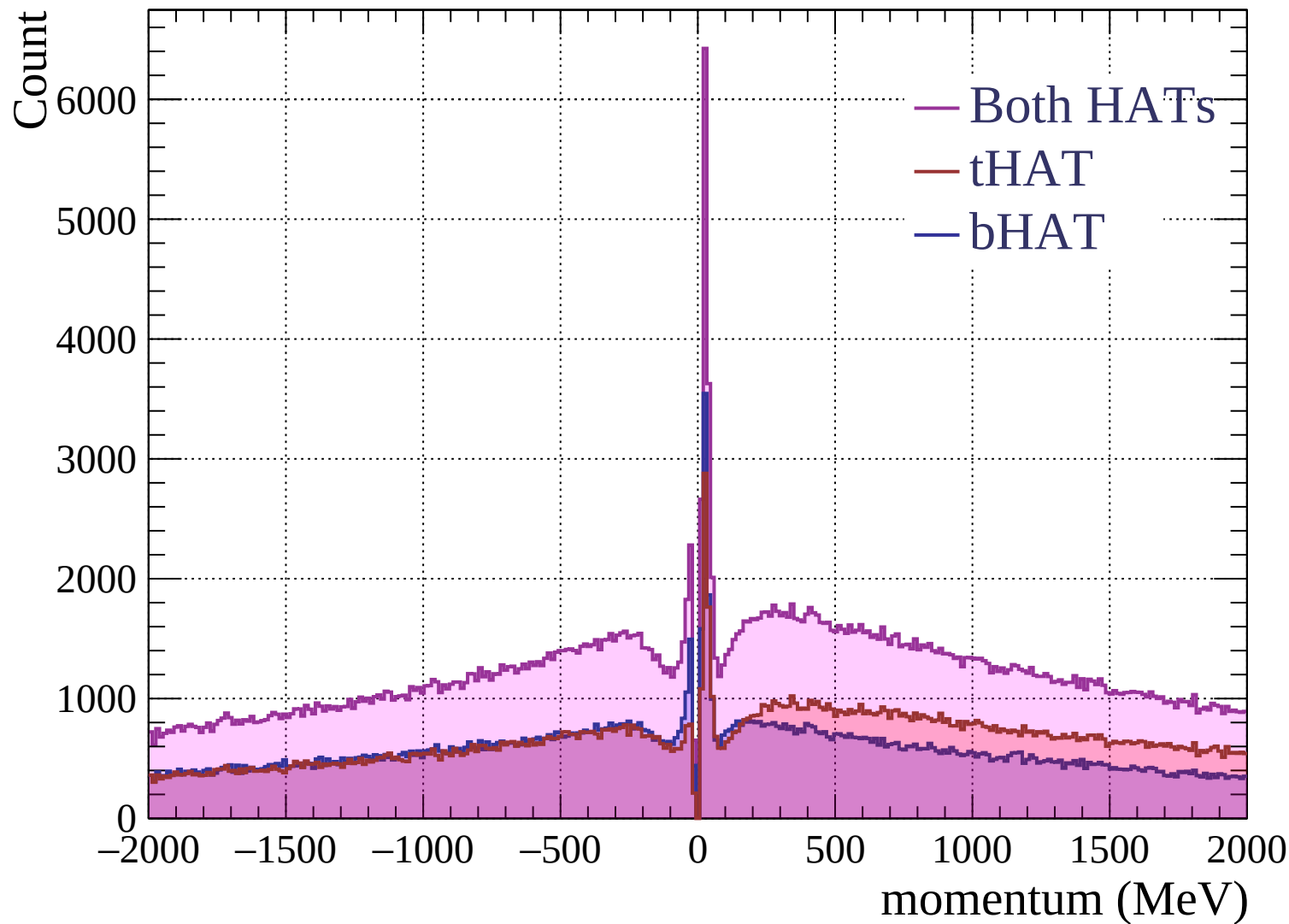


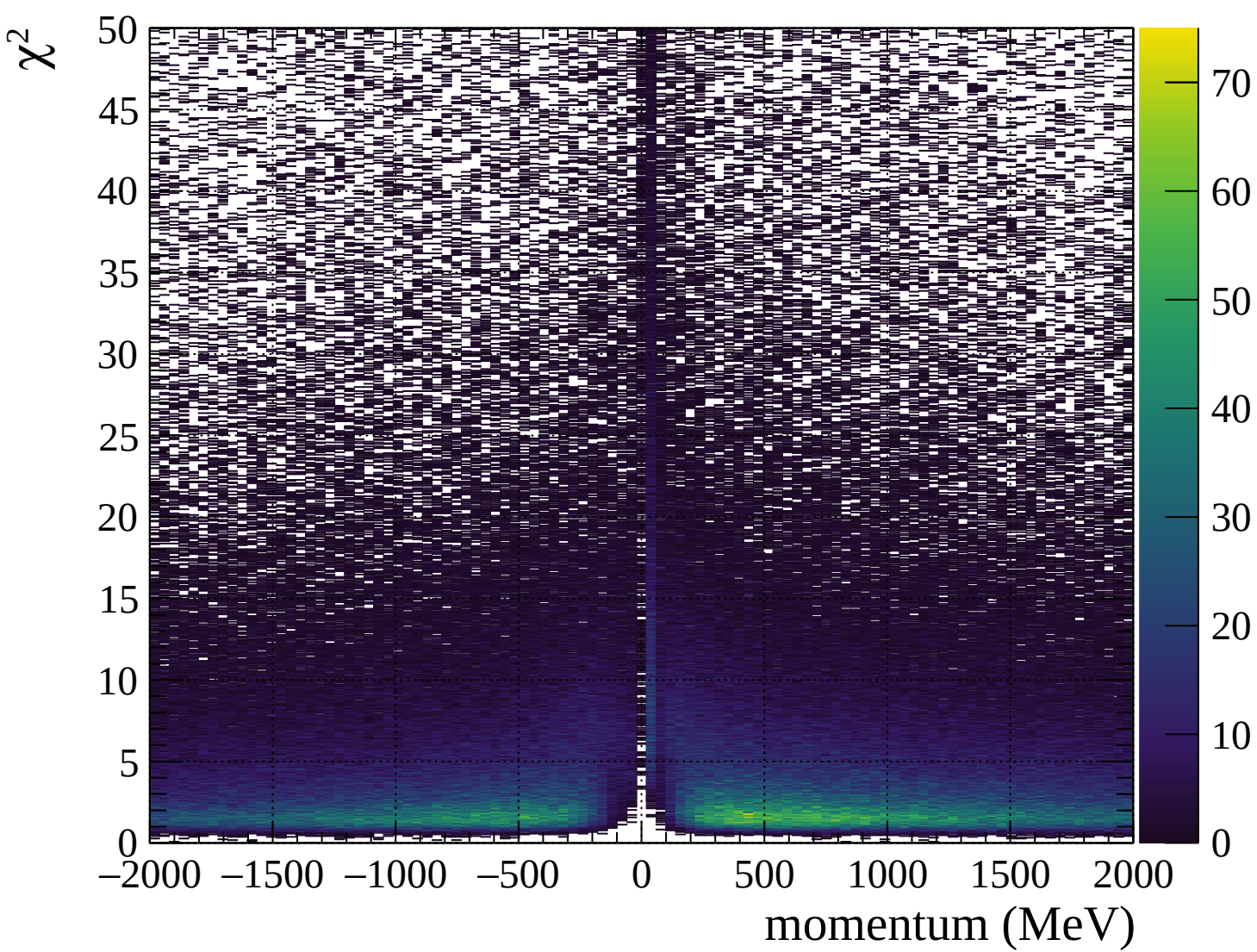




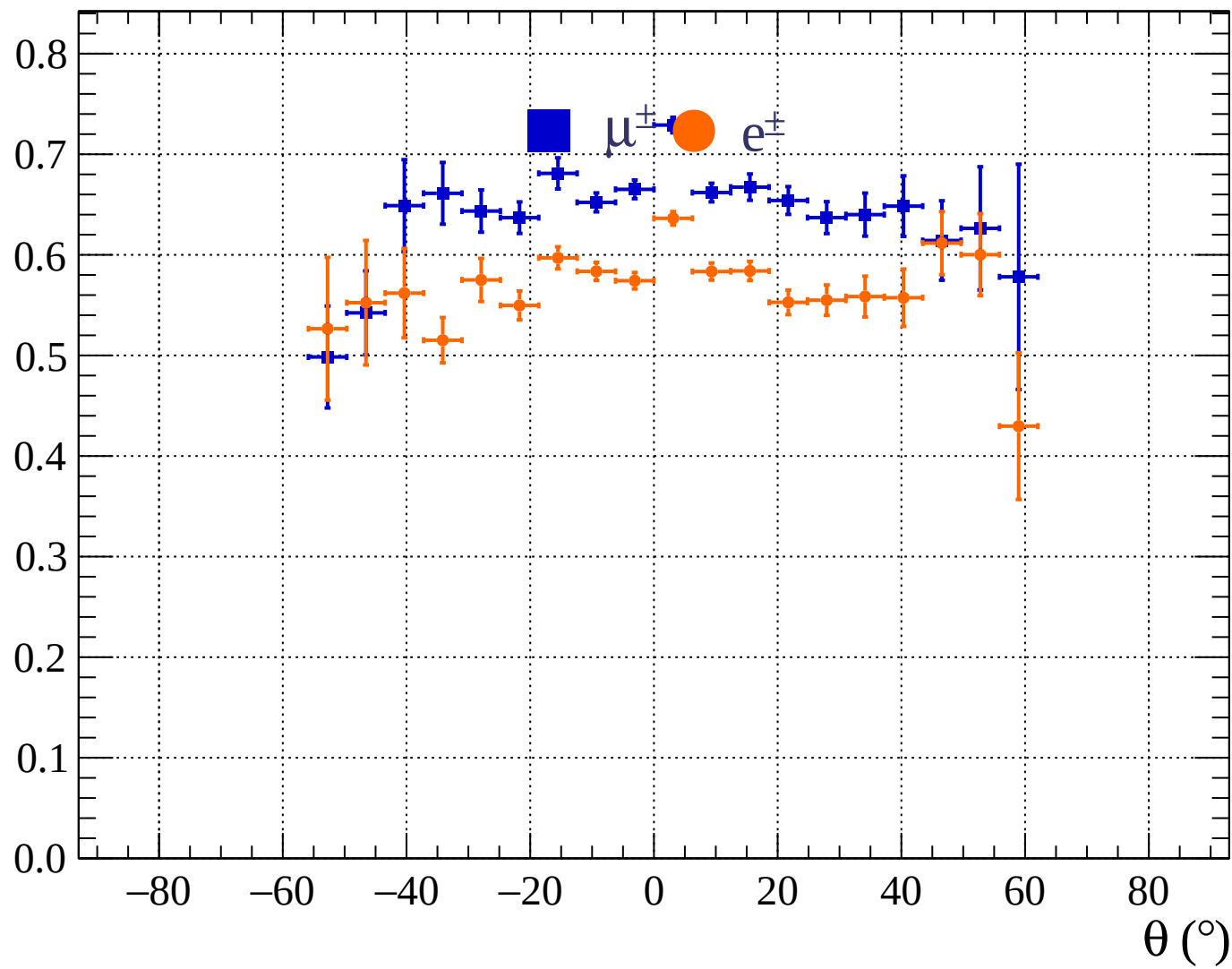




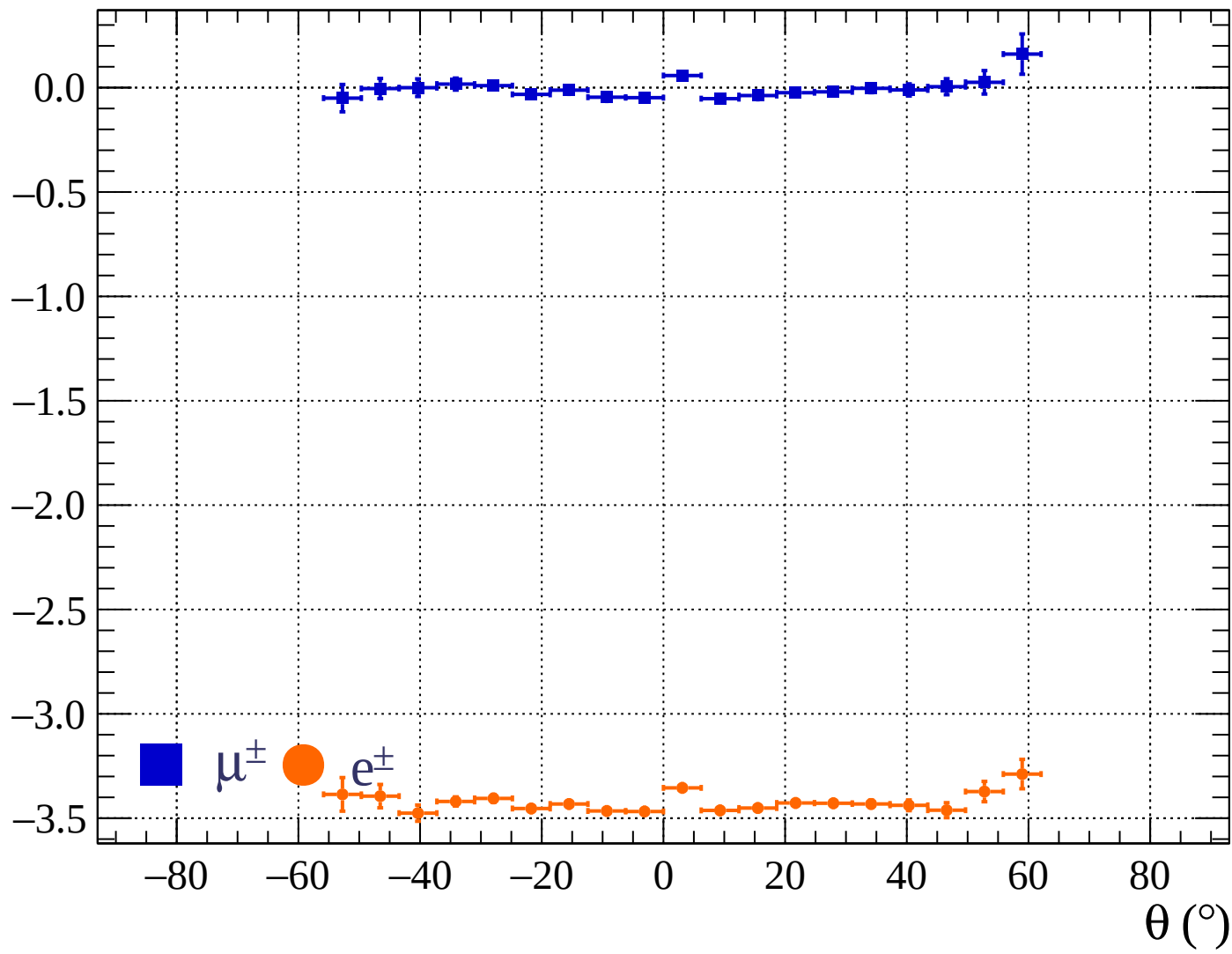




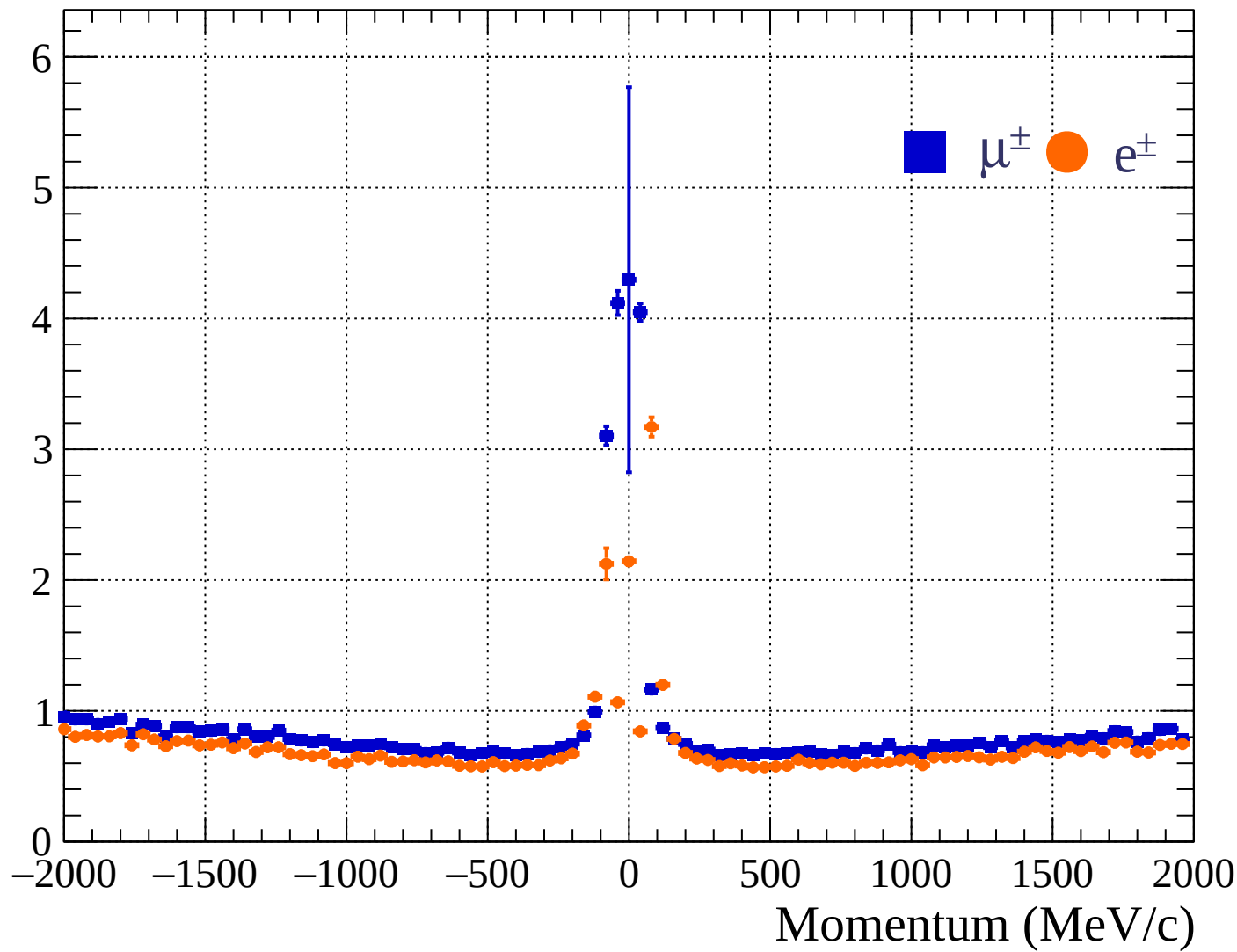
Pulls standard deviation



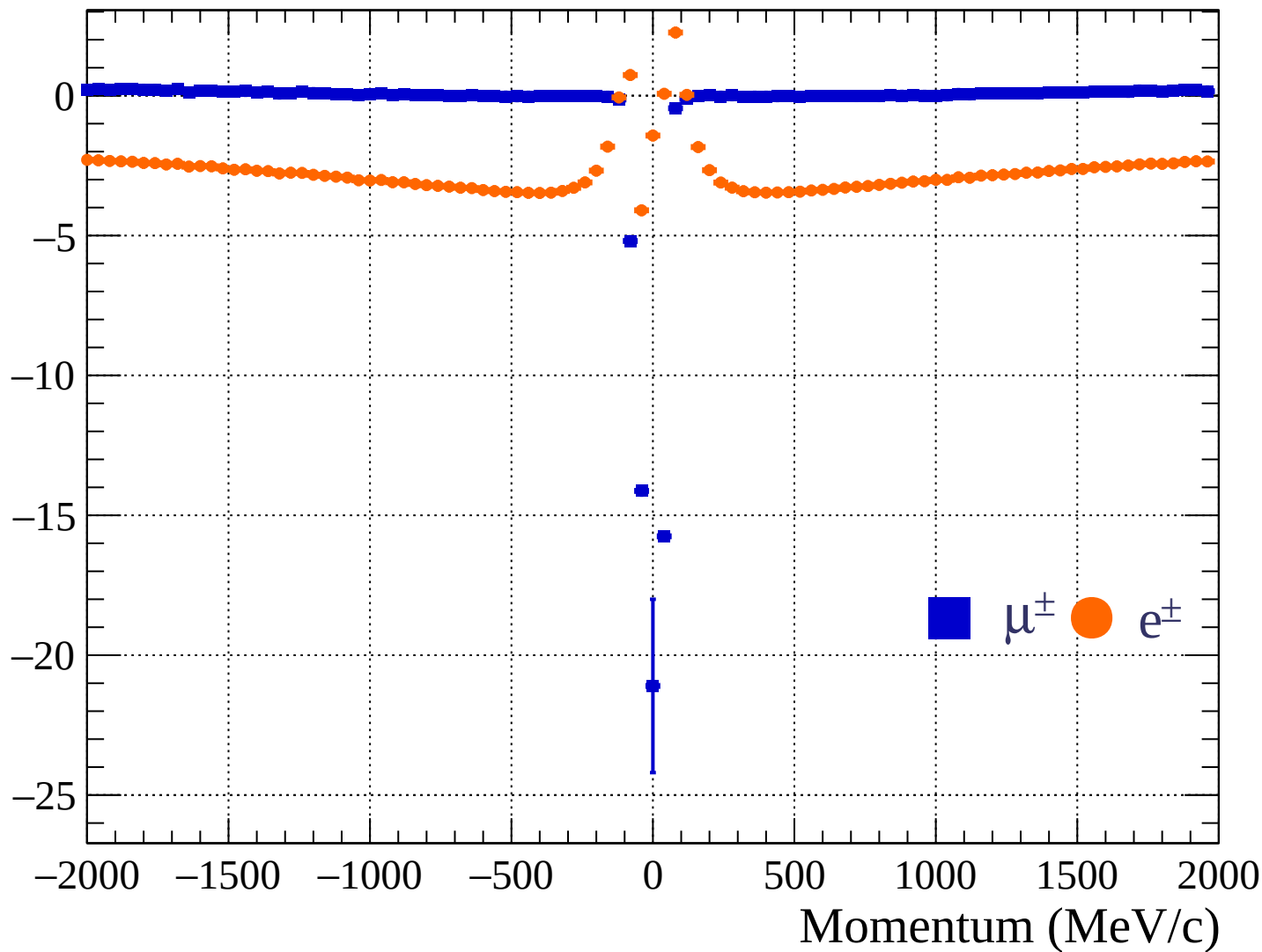
Pulls mean

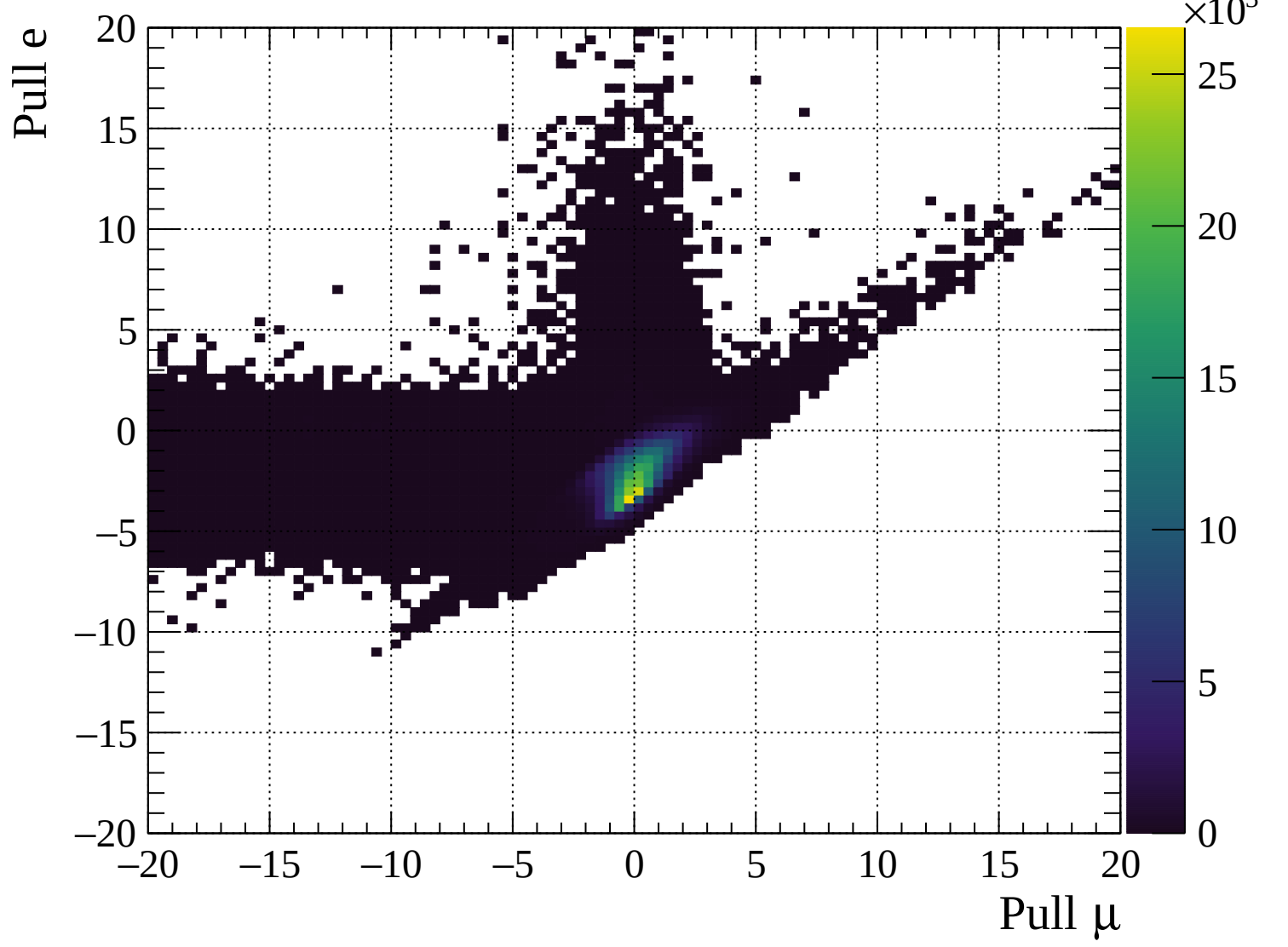


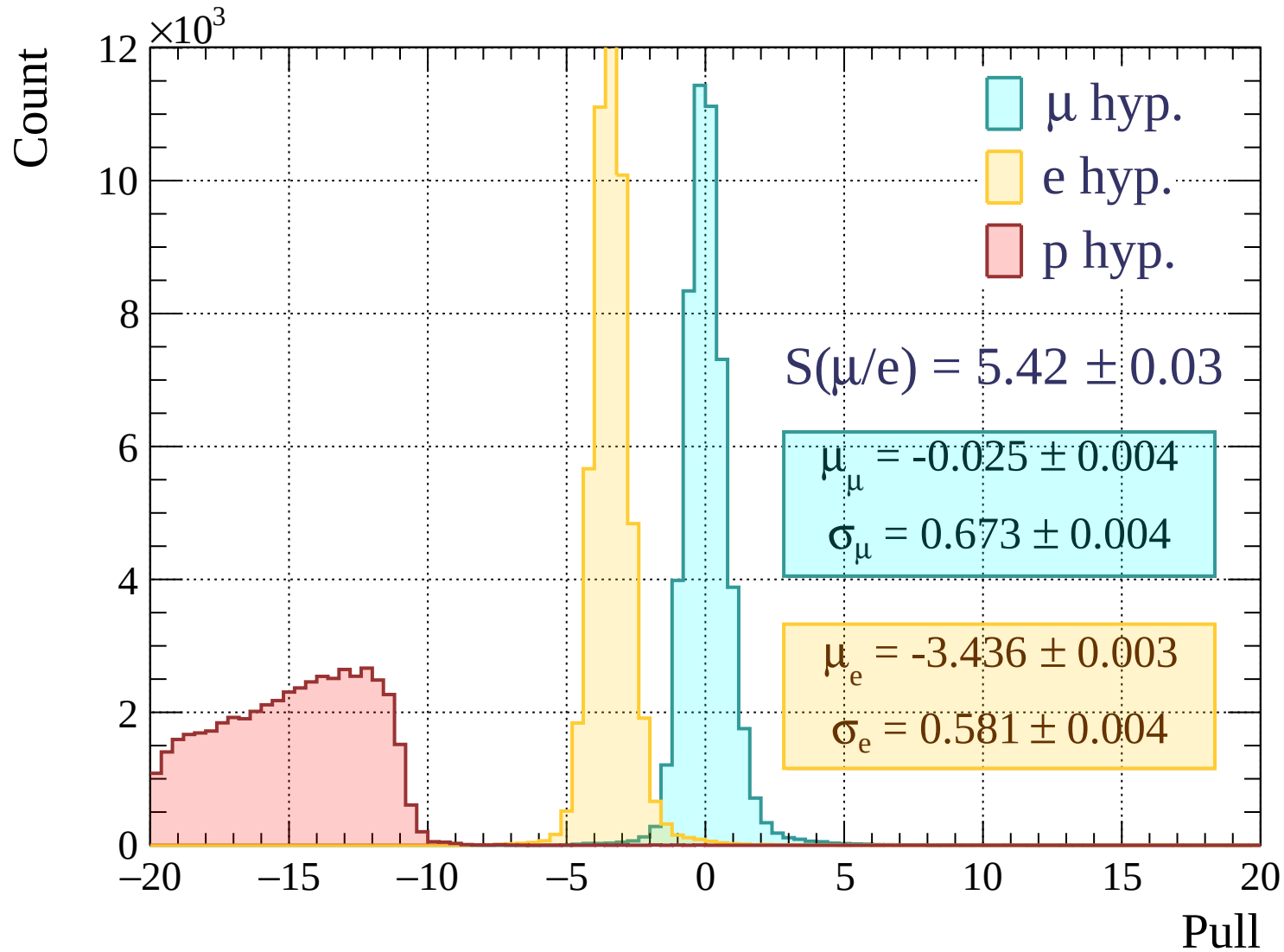
Pulls standard deviation

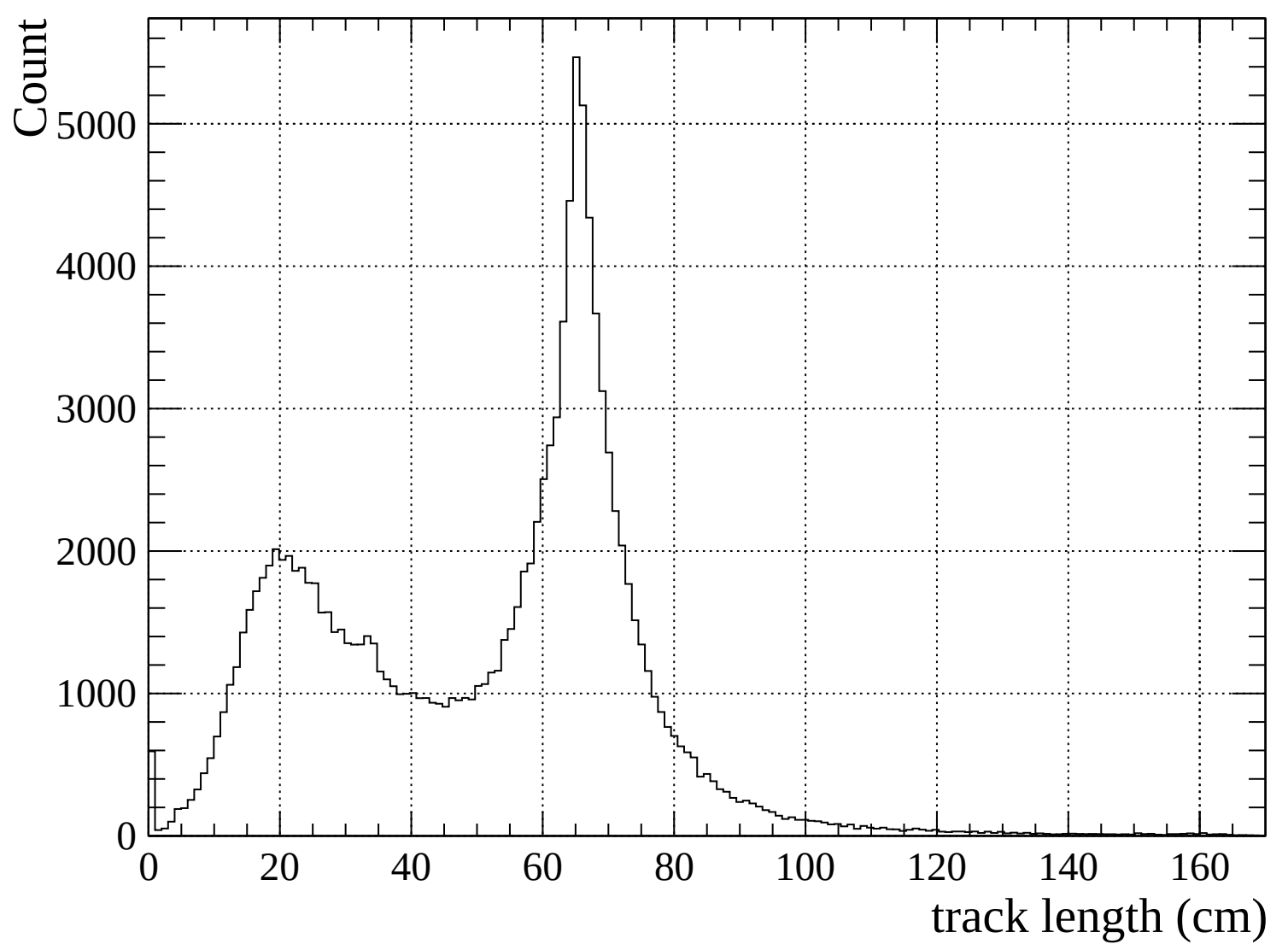


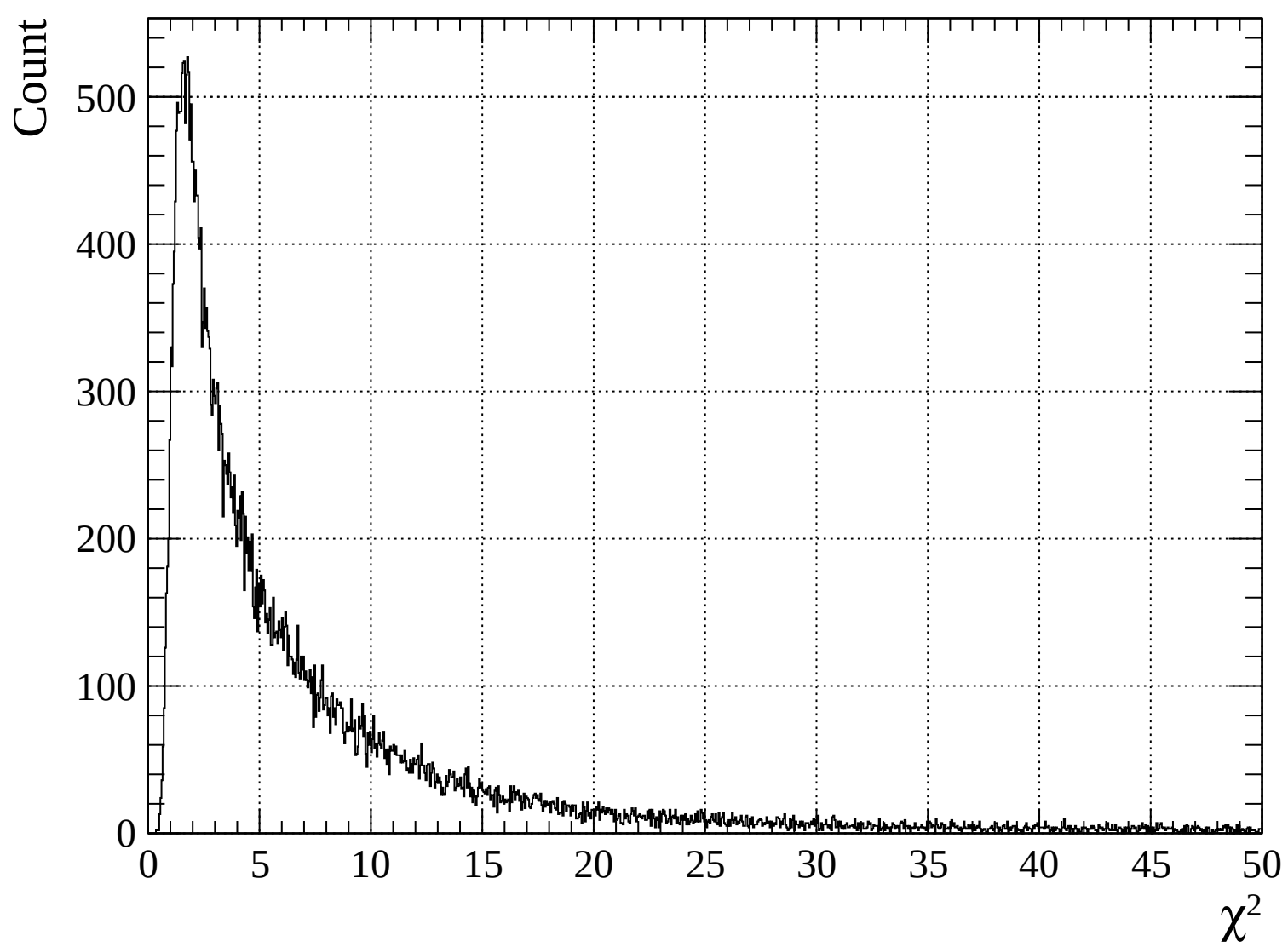
Pulls mean

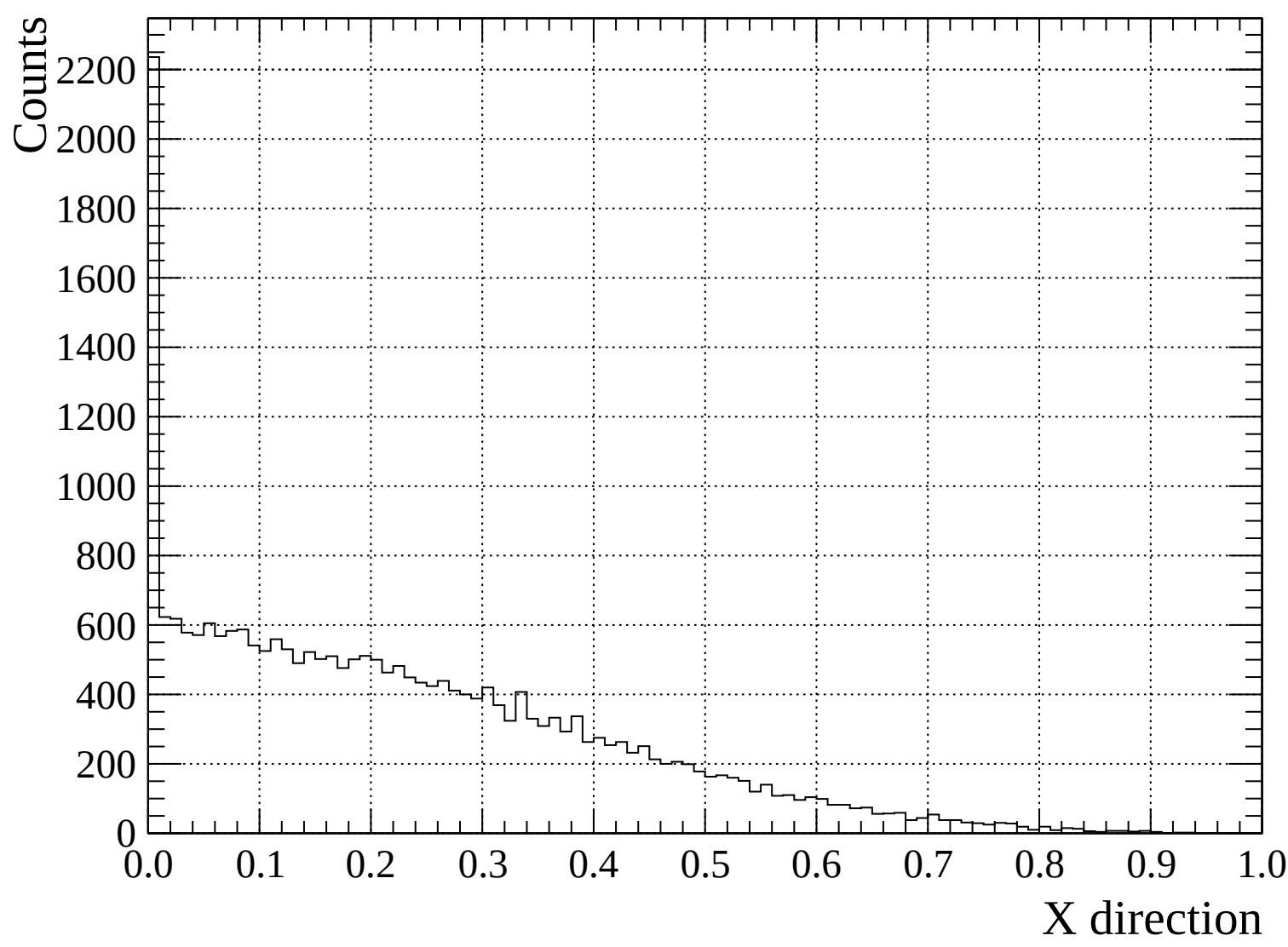


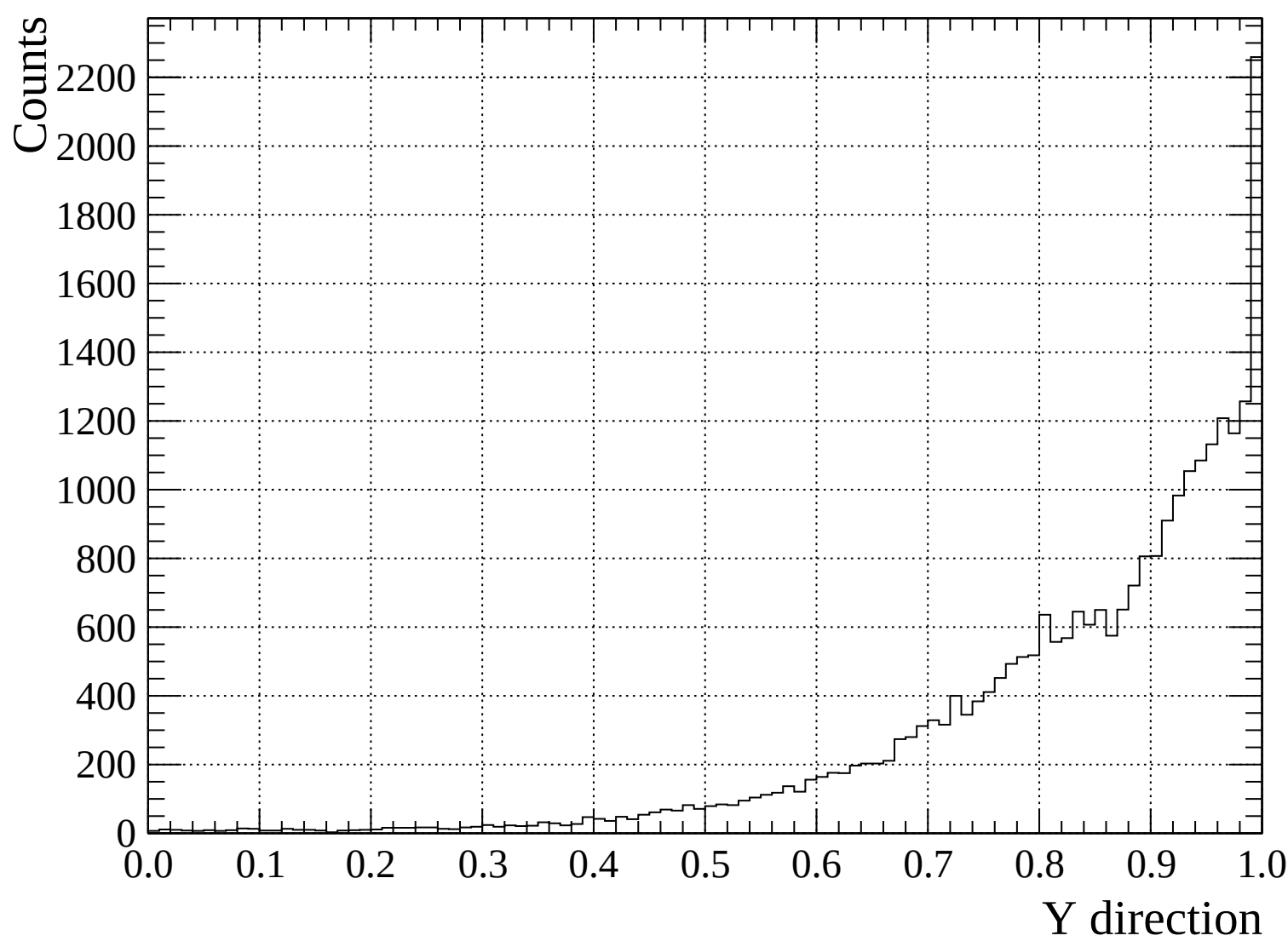


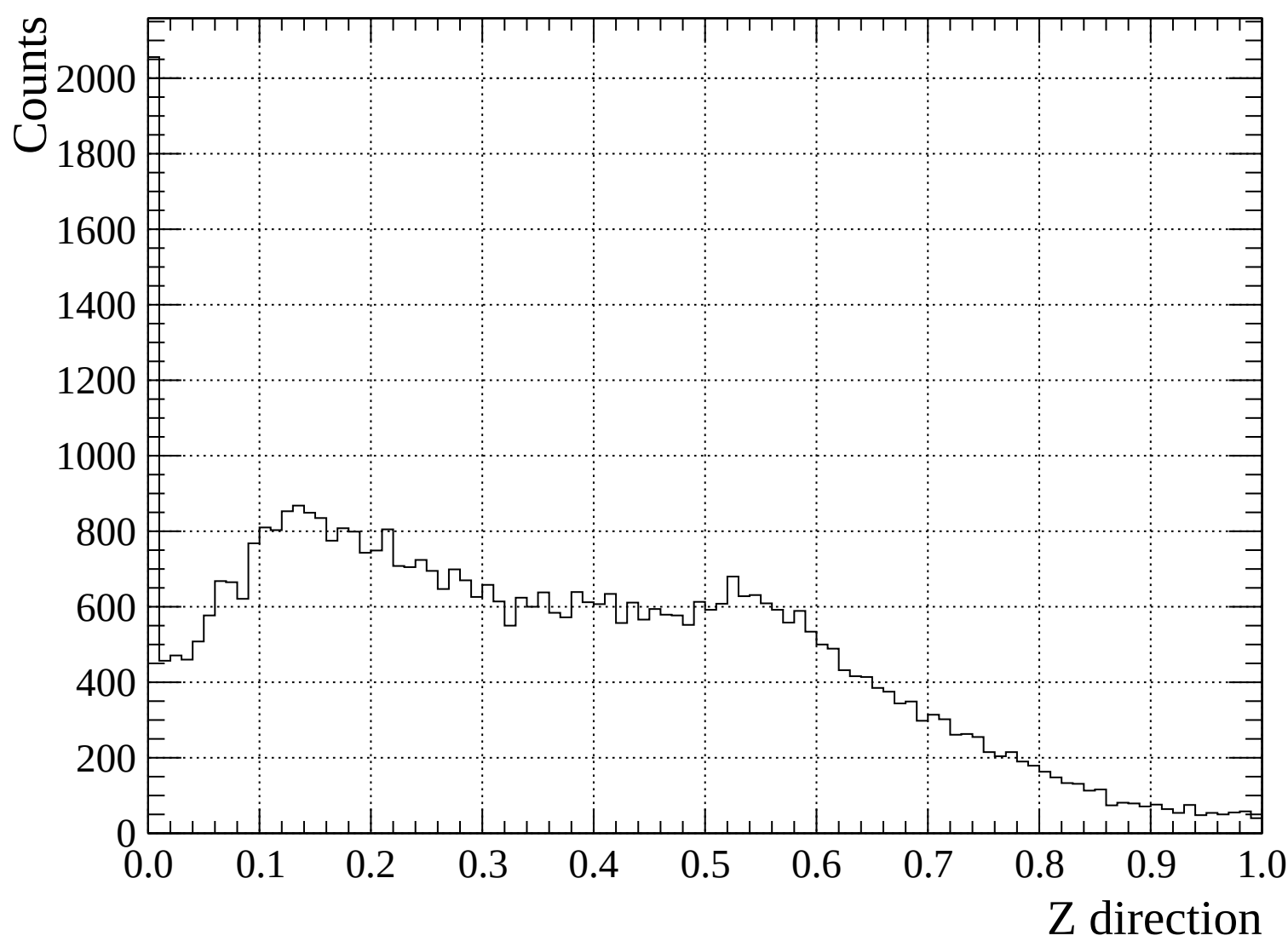


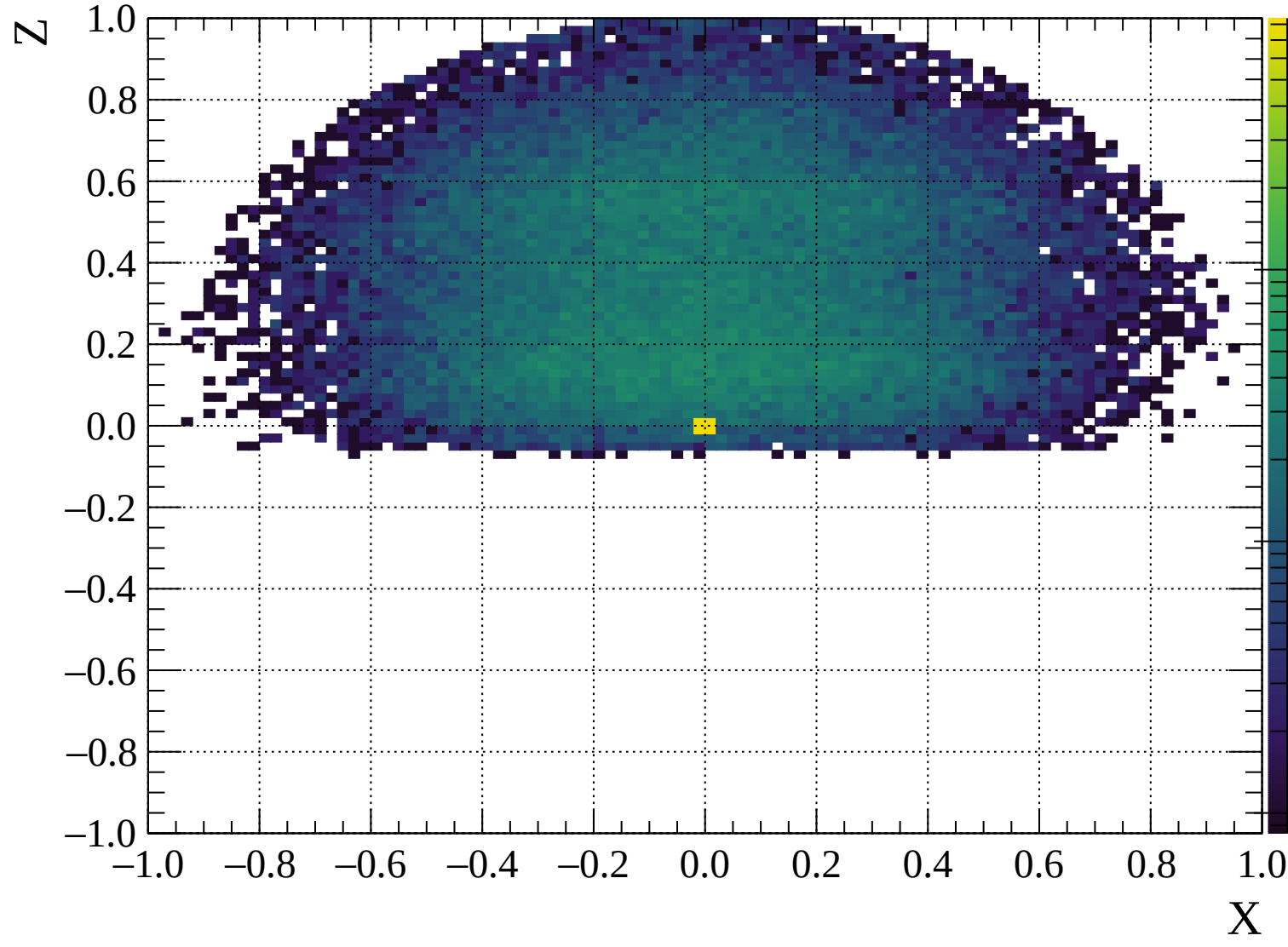


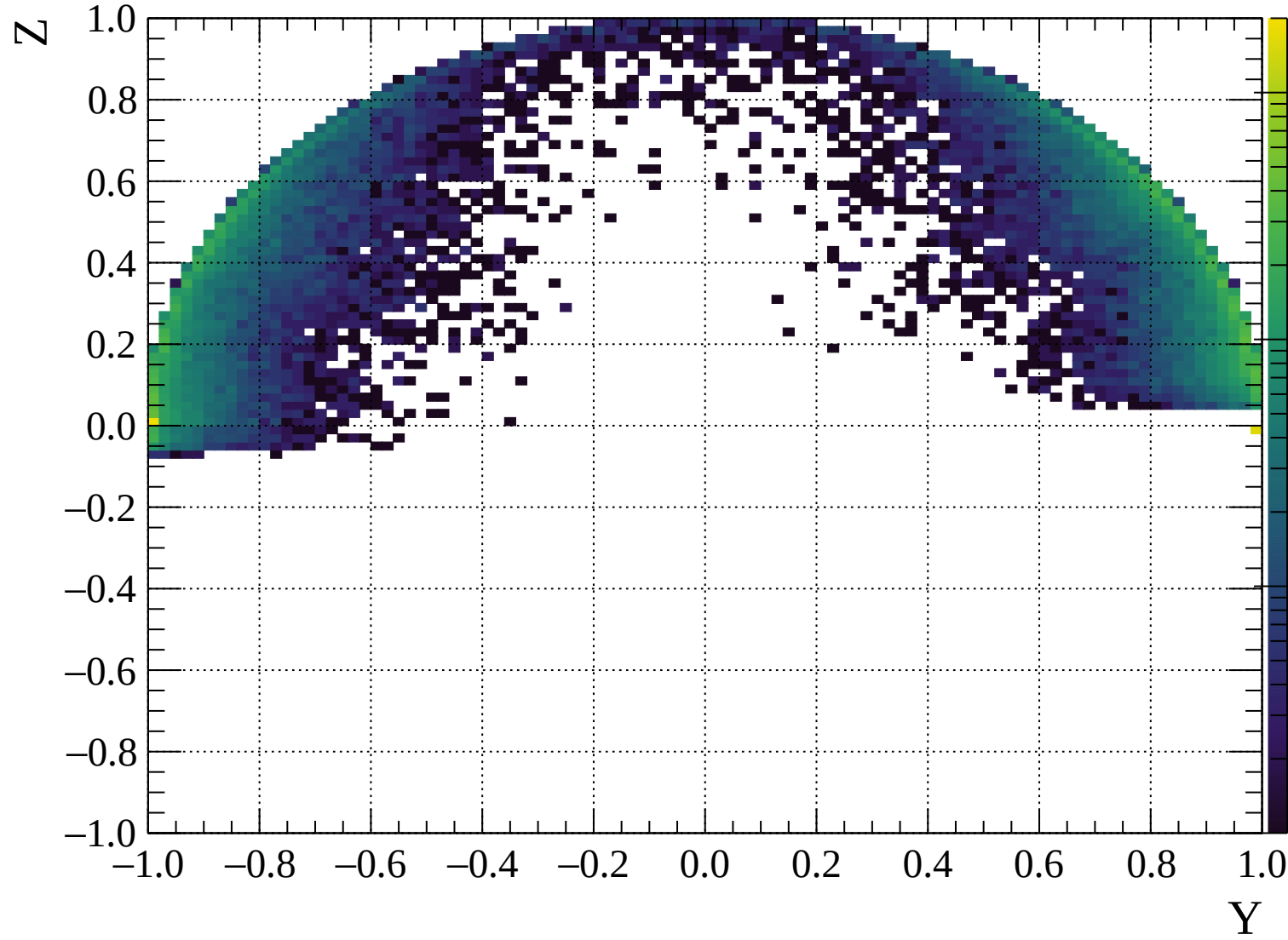


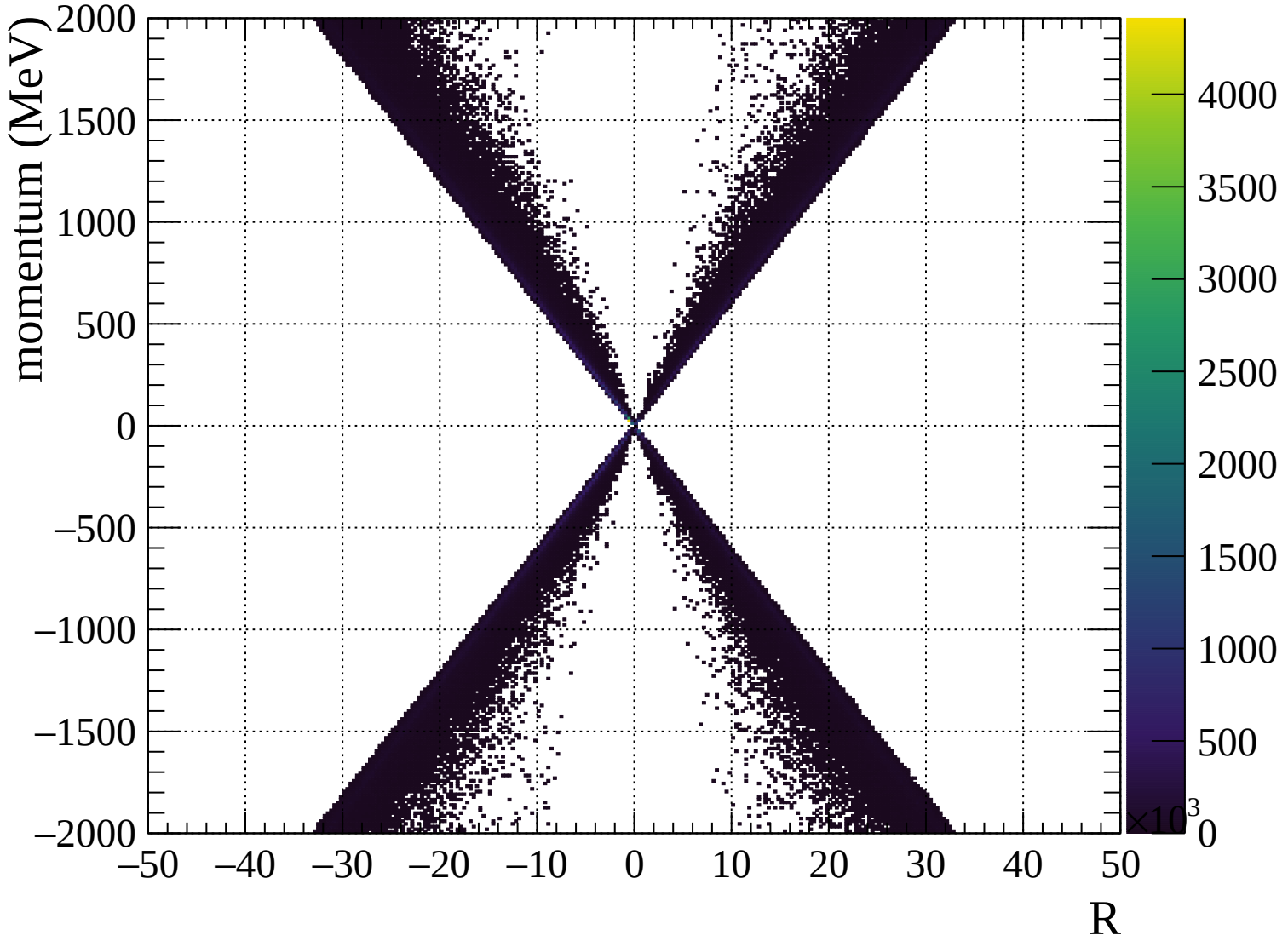


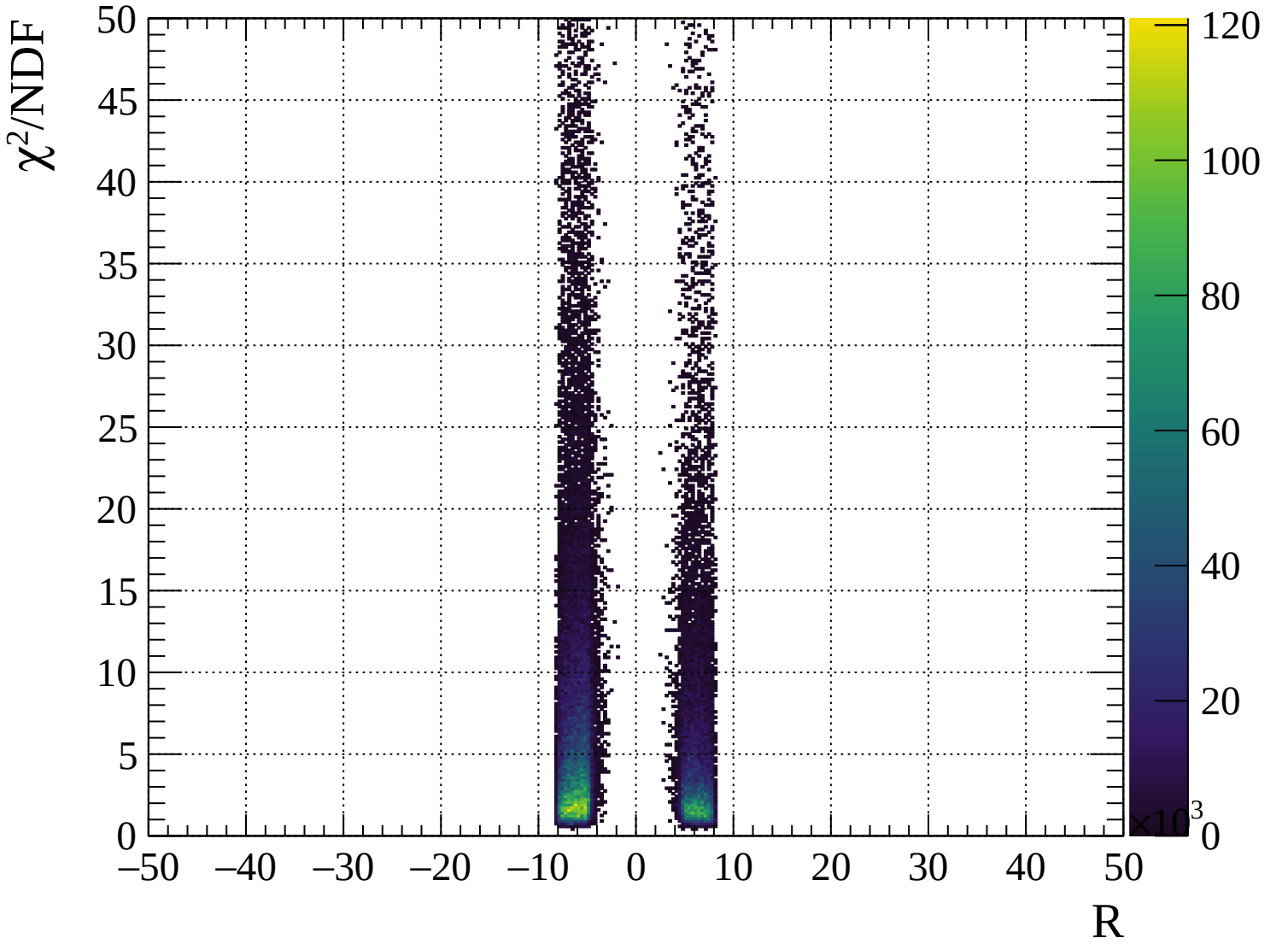




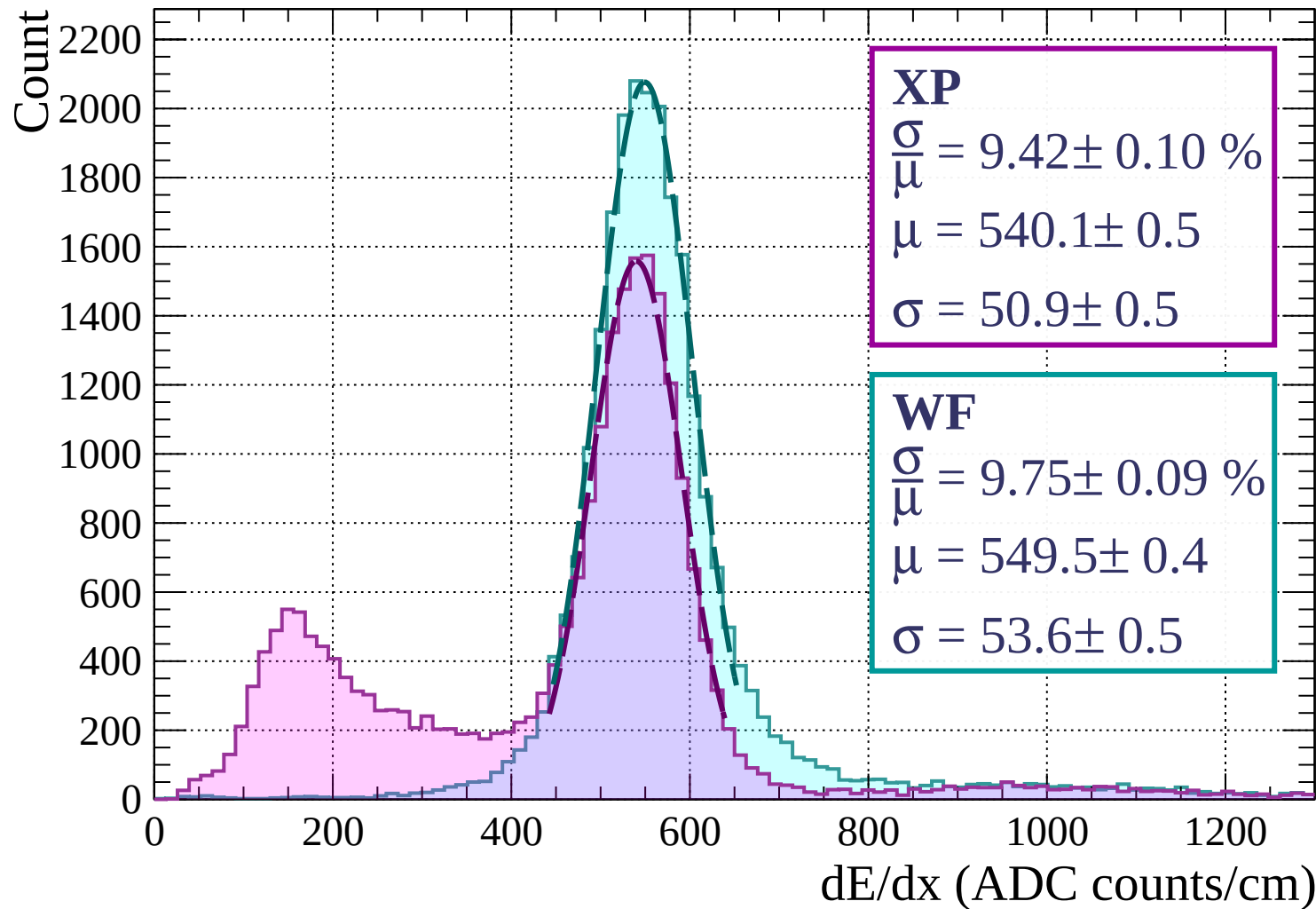








Energy loss | $0 < p < 80$



Energy loss | $80 < p < 160$

Count

1000

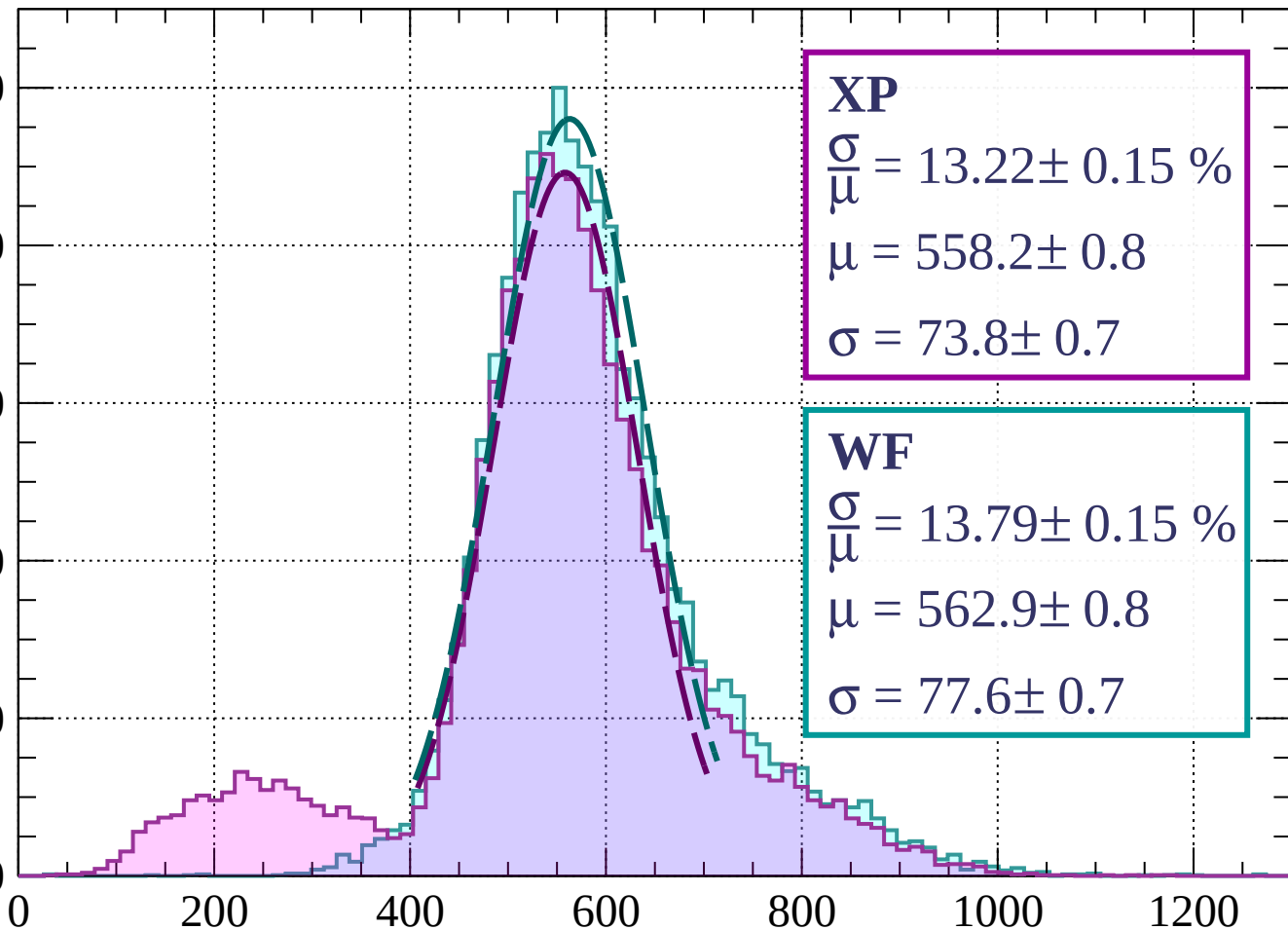
800

600

400

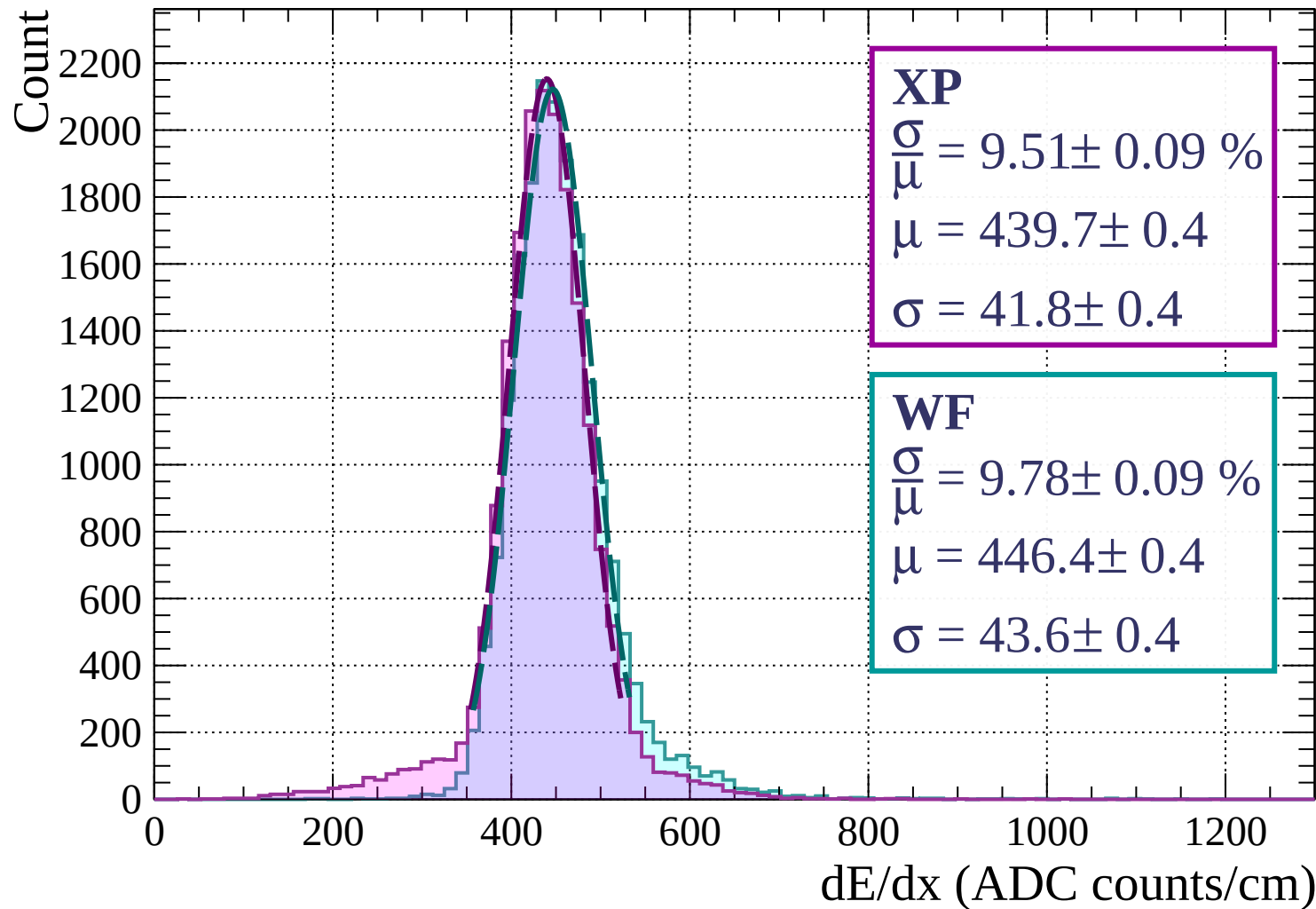
200

0

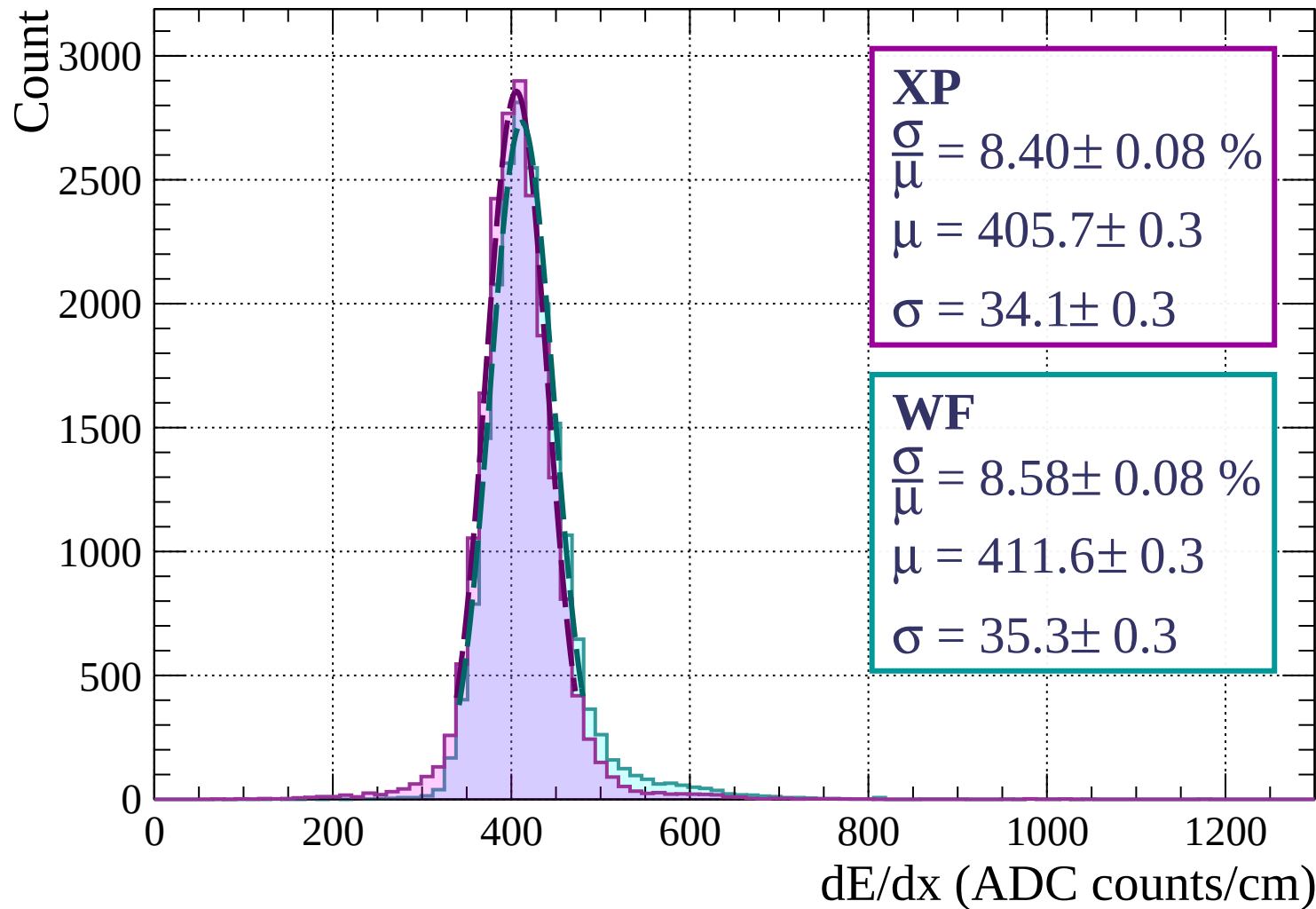


dE/dx (ADC counts/cm)

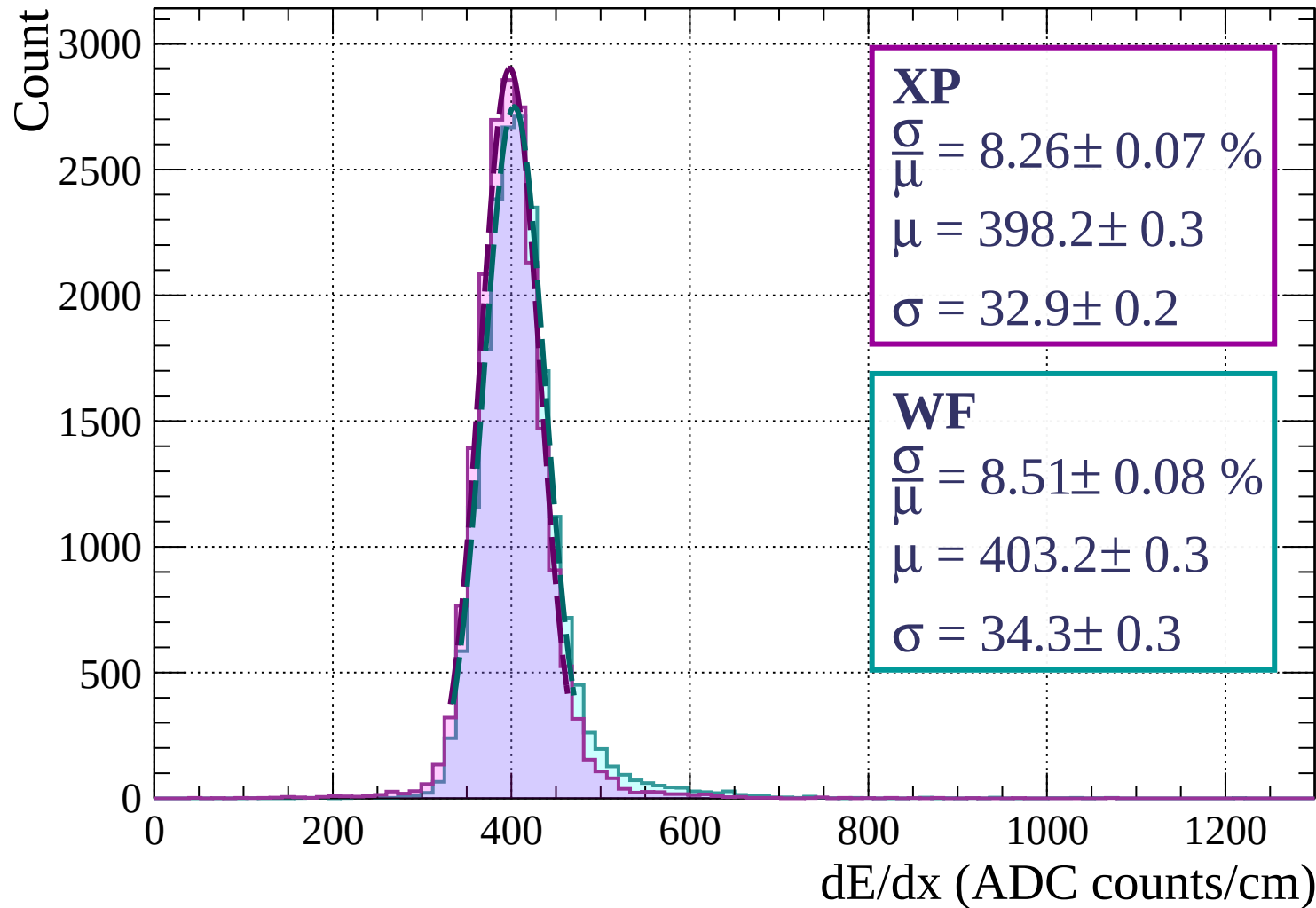
Energy loss | $160 < p < 240$



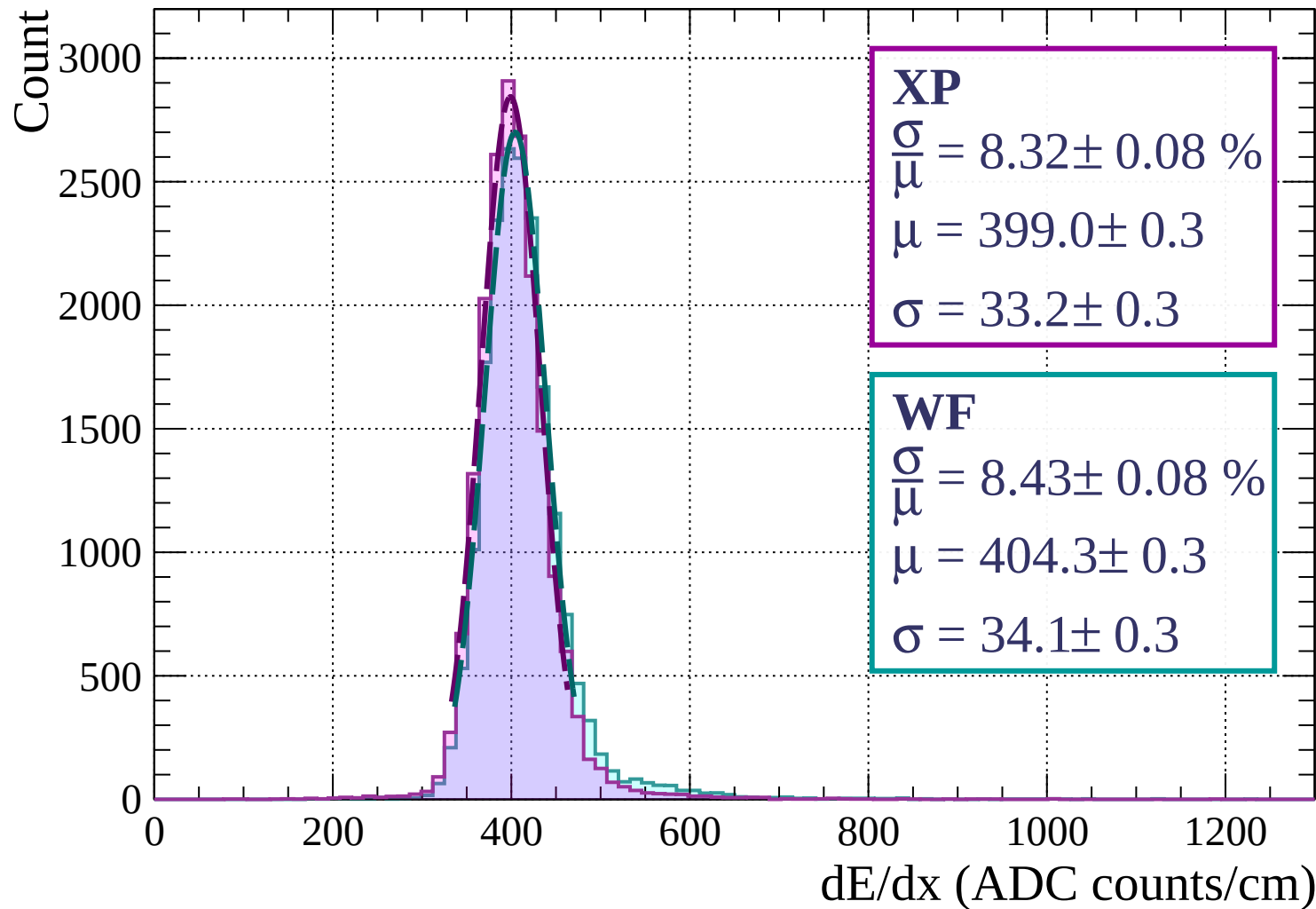
Energy loss | $240 < p < 320$



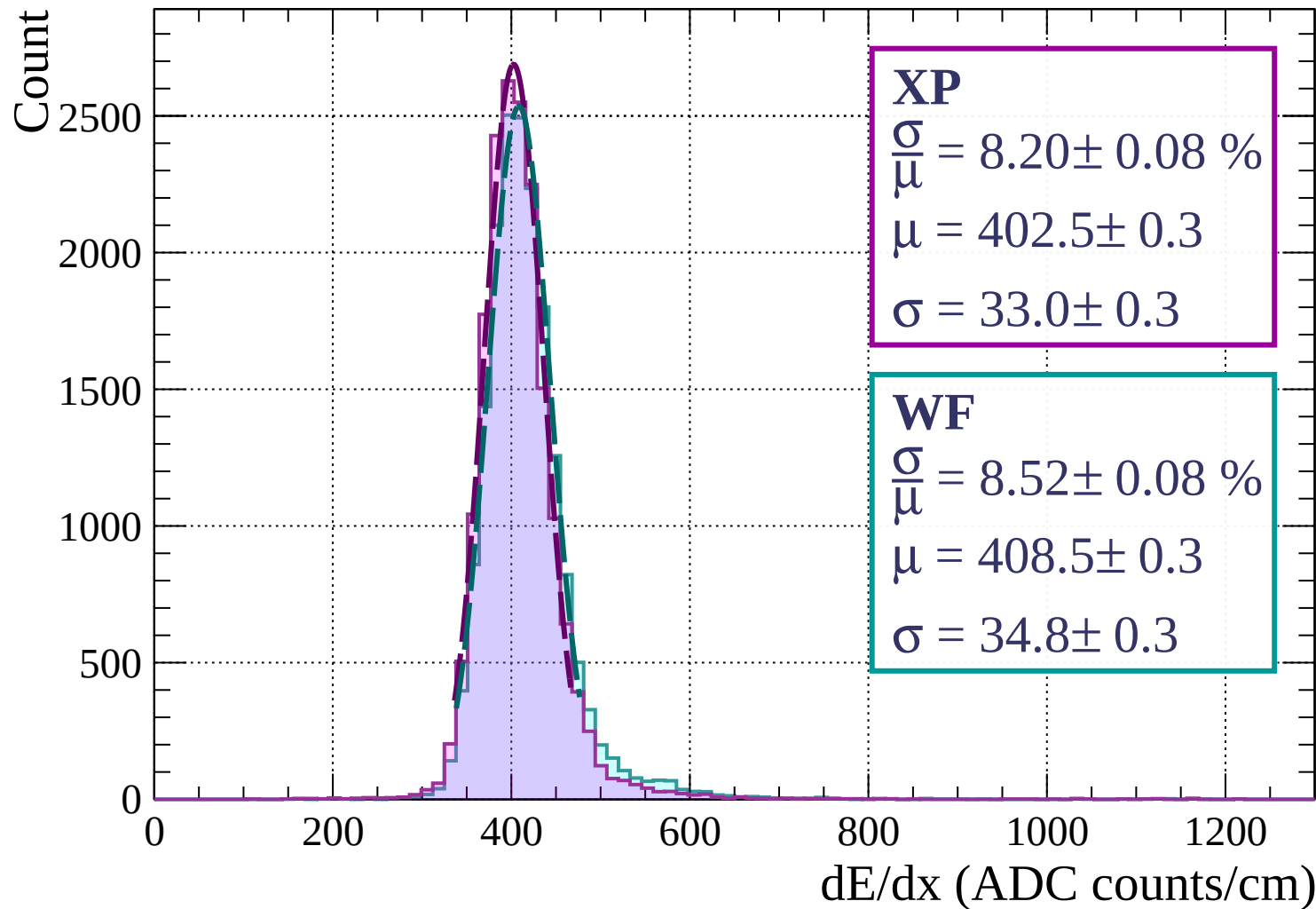
Energy loss | $320 < p < 400$



Energy loss | $400 < p < 480$



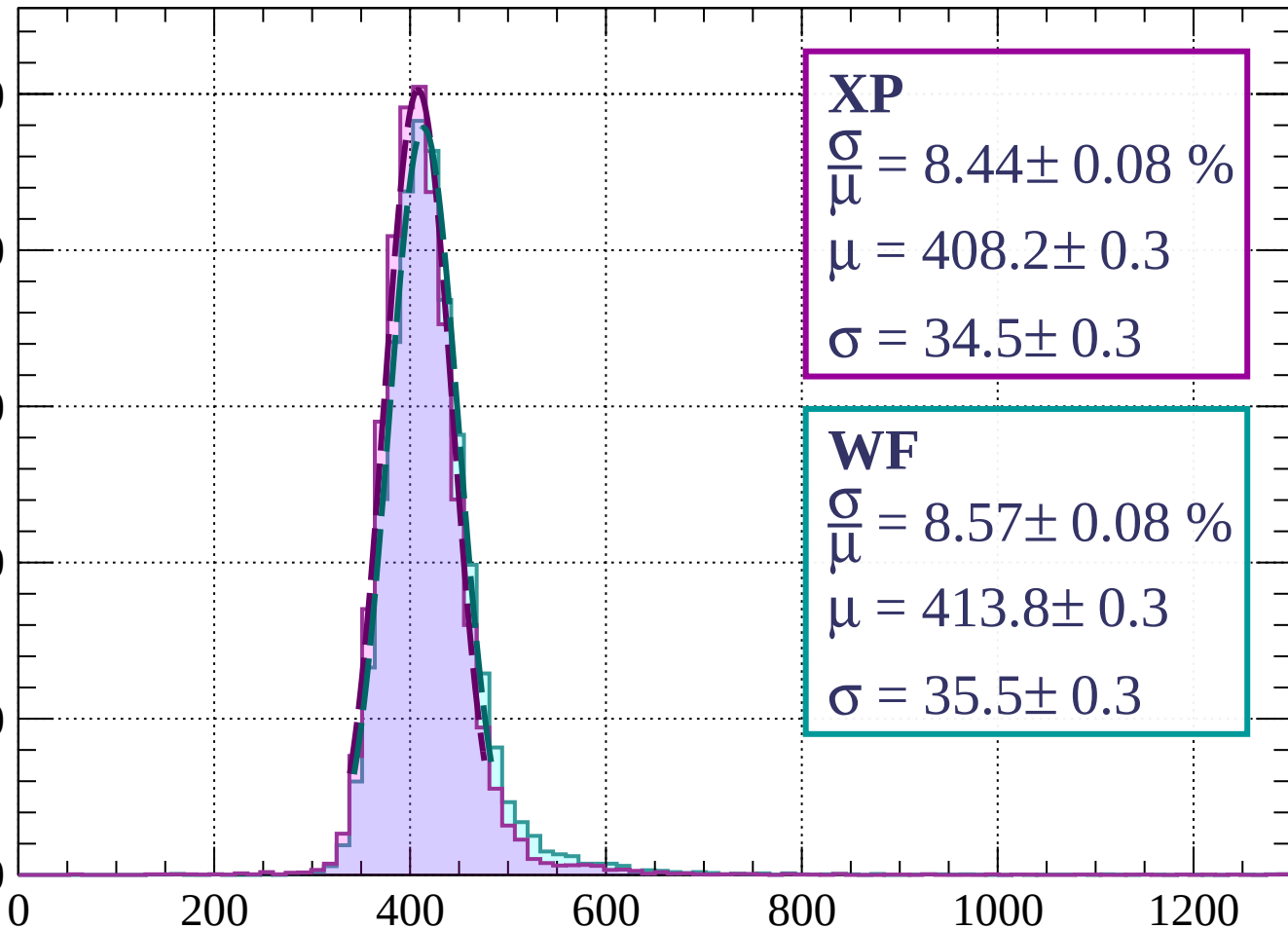
Energy loss | $480 < p < 560$



Energy loss | $560 < p < 640$

Count

2500
2000
1500
1000
500
0



XP

$$\frac{\sigma}{\mu} = 8.44 \pm 0.08 \%$$

$$\mu = 408.2 \pm 0.3$$

$$\sigma = 34.5 \pm 0.3$$

WF

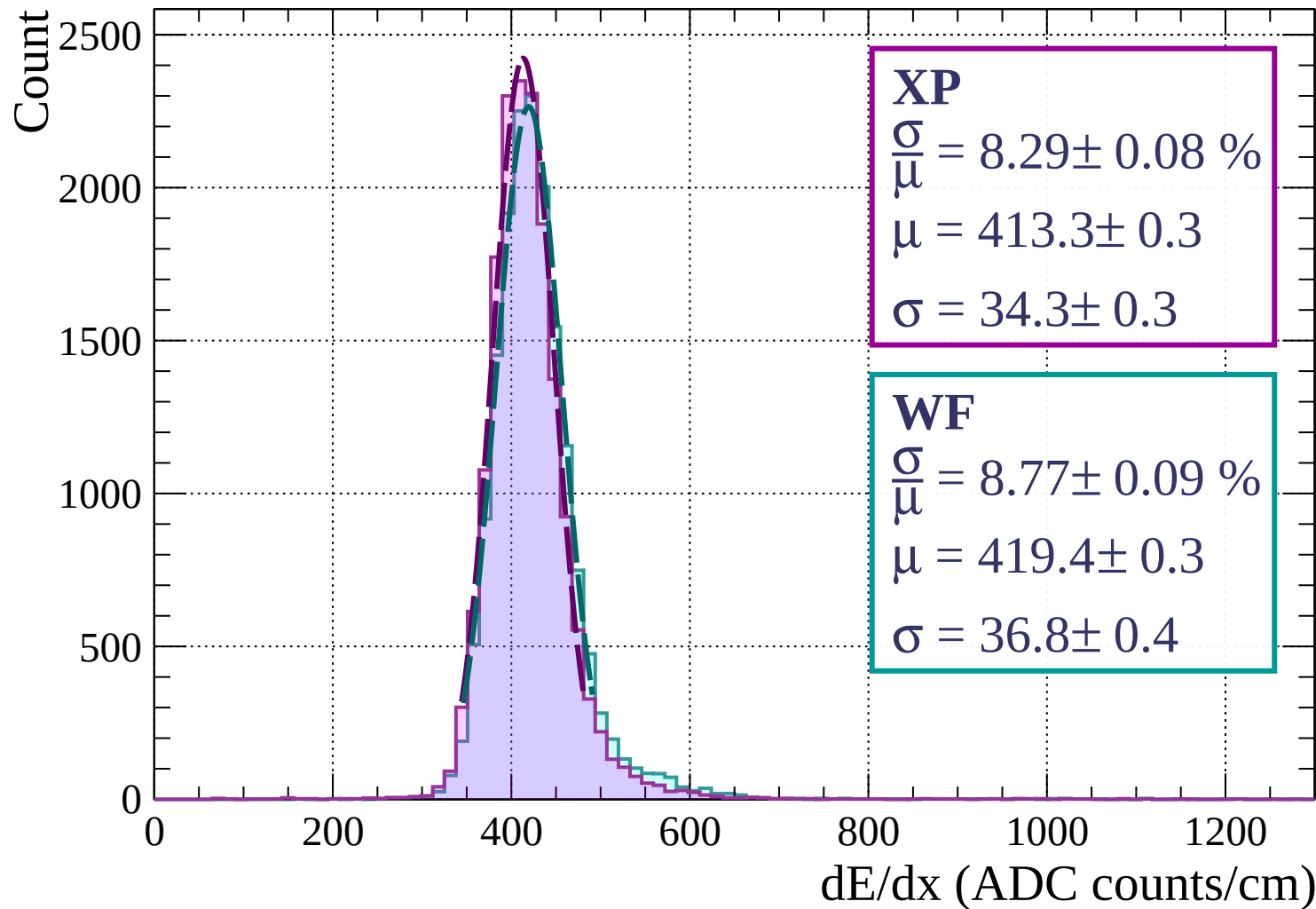
$$\frac{\sigma}{\mu} = 8.57 \pm 0.08 \%$$

$$\mu = 413.8 \pm 0.3$$

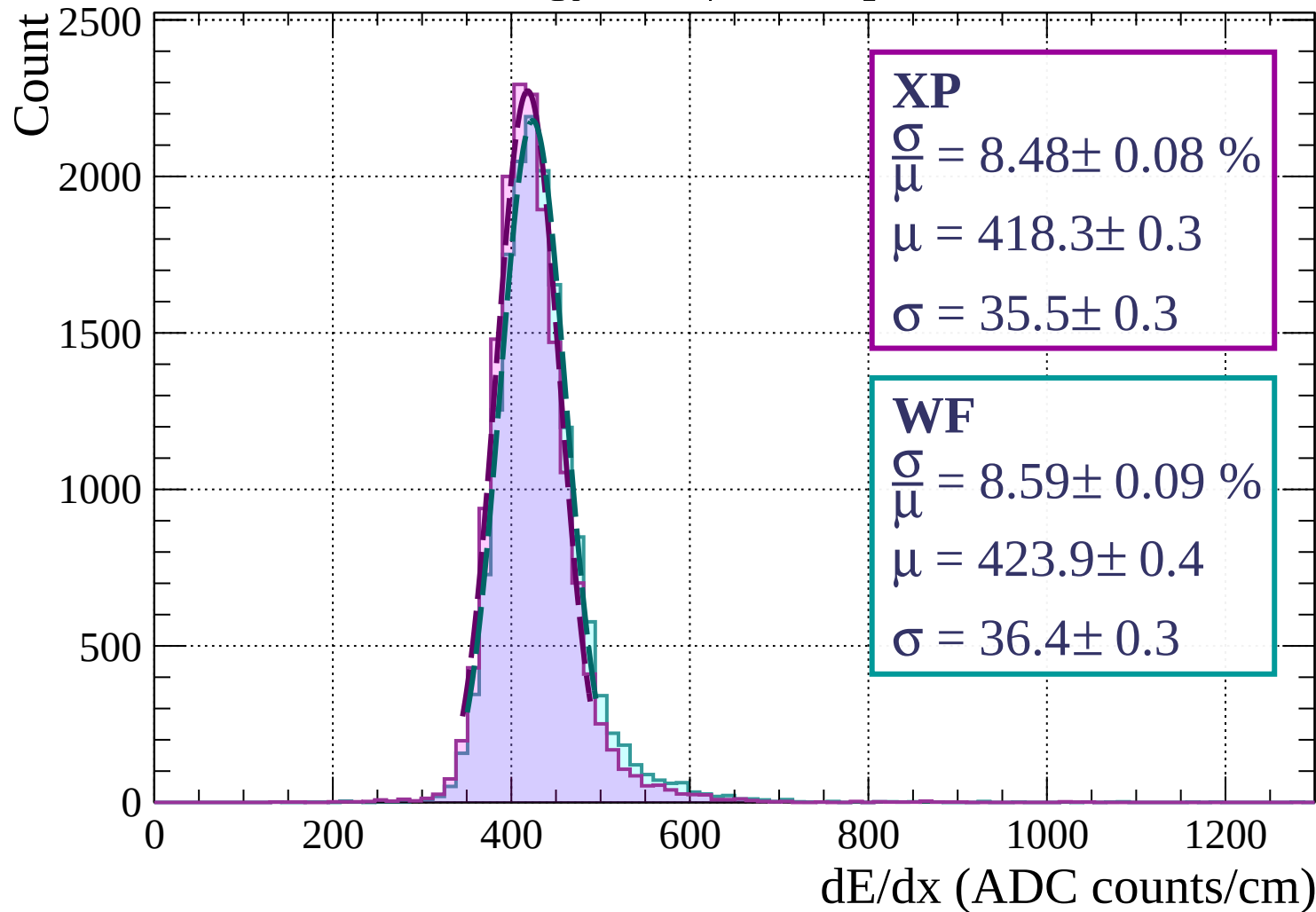
$$\sigma = 35.5 \pm 0.3$$

dE/dx (ADC counts/cm)

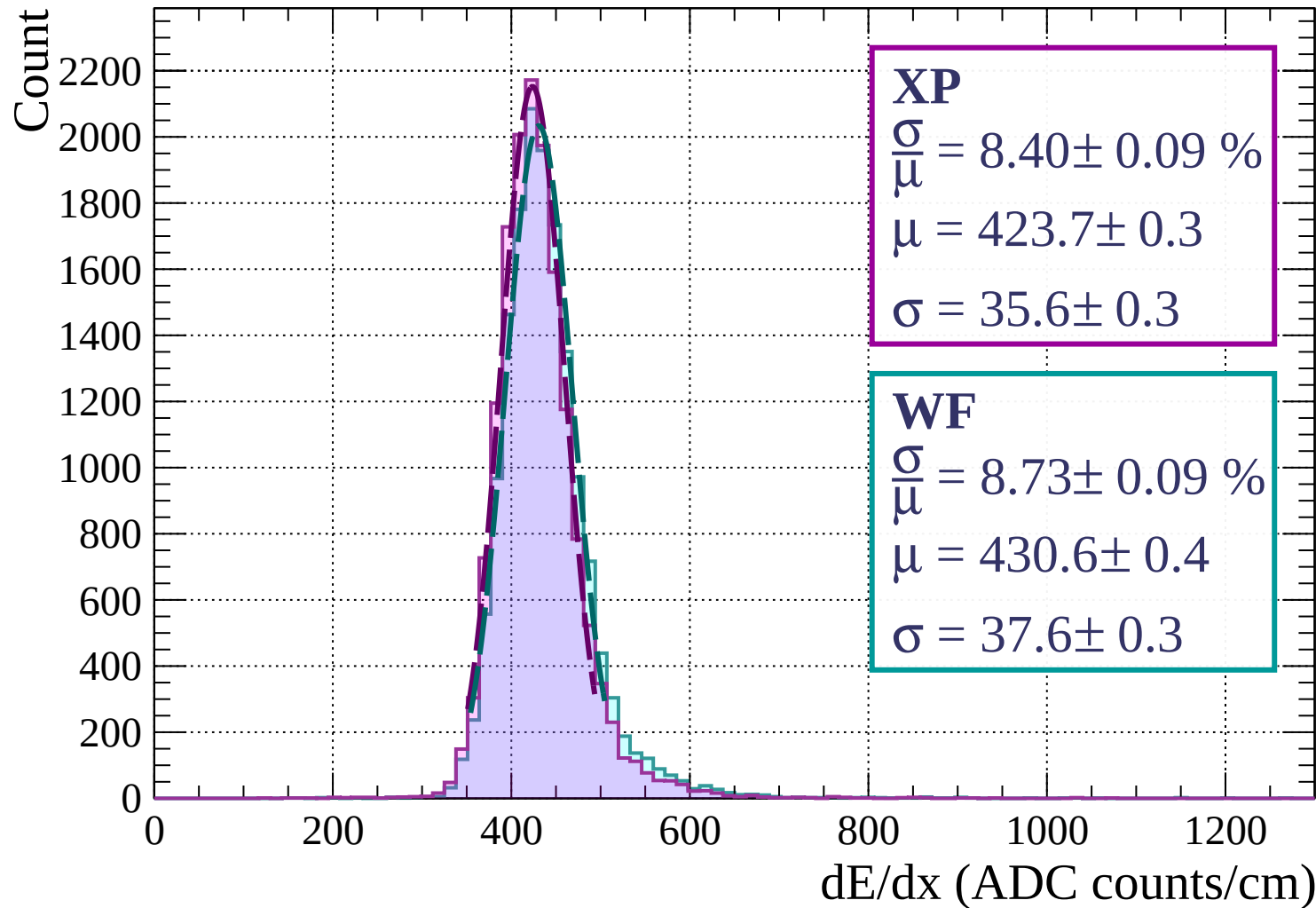
Energy loss | $640 < p < 720$



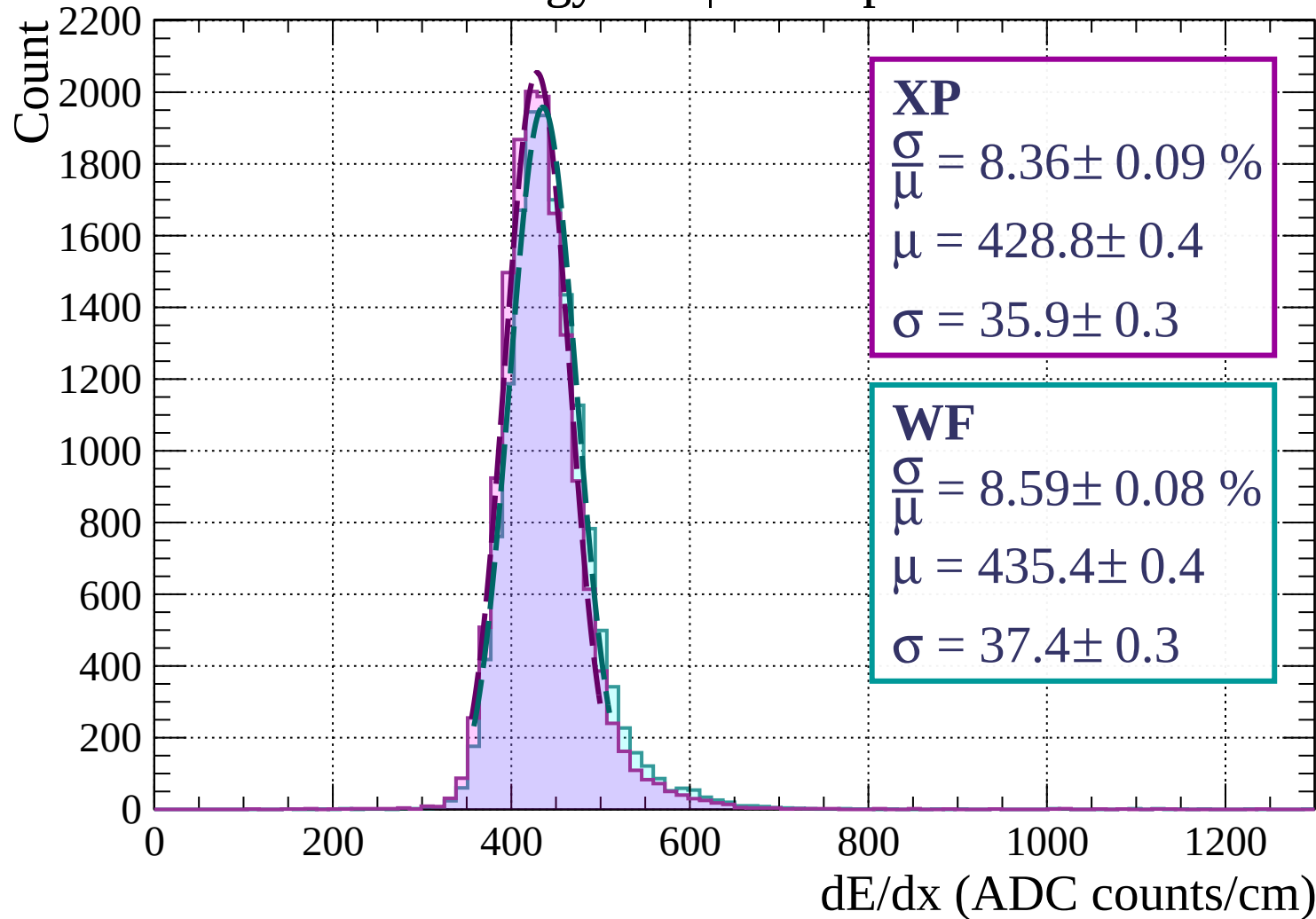
Energy loss | $720 < p < 800$



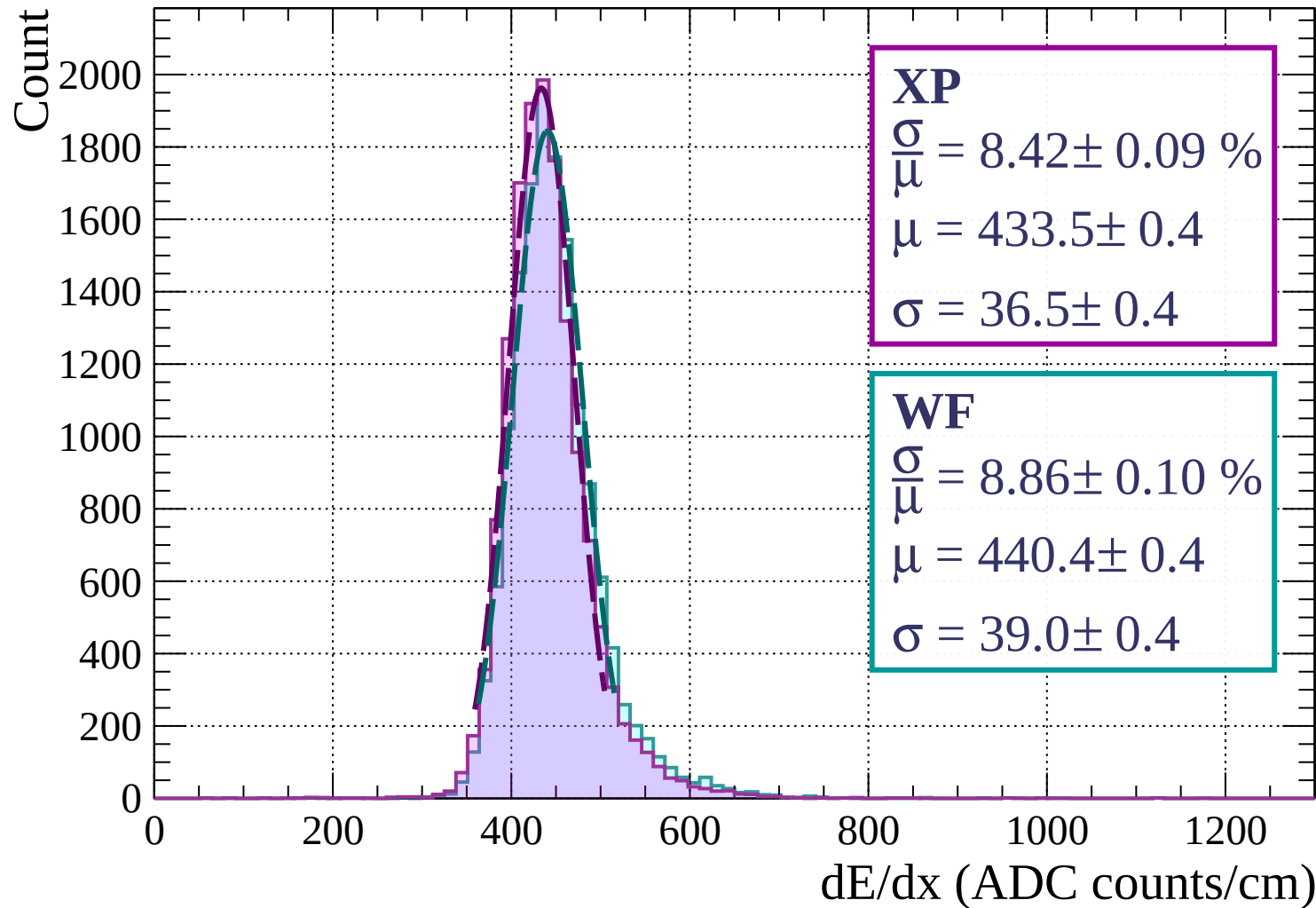
Energy loss | $800 < p < 880$



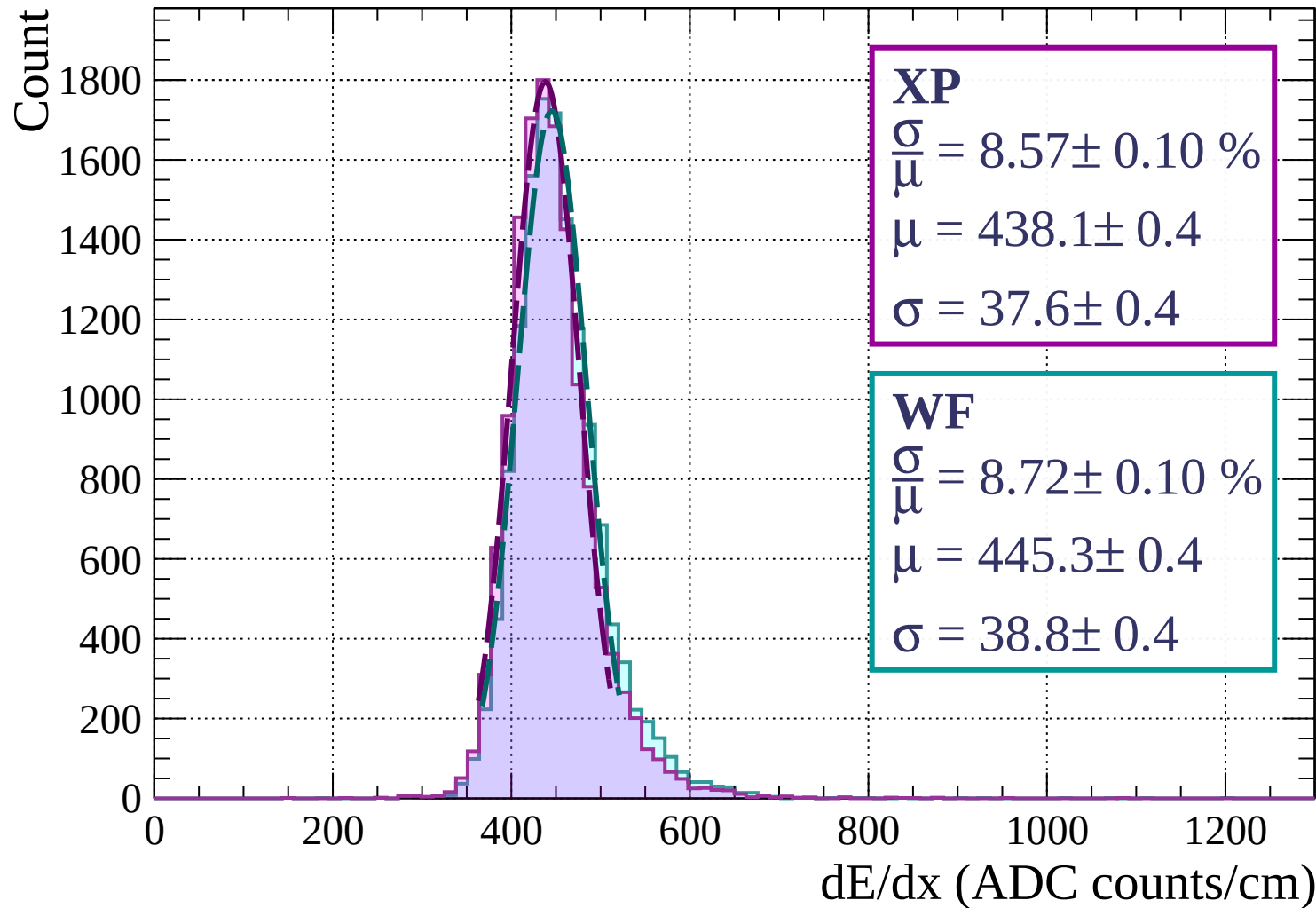
Energy loss | $880 < p < 960$



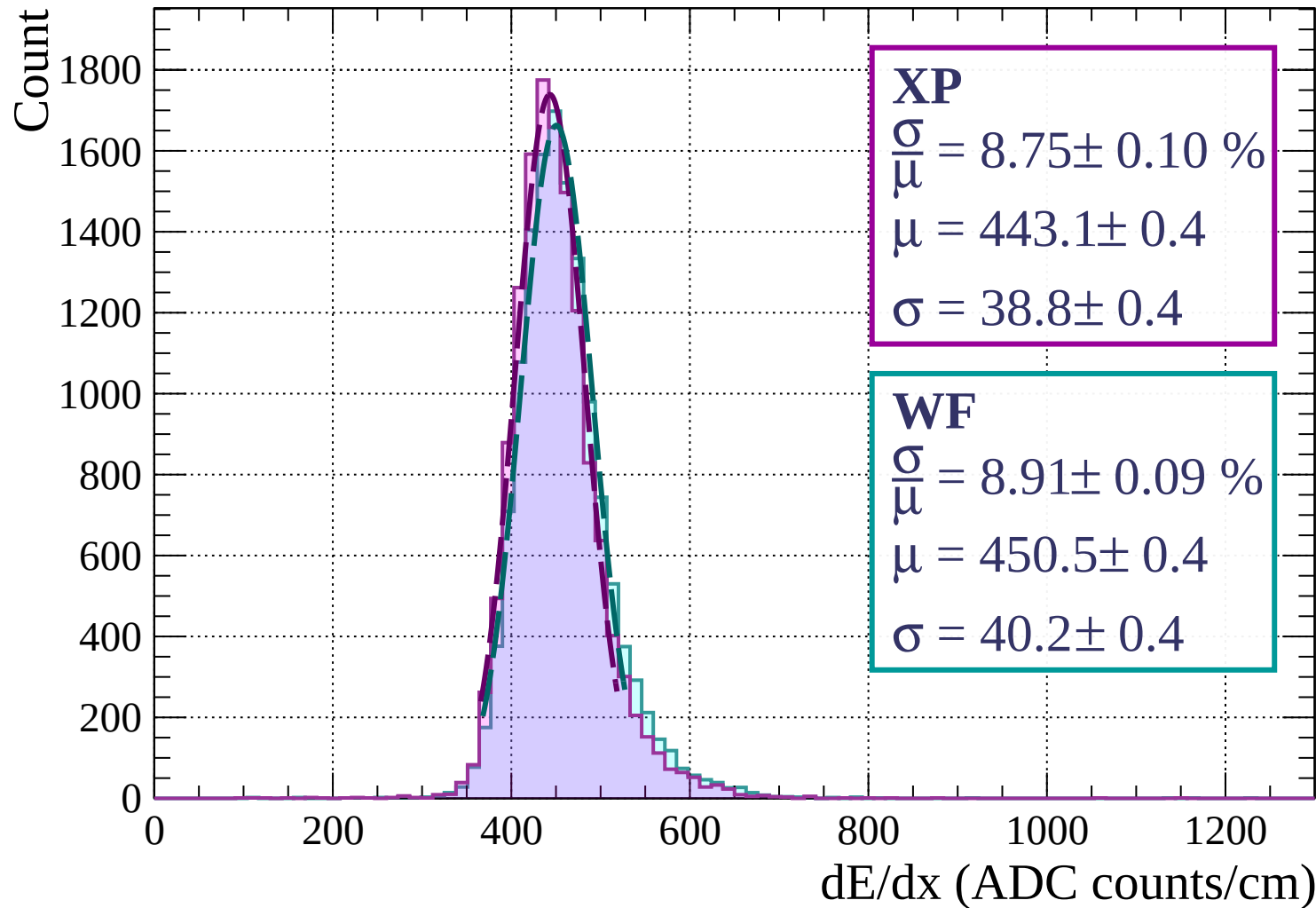
Energy loss | $960 < p < 1040$



Energy loss | $1040 < p < 1120$



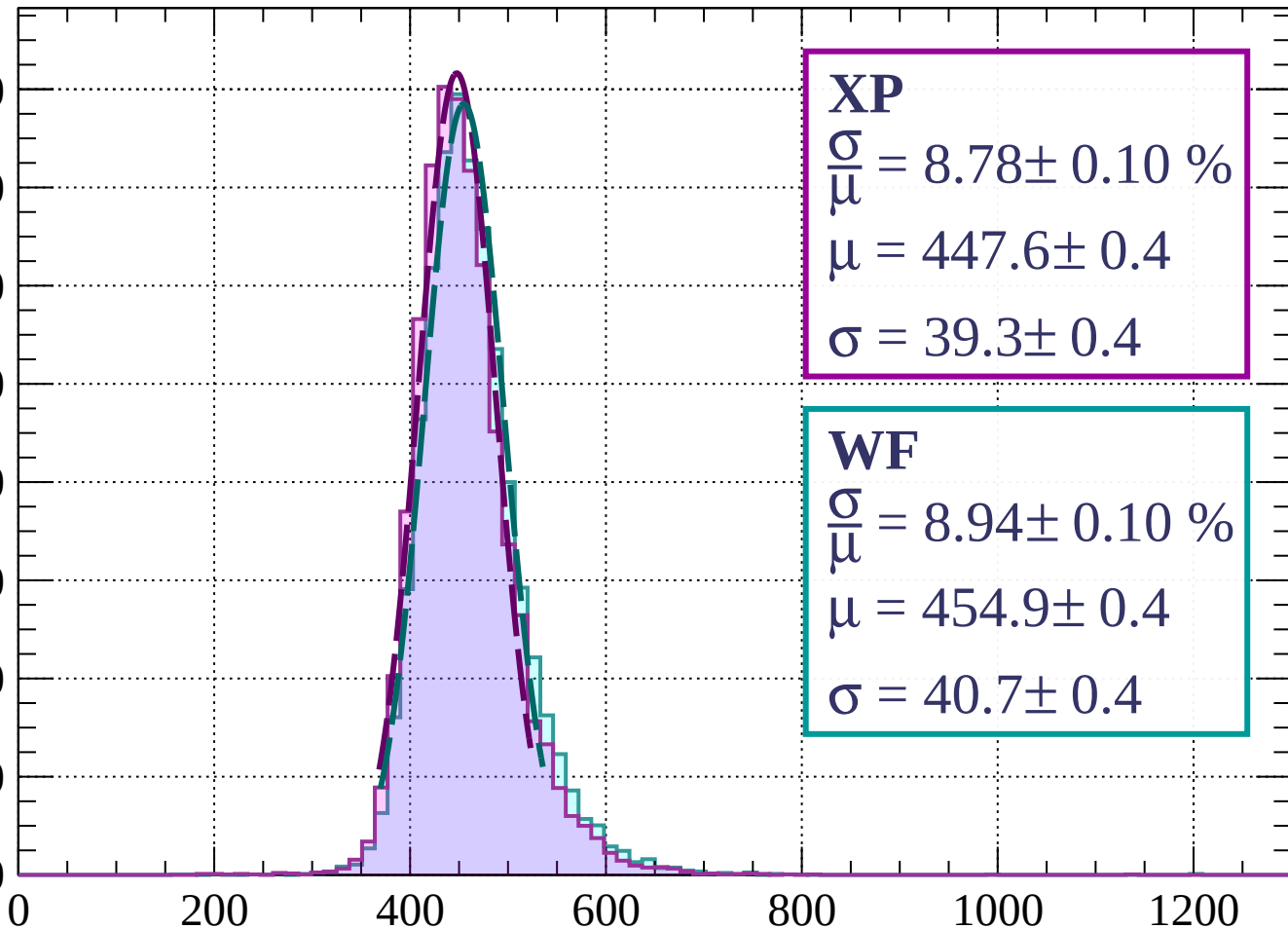
Energy loss | $1120 < p < 1200$



Energy loss | $1200 < p < 1280$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 8.78 \pm 0.10 \%$$

$$\mu = 447.6 \pm 0.4$$

$$\sigma = 39.3 \pm 0.4$$

WF

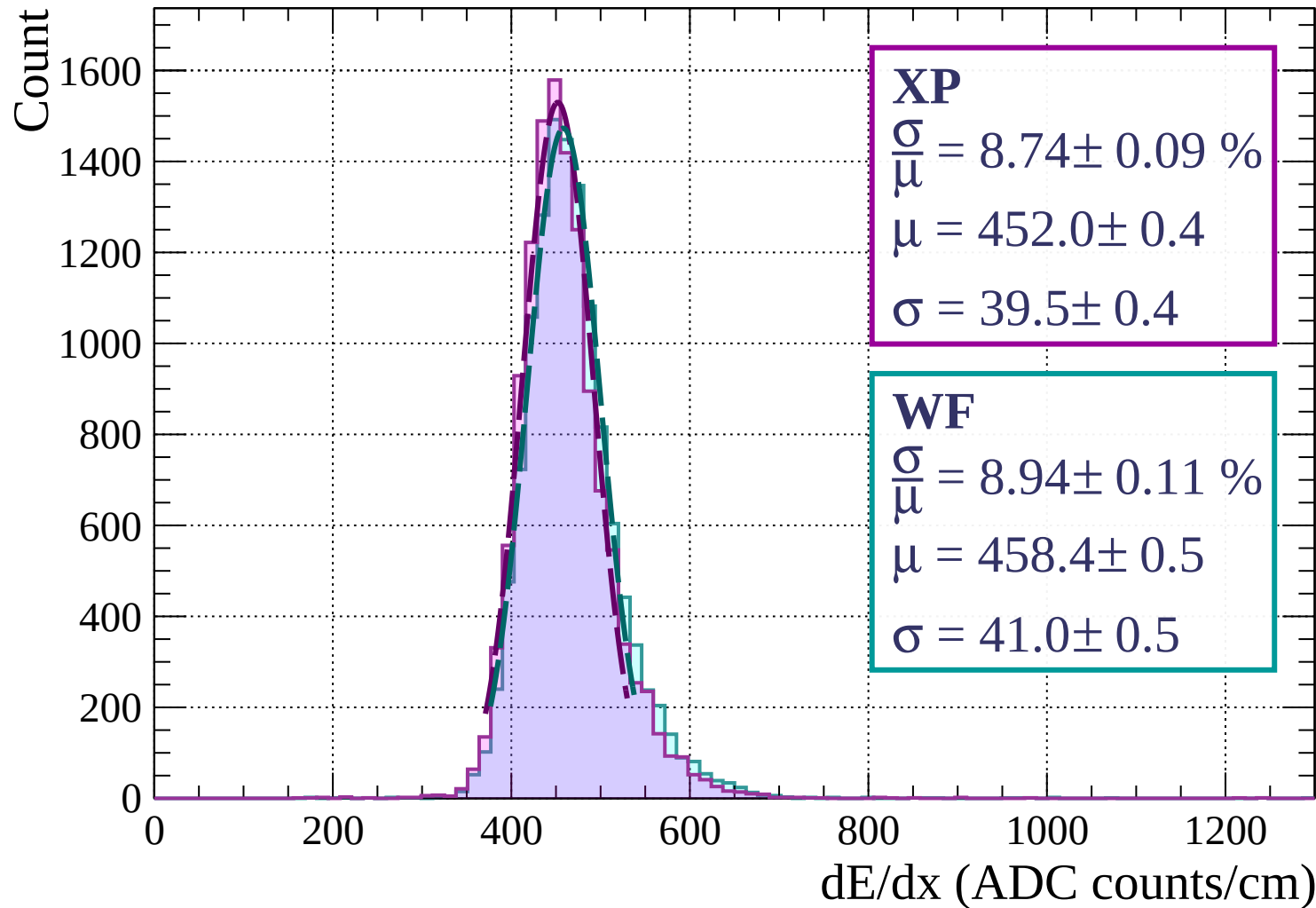
$$\frac{\sigma}{\mu} = 8.94 \pm 0.10 \%$$

$$\mu = 454.9 \pm 0.4$$

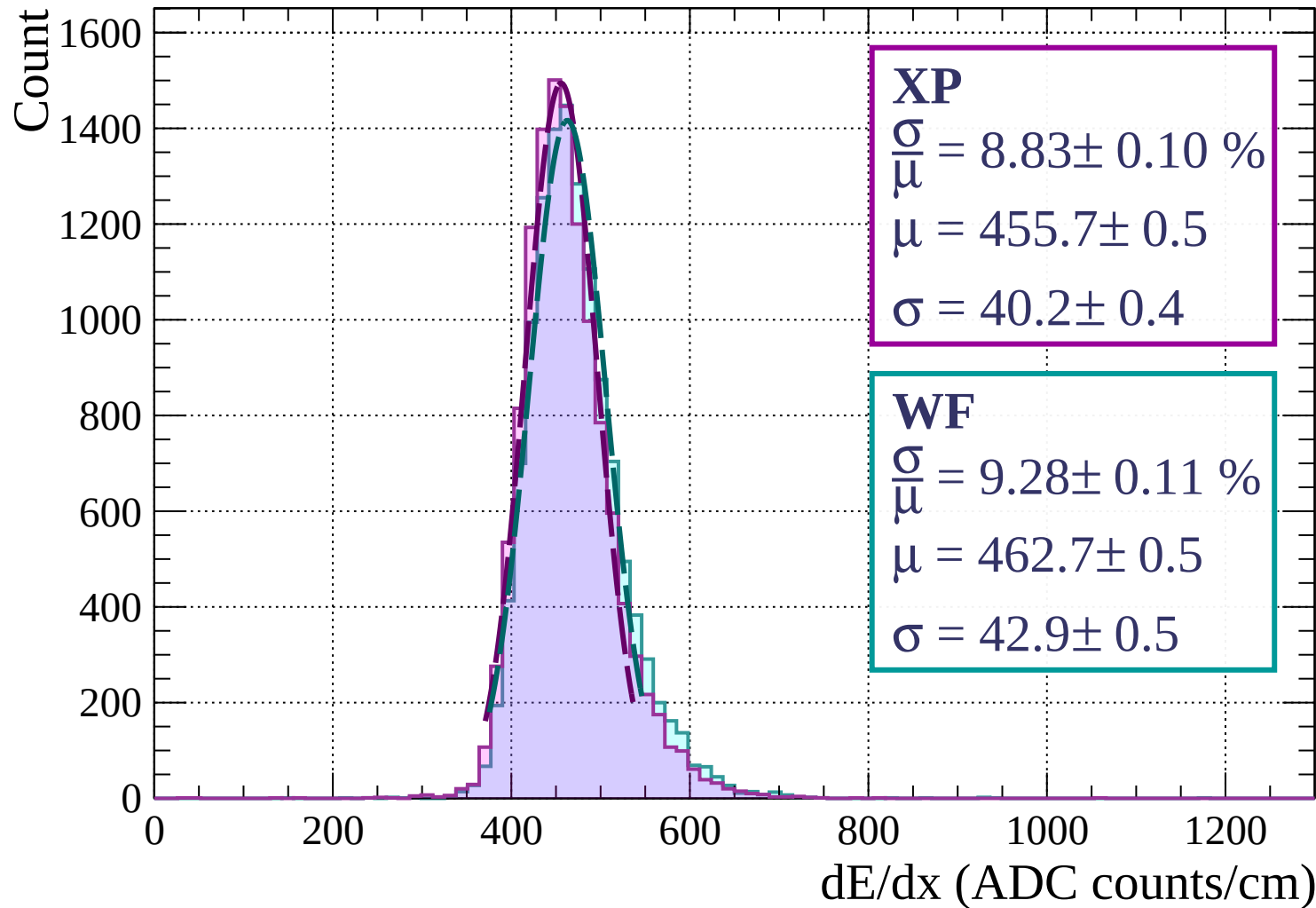
$$\sigma = 40.7 \pm 0.4$$

dE/dx (ADC counts/cm)

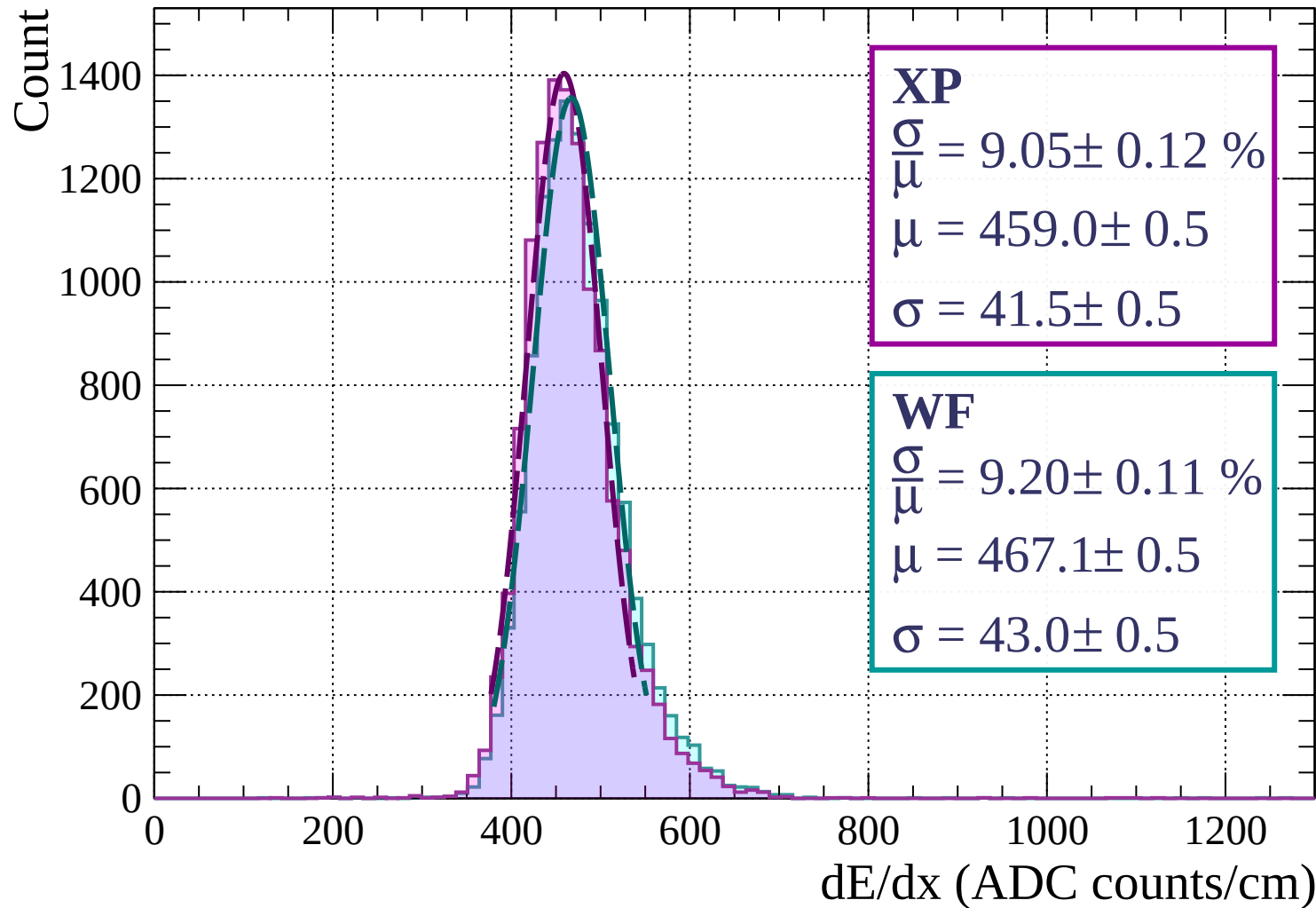
Energy loss | $1280 < p < 1360$



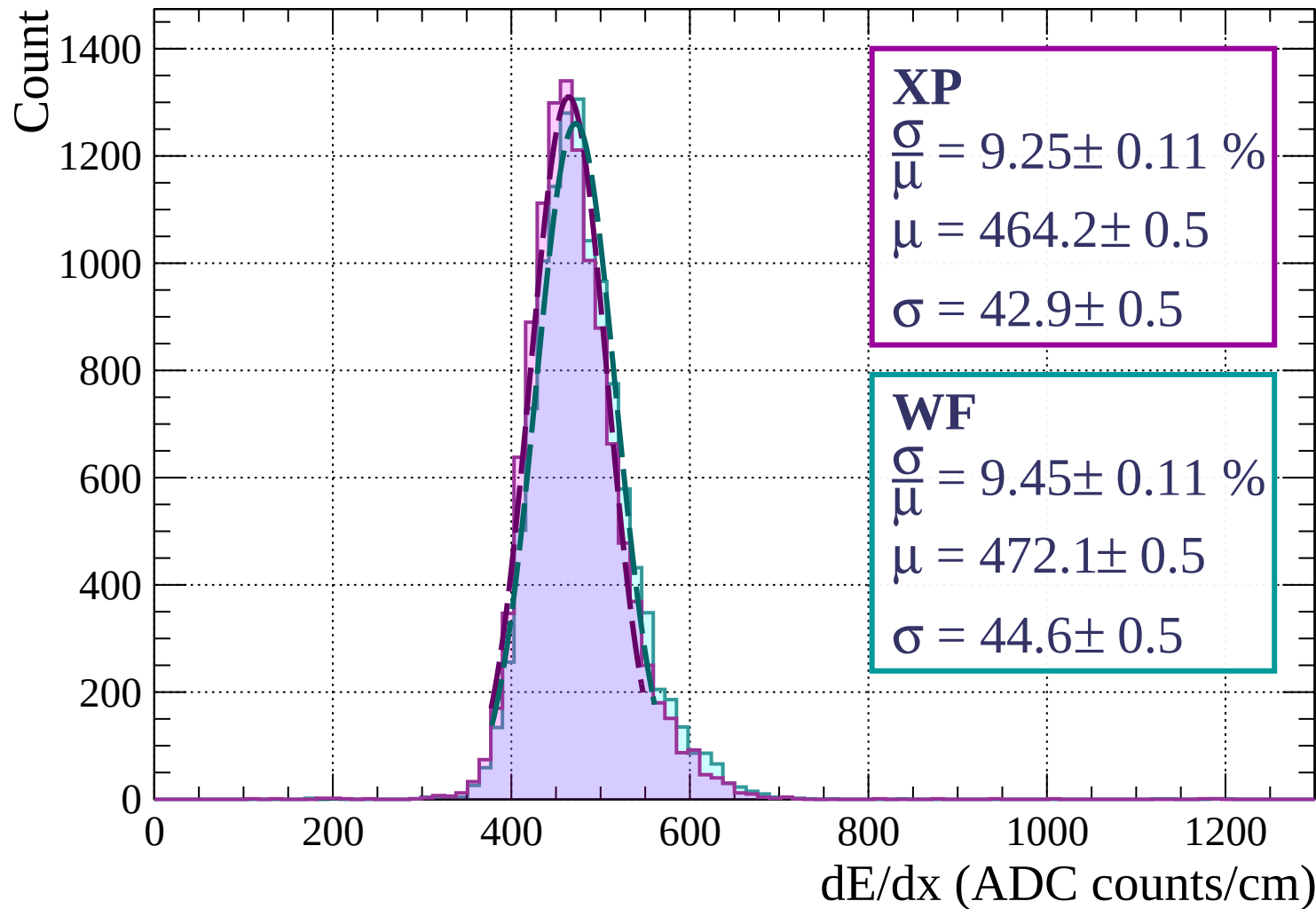
Energy loss | $1360 < p < 1440$



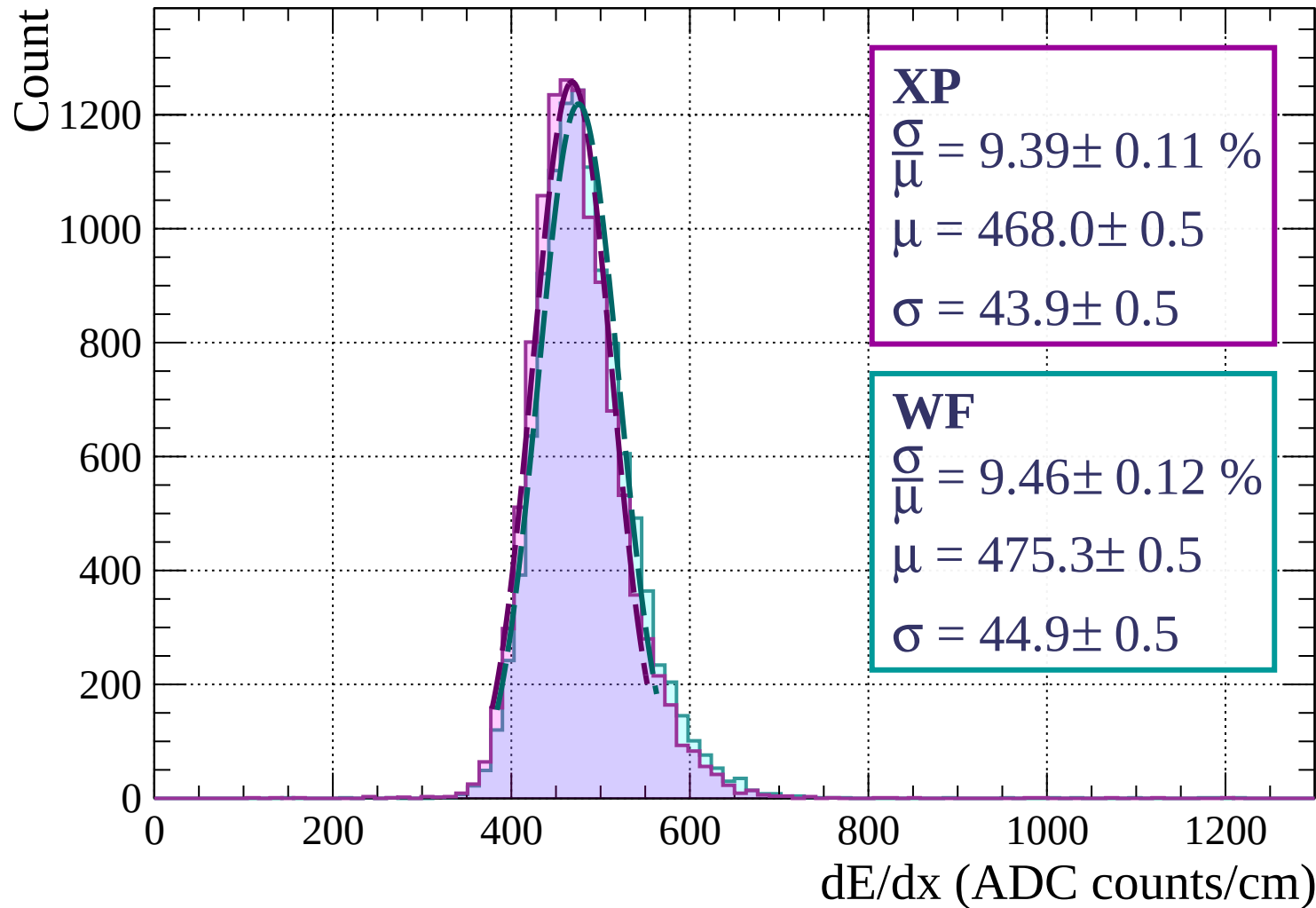
Energy loss | $1440 < p < 1520$



Energy loss | $1520 < p < 1600$



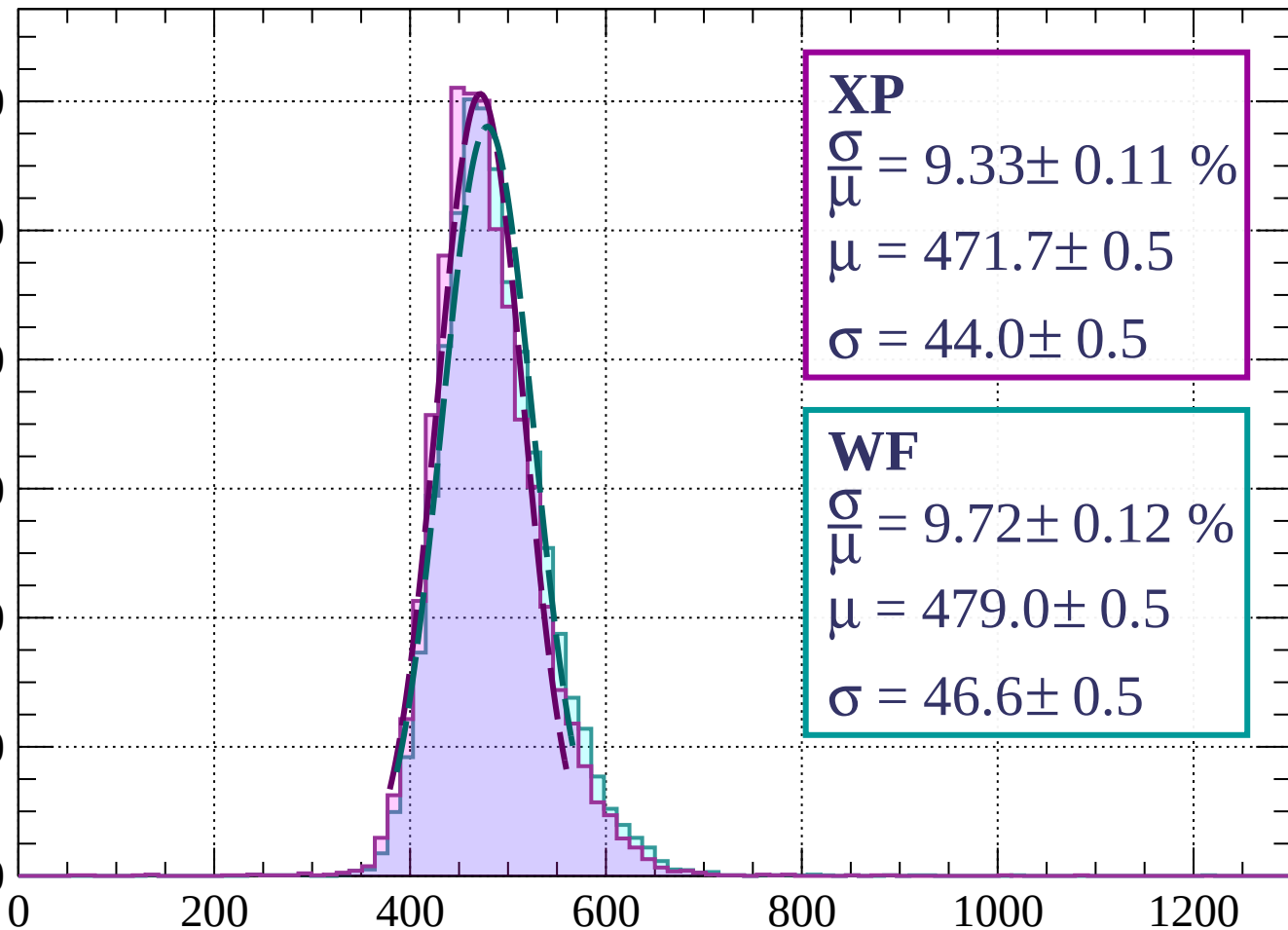
Energy loss | $1600 < p < 1680$



Energy loss | $1680 < p < 1760$

Count

1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 9.33 \pm 0.11 \%$$

$$\mu = 471.7 \pm 0.5$$

$$\sigma = 44.0 \pm 0.5$$

WF

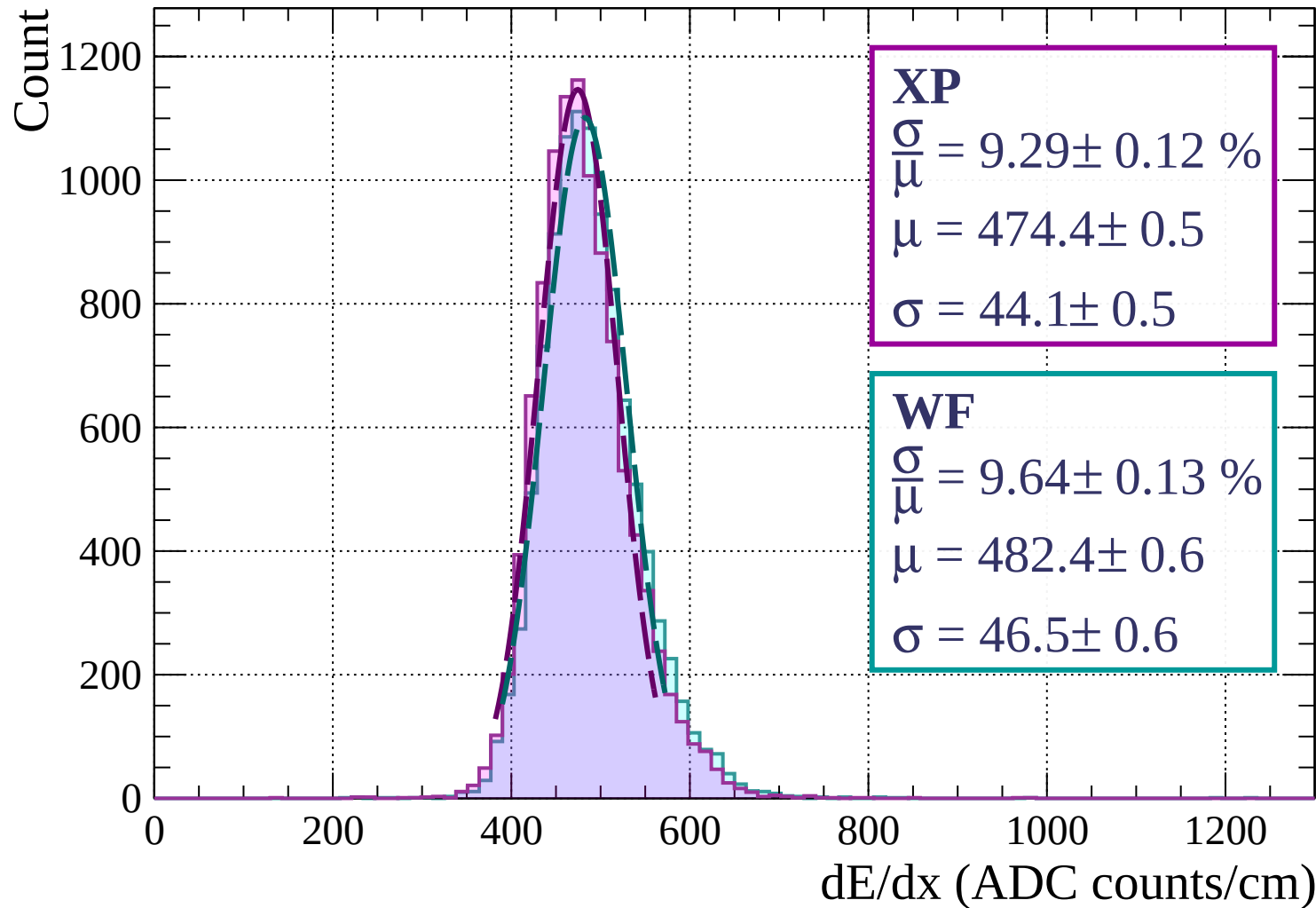
$$\frac{\sigma}{\mu} = 9.72 \pm 0.12 \%$$

$$\mu = 479.0 \pm 0.5$$

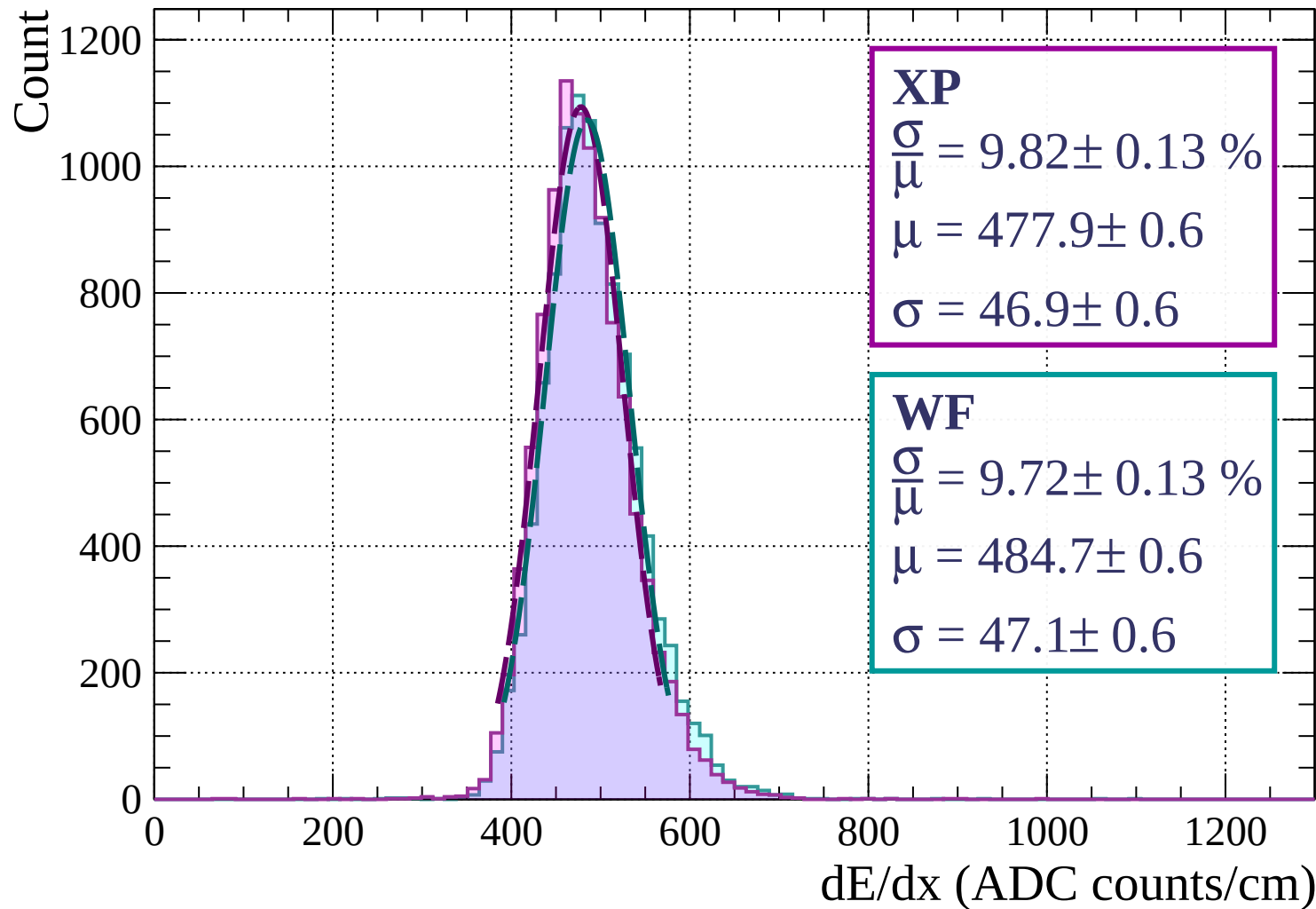
$$\sigma = 46.6 \pm 0.5$$

dE/dx (ADC counts/cm)

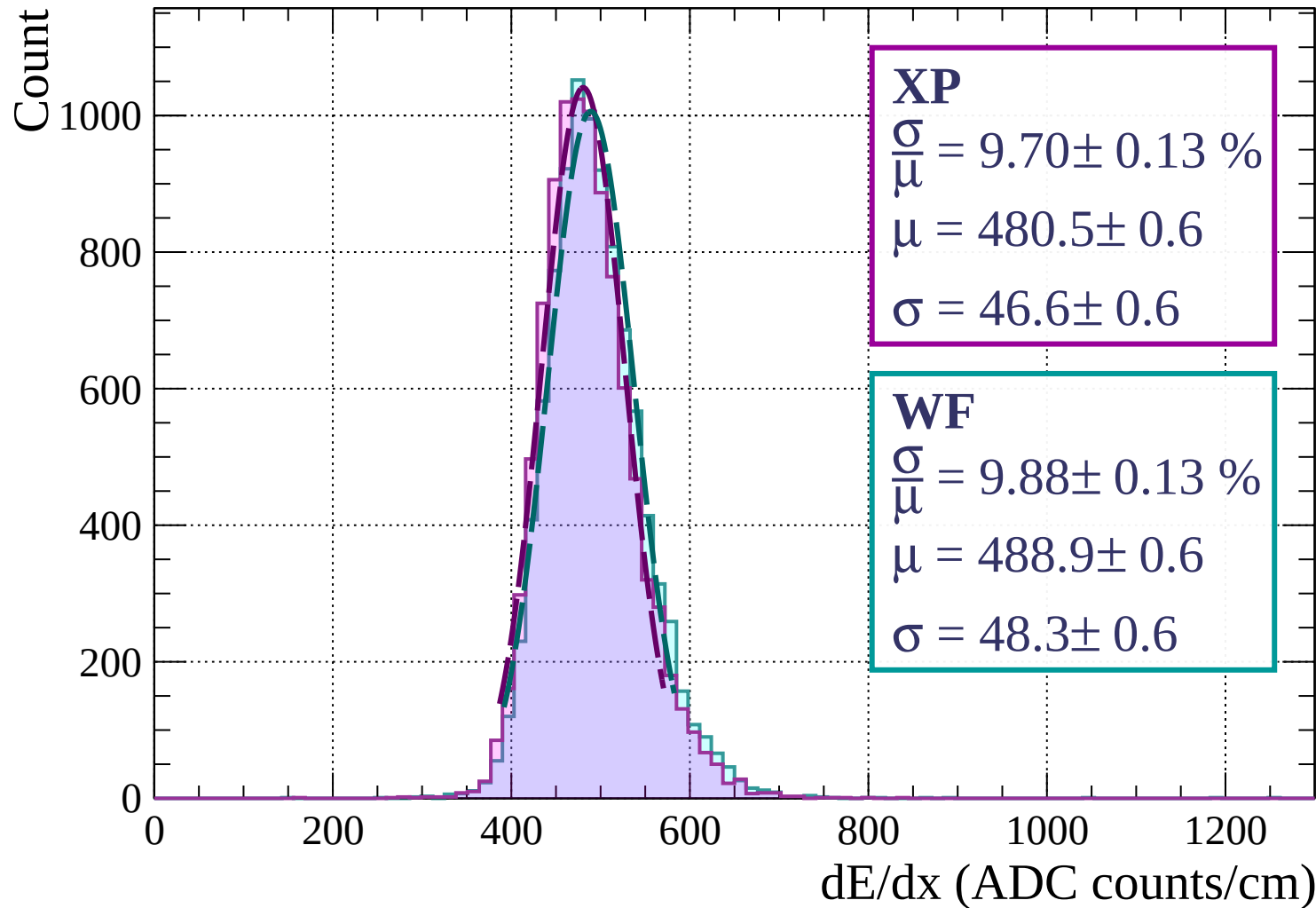
Energy loss | $1760 < p < 1840$



Energy loss | $1840 < p < 1920$

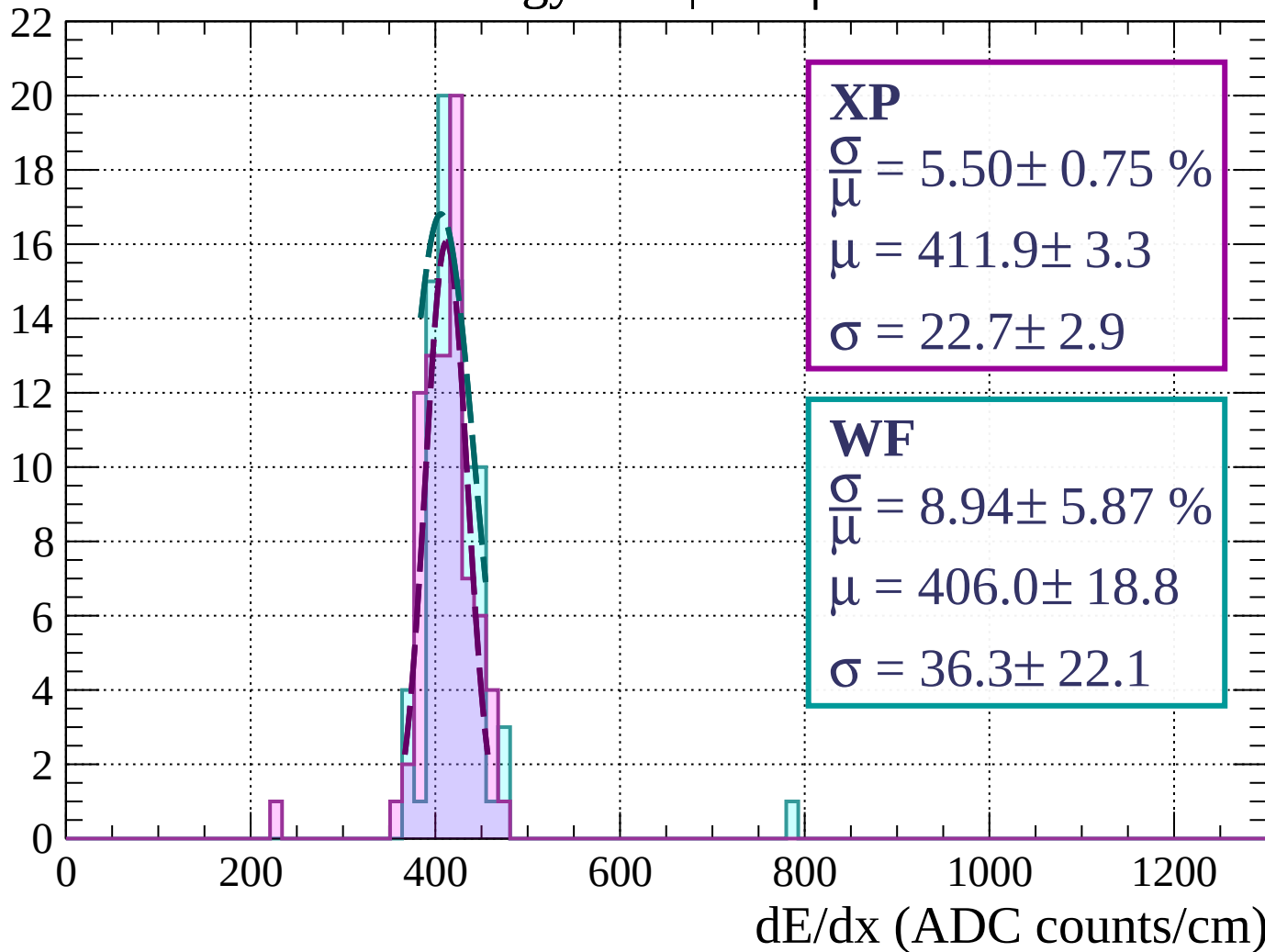


Energy loss | $1920 < p < 2000$



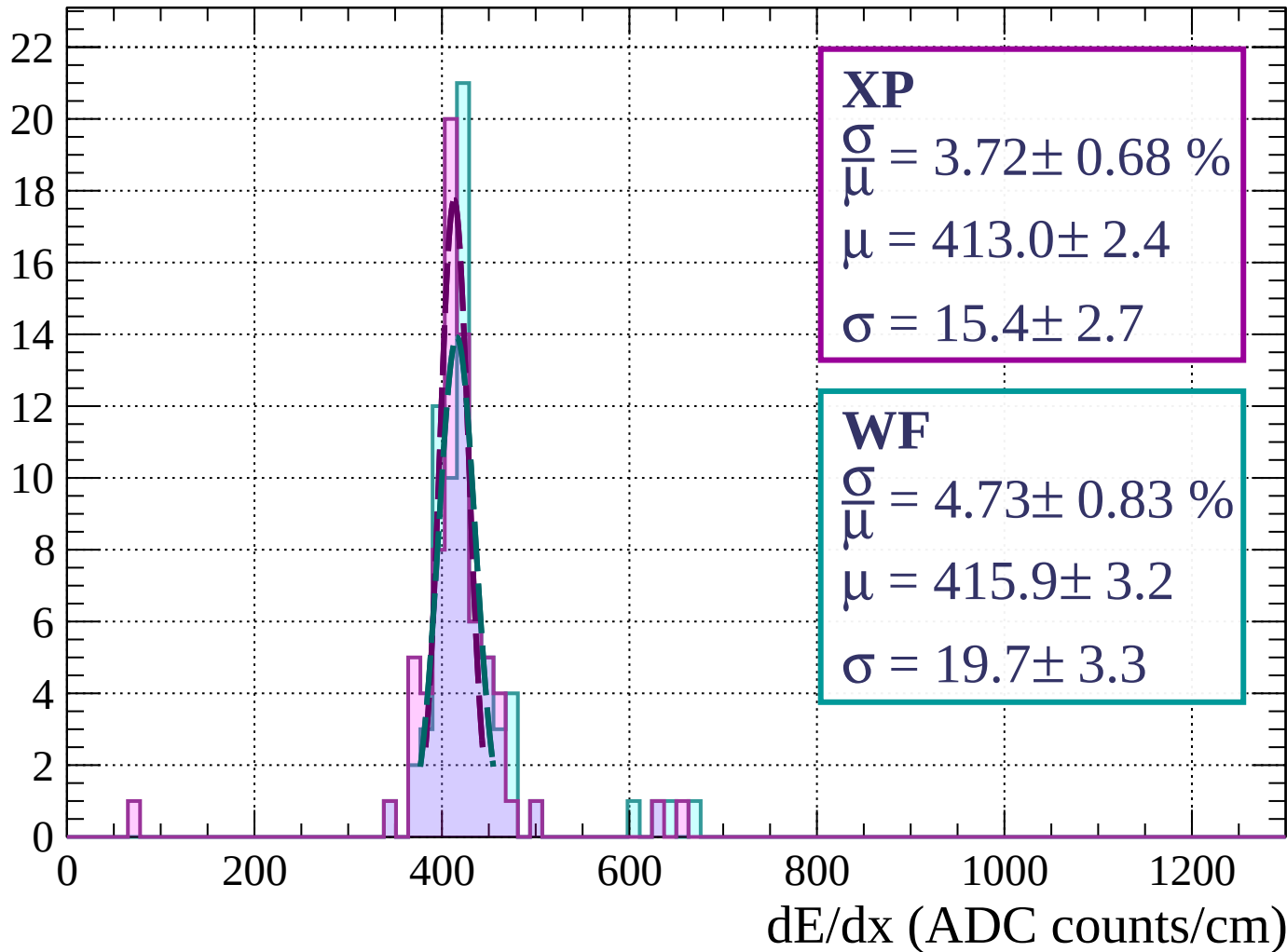
Energy loss | $0 < \phi < 3$

Count



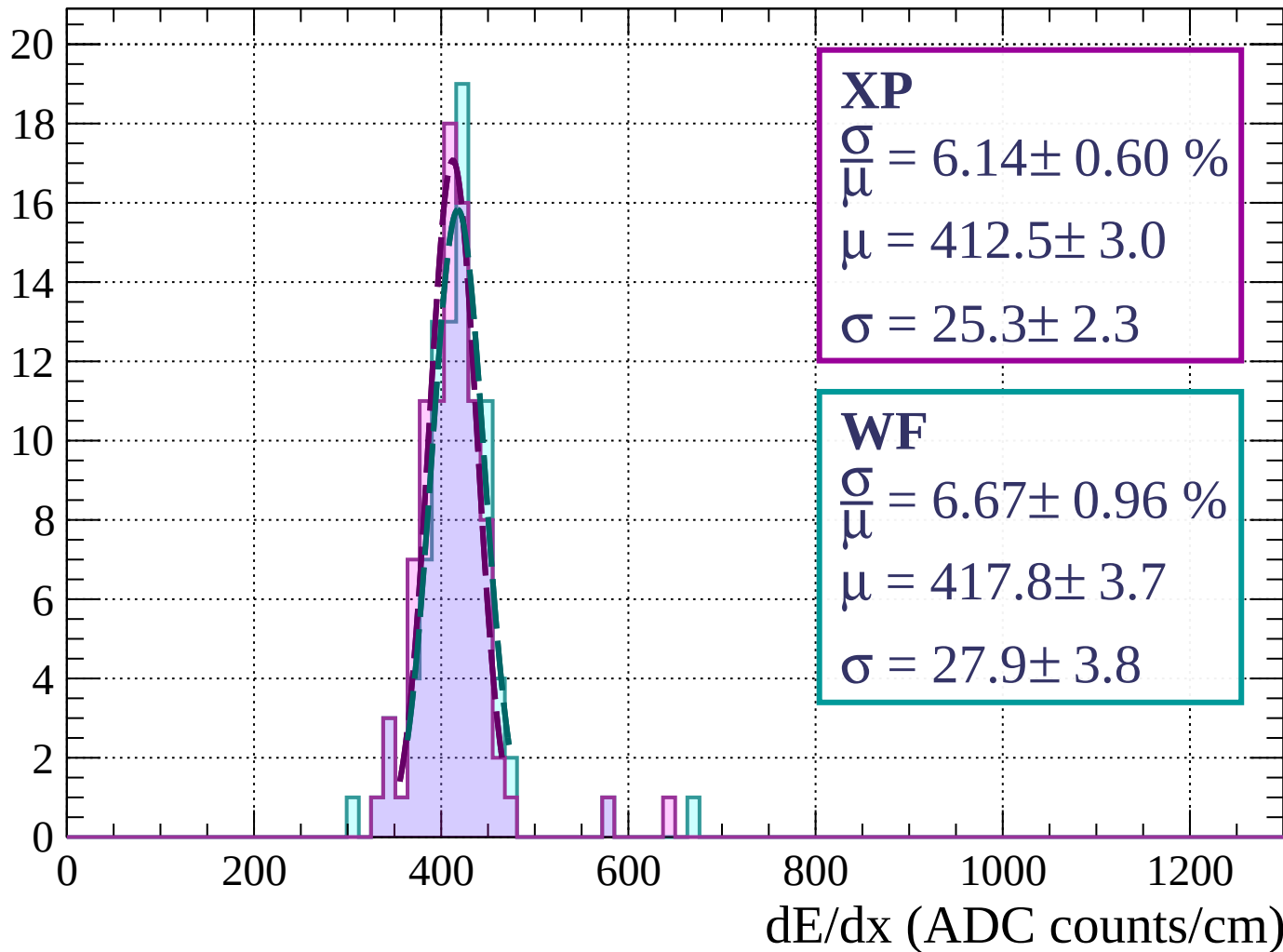
Energy loss | $3 < \phi < 6$

Count



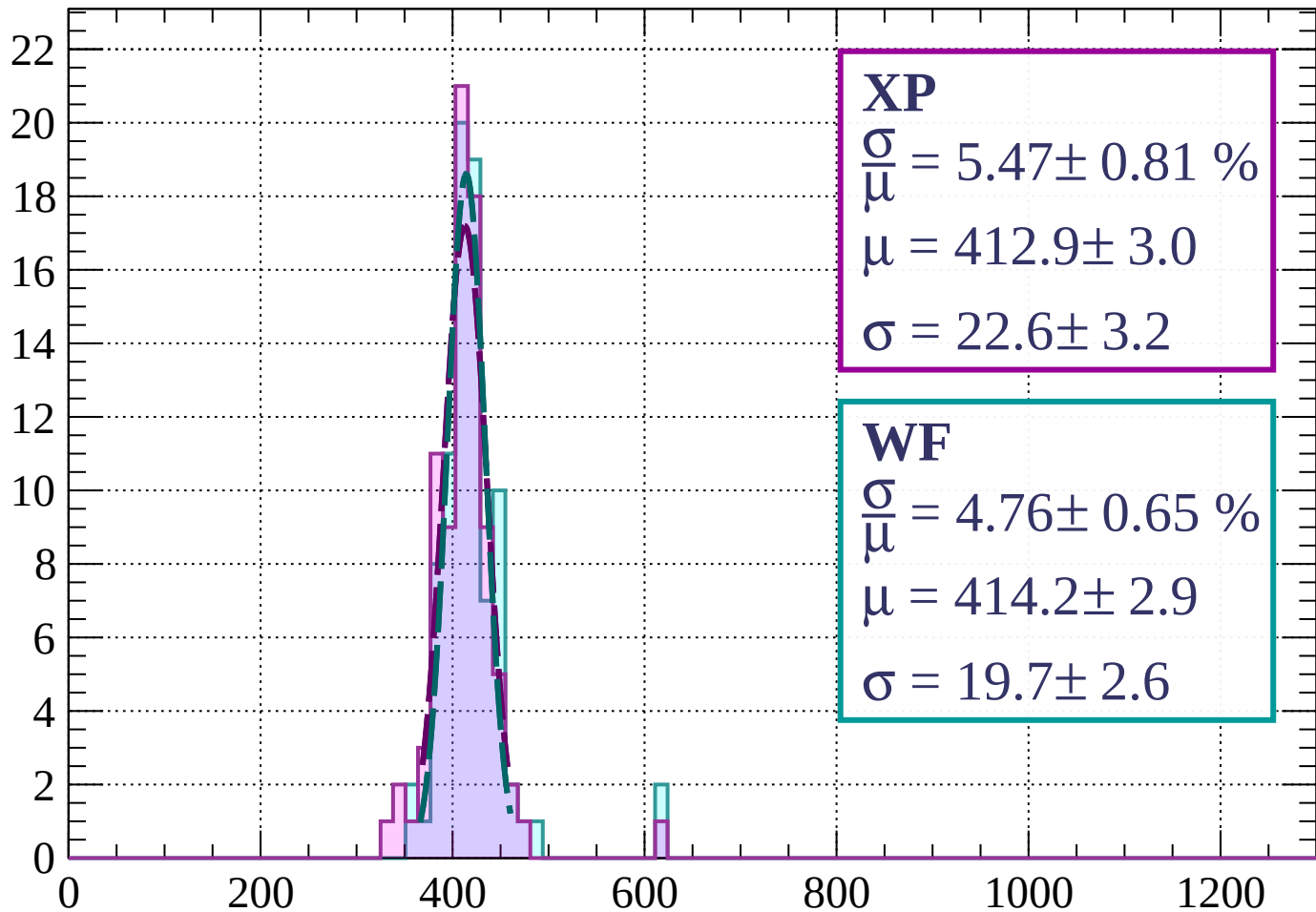
Energy loss | $6 < \phi < 9$

Count



Energy loss | $9 < \phi < 12$

Count



XP

$$\frac{\sigma}{\mu} = 5.47 \pm 0.81 \%$$

$$\mu = 412.9 \pm 3.0$$

$$\sigma = 22.6 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 4.76 \pm 0.65 \%$$

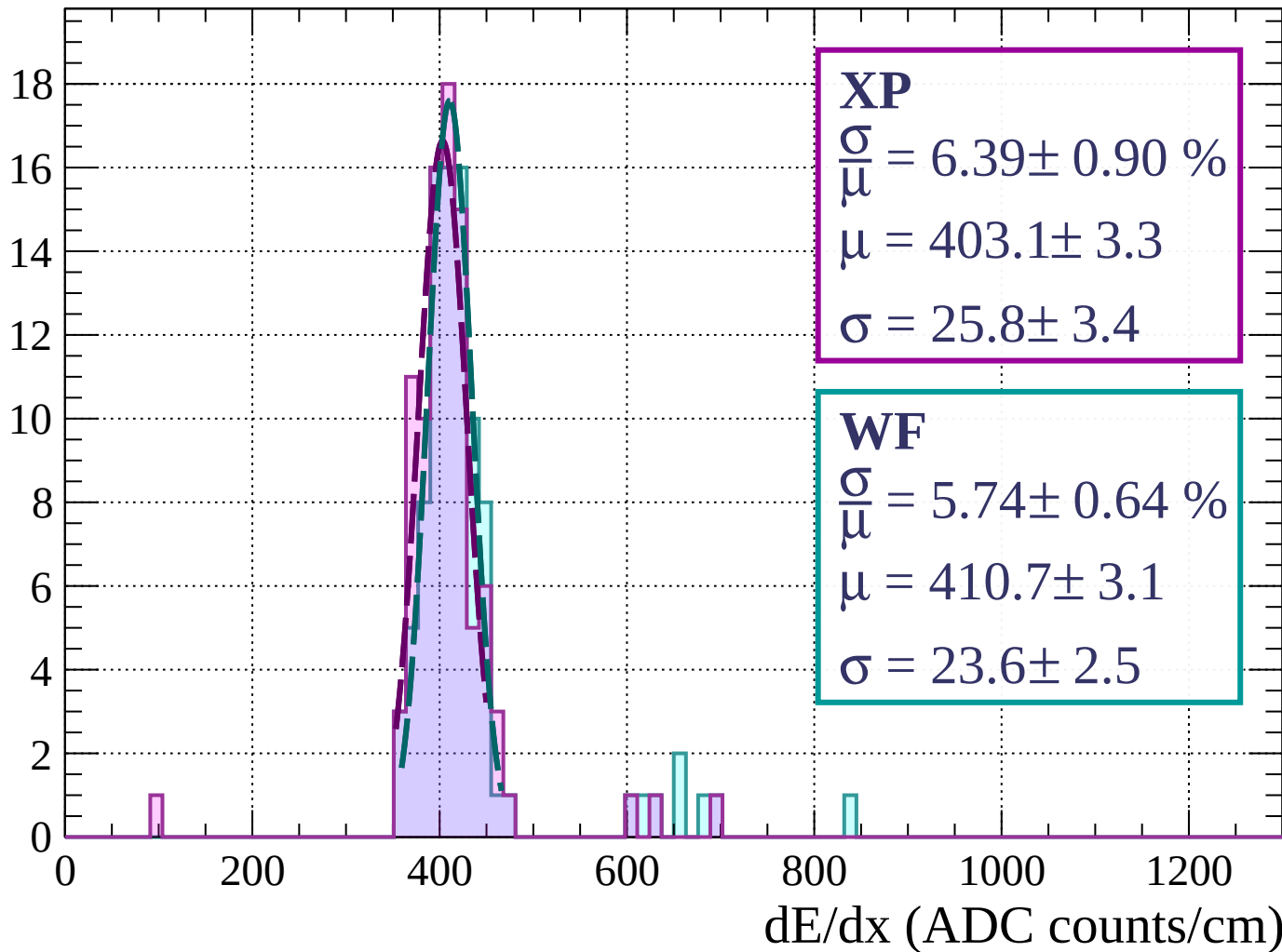
$$\mu = 414.2 \pm 2.9$$

$$\sigma = 19.7 \pm 2.6$$

dE/dx (ADC counts/cm)

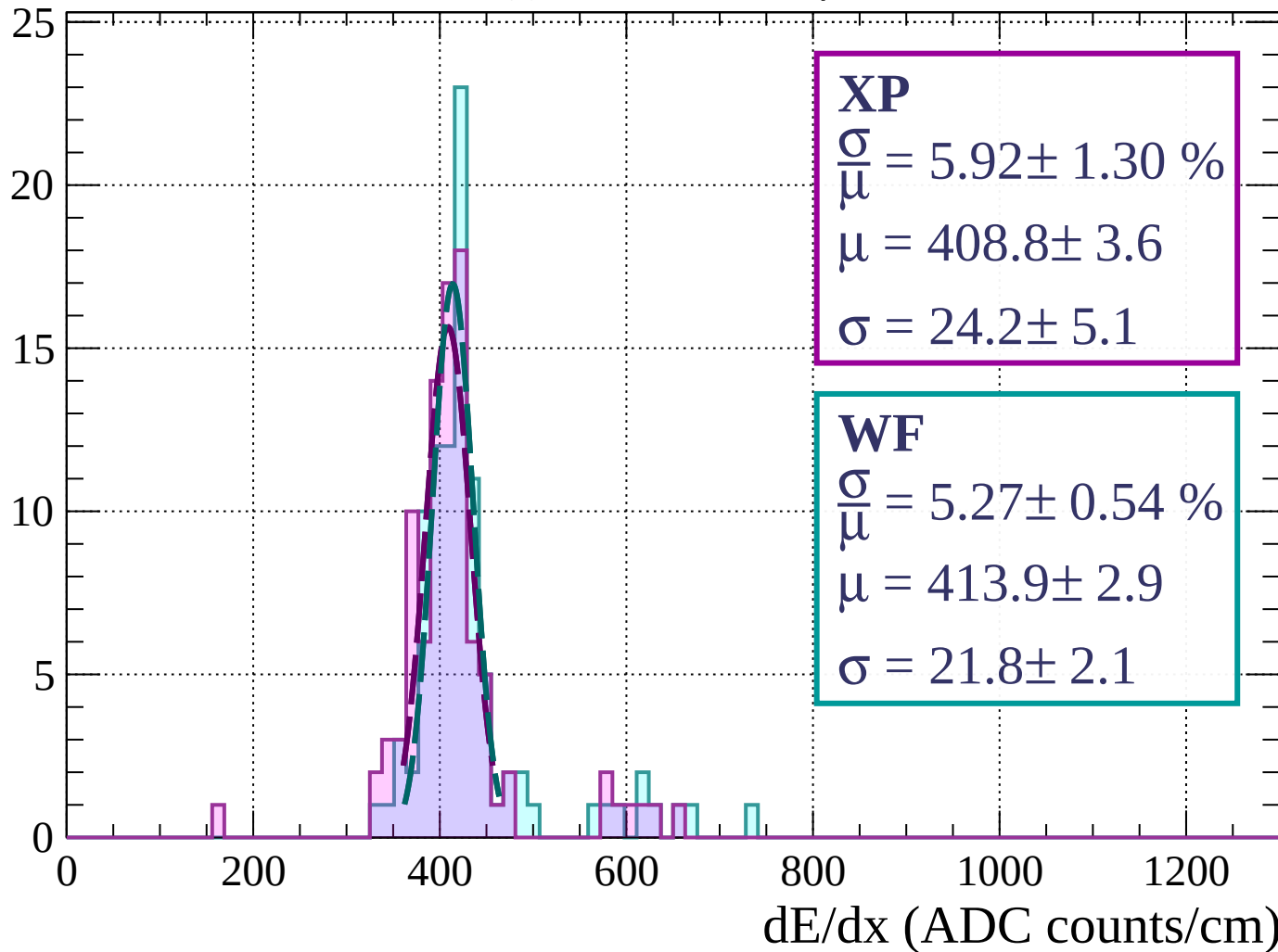
Energy loss | $12 < \phi < 15$

Count



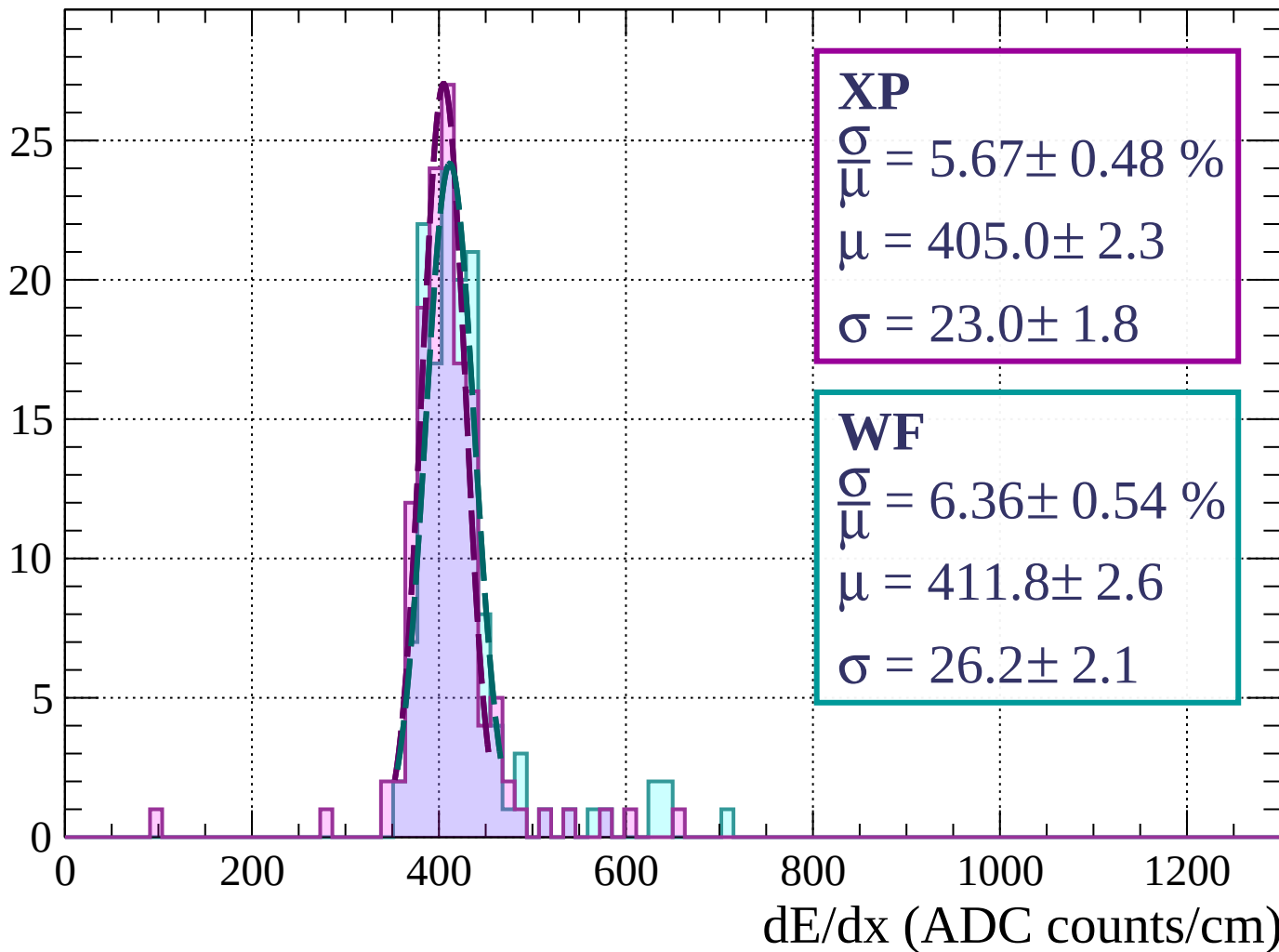
Energy loss | $15 < \phi < 18$

Count



Energy loss | $18 < \phi < 21$

Count



Energy loss | $21 < \phi < 24$

Count

XP

$$\frac{\sigma}{\mu} = 6.32 \pm 0.72 \%$$

$$\mu = 408.4 \pm 2.5$$

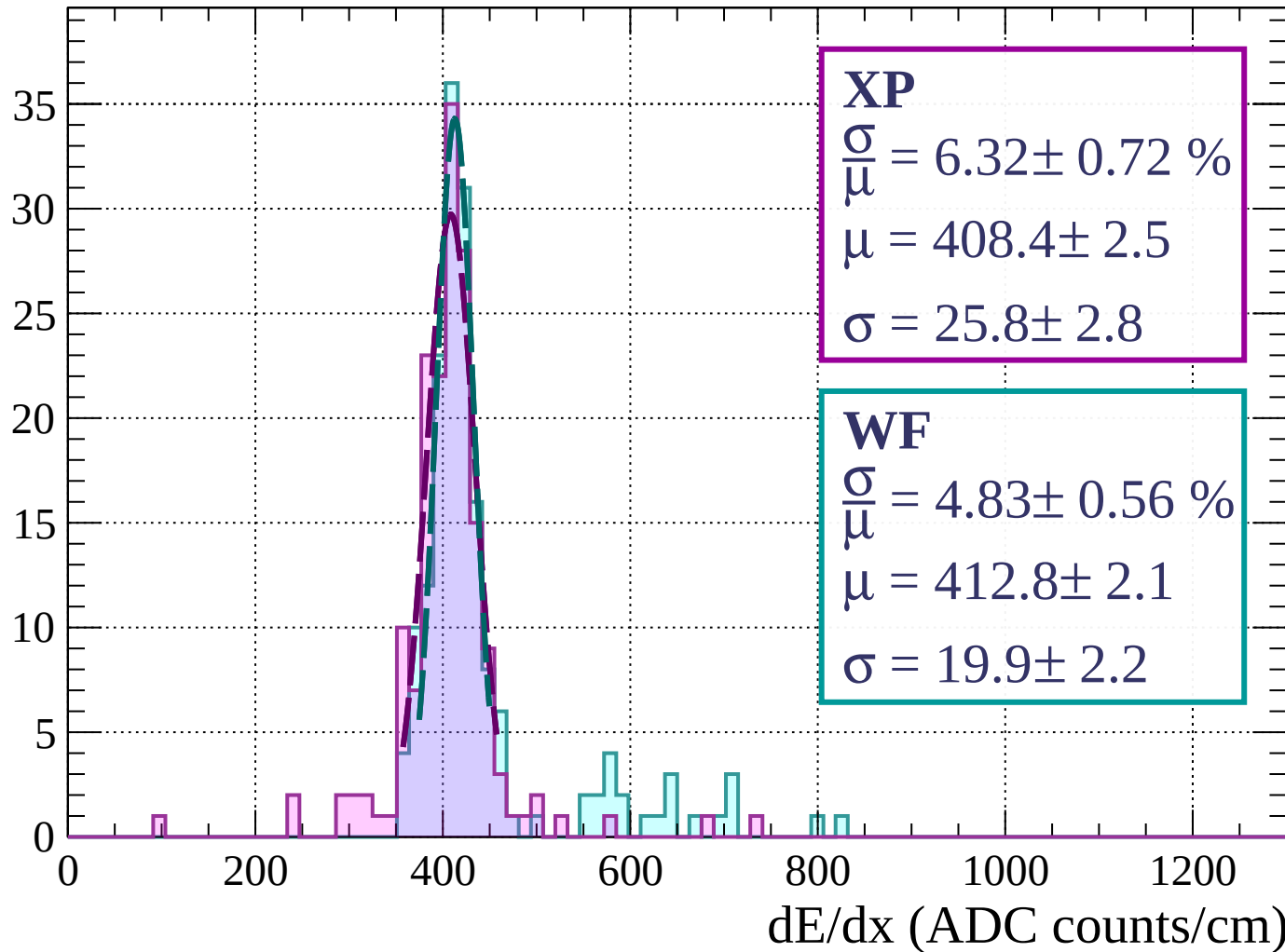
$$\sigma = 25.8 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 4.83 \pm 0.56 \%$$

$$\mu = 412.8 \pm 2.1$$

$$\sigma = 19.9 \pm 2.2$$



Energy loss | $24 < \phi < 27$

Count

XP

$$\frac{\sigma}{\mu} = 6.57 \pm 0.58 \%$$

$$\mu = 404.1 \pm 2.2$$

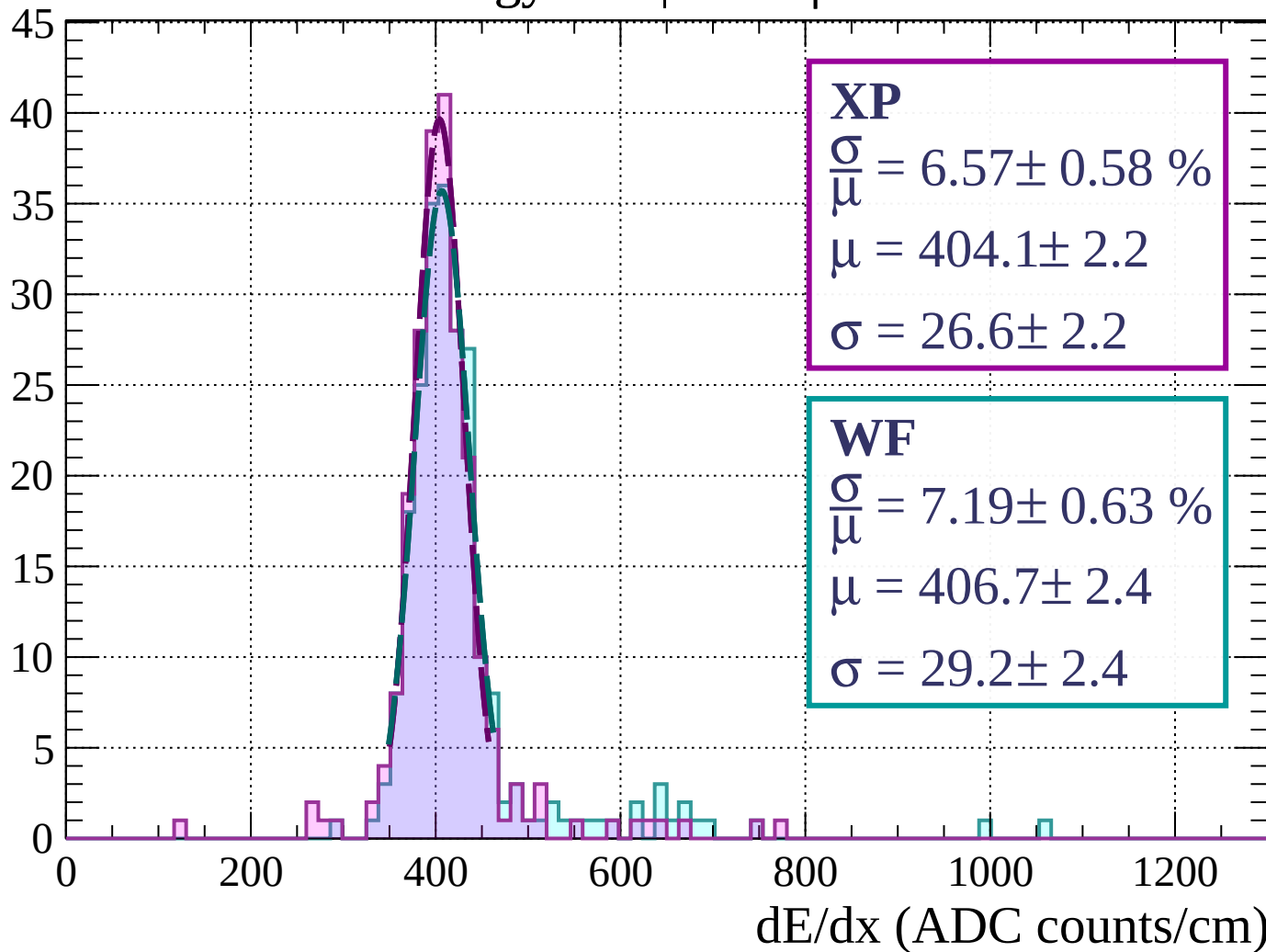
$$\sigma = 26.6 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 7.19 \pm 0.63 \%$$

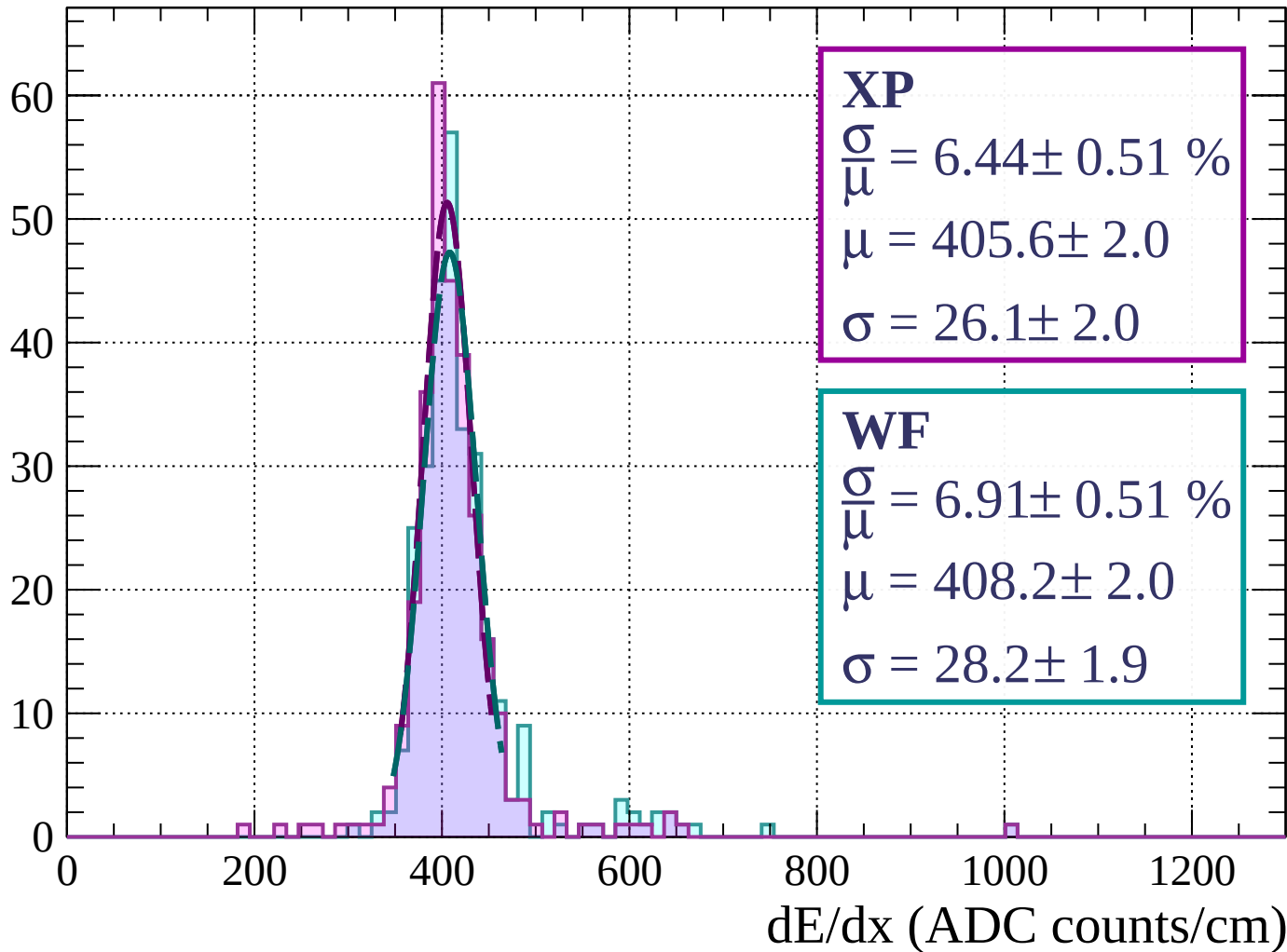
$$\mu = 406.7 \pm 2.4$$

$$\sigma = 29.2 \pm 2.4$$



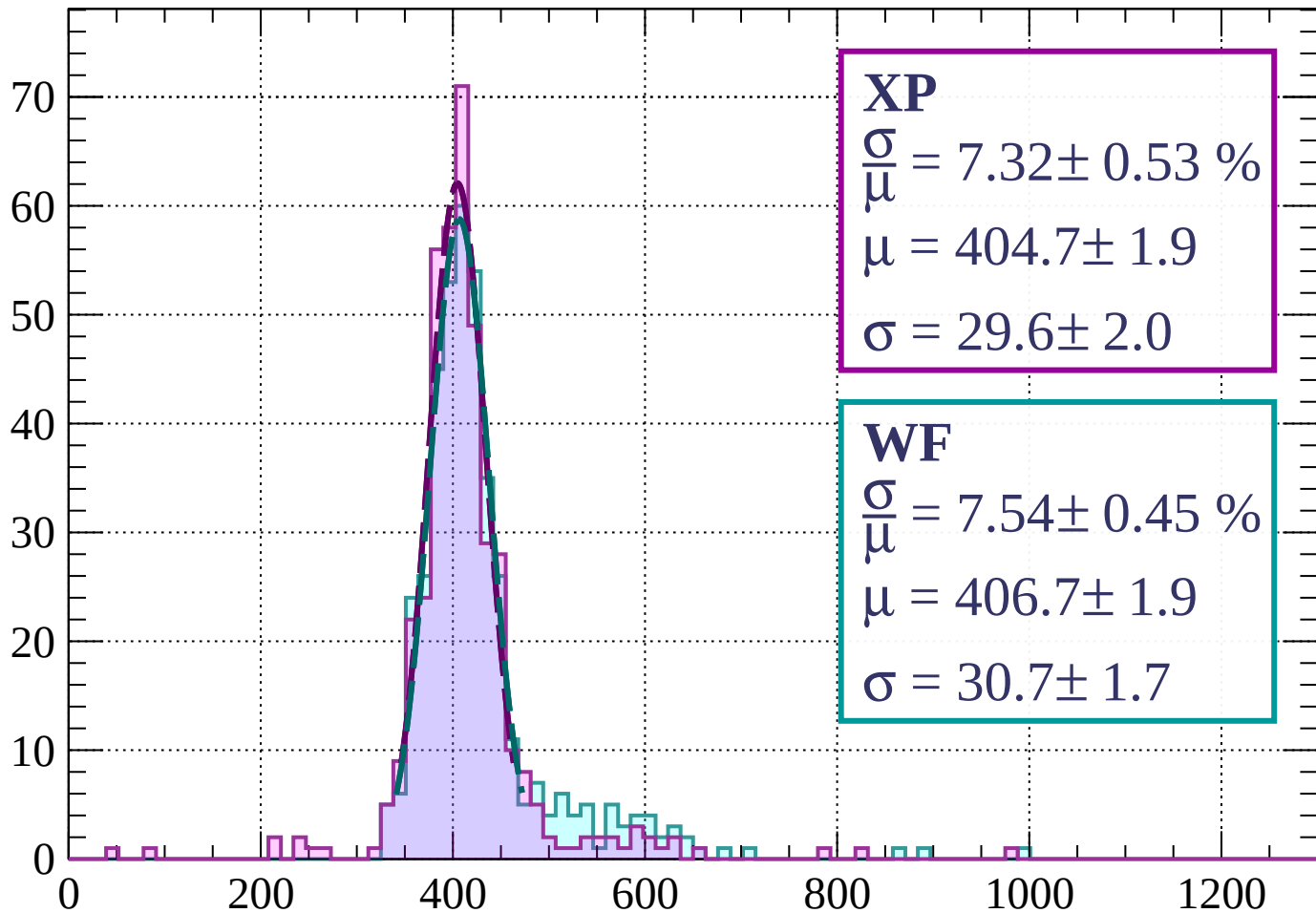
Energy loss | $27 < \phi < 30$

Count



Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 7.32 \pm 0.53 \%$$

$$\mu = 404.7 \pm 1.9$$

$$\sigma = 29.6 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 7.54 \pm 0.45 \%$$

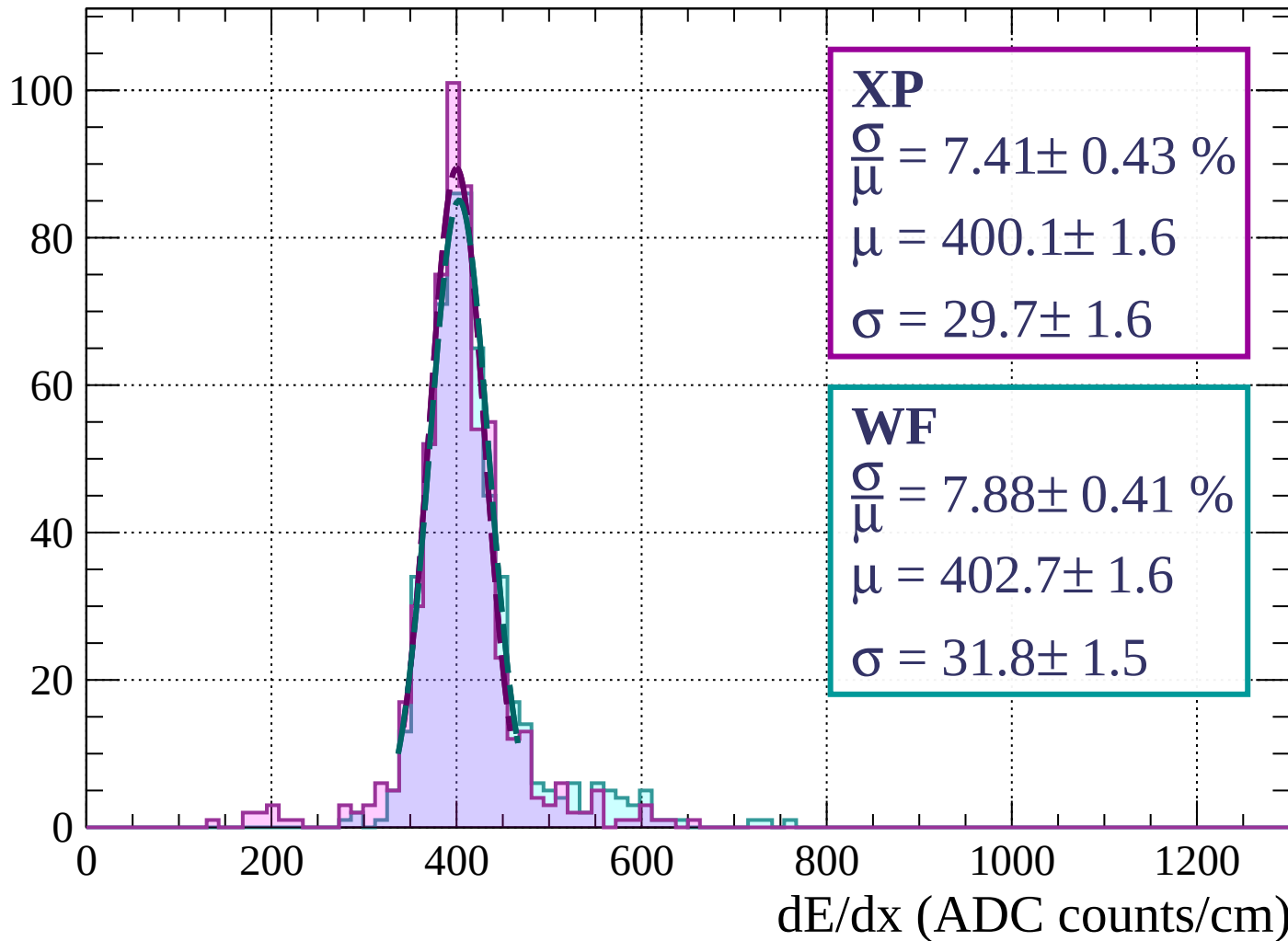
$$\mu = 406.7 \pm 1.9$$

$$\sigma = 30.7 \pm 1.7$$

dE/dx (ADC counts/cm)

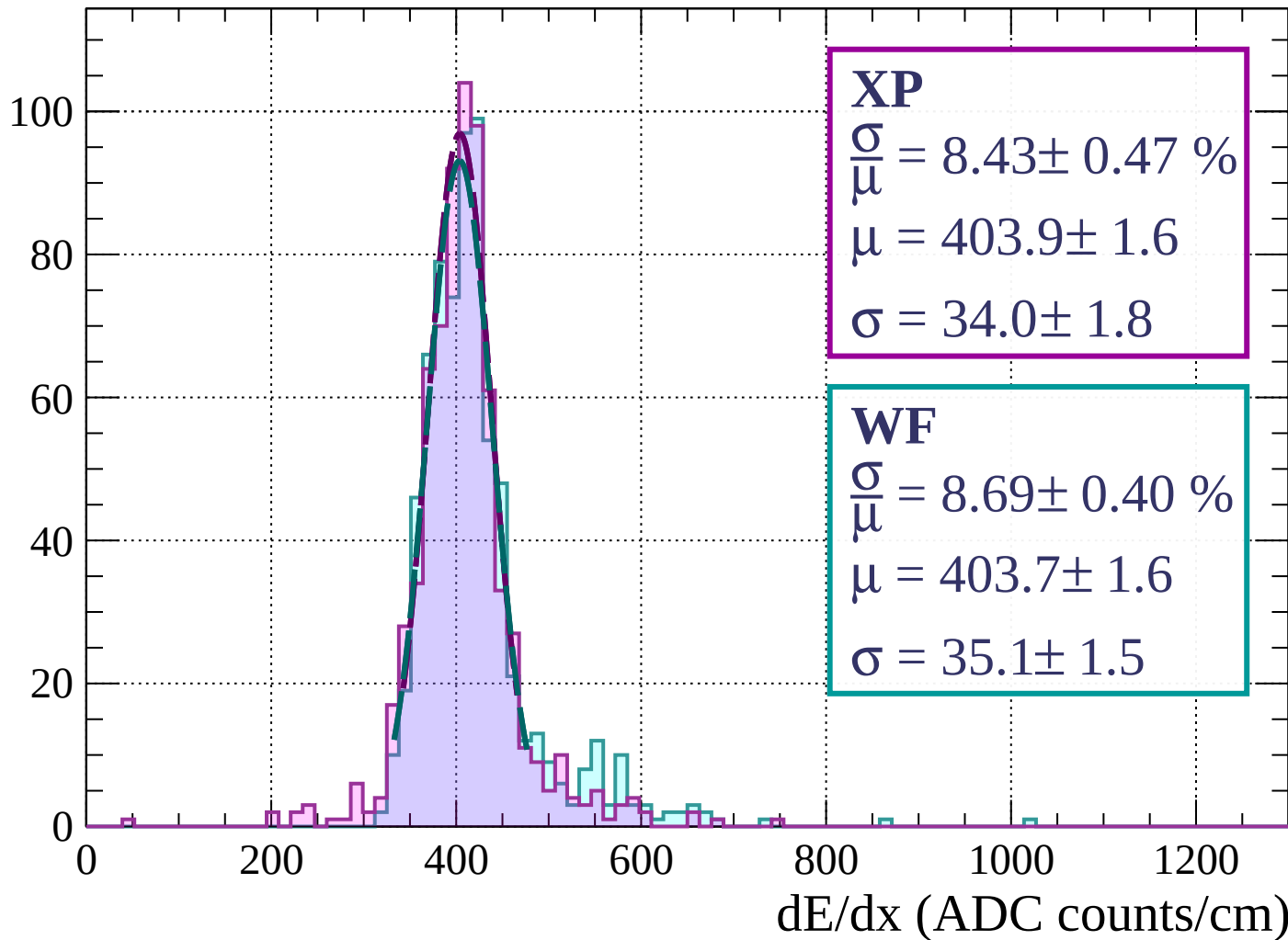
Energy loss | $33 < \phi < 36$

Count



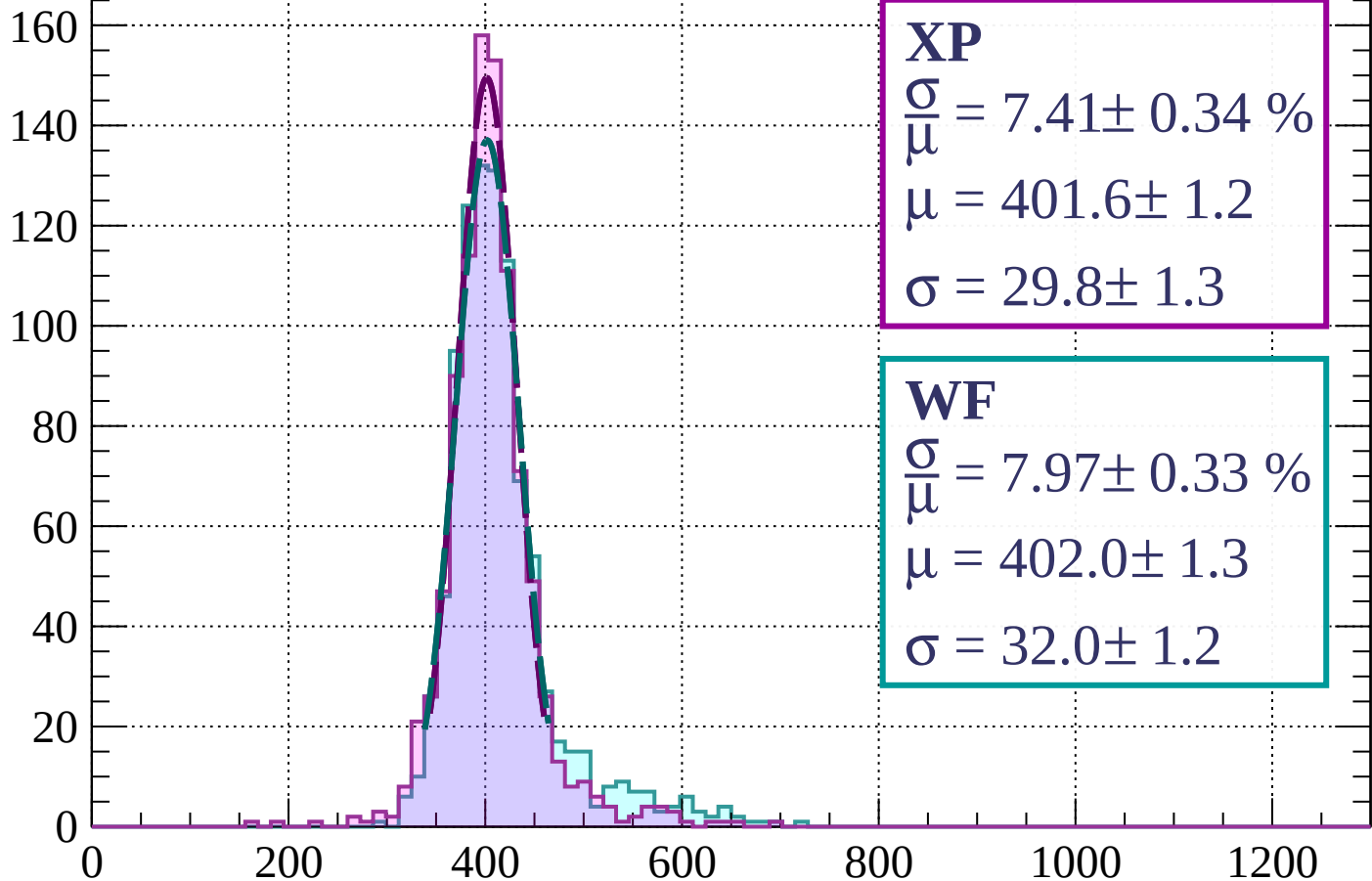
Energy loss | $36 < \phi < 39$

Count



Energy loss | $39 < \phi < 42$

Count



XP

$$\frac{\sigma}{\mu} = 7.41 \pm 0.34 \%$$

$$\mu = 401.6 \pm 1.2$$

$$\sigma = 29.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 7.97 \pm 0.33 \%$$

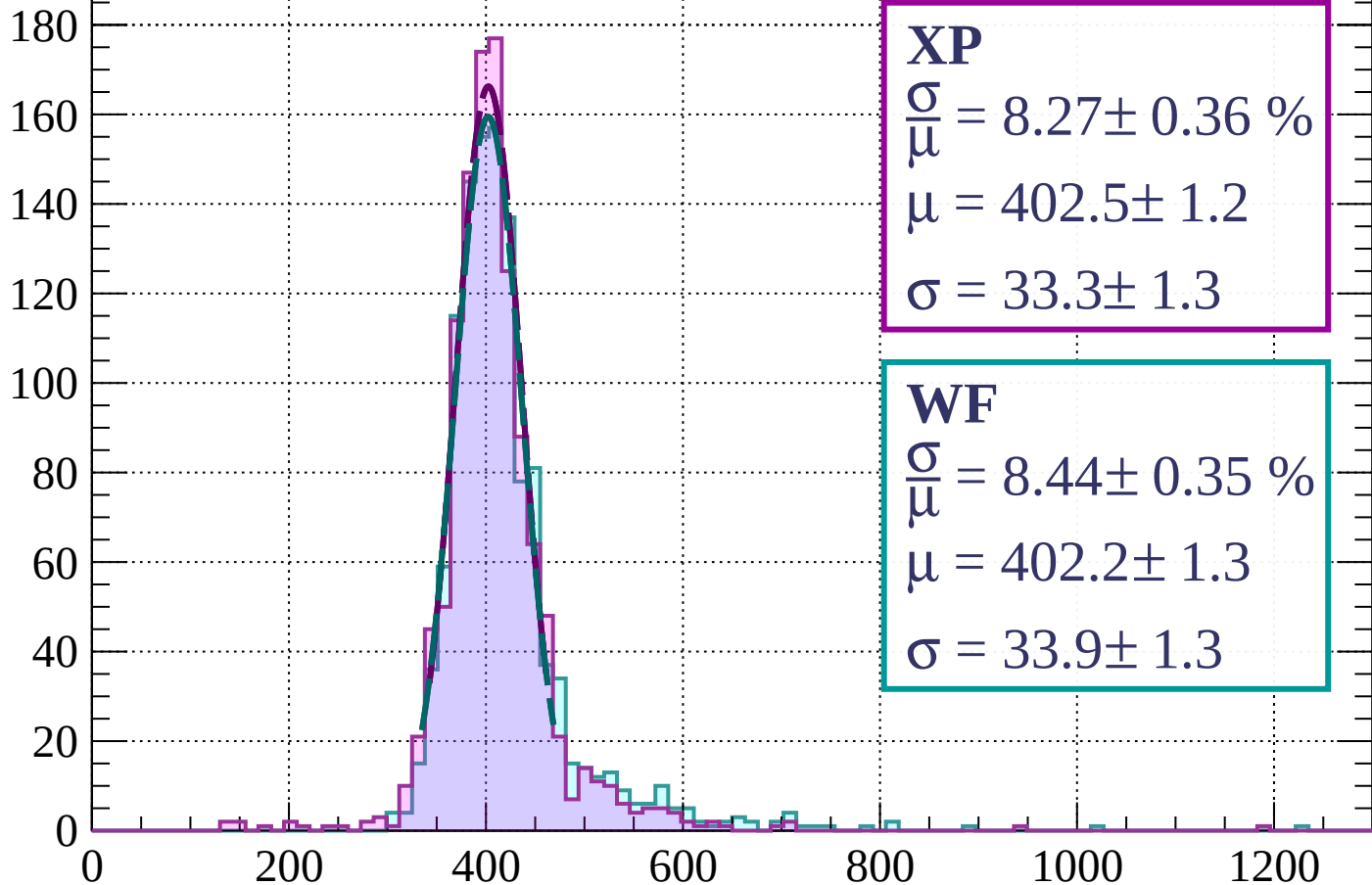
$$\mu = 402.0 \pm 1.3$$

$$\sigma = 32.0 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $42 < \phi < 45$

Count



XP

$$\frac{\sigma}{\mu} = 8.27 \pm 0.36 \%$$

$$\mu = 402.5 \pm 1.2$$

$$\sigma = 33.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.44 \pm 0.35 \%$$

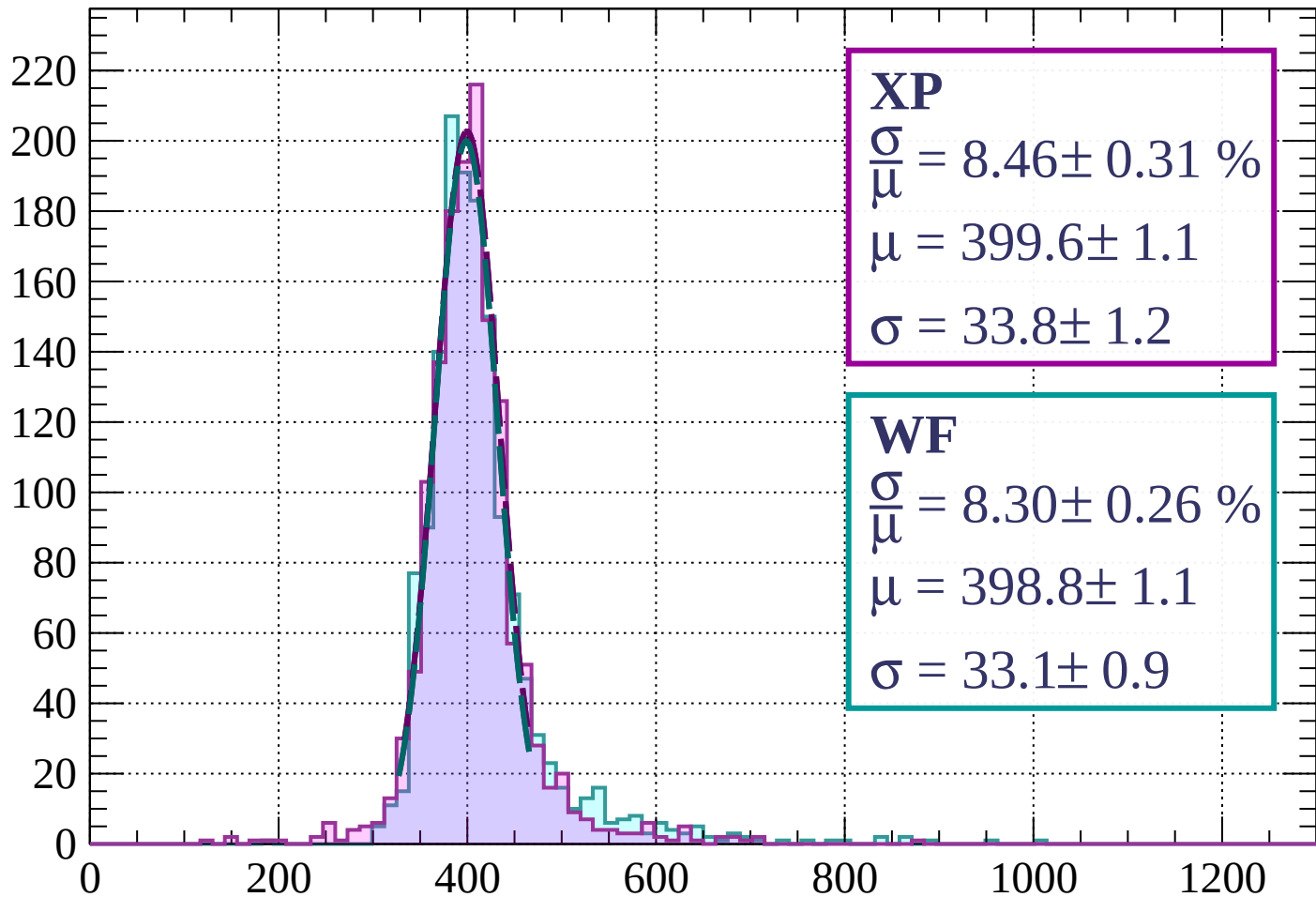
$$\mu = 402.2 \pm 1.3$$

$$\sigma = 33.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $45 < \phi < 48$

Count



XP

$$\frac{\sigma}{\mu} = 8.46 \pm 0.31 \%$$

$$\mu = 399.6 \pm 1.1$$

$$\sigma = 33.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 8.30 \pm 0.26 \%$$

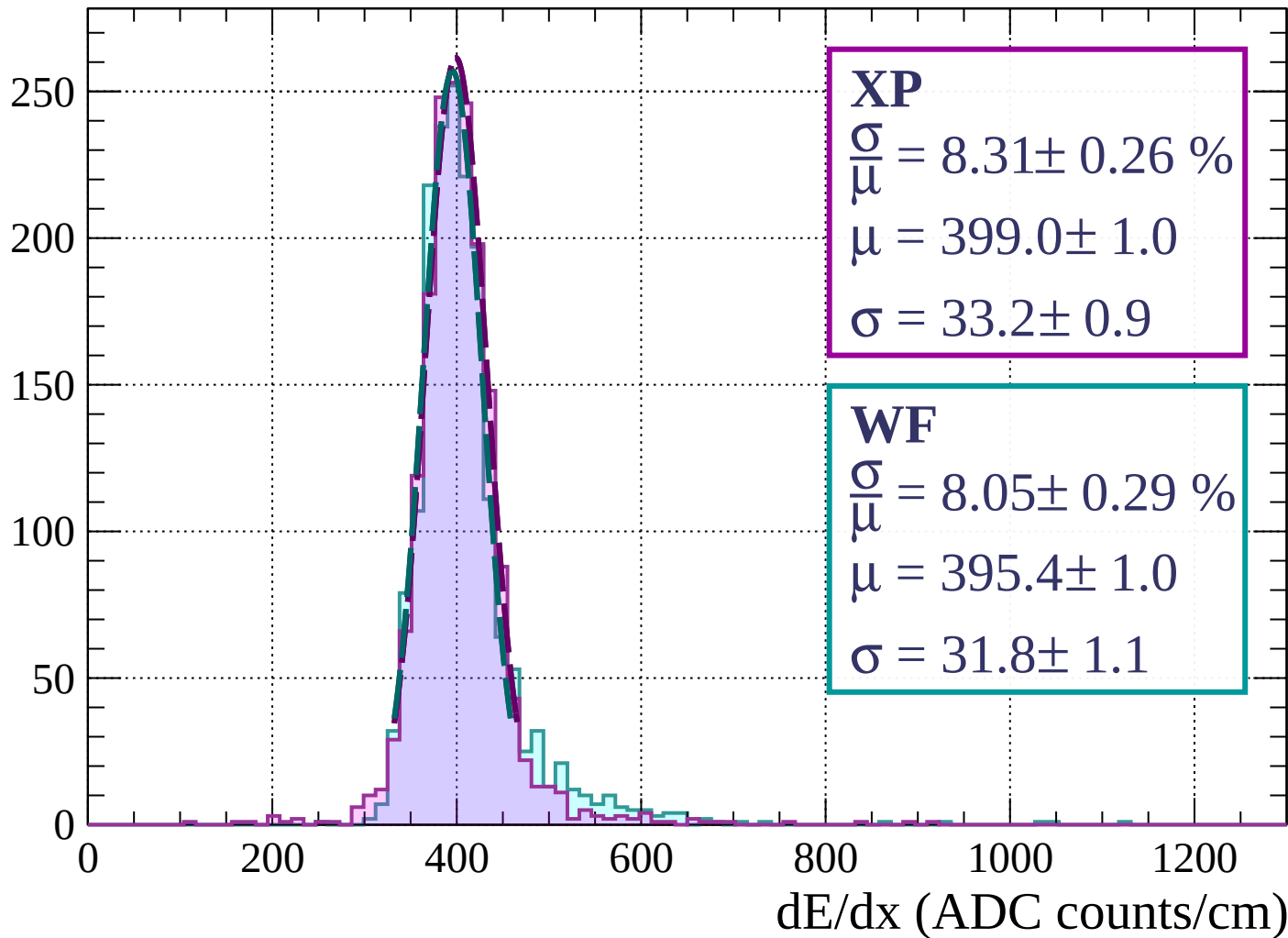
$$\mu = 398.8 \pm 1.1$$

$$\sigma = 33.1 \pm 0.9$$

dE/dx (ADC counts/cm)

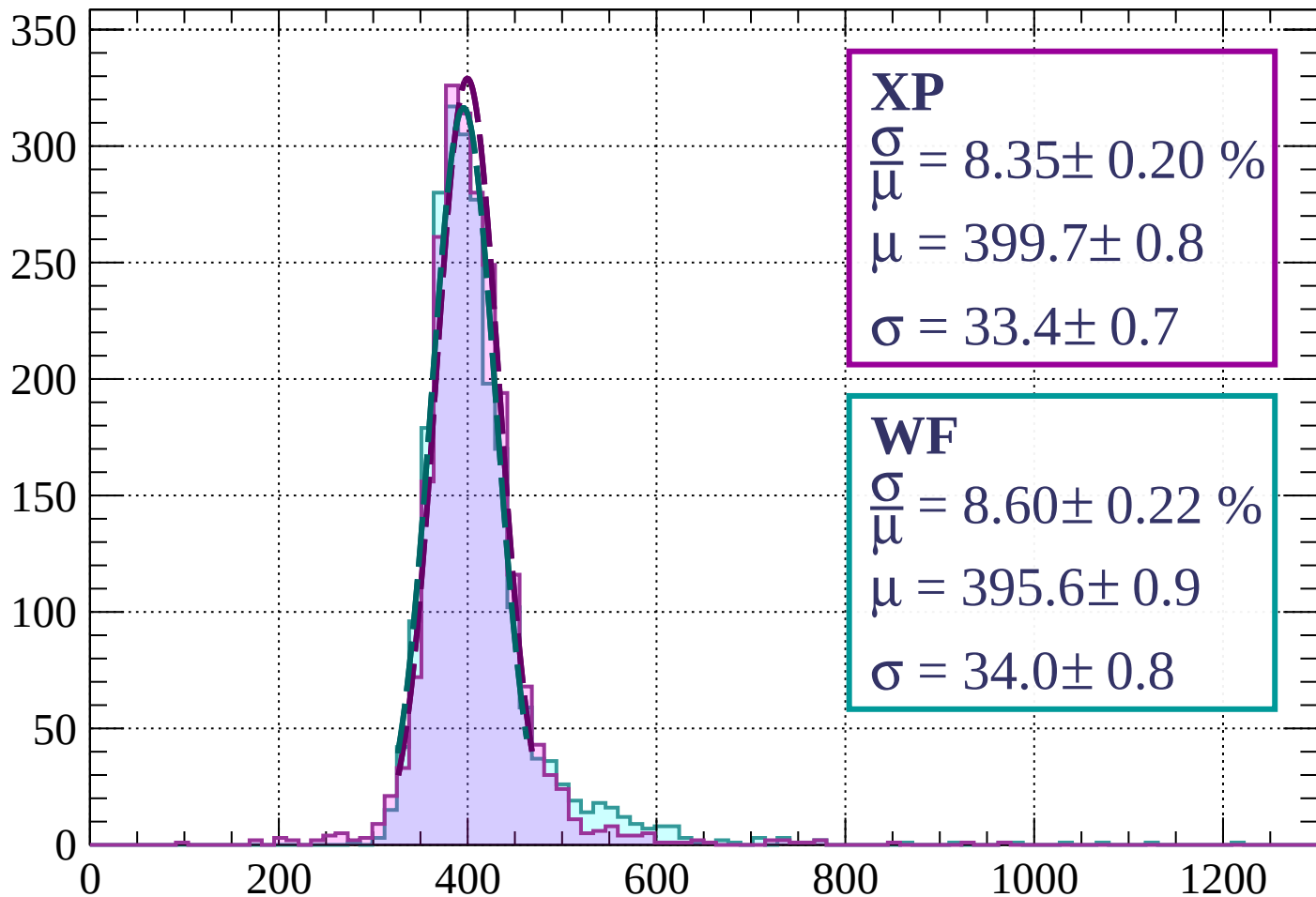
Energy loss | $48 < \phi < 51$

Count



Energy loss | $51 < \phi < 54$

Count



XP

$$\frac{\sigma}{\mu} = 8.35 \pm 0.20 \%$$

$$\mu = 399.7 \pm 0.8$$

$$\sigma = 33.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.60 \pm 0.22 \%$$

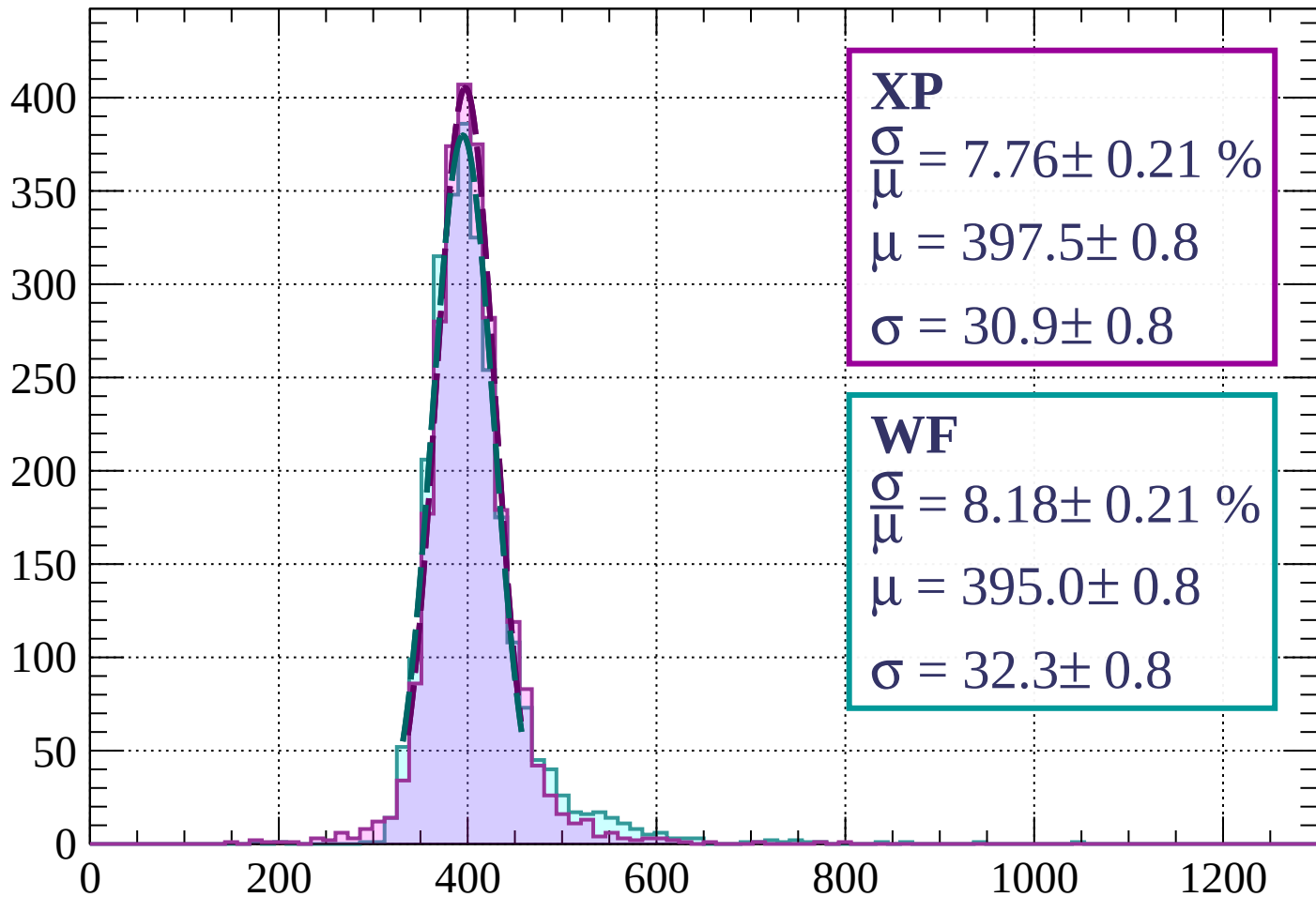
$$\mu = 395.6 \pm 0.9$$

$$\sigma = 34.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $54 < \phi < 57$

Count



XP

$$\frac{\sigma}{\mu} = 7.76 \pm 0.21 \%$$

$$\mu = 397.5 \pm 0.8$$

$$\sigma = 30.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.18 \pm 0.21 \%$$

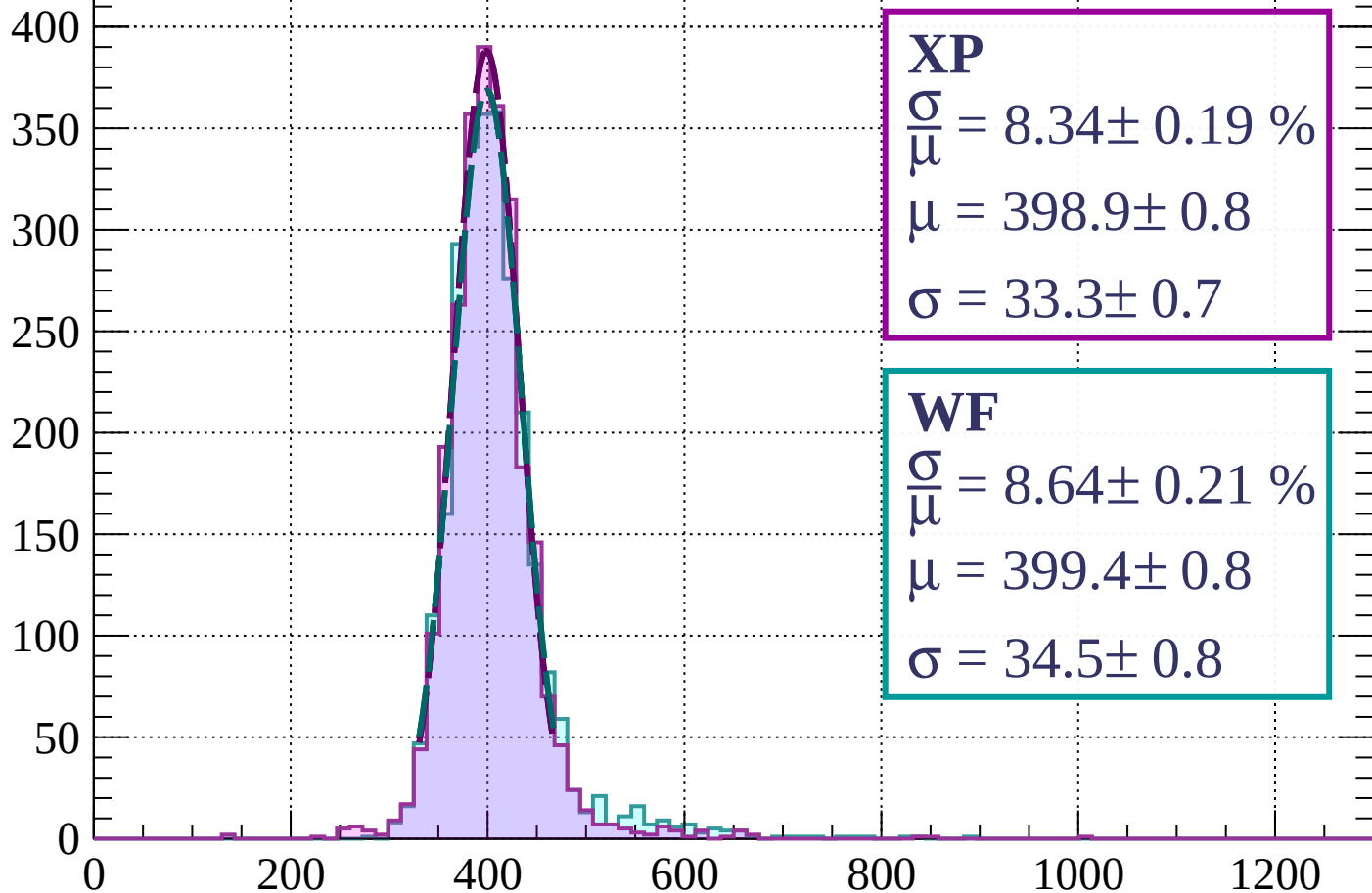
$$\mu = 395.0 \pm 0.8$$

$$\sigma = 32.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $57 < \phi < 60$

Count



XP

$$\frac{\sigma}{\mu} = 8.34 \pm 0.19 \%$$

$$\mu = 398.9 \pm 0.8$$

$$\sigma = 33.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.64 \pm 0.21 \%$$

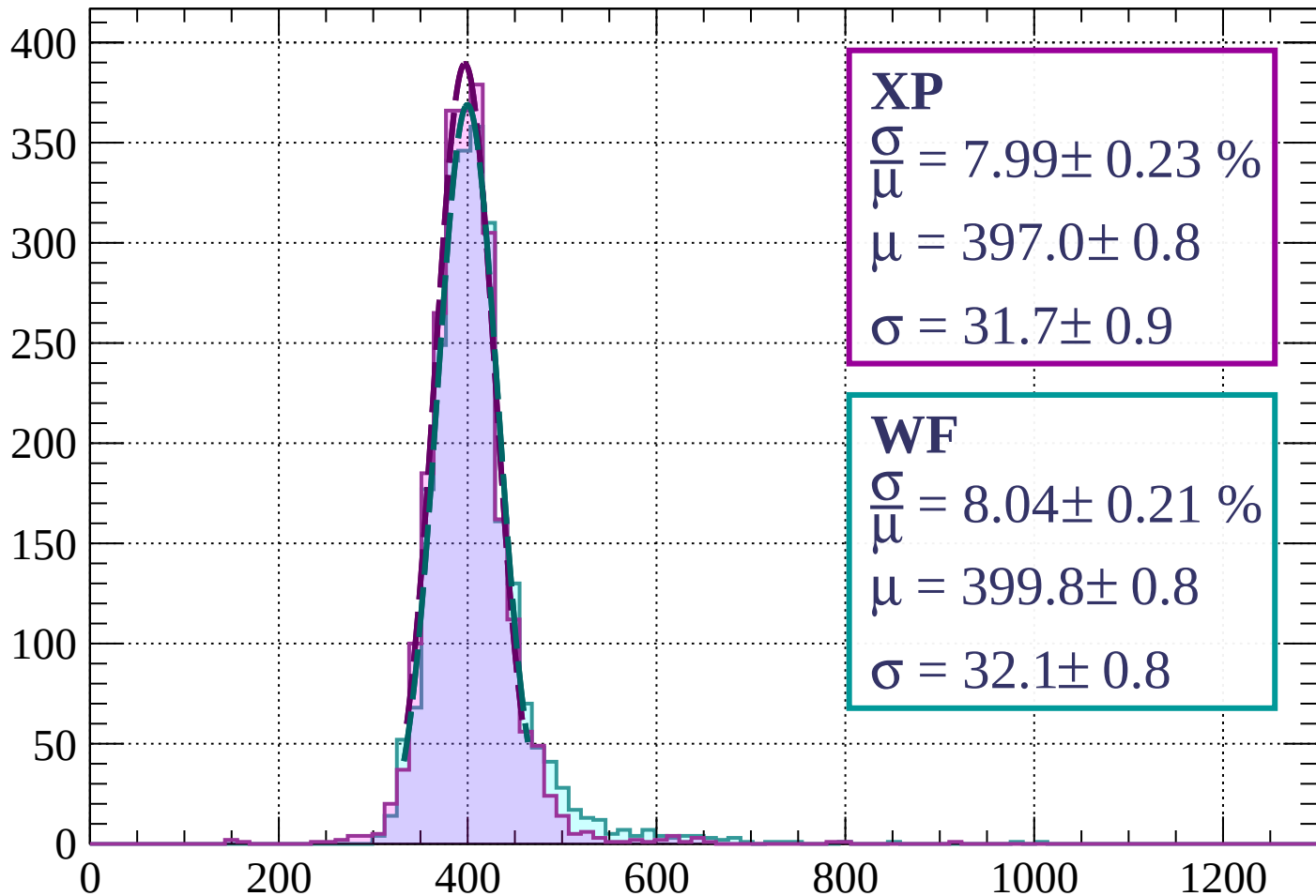
$$\mu = 399.4 \pm 0.8$$

$$\sigma = 34.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $60 < \phi < 63$

Count



XP

$$\frac{\sigma}{\mu} = 7.99 \pm 0.23 \%$$

$$\mu = 397.0 \pm 0.8$$

$$\sigma = 31.7 \pm 0.9$$

WF

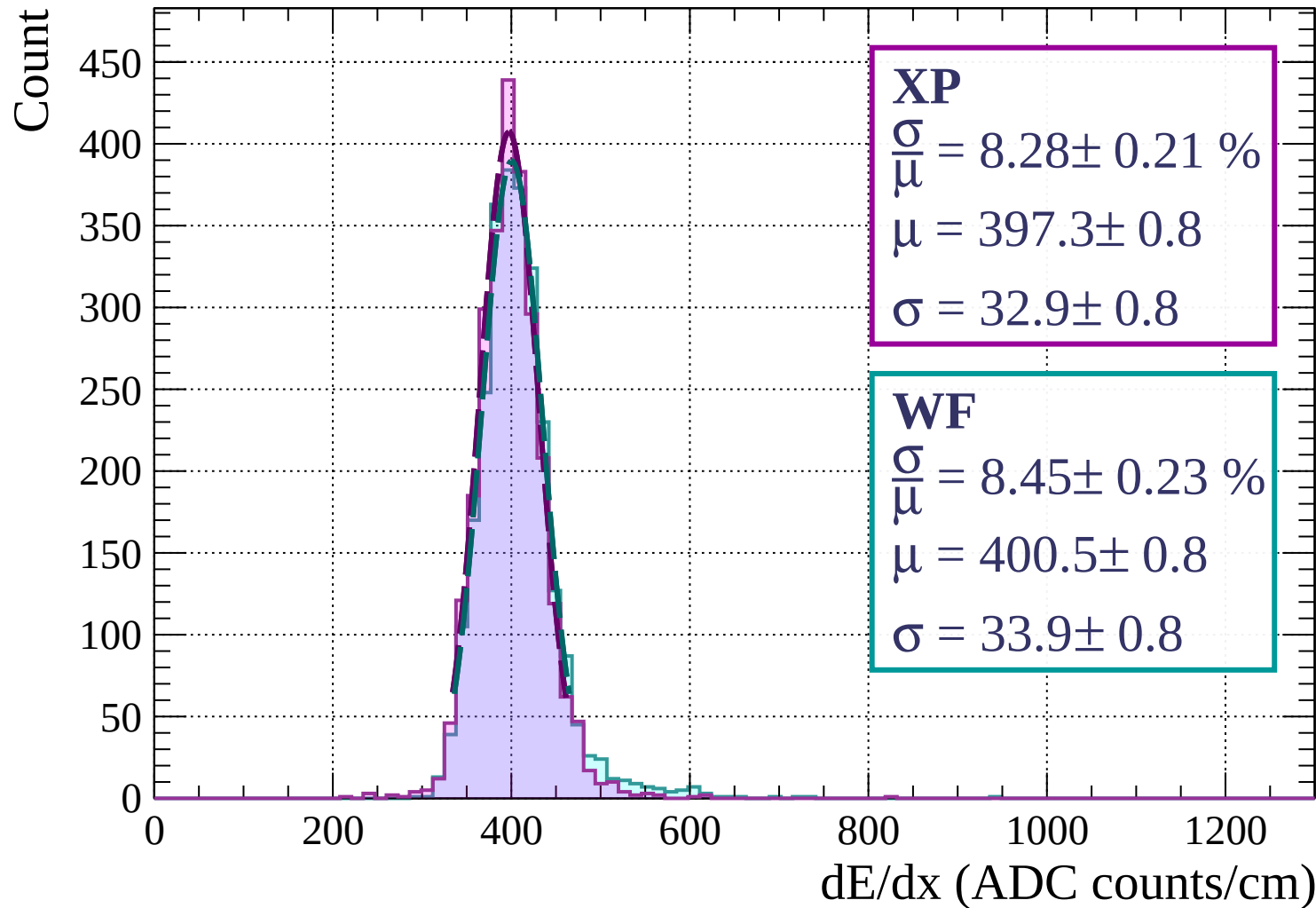
$$\frac{\sigma}{\mu} = 8.04 \pm 0.21 \%$$

$$\mu = 399.8 \pm 0.8$$

$$\sigma = 32.1 \pm 0.8$$

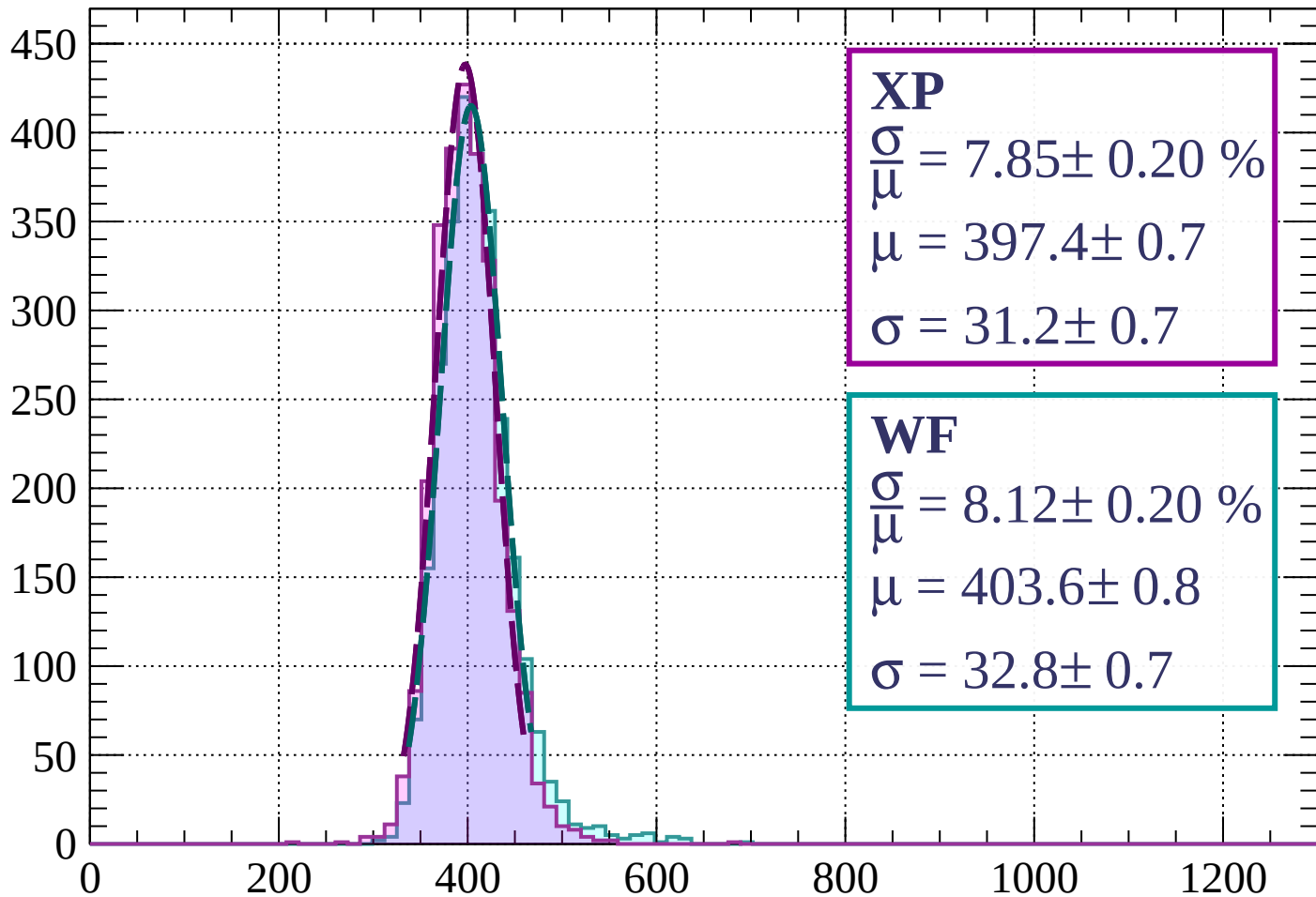
dE/dx (ADC counts/cm)

Energy loss | $63 < \phi < 66$



Energy loss | $66 < \phi < 69$

Count



XP

$$\frac{\sigma}{\mu} = 7.85 \pm 0.20 \%$$

$$\mu = 397.4 \pm 0.7$$

$$\sigma = 31.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.12 \pm 0.20 \%$$

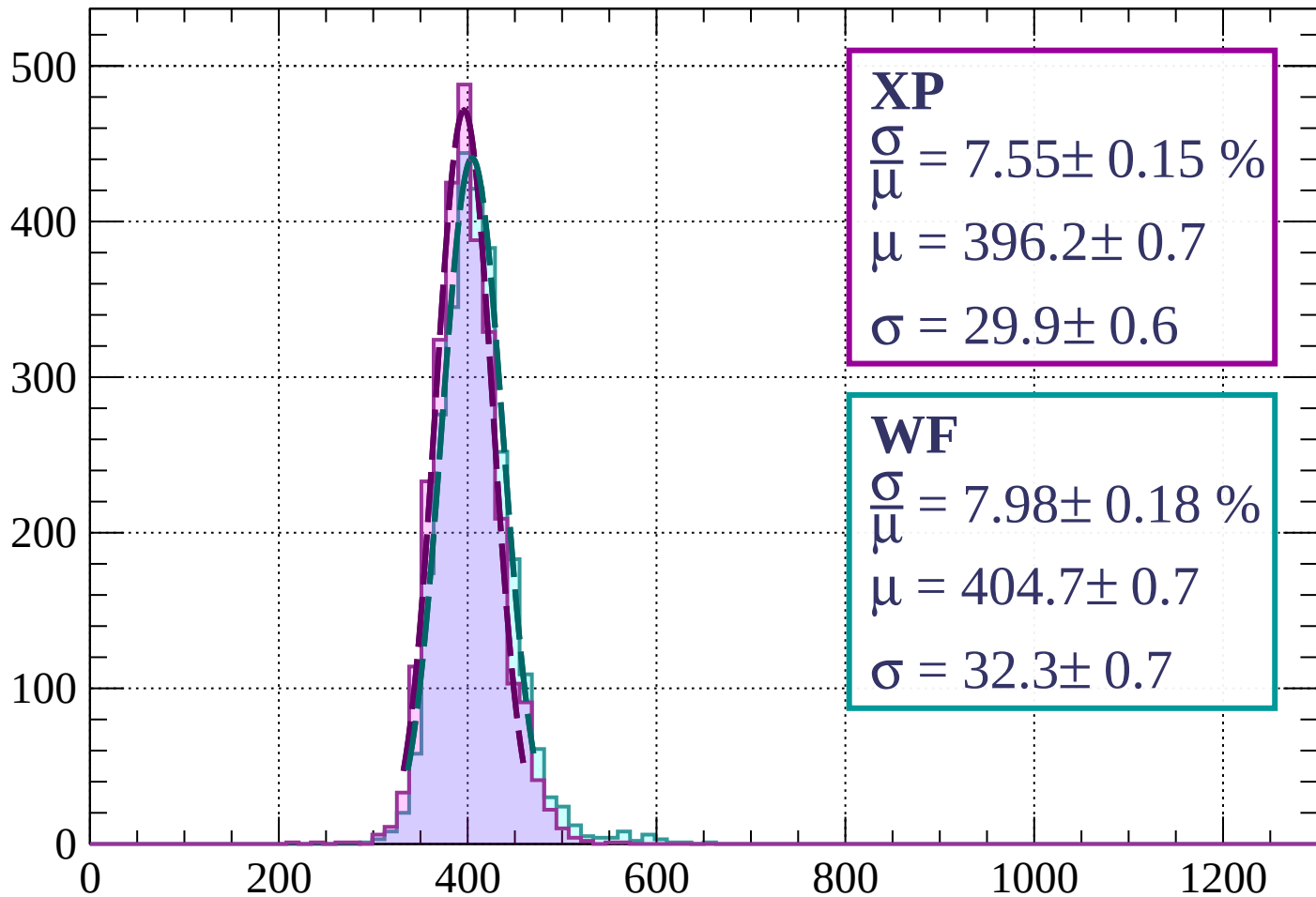
$$\mu = 403.6 \pm 0.8$$

$$\sigma = 32.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $69 < \phi < 72$

Count



XP

$$\frac{\sigma}{\mu} = 7.55 \pm 0.15 \%$$

$$\mu = 396.2 \pm 0.7$$

$$\sigma = 29.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.98 \pm 0.18 \%$$

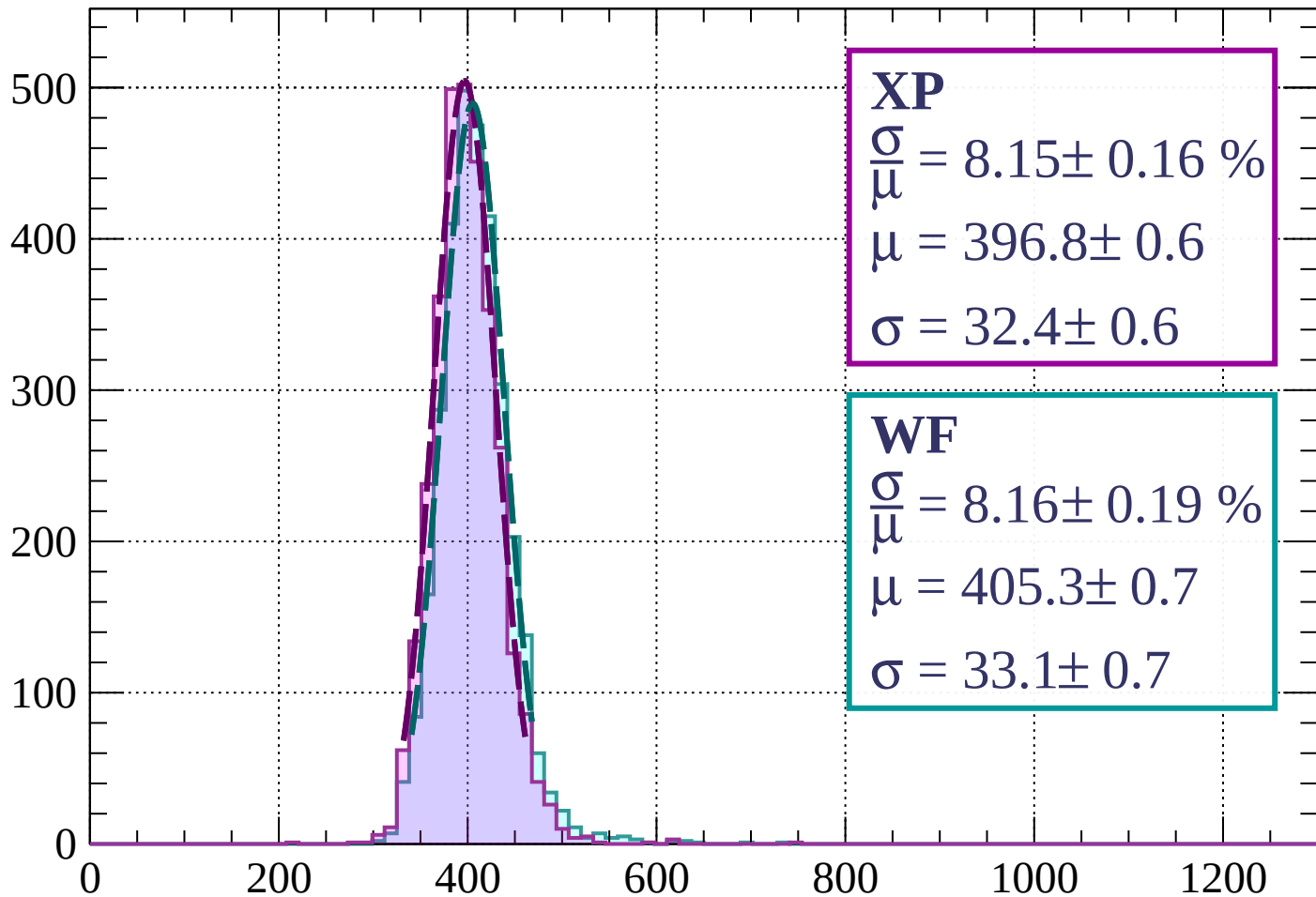
$$\mu = 404.7 \pm 0.7$$

$$\sigma = 32.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $72 < \phi < 75$

Count



XP

$$\frac{\sigma}{\mu} = 8.15 \pm 0.16 \%$$

$$\mu = 396.8 \pm 0.6$$

$$\sigma = 32.4 \pm 0.6$$

WF

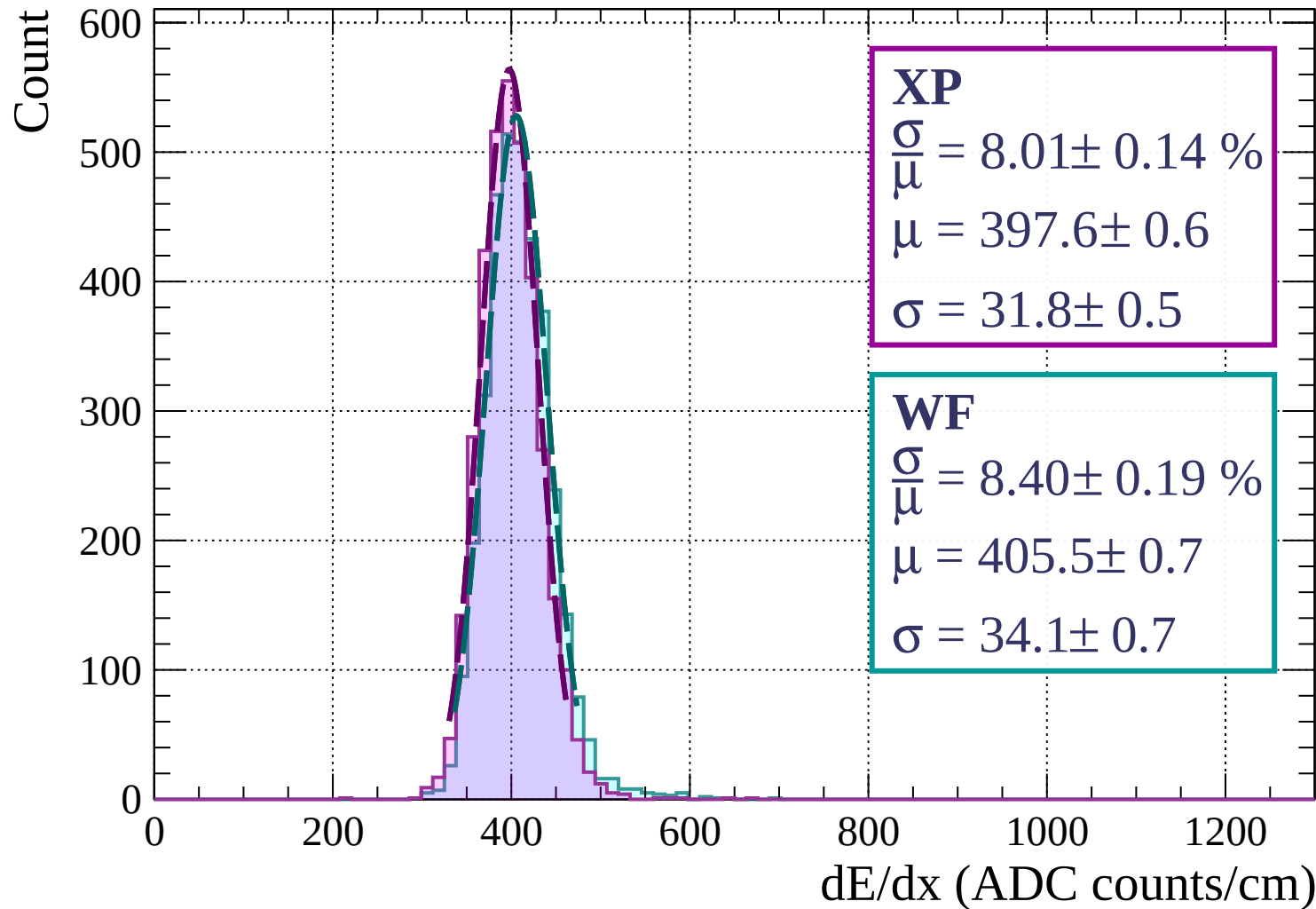
$$\frac{\sigma}{\mu} = 8.16 \pm 0.19 \%$$

$$\mu = 405.3 \pm 0.7$$

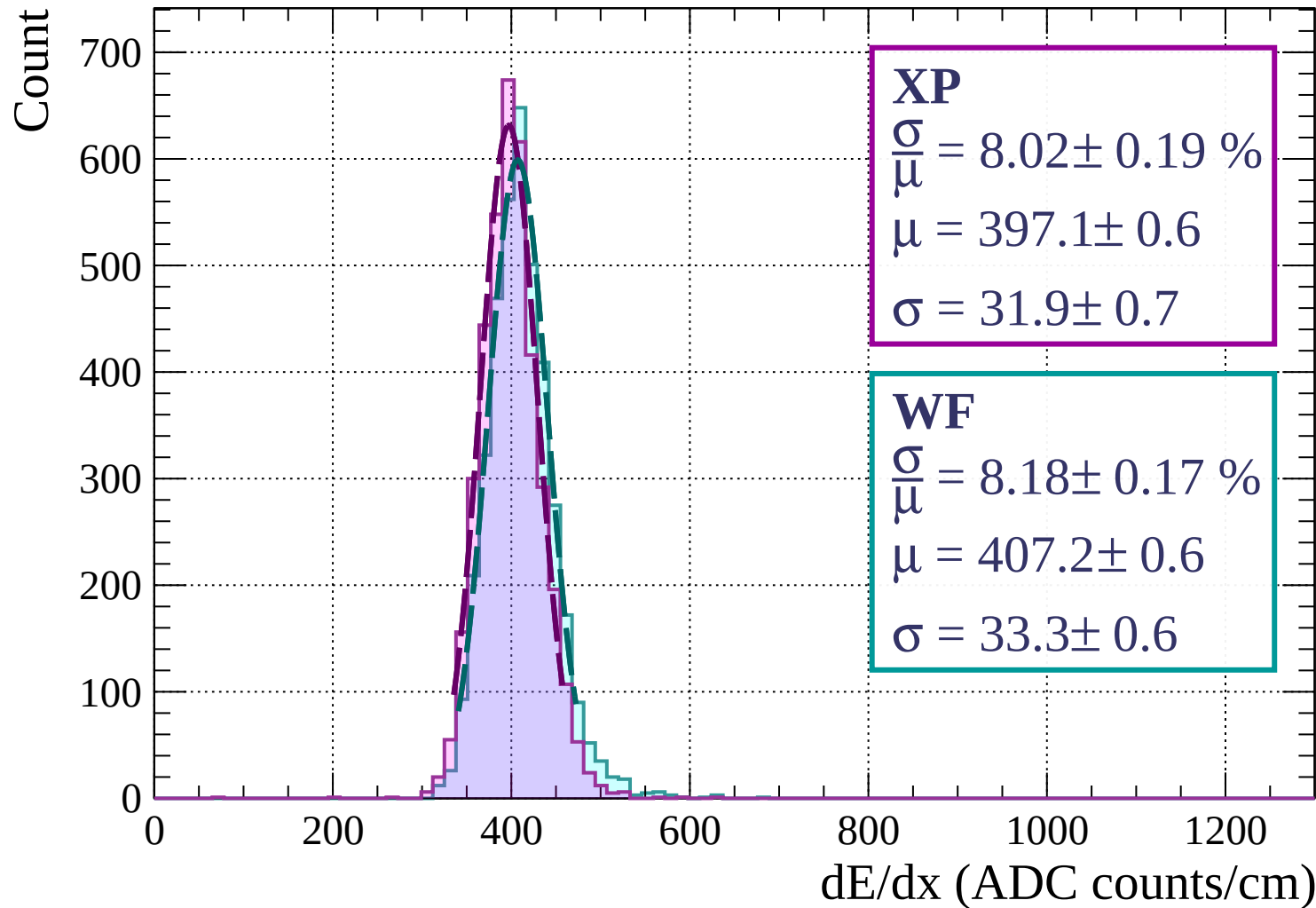
$$\sigma = 33.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $75 < \phi < 78$

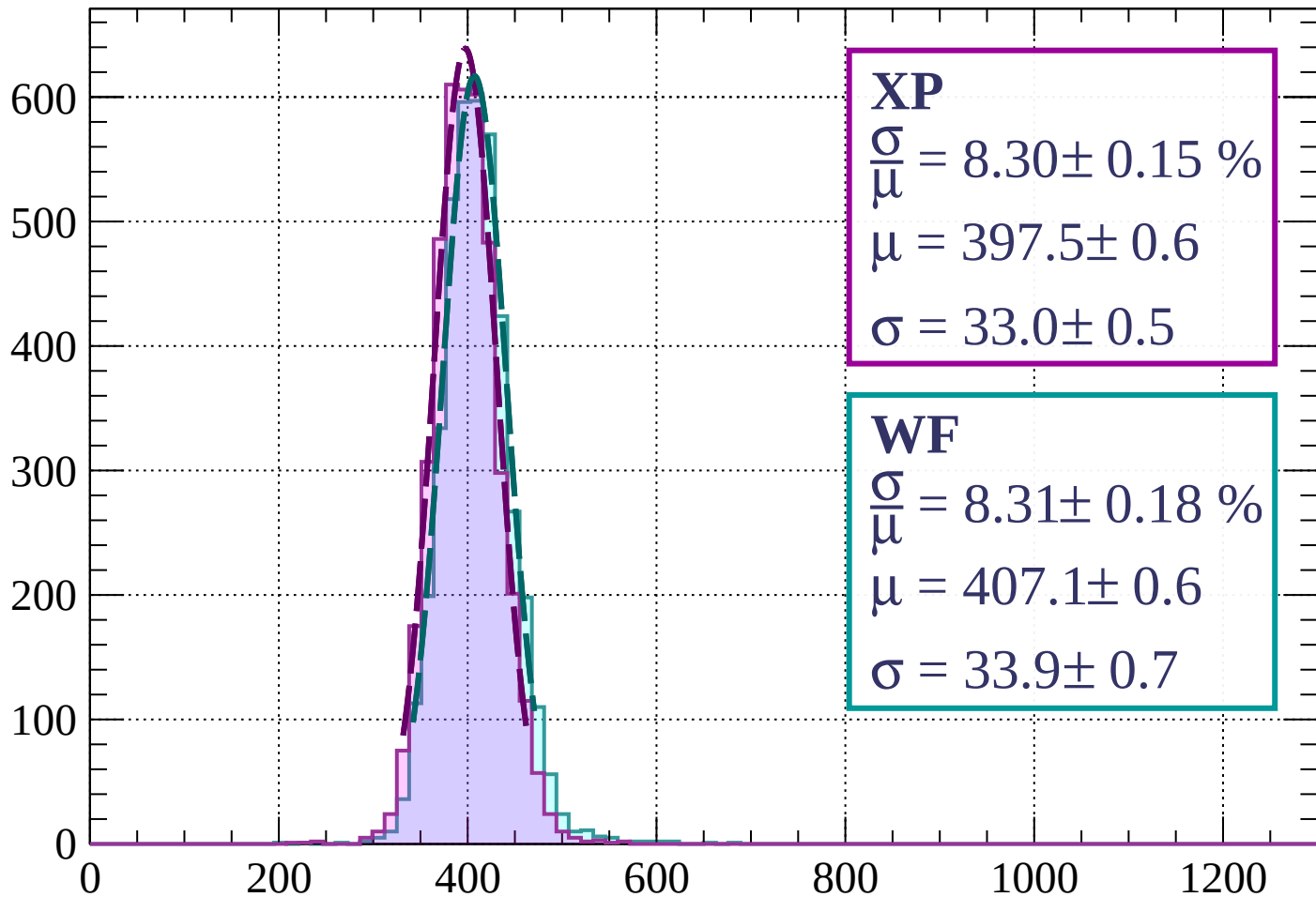


Energy loss | $78 < \phi < 81$



Energy loss | $81 < \phi < 84$

Count



XP

$$\frac{\sigma}{\mu} = 8.30 \pm 0.15 \%$$

$$\mu = 397.5 \pm 0.6$$

$$\sigma = 33.0 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.31 \pm 0.18 \%$$

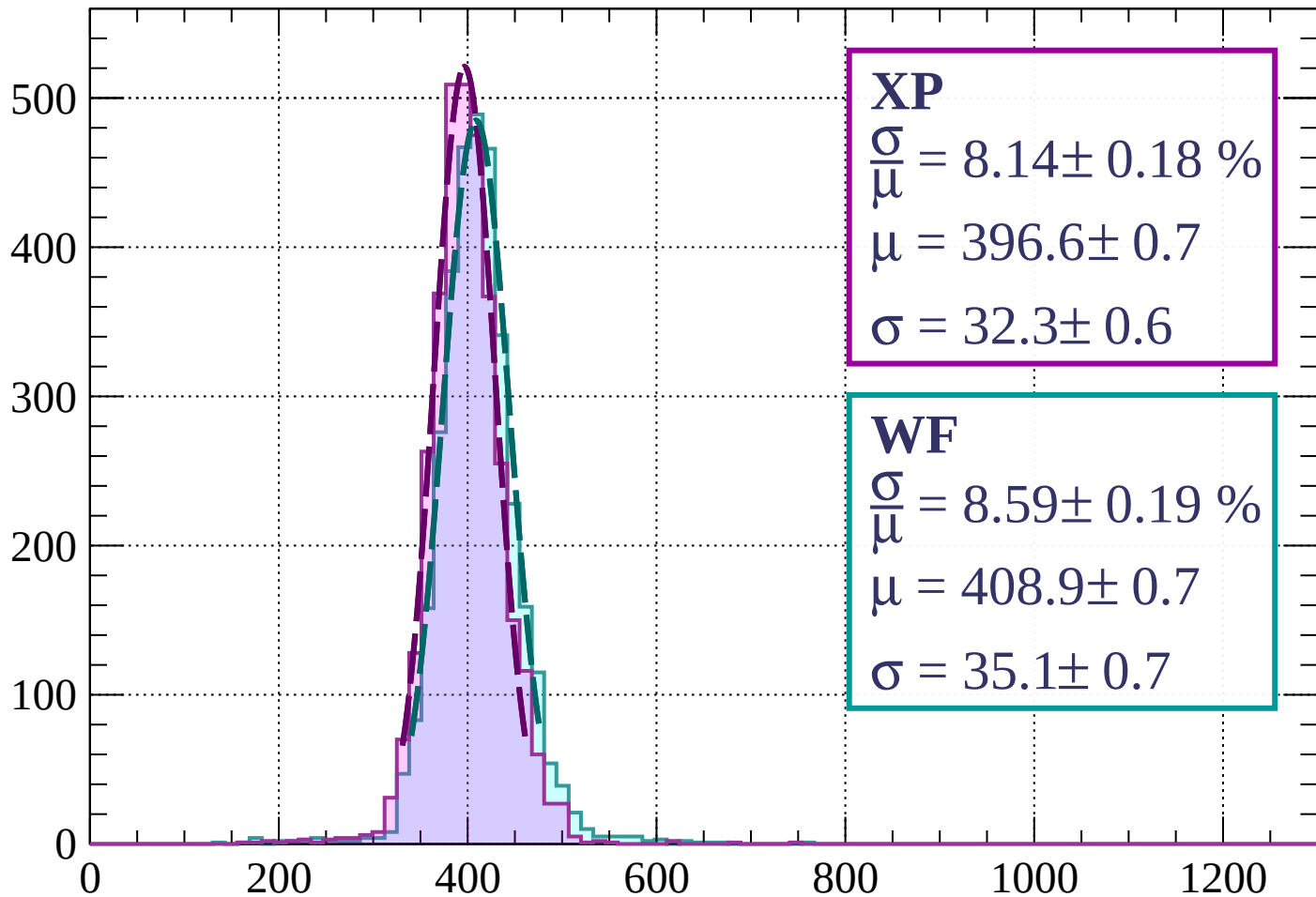
$$\mu = 407.1 \pm 0.6$$

$$\sigma = 33.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $84 < \phi < 87$

Count



XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.18 \%$$

$$\mu = 396.6 \pm 0.7$$

$$\sigma = 32.3 \pm 0.6$$

WF

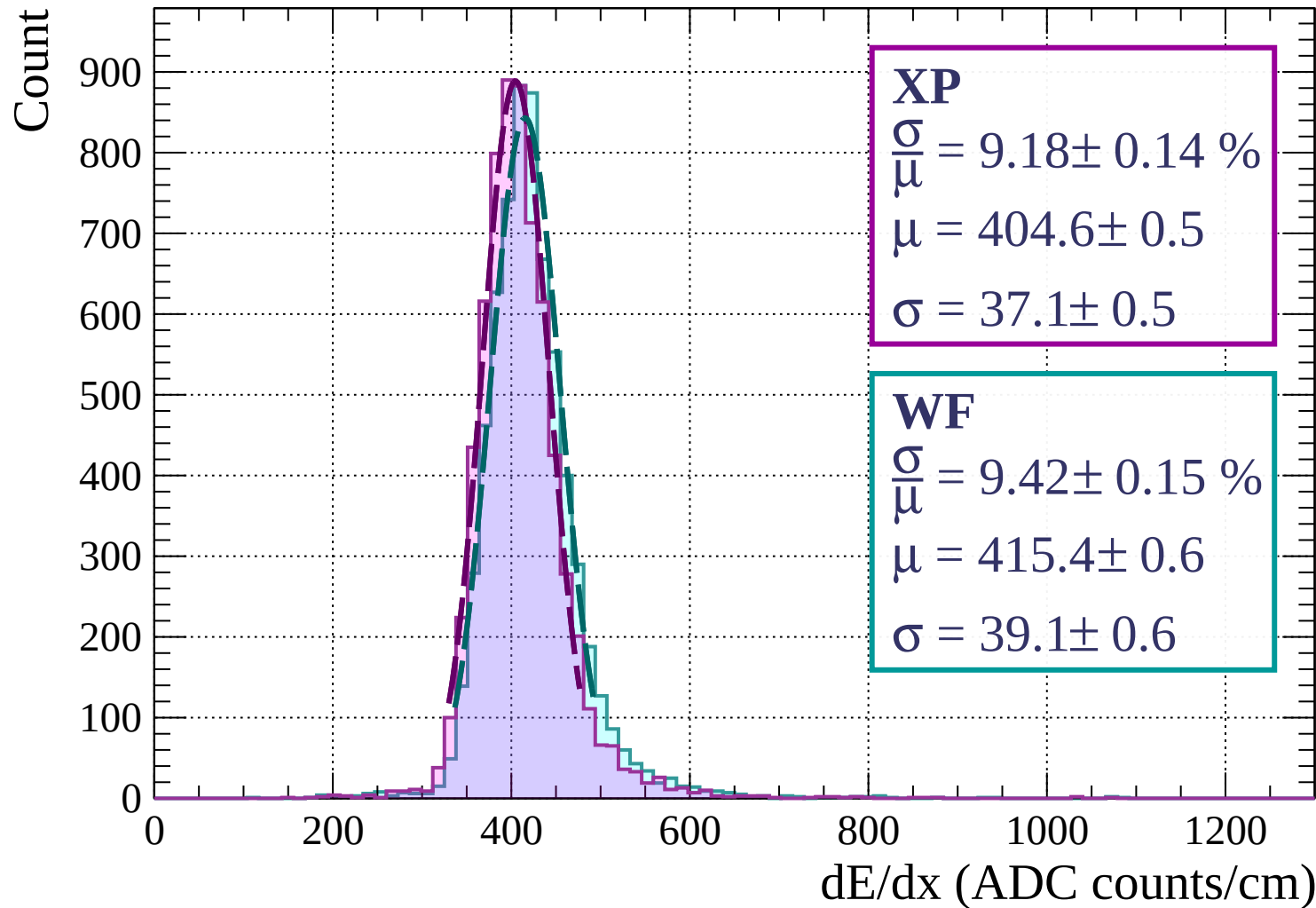
$$\frac{\sigma}{\mu} = 8.59 \pm 0.19 \%$$

$$\mu = 408.9 \pm 0.7$$

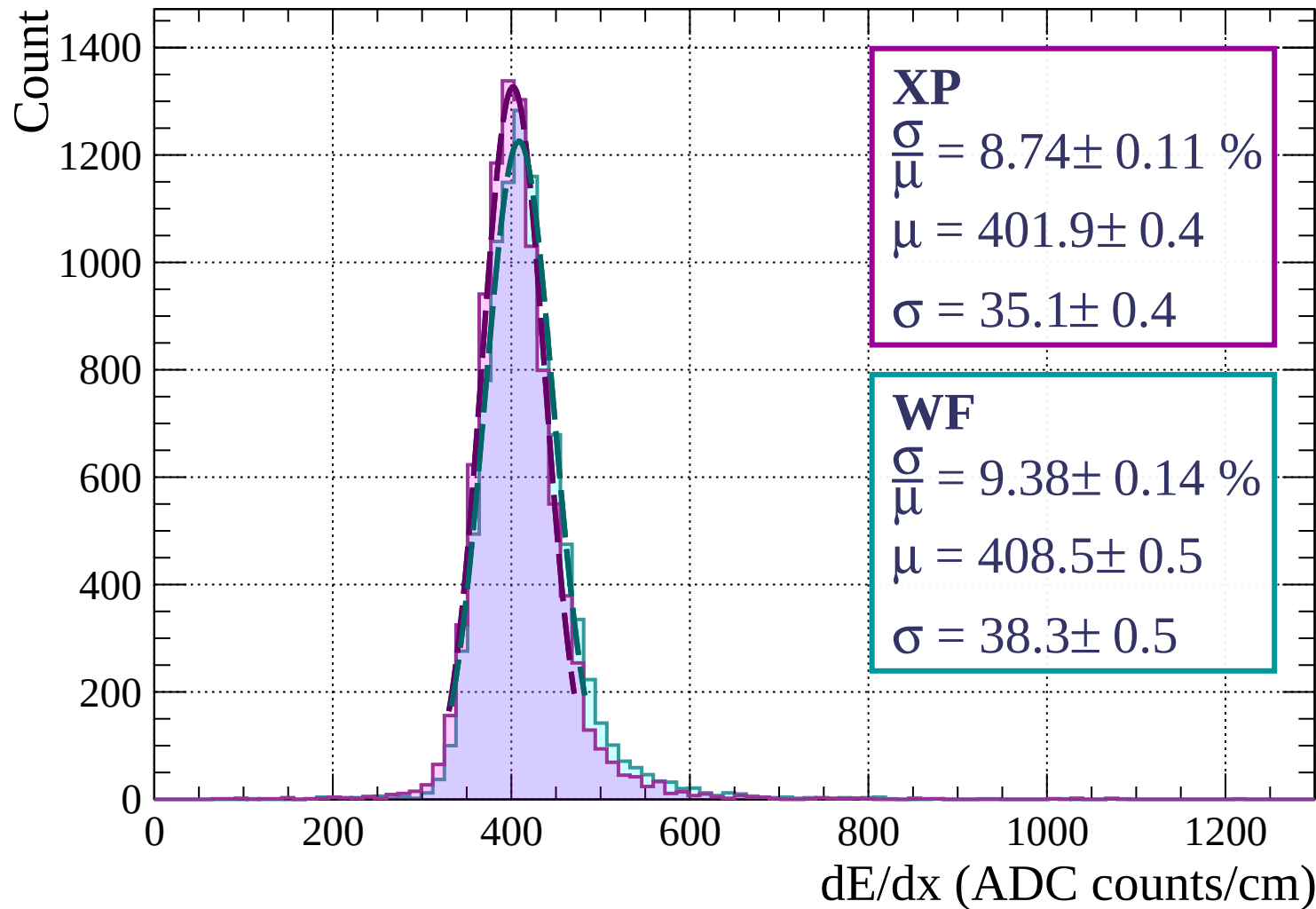
$$\sigma = 35.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $87 < \phi < 90$

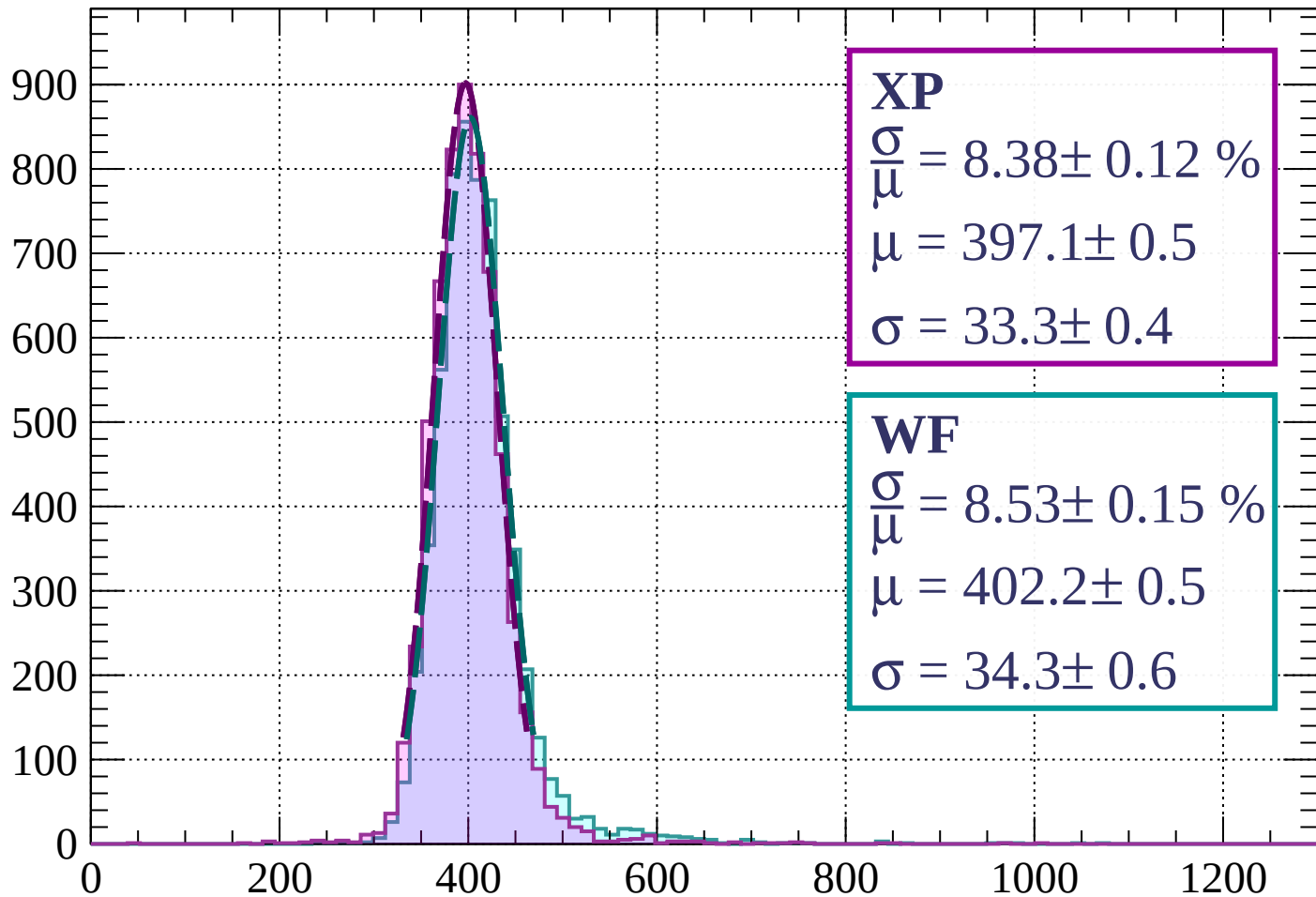


Energy loss | $0 < \theta < 3$



Energy loss | $3 < \theta < 6$

Count



XP

$$\frac{\sigma}{\mu} = 8.38 \pm 0.12 \%$$

$$\mu = 397.1 \pm 0.5$$

$$\sigma = 33.3 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 8.53 \pm 0.15 \%$$

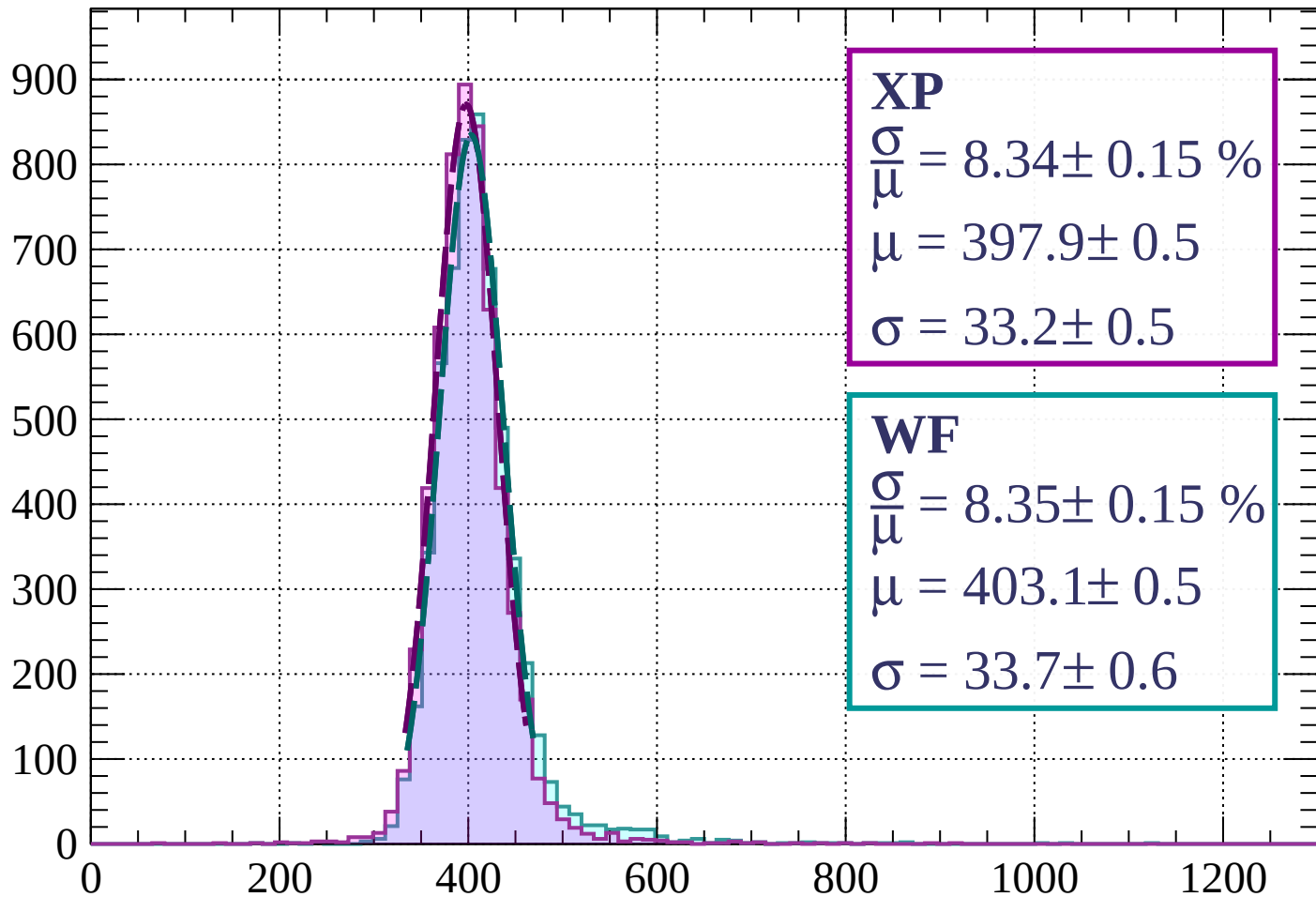
$$\mu = 402.2 \pm 0.5$$

$$\sigma = 34.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 9$

Count



XP

$$\frac{\sigma}{\mu} = 8.34 \pm 0.15 \%$$

$$\mu = 397.9 \pm 0.5$$

$$\sigma = 33.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 8.35 \pm 0.15 \%$$

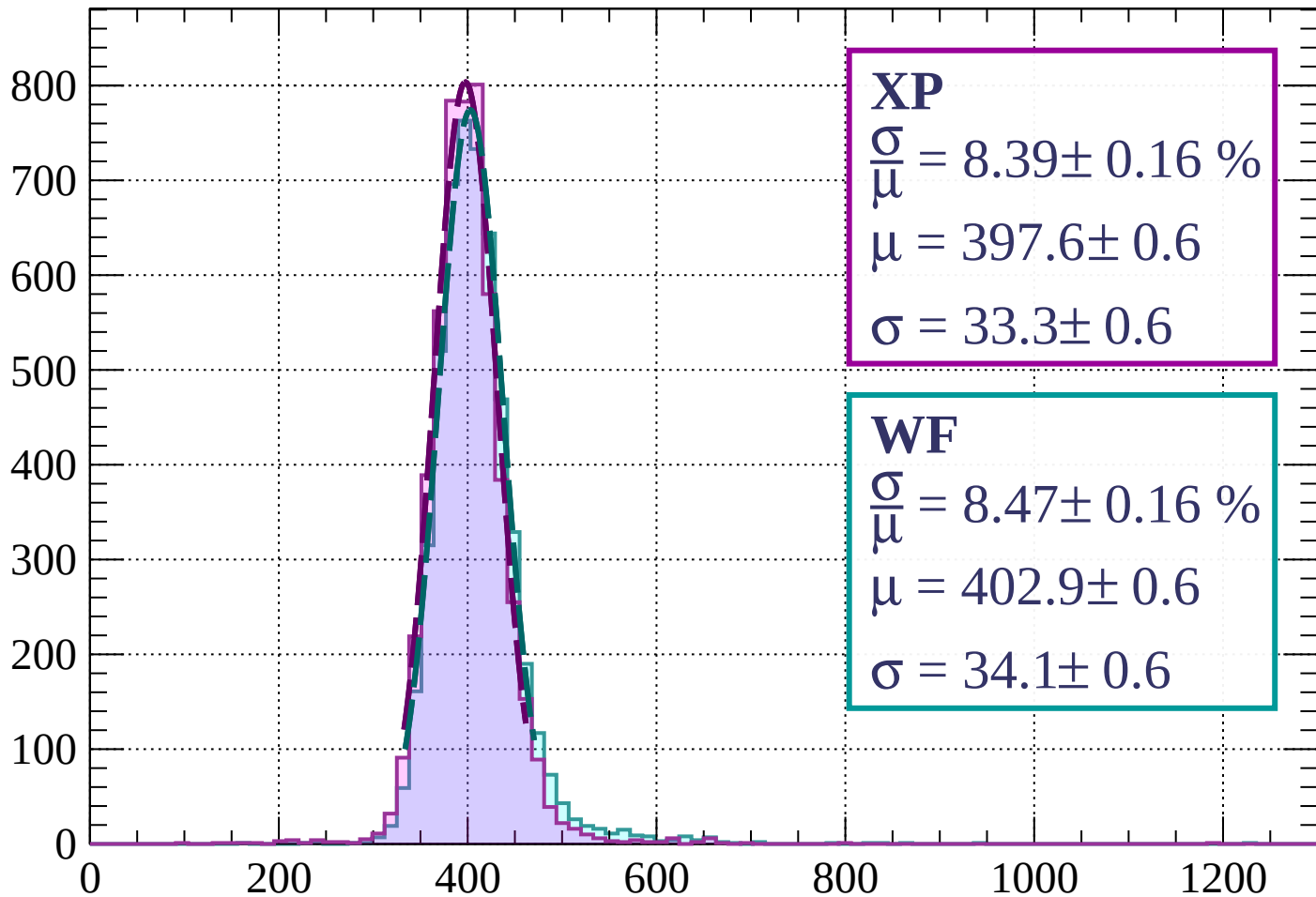
$$\mu = 403.1 \pm 0.5$$

$$\sigma = 33.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $9 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 8.39 \pm 0.16 \%$$

$$\mu = 397.6 \pm 0.6$$

$$\sigma = 33.3 \pm 0.6$$

WF

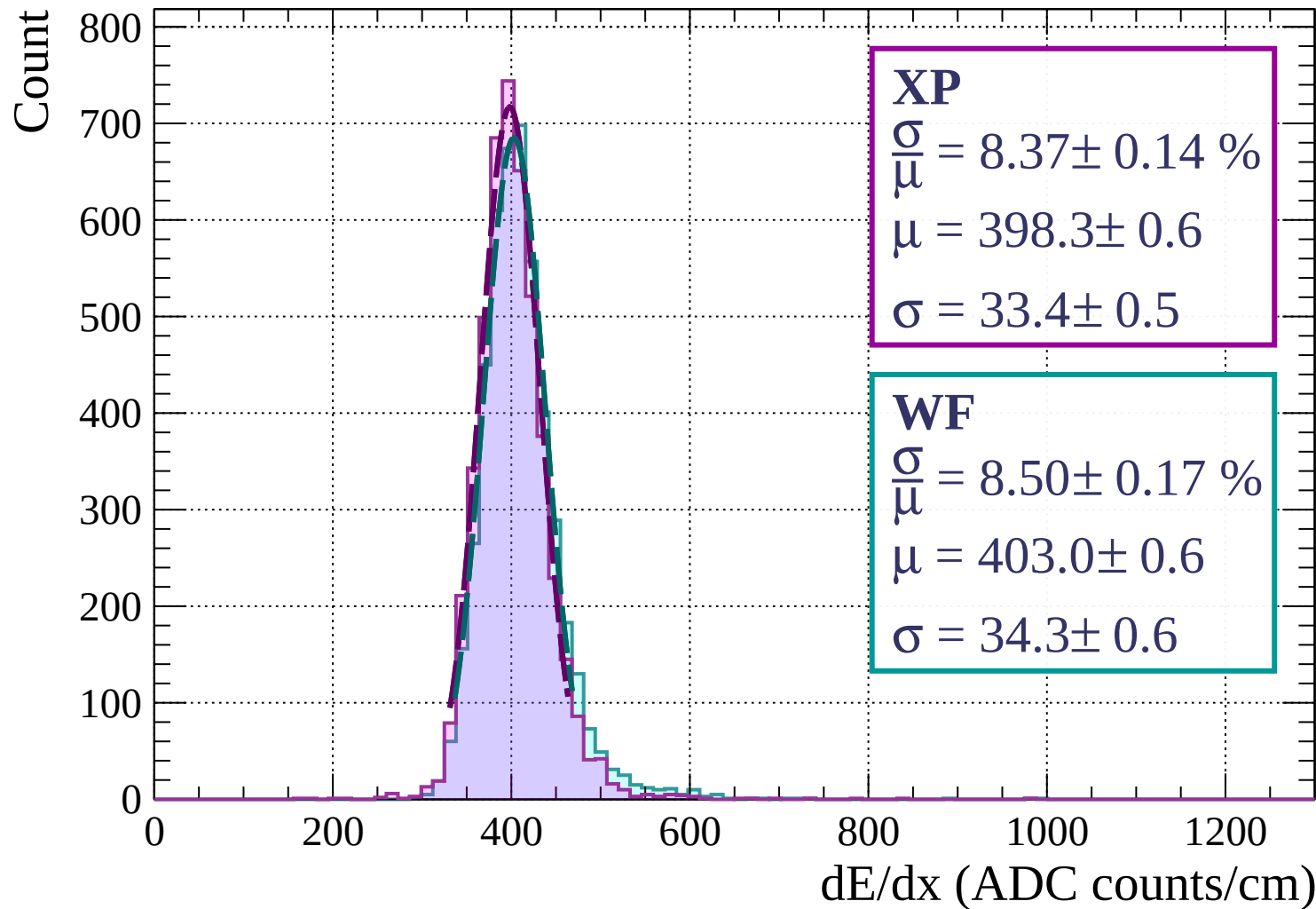
$$\frac{\sigma}{\mu} = 8.47 \pm 0.16 \%$$

$$\mu = 402.9 \pm 0.6$$

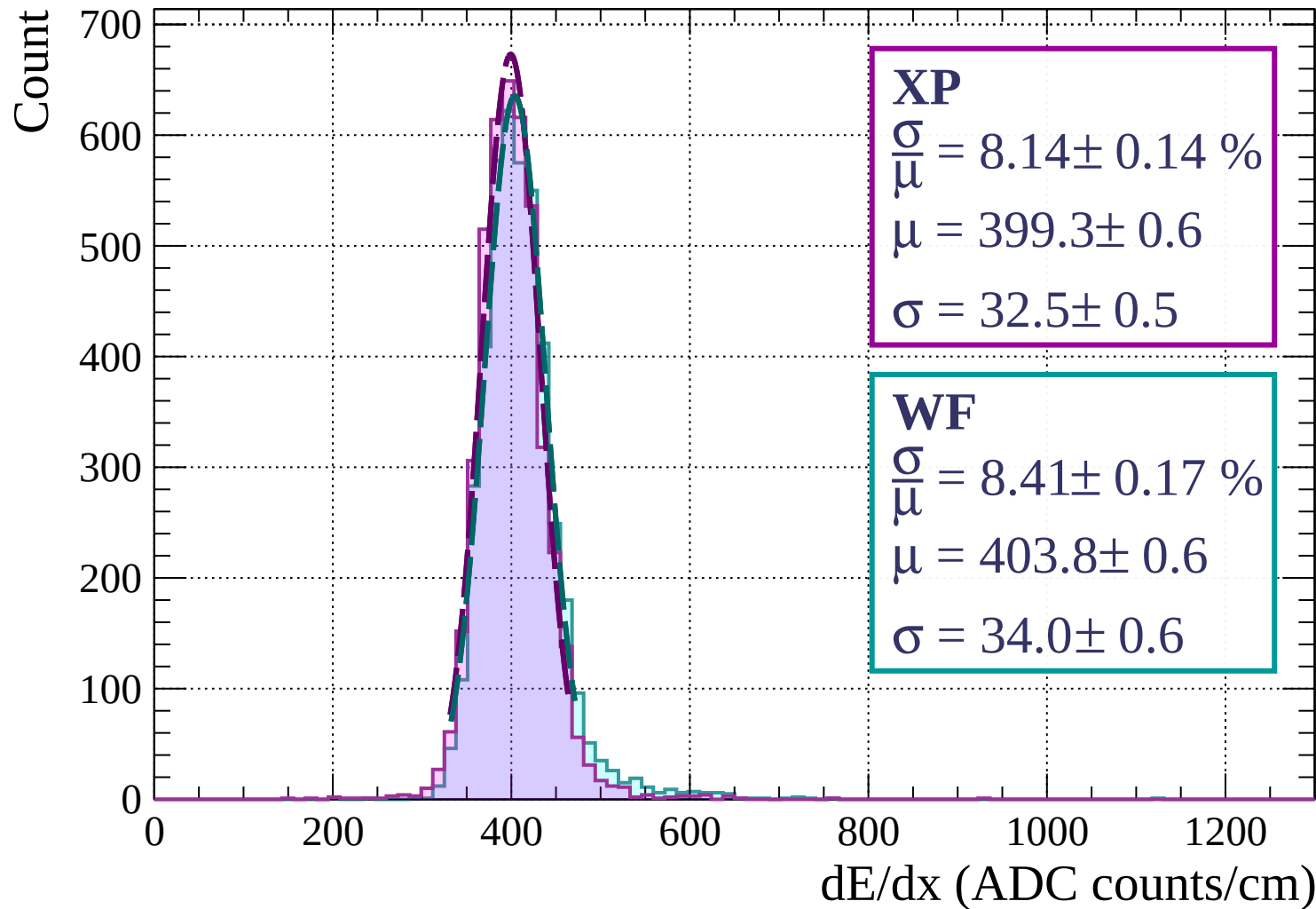
$$\sigma = 34.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 15$

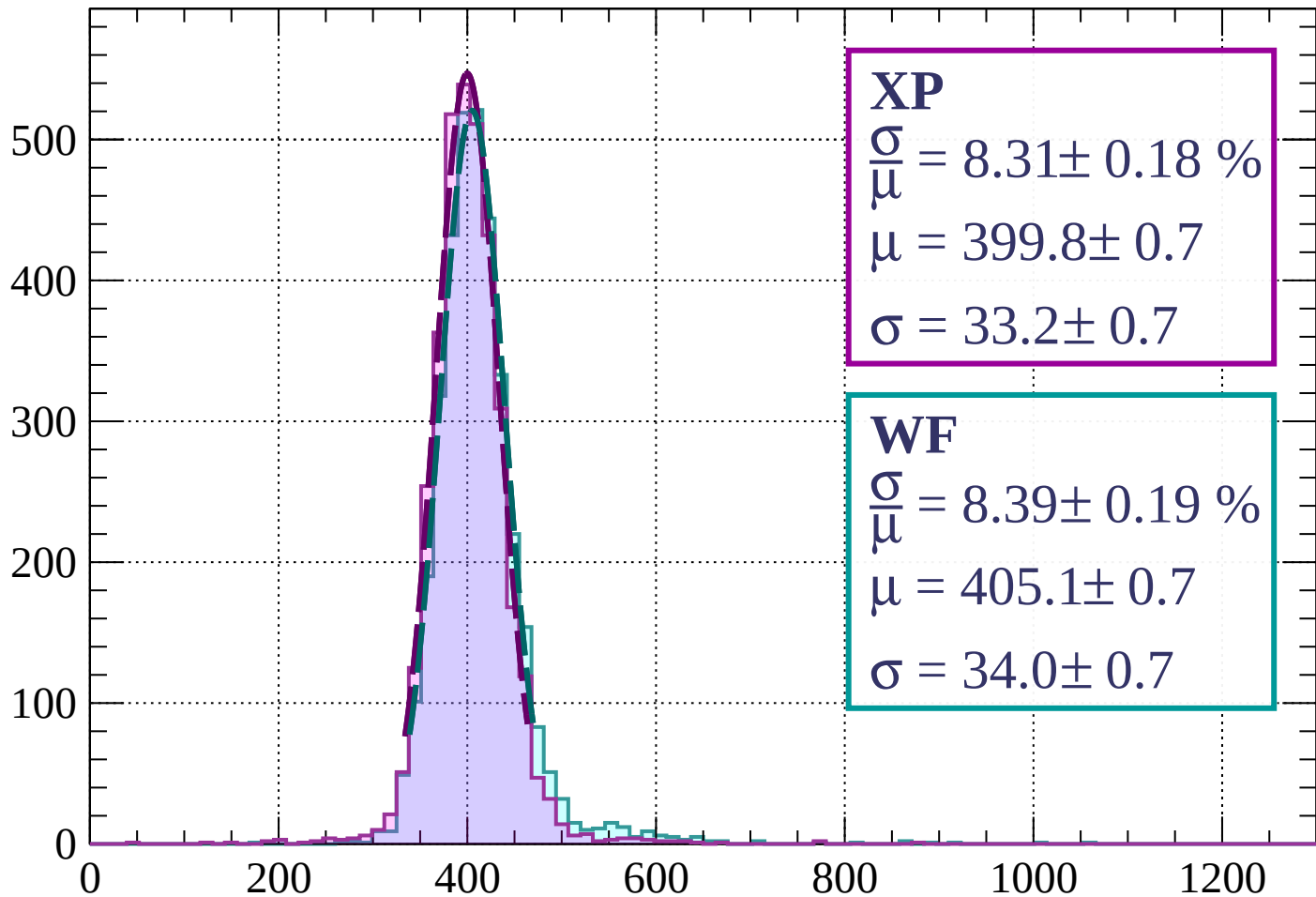


Energy loss | $15 < \theta < 18$



Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\mu} = 8.31 \pm 0.18 \%$$

$$\mu = 399.8 \pm 0.7$$

$$\sigma = 33.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.39 \pm 0.19 \%$$

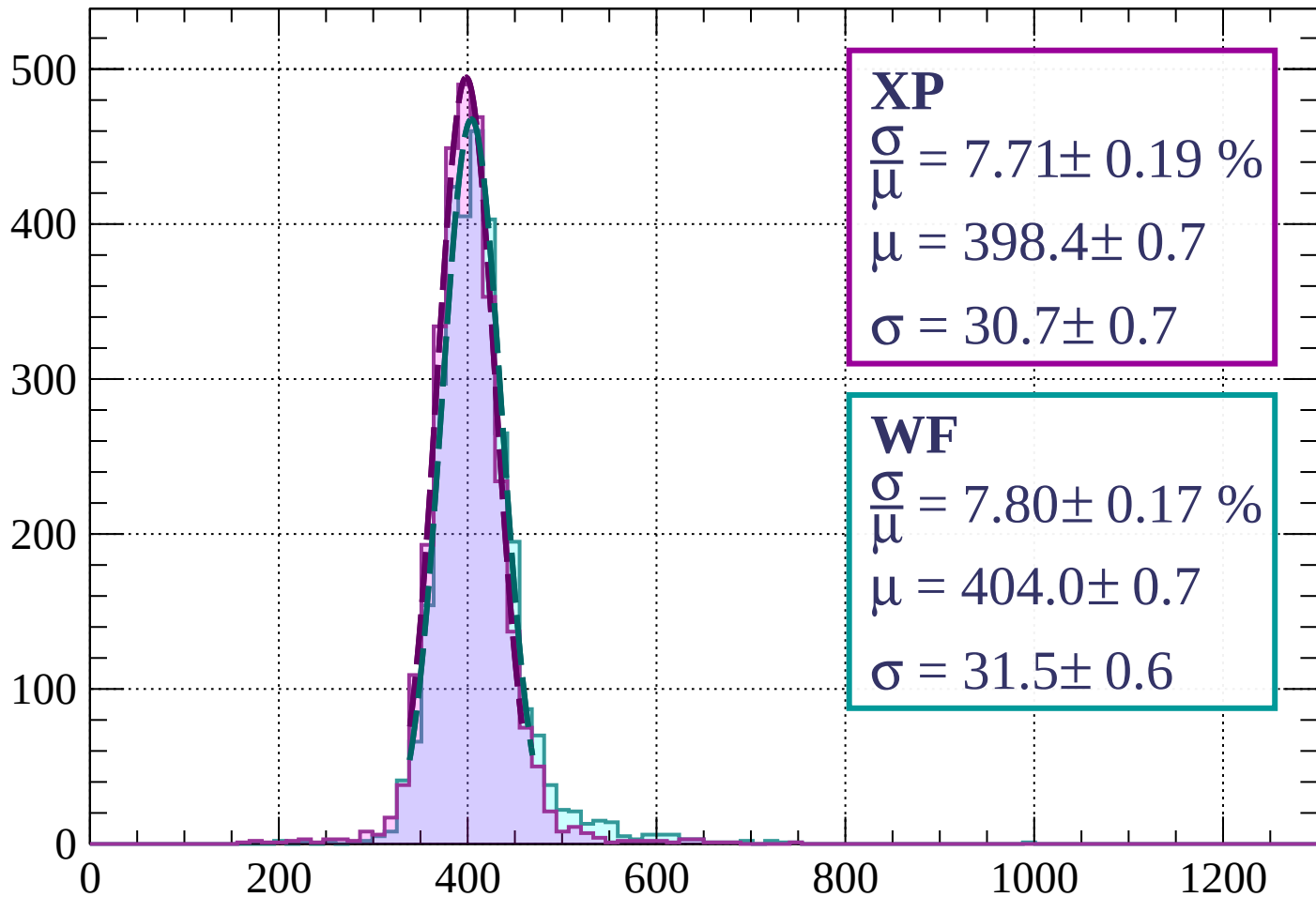
$$\mu = 405.1 \pm 0.7$$

$$\sigma = 34.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $21 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 7.71 \pm 0.19 \%$$

$$\mu = 398.4 \pm 0.7$$

$$\sigma = 30.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.80 \pm 0.17 \%$$

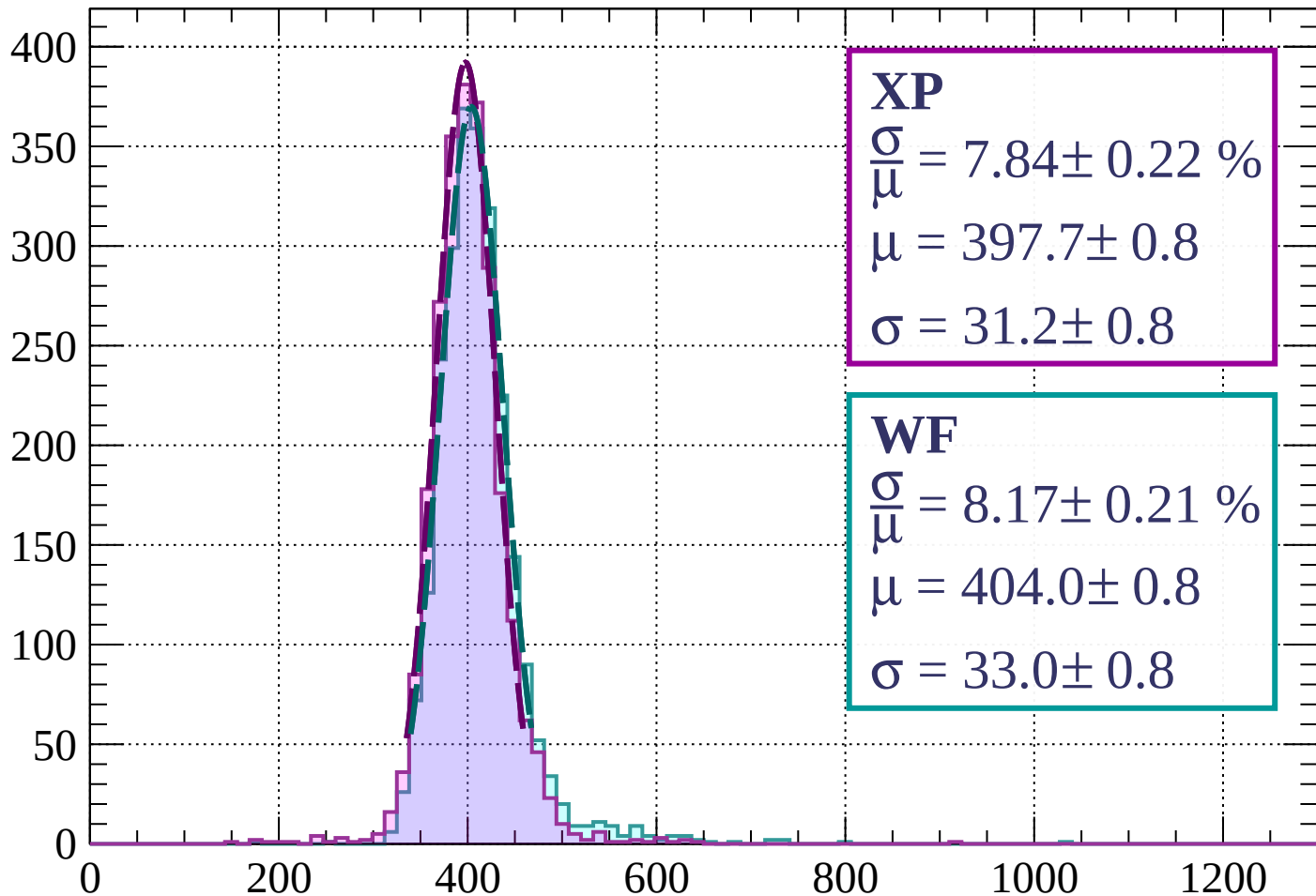
$$\mu = 404.0 \pm 0.7$$

$$\sigma = 31.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $24 < \theta < 27$

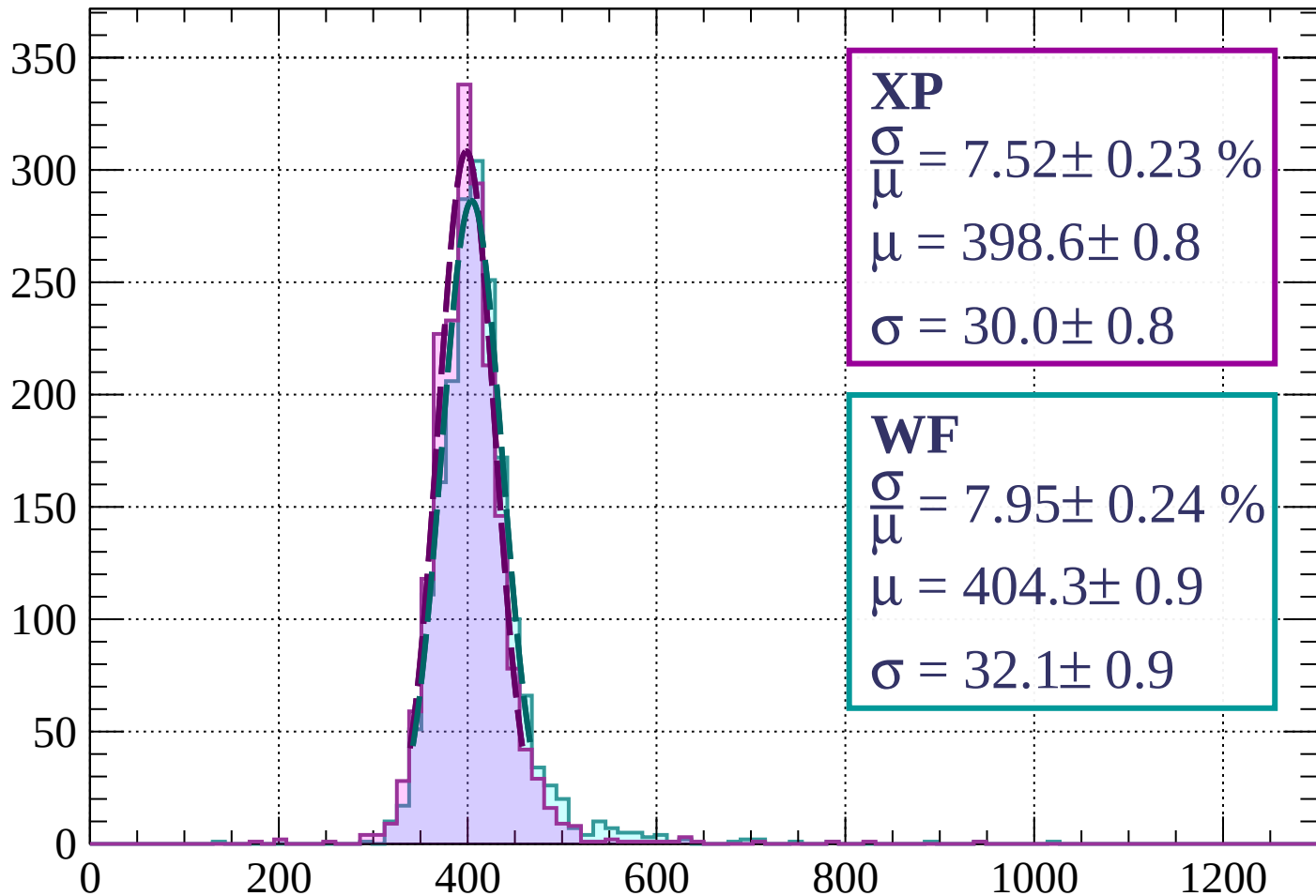
Count



dE/dx (ADC counts/cm)

Energy loss | $27 < \theta < 30$

Count



XP

$$\frac{\sigma}{\mu} = 7.52 \pm 0.23 \%$$

$$\mu = 398.6 \pm 0.8$$

$$\sigma = 30.0 \pm 0.8$$

WF

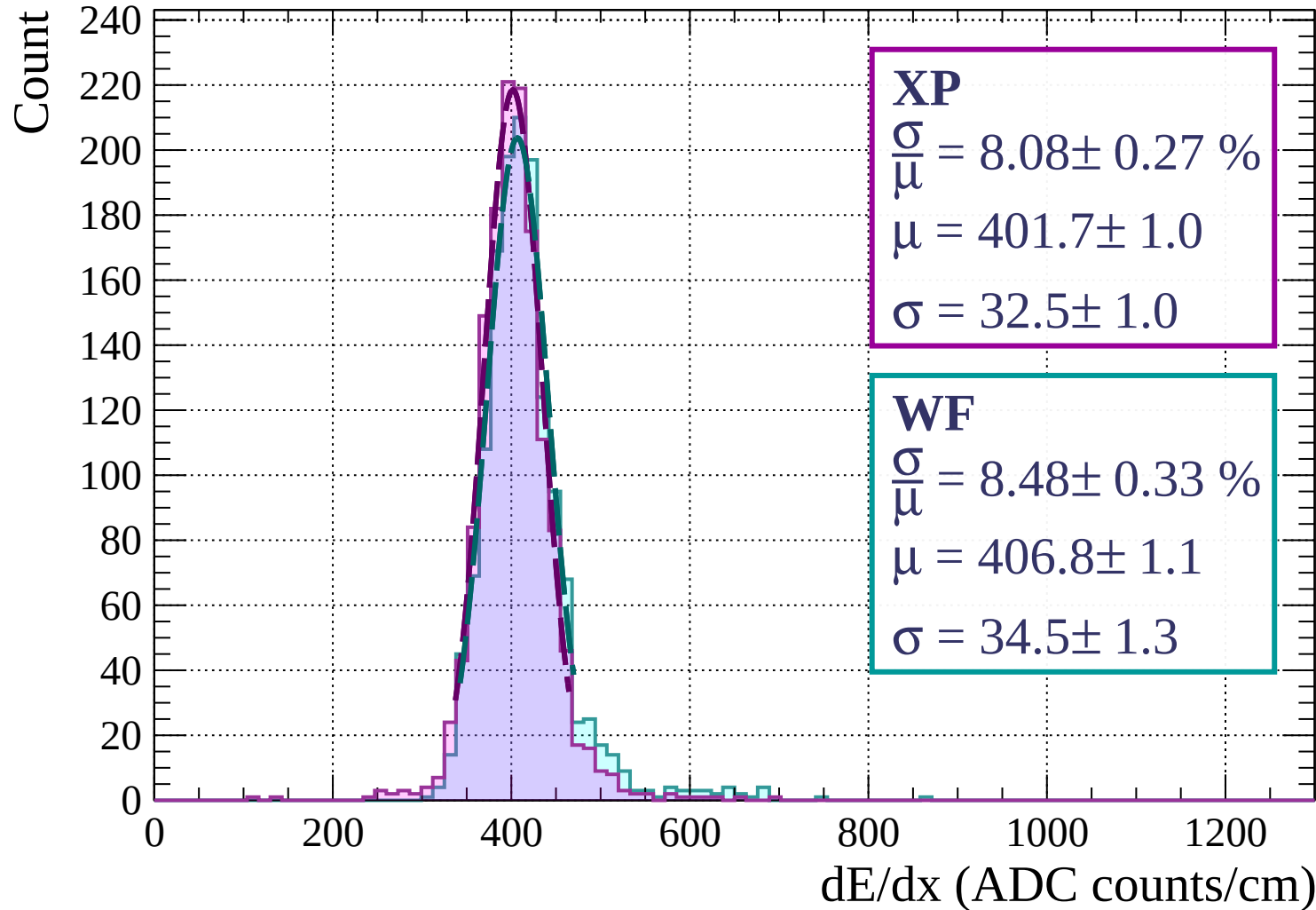
$$\frac{\sigma}{\mu} = 7.95 \pm 0.24 \%$$

$$\mu = 404.3 \pm 0.9$$

$$\sigma = 32.1 \pm 0.9$$

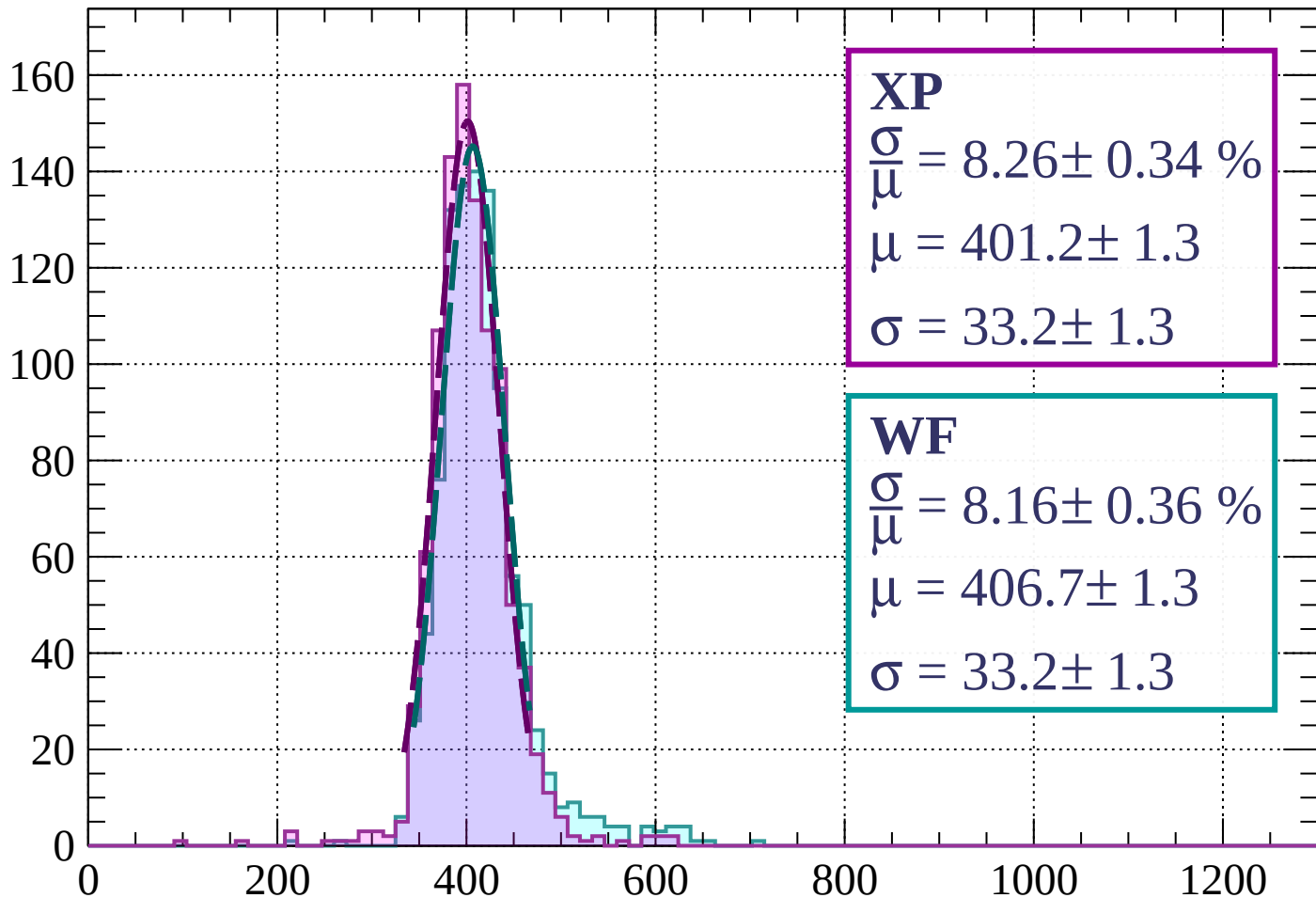
dE/dx (ADC counts/cm)

Energy loss | $30 < \theta < 33$



Energy loss | $33 < \theta < 36$

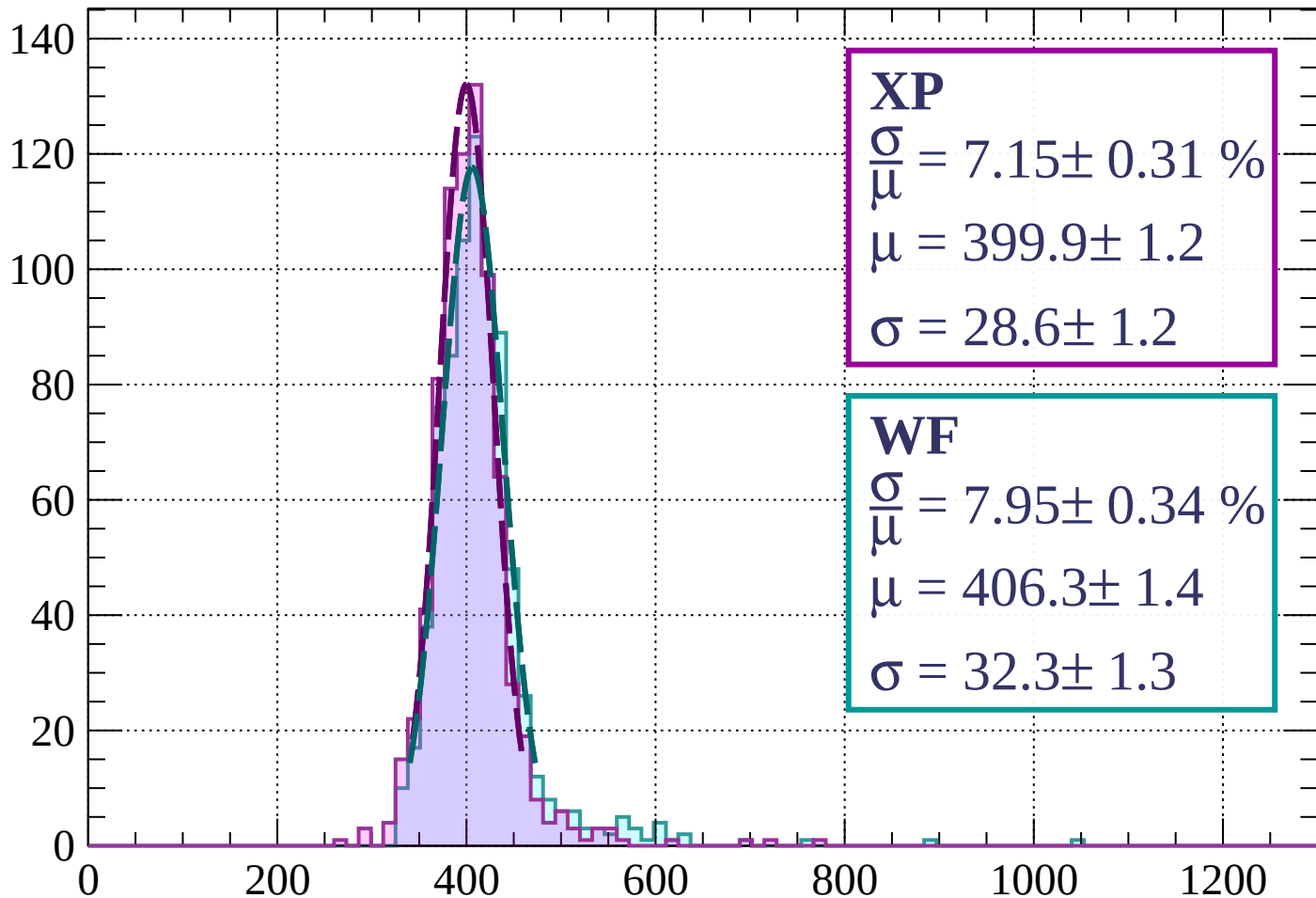
Count



dE/dx (ADC counts/cm)

Energy loss | $36 < \theta < 39$

Count



XP

$$\frac{\sigma}{\mu} = 7.15 \pm 0.31 \%$$

$$\mu = 399.9 \pm 1.2$$

$$\sigma = 28.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.95 \pm 0.34 \%$$

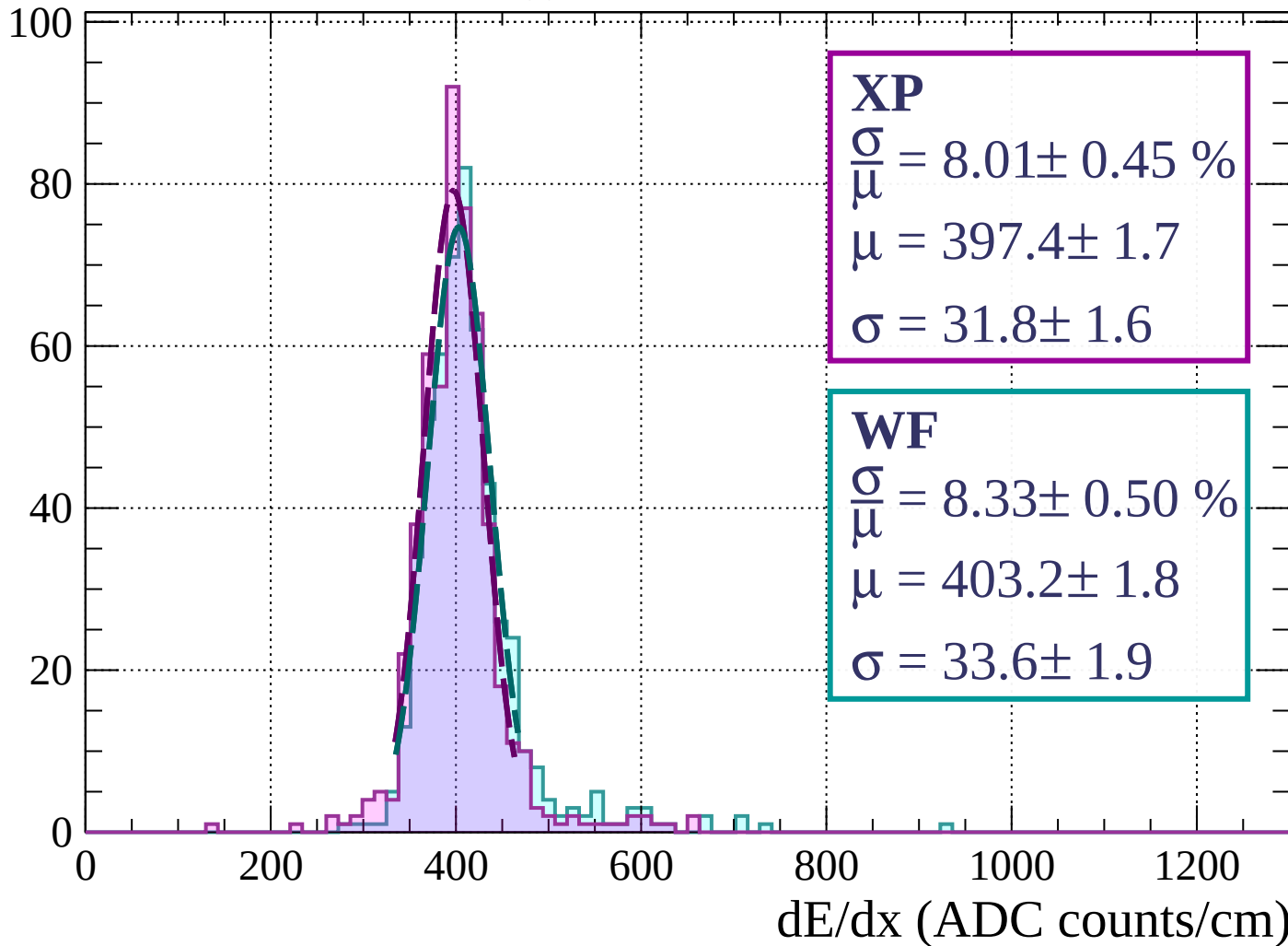
$$\mu = 406.3 \pm 1.4$$

$$\sigma = 32.3 \pm 1.3$$

dE/dx (ADC counts/cm)

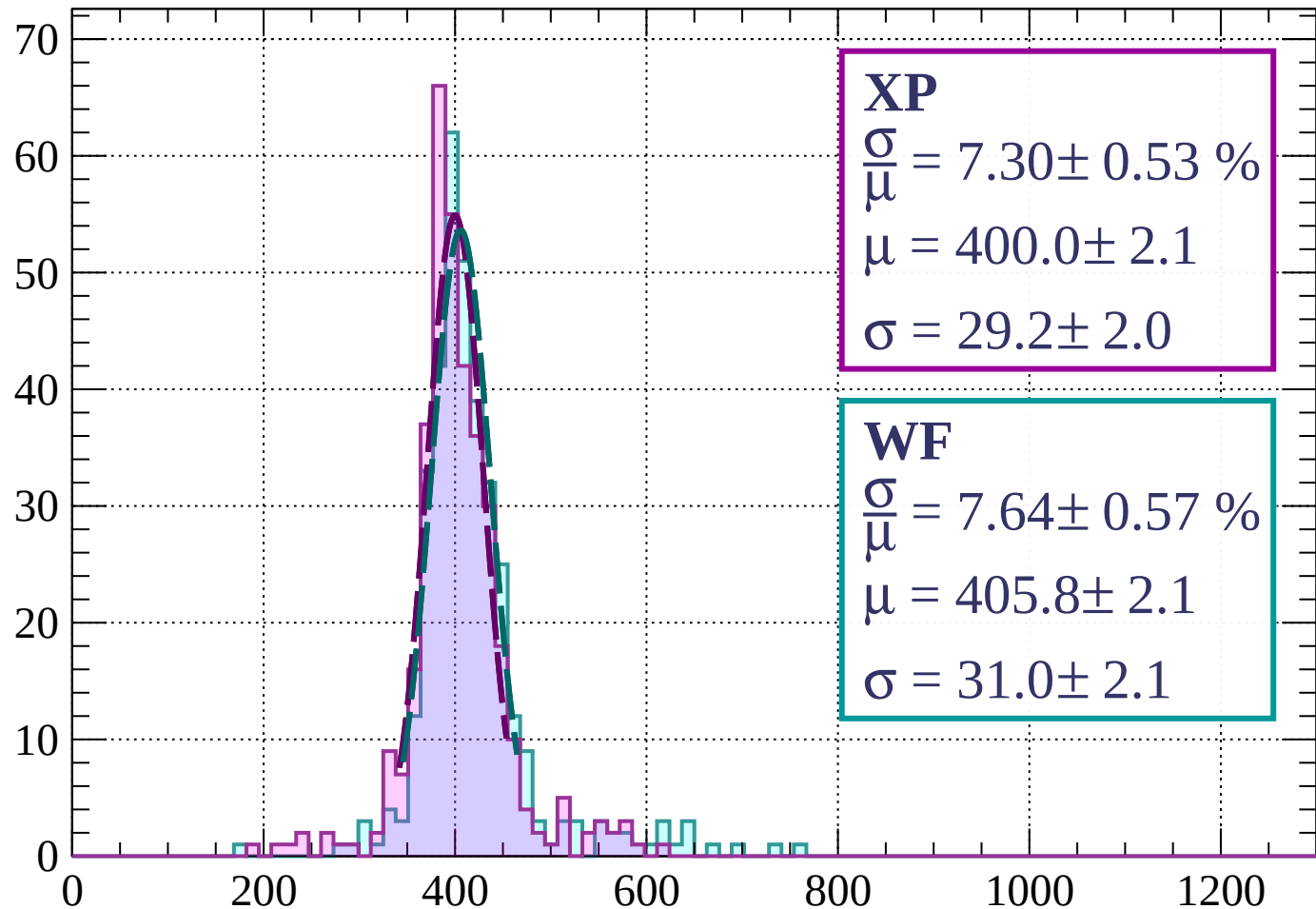
Energy loss | $39 < \theta < 42$

Count



Energy loss | $42 < \theta < 45$

Count



XP

$$\frac{\sigma}{\mu} = 7.30 \pm 0.53 \%$$

$$\mu = 400.0 \pm 2.1$$

$$\sigma = 29.2 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 7.64 \pm 0.57 \%$$

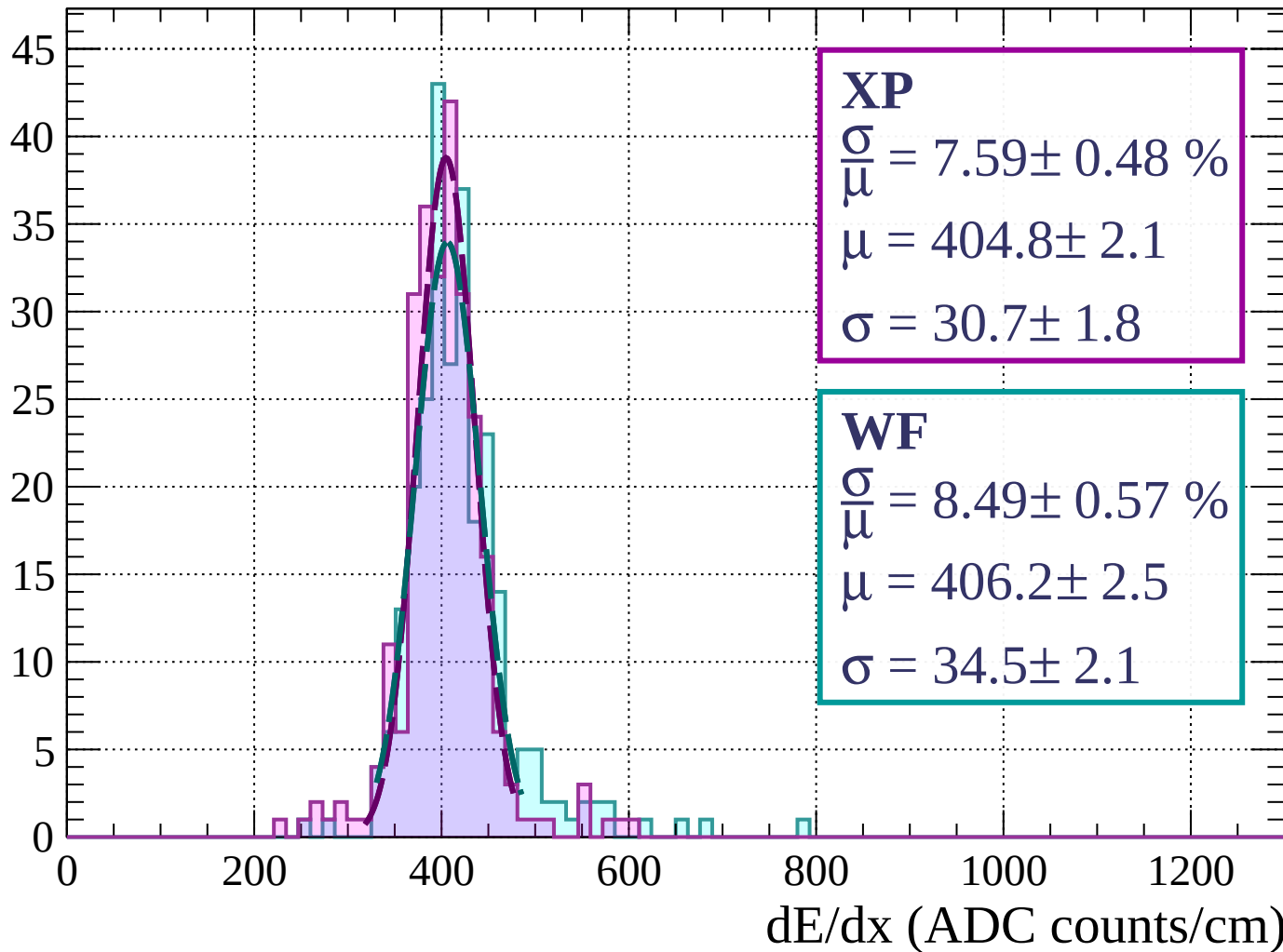
$$\mu = 405.8 \pm 2.1$$

$$\sigma = 31.0 \pm 2.1$$

dE/dx (ADC counts/cm)

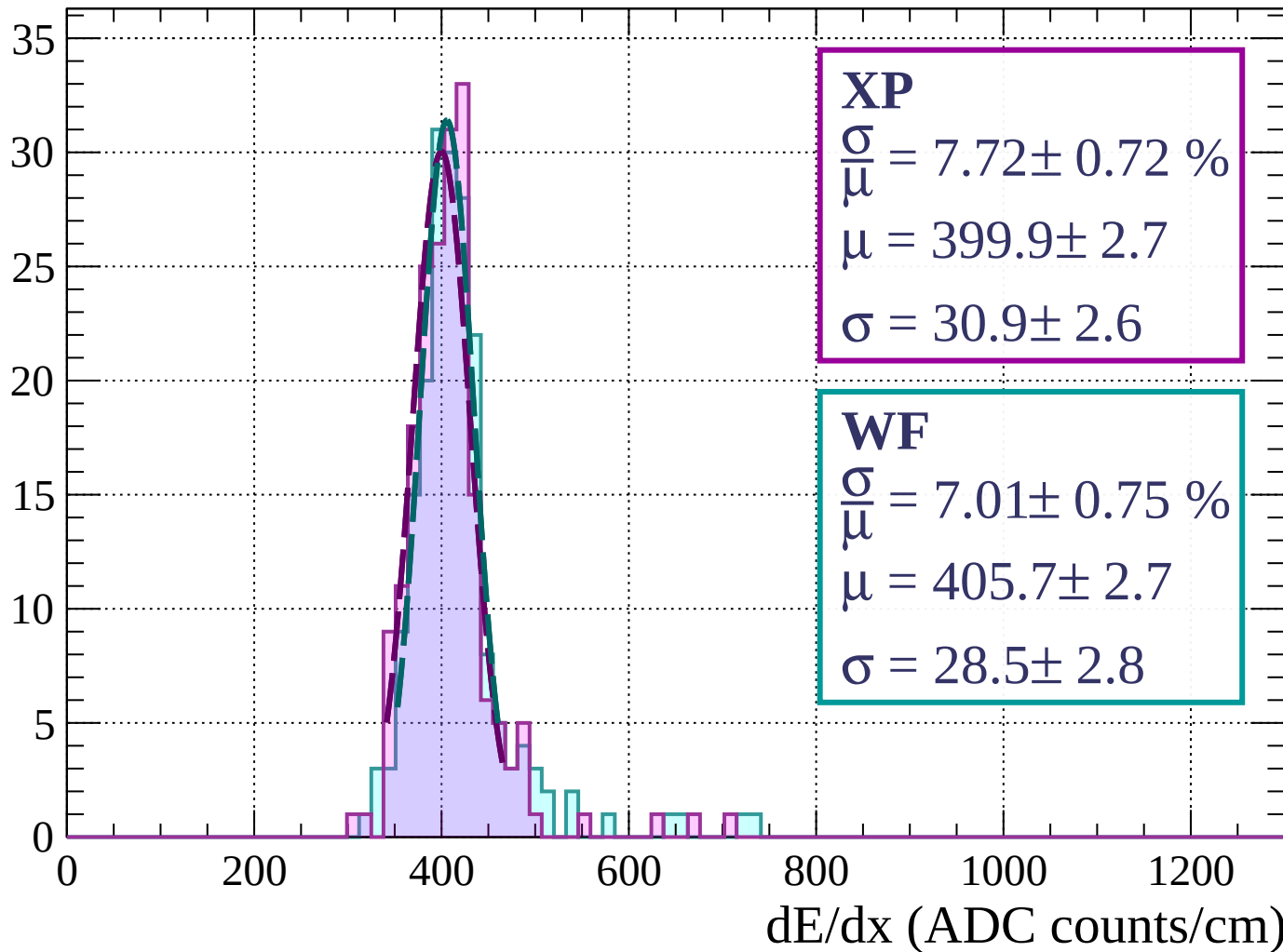
Energy loss | $45 < \theta < 48$

Count



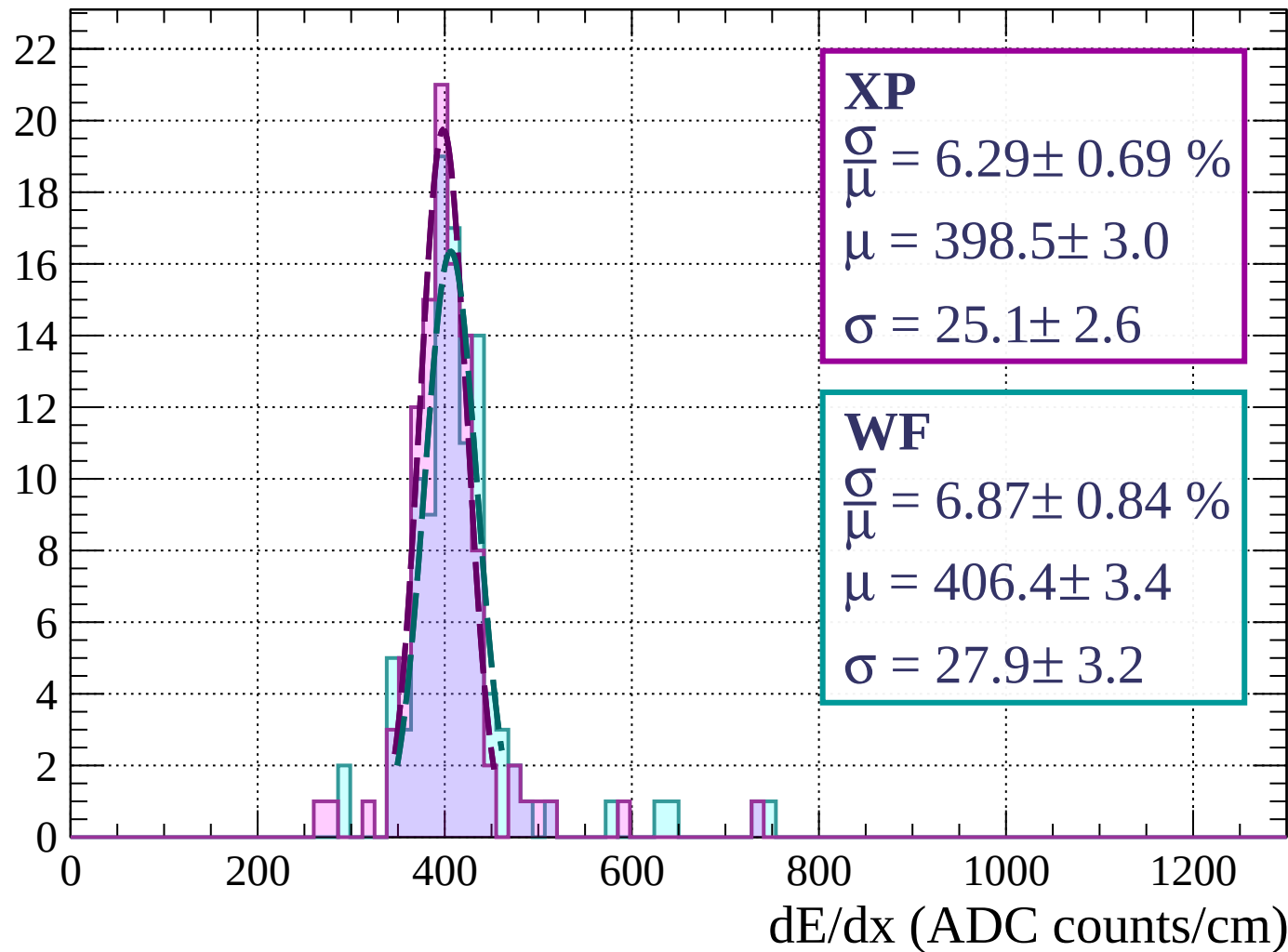
Energy loss | $48 < \theta < 51$

Count



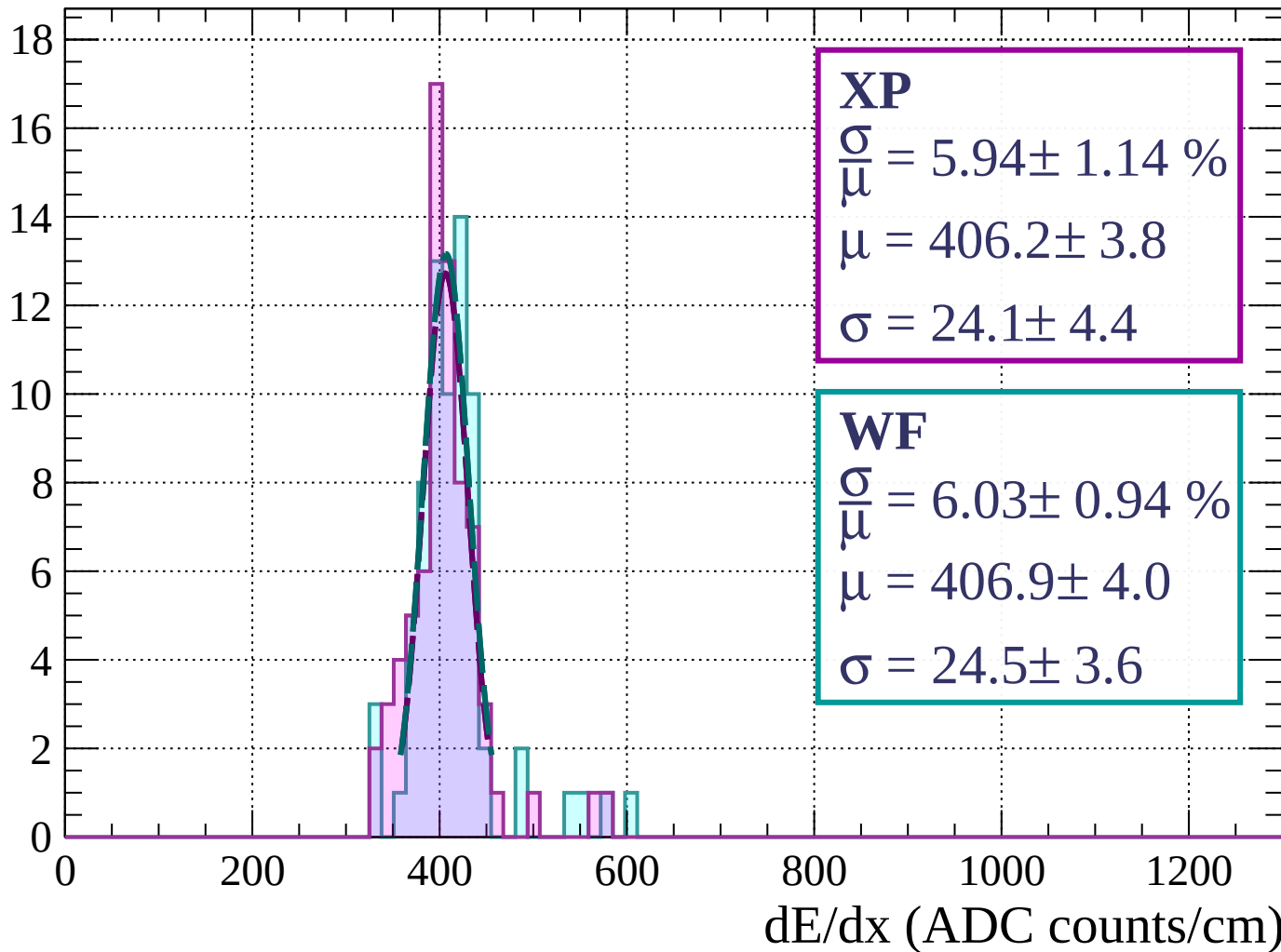
Energy loss | $51 < \theta < 54$

Count



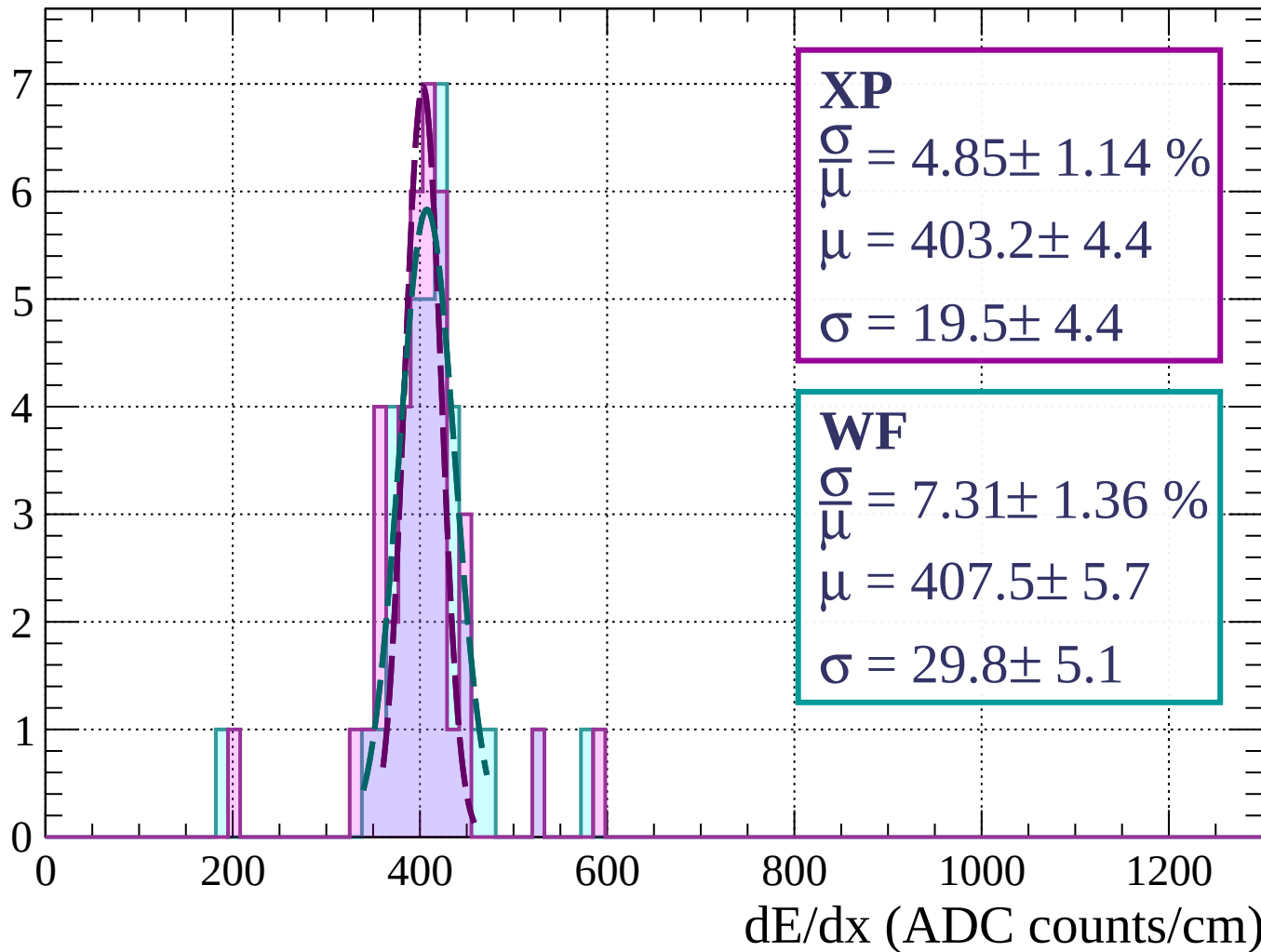
Energy loss | $54 < \theta < 57$

Count



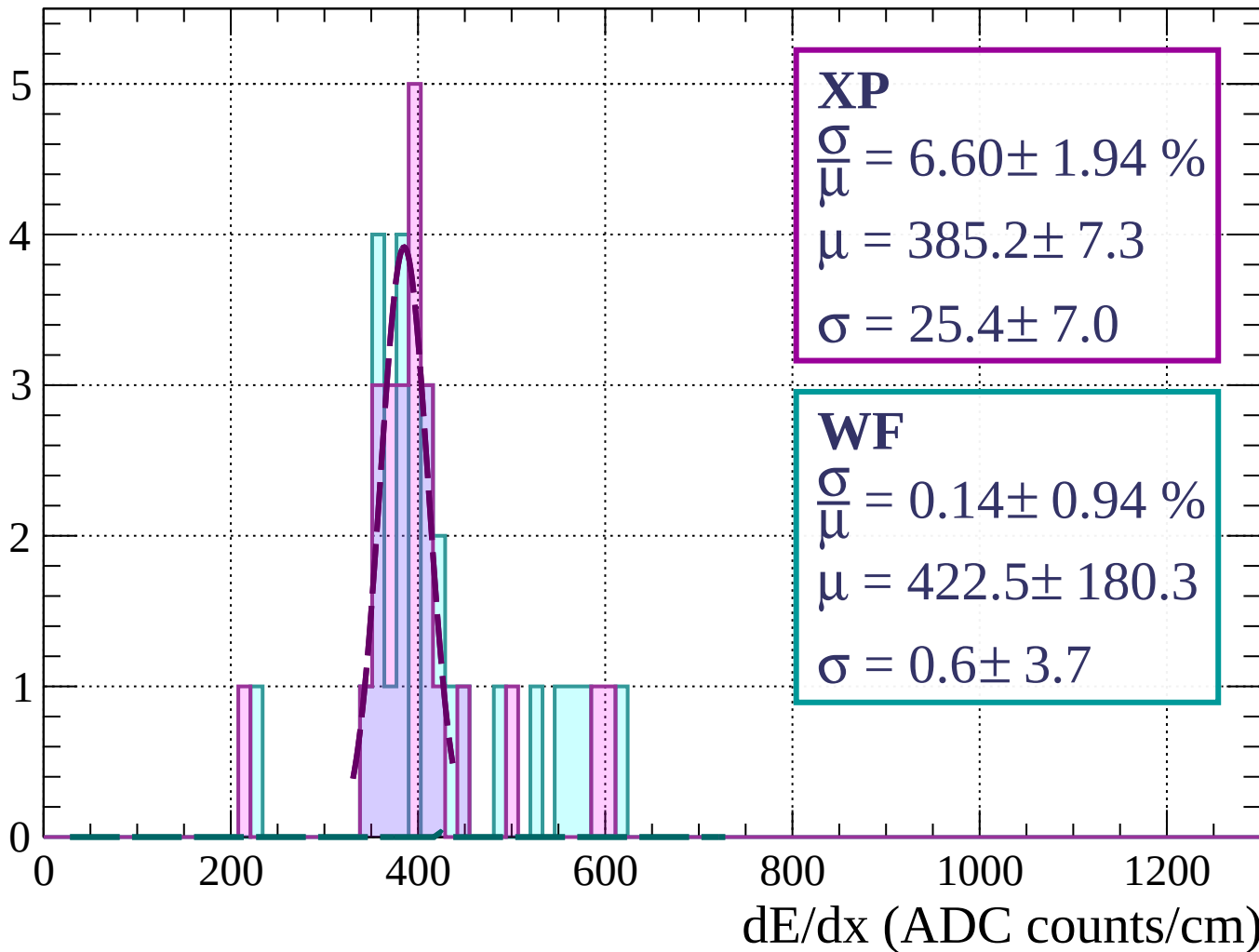
Energy loss | $57 < \theta < 60$

Count



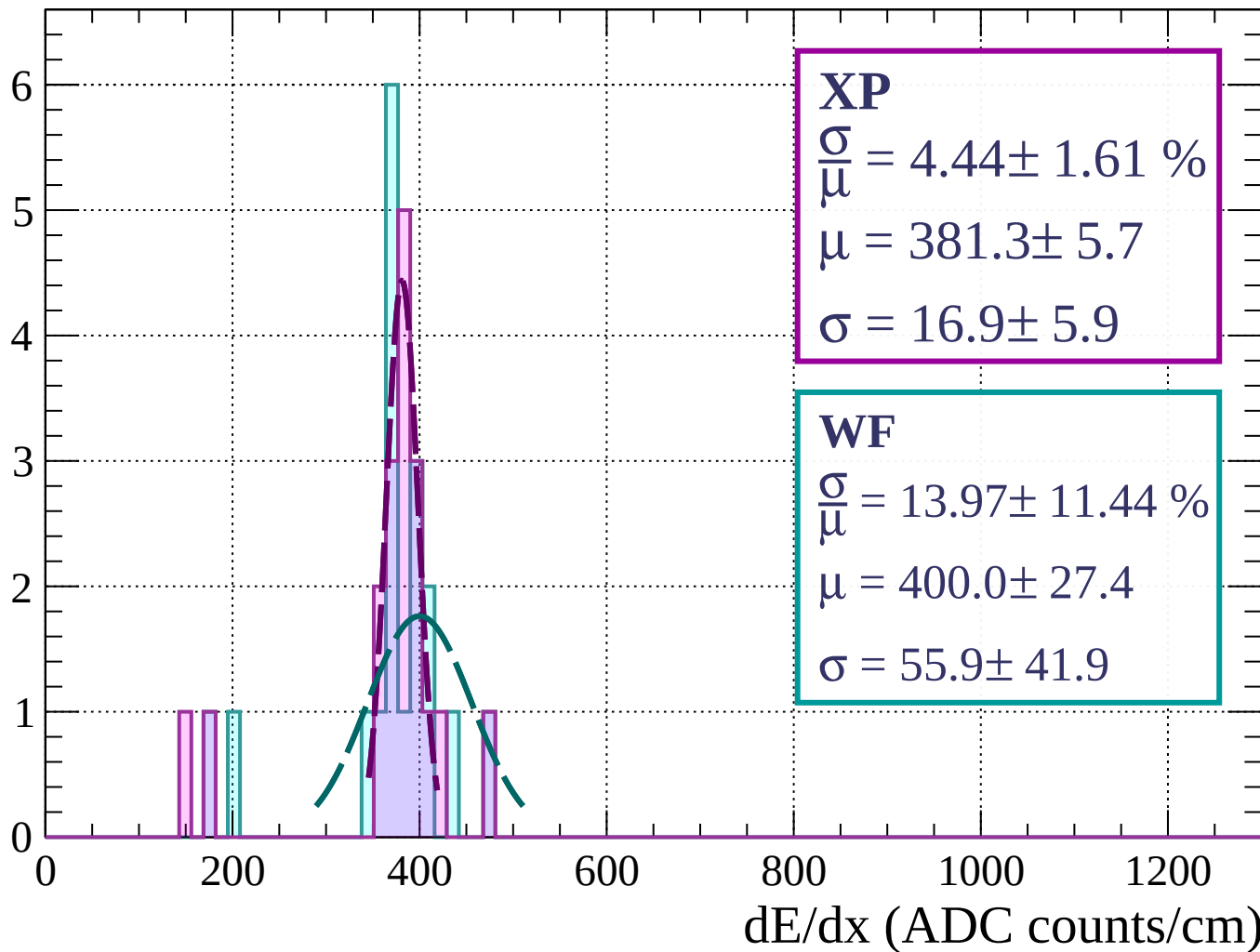
Energy loss | $60 < \theta < 63$

Count



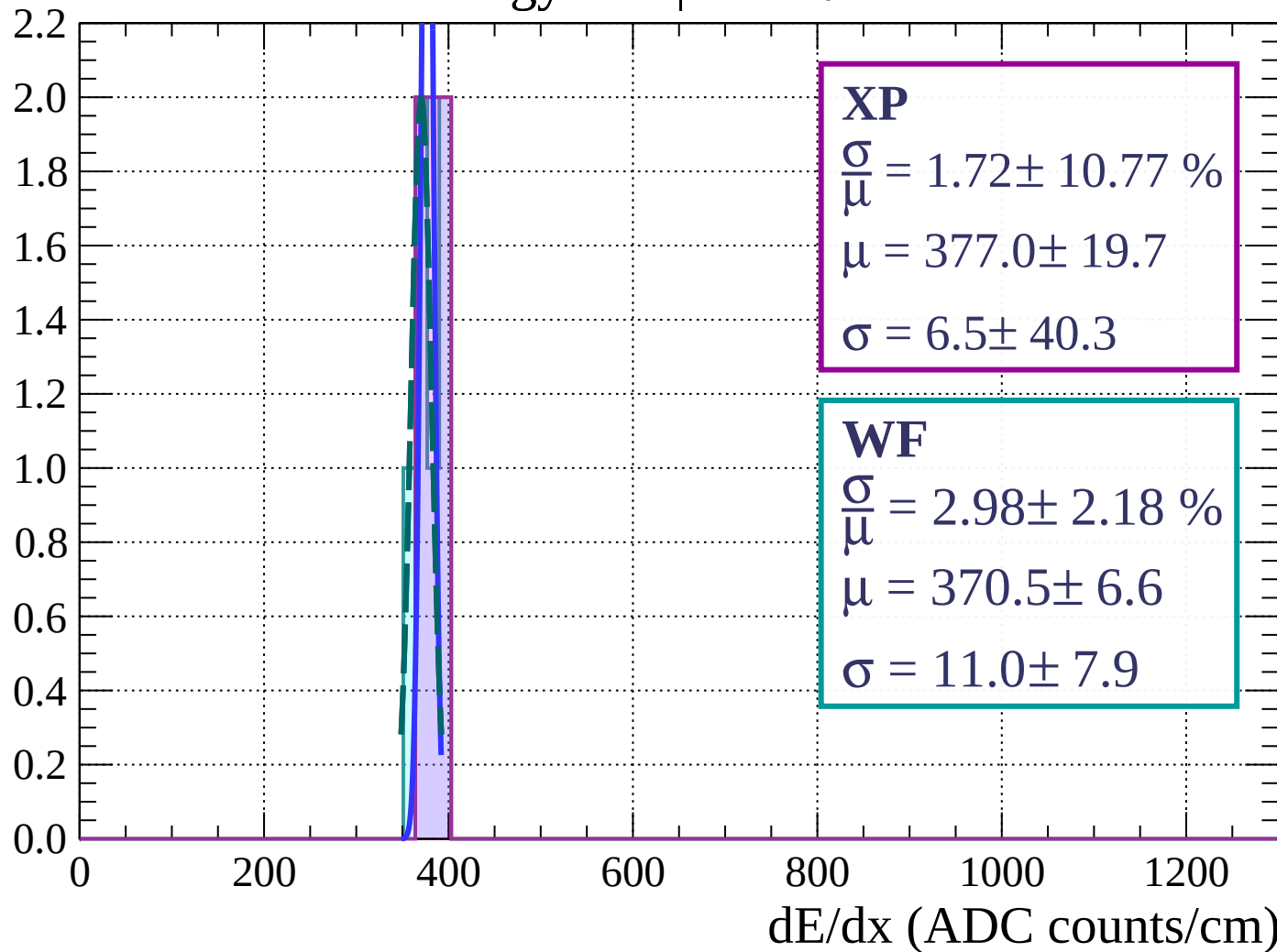
Energy loss | $63 < \theta < 66$

Count



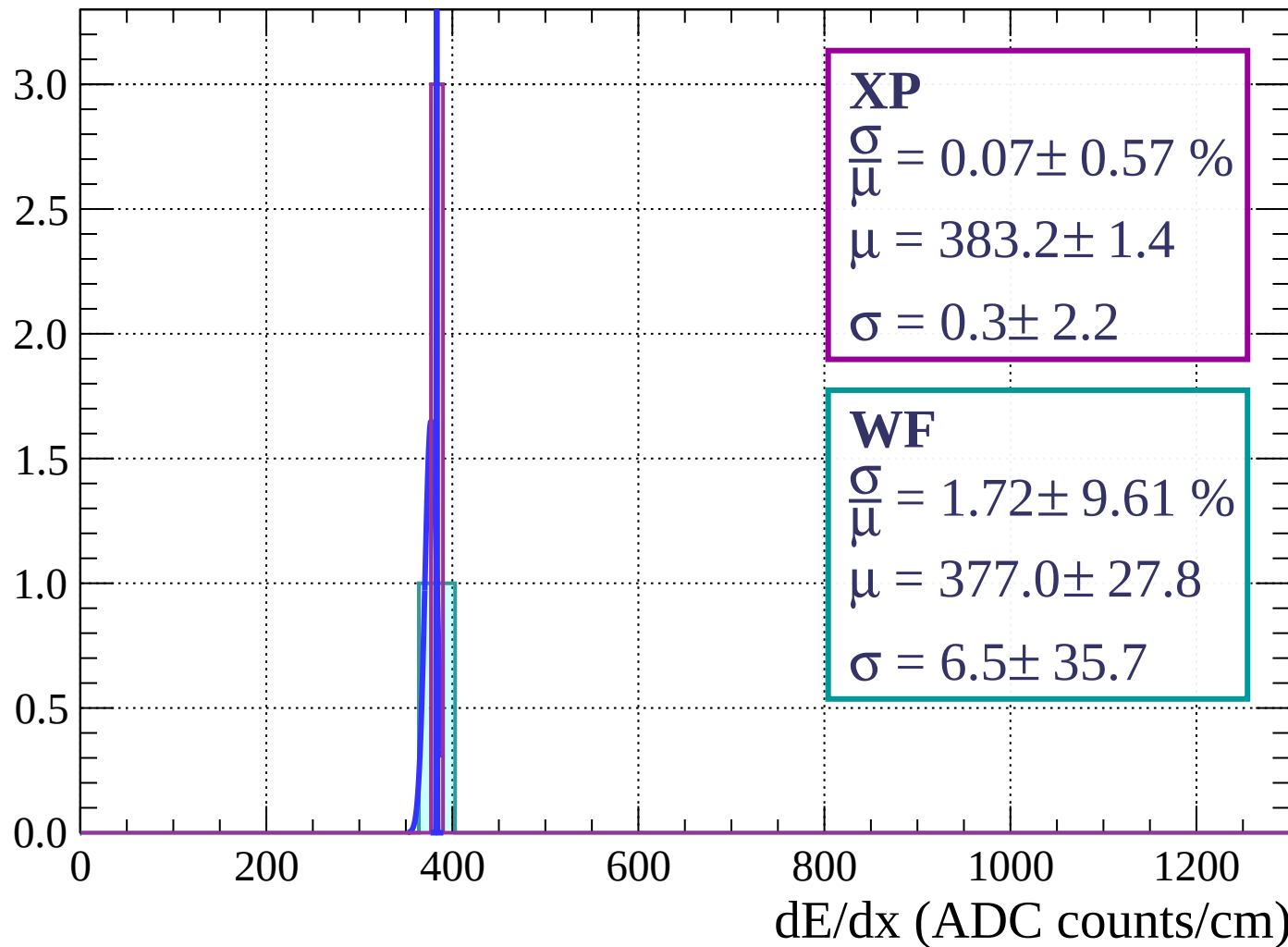
Energy loss | $66 < \theta < 69$

Count



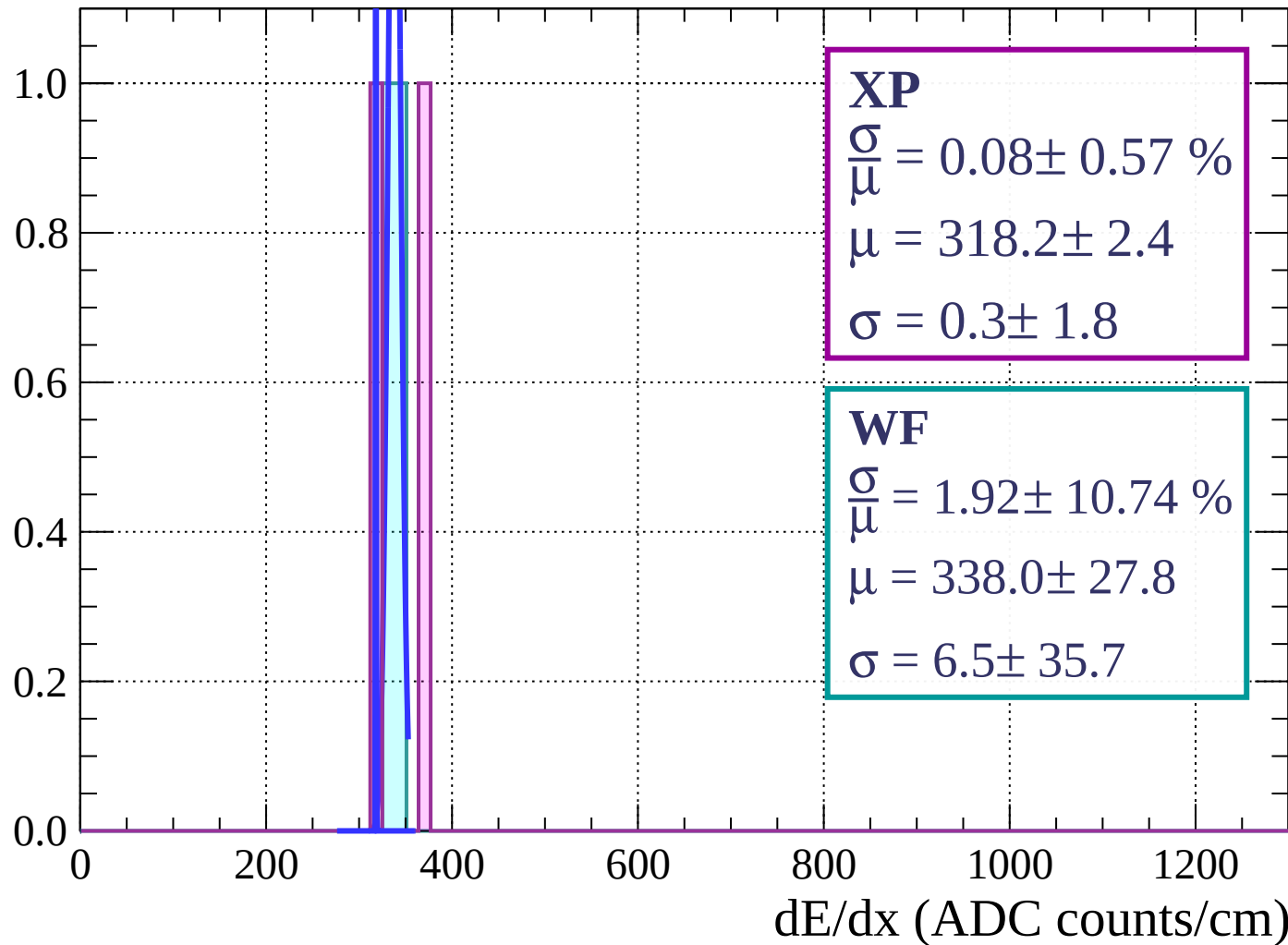
Energy loss | $69 < \theta < 72$

Count



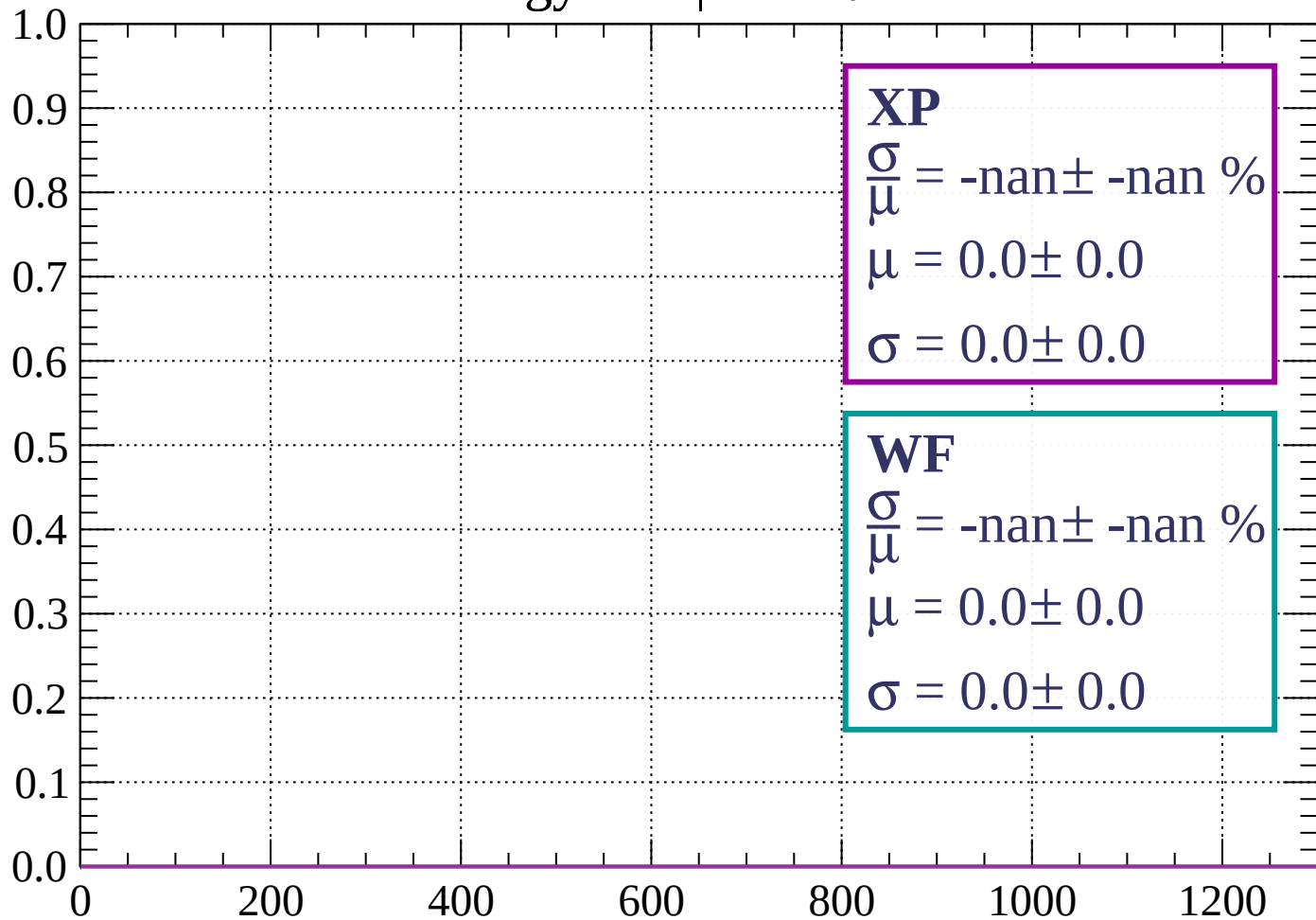
Energy loss | $72 < \theta < 75$

Count



Energy loss | $75 < \theta < 78$

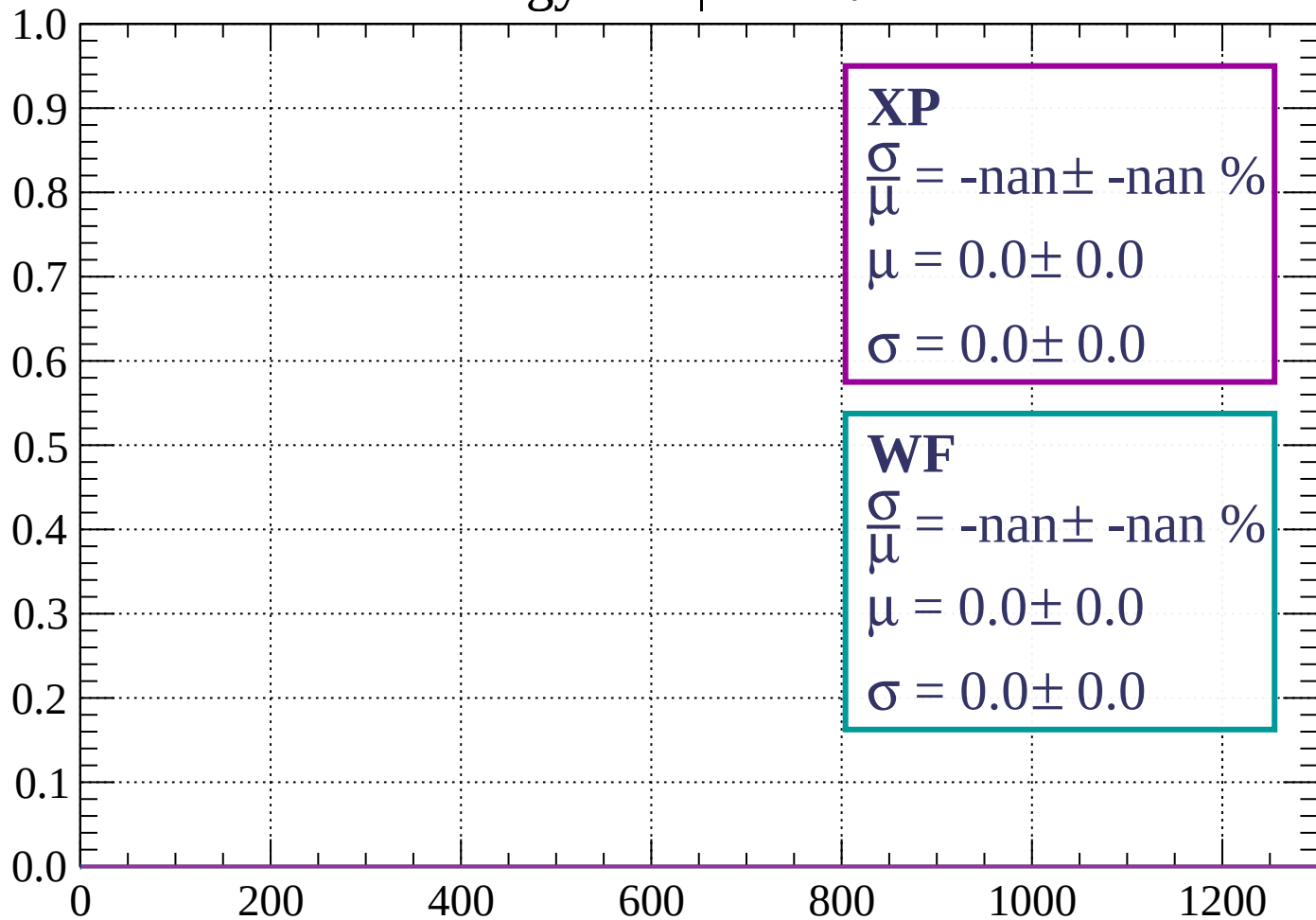
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

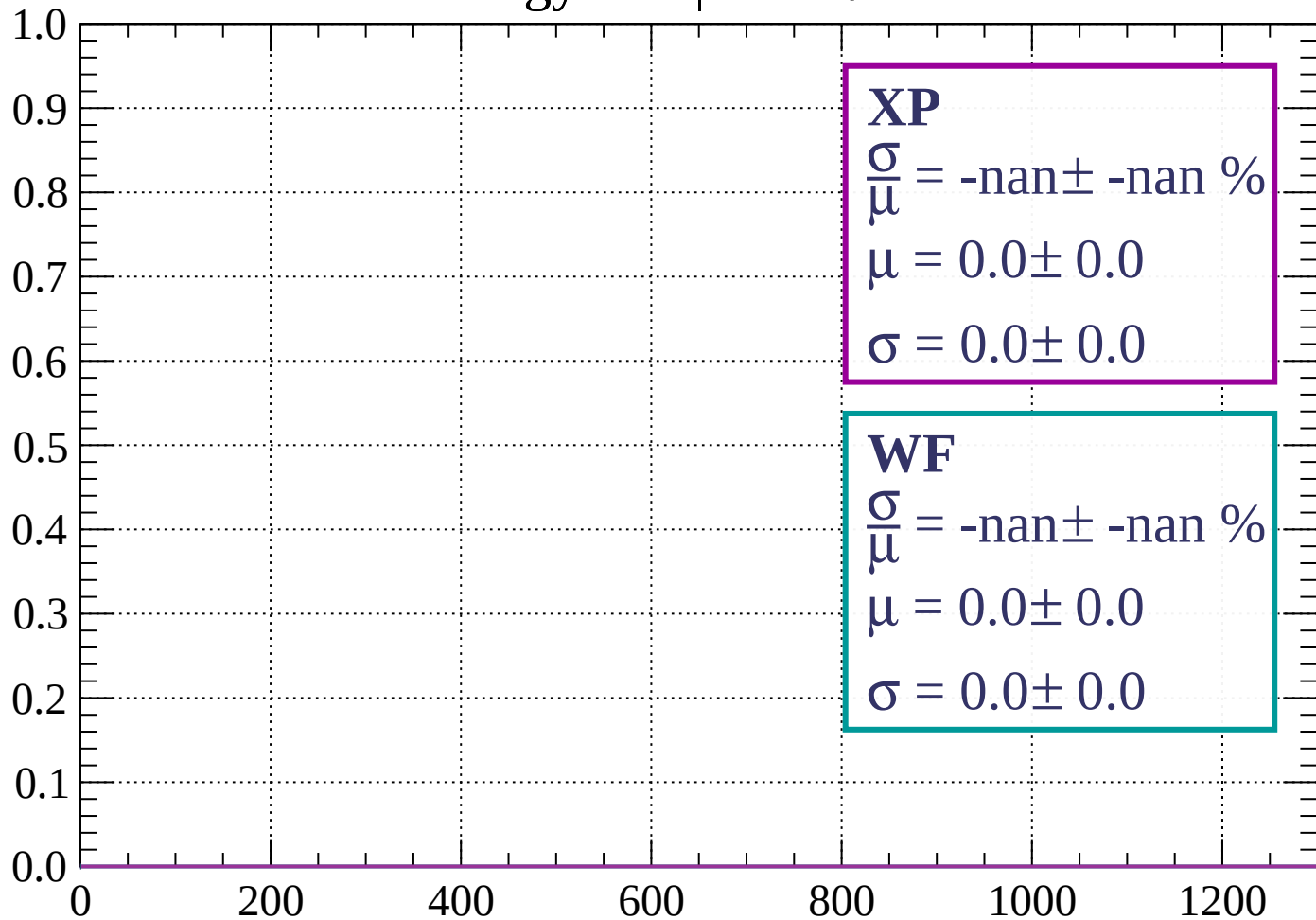
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

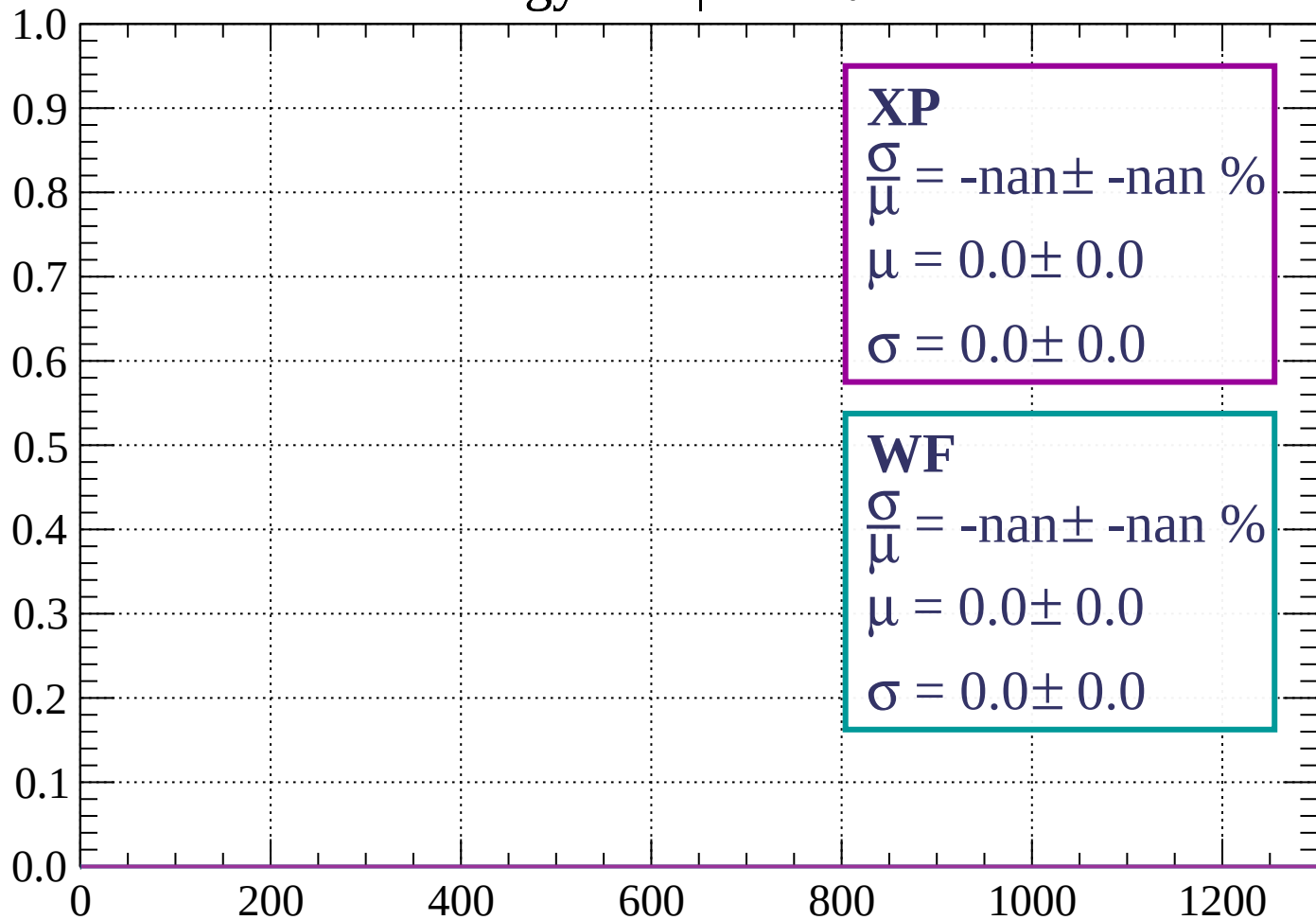
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

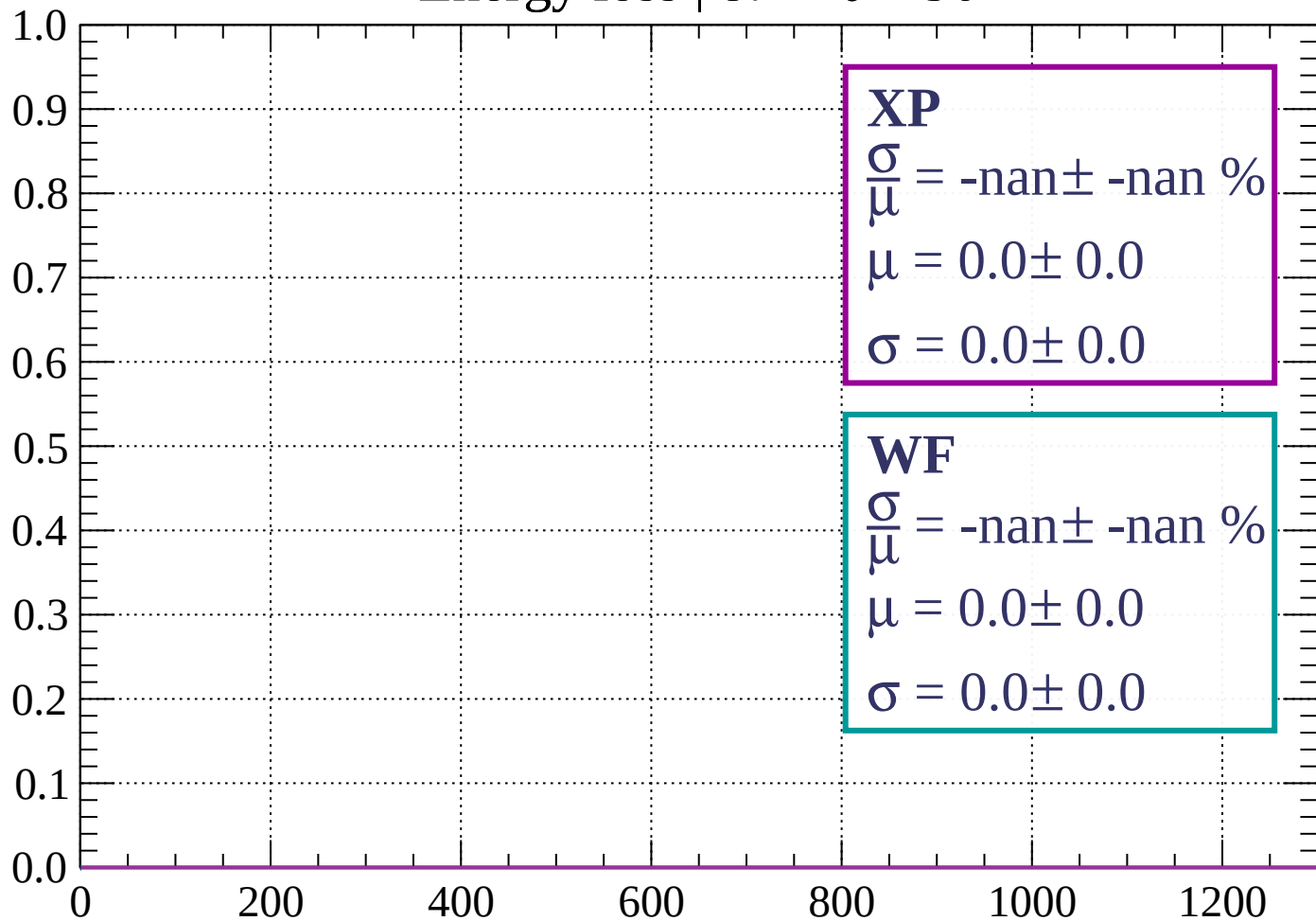
Count



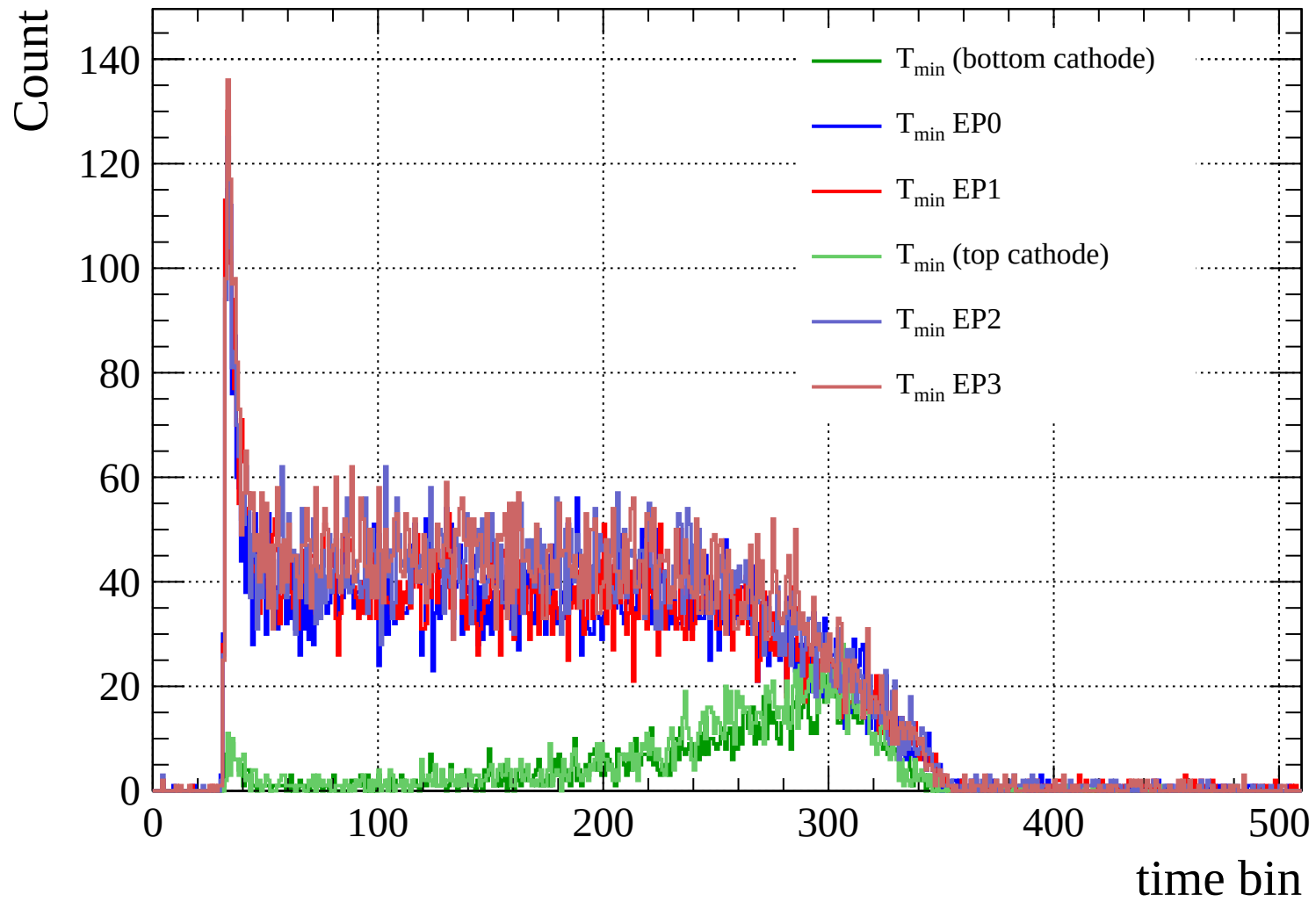
dE/dx (ADC counts/cm)

Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)



Count

